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(54) SEMICONDUCTOR MODULE

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(21) Appl. No.: **19/219,204**

(22) Filed: **May 27, 2025**

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H01L 23/58	(2006.01)
H10B 80/00	(2023.01)

(52) **U.S. Cl.**

CPC ***H01L 25/16*** (2013.01); ***H01L 23/5286*** (2013.01); ***H01L 23/585*** (2013.01); ***H01L 24/16*** (2013.01); ***H10B 80/00*** (2023.02); *H01L 2224/16225* (2013.01)

(57) **ABSTRACT**

A semiconductor module includes a plurality of memory chips in a memory cube are stacked along a first direction parallel to a plane formed by a second direction and a third direction perpendicular to the first direction. The memory cube includes first and third outermost surfaces in the third direction and second and fourth outermost surfaces in the second direction. Each of the memory chips includes a substrate and a wiring layer. The substrate includes first to fourth side surfaces along each of the first to fourth outermost surfaces. A first insulating film is included between the first to fourth outermost surfaces and the corresponding first to fourth side surfaces. Internal wirings included in the wiring layer are connected to electrodes provided on at least one of the second to fourth outermost surfaces. The memory cube is spaced apart from the semiconductor chips and performs signal transmission.

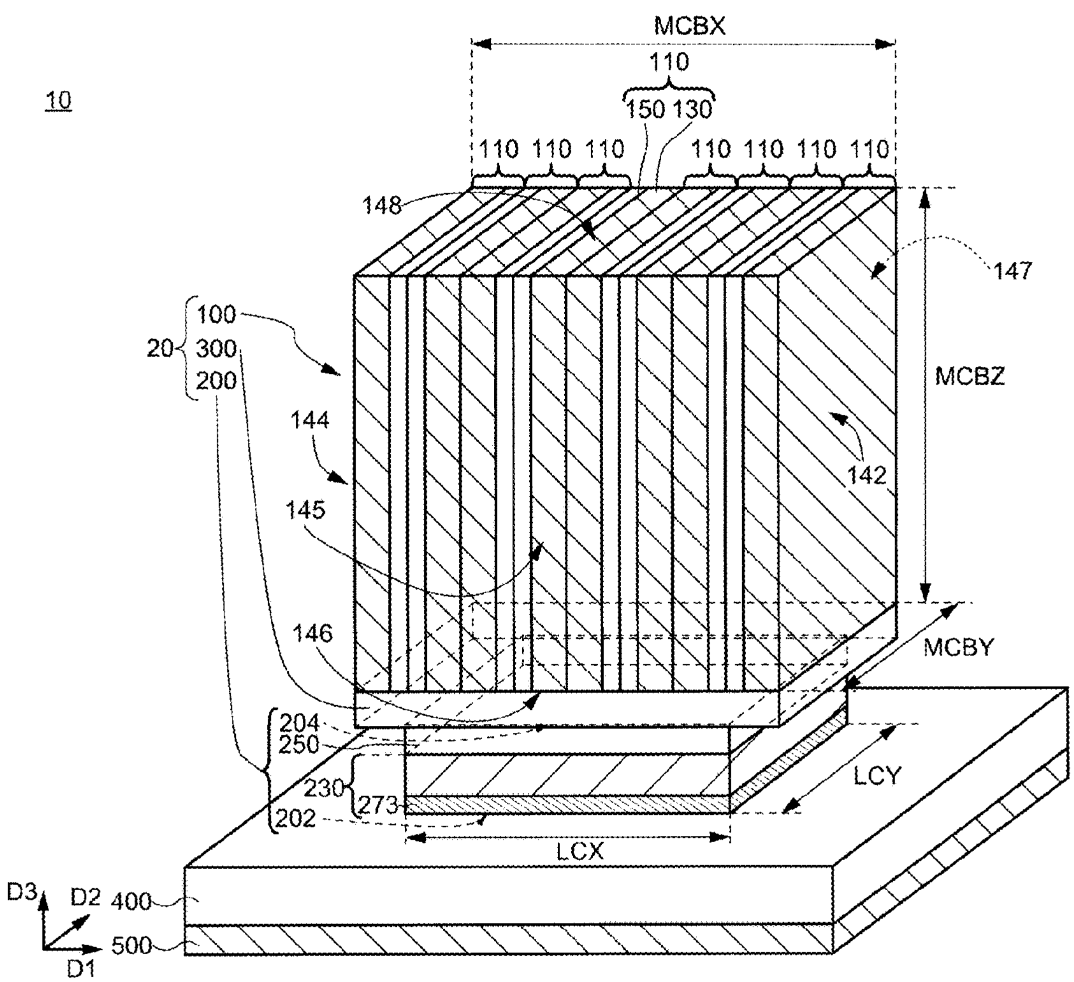


FIG. 1

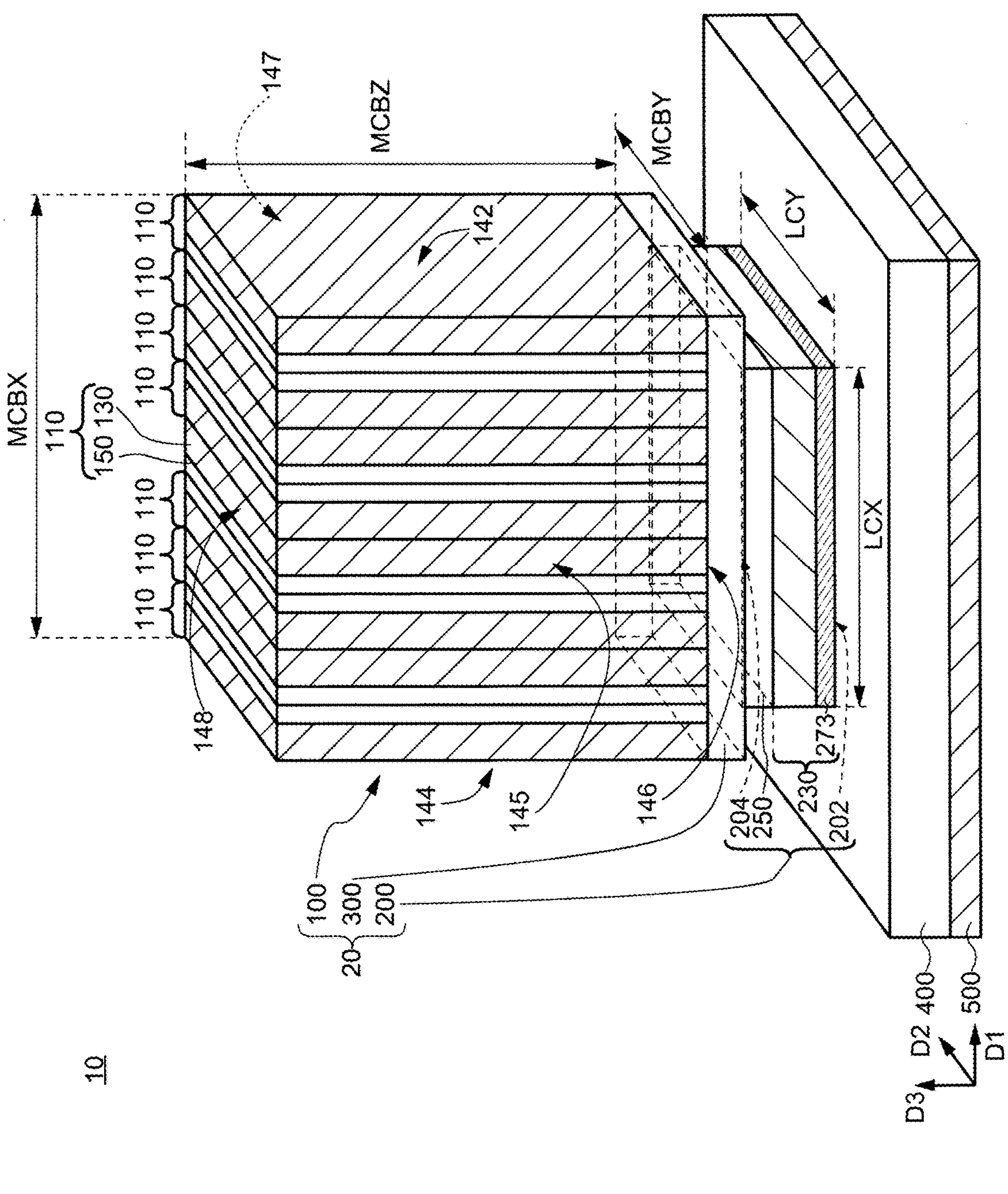

10
MCBX
MCBZ
MCBY
147
142
110
150
130
148
144
145
146
100
300
200
20
204
250
230
202
273
LCX
LCY
400
500
D1
D2
D3

FIG. 2

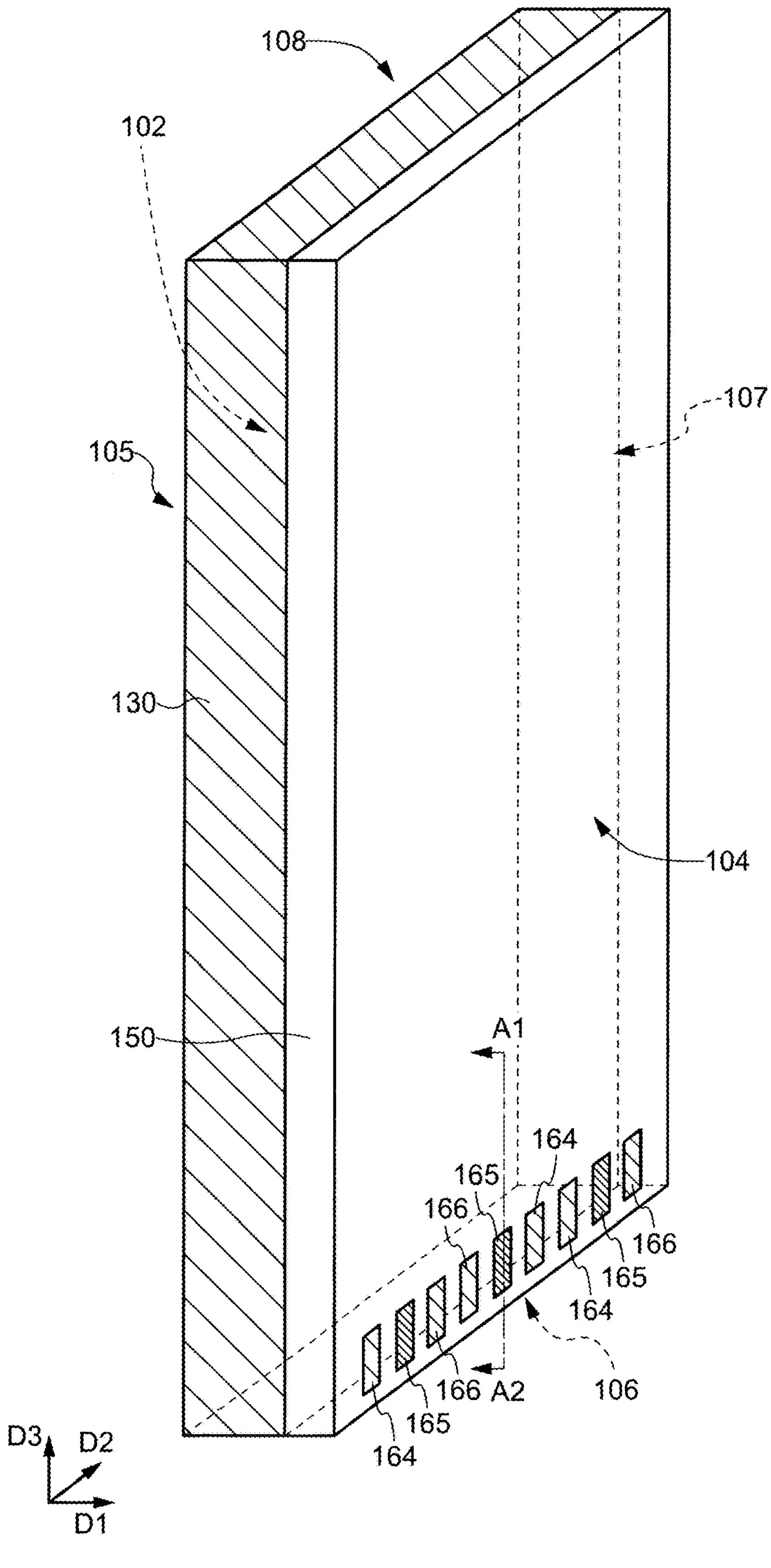

108
102
107
105
130
104
150
A1
164
165
166
164
166
165
164
165
166 A2
106
165
164
D3
D2
D1

FIG. 3

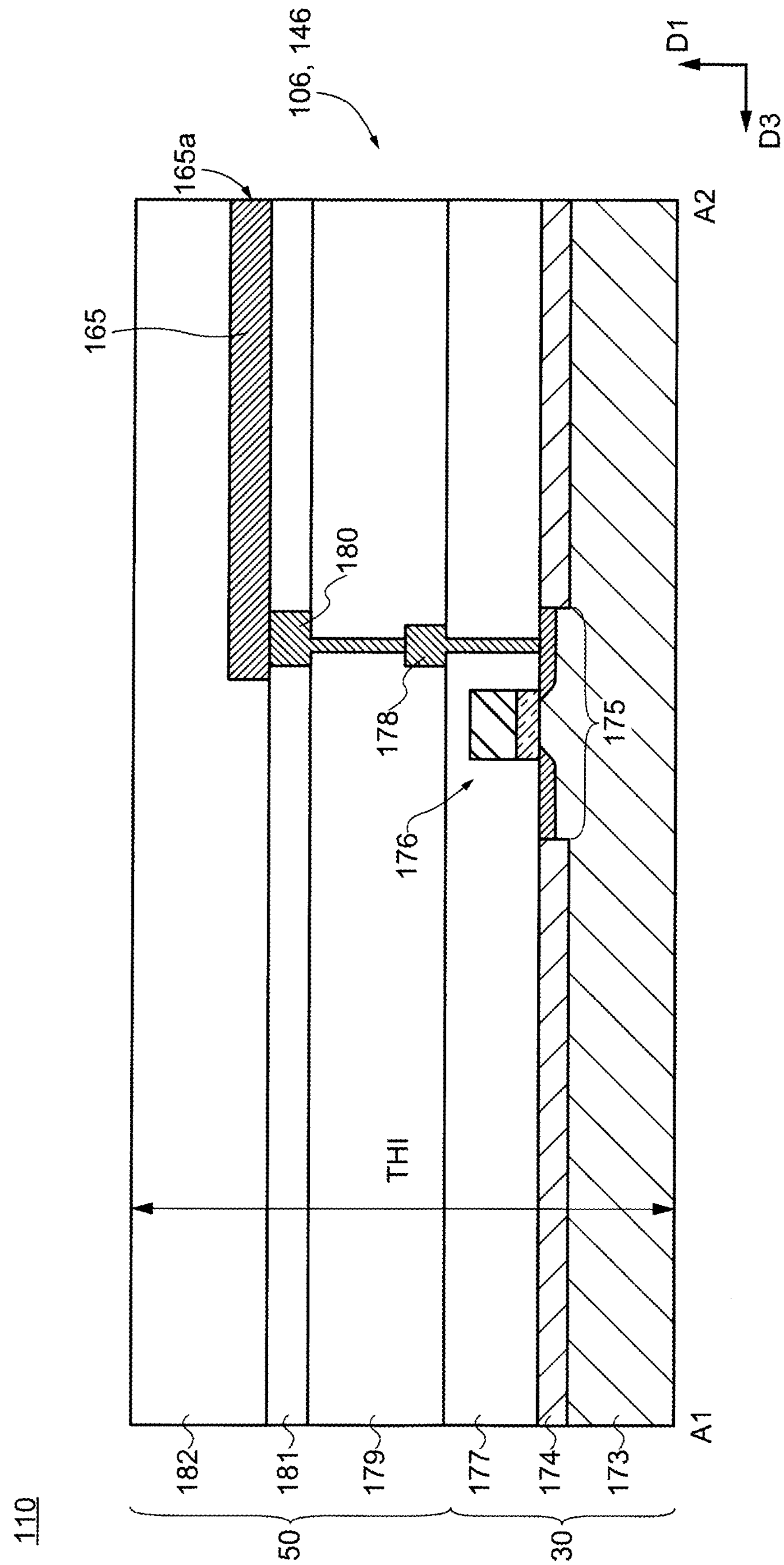

106, 146
D1
D3
165a
165
A2
180
178
175
176
THI
A1
182
181
179
177
174
173
110
150
130

FIG. 4

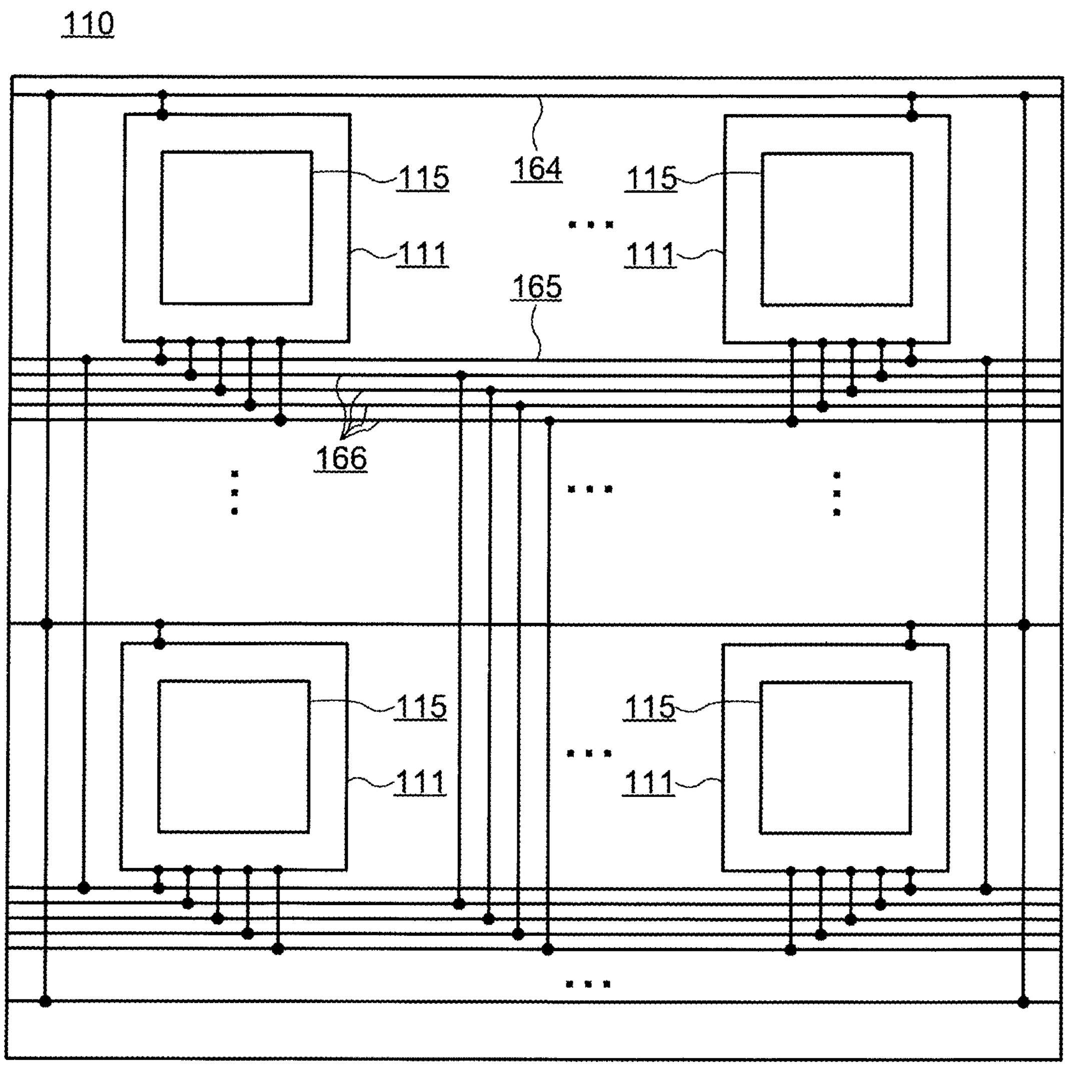


FIG. 5A

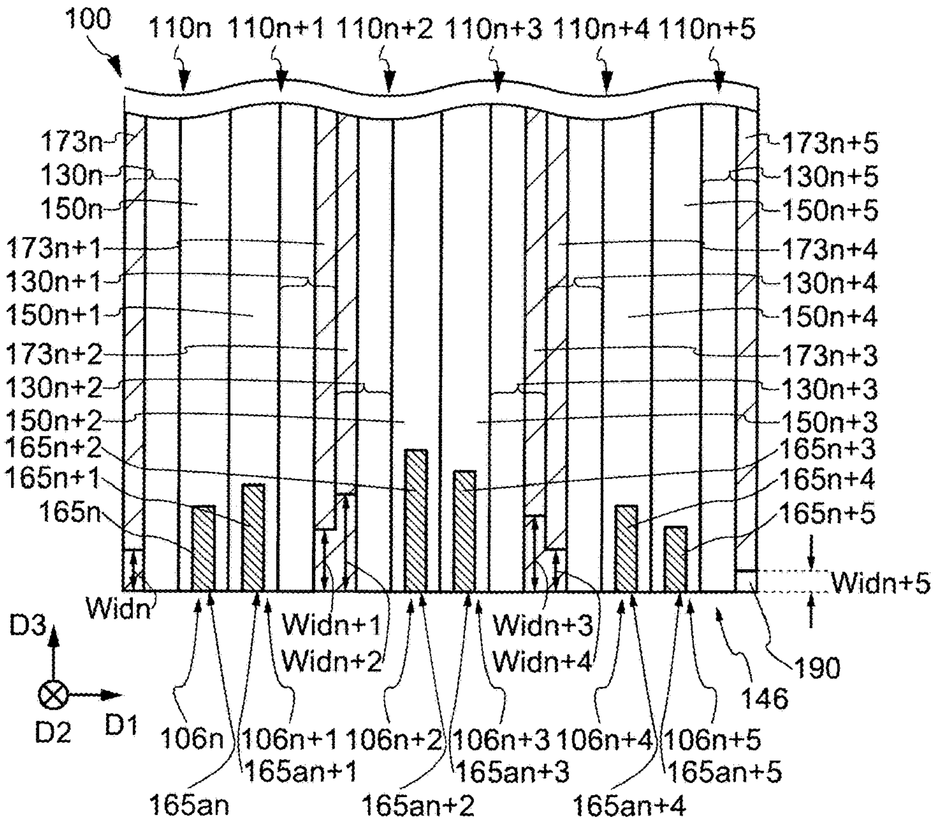

100
110n
110n+1
110n+2
110n+3
110n+4
110n+5
173n
130n
150n
173n+1
130n+1
150n+1
173n+2
130n+2
150n+2
165n+2
165n+1
165n
173n+5
130n+5
150n+5
173n+4
130n+4
150n+4
173n+3
130n+3
150n+3
165n+3
165n+4
165n+5
Widn
Widn+1
Widn+2
Widn+3
Widn+4
Widn+5
190
146
D3
D2
D1
106n
106n+1
106n+2
106n+3
106n+4
106n+5
165an+1
165an+3
165an+5
165an
165an+2
165an+4

FIG. 5B

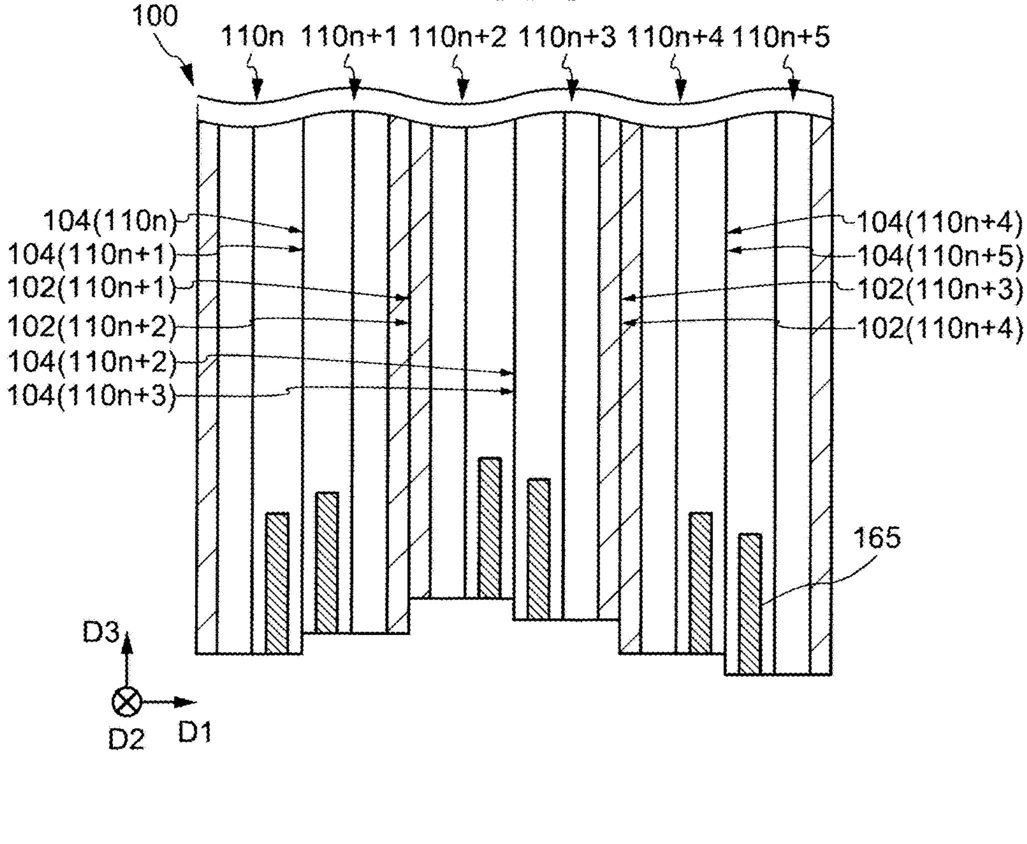

100
110n
110n+1
110n+2
110n+3
110n+4
110n+5
104(110n)
104(110n+1)
102(110n+1)
102(110n+2)
104(110n+2)
104(110n+3)
104(110n+4)
104(110n+5)
102(110n+3)
102(110n+4)
165
D3
D2
D1

FIG. 6A

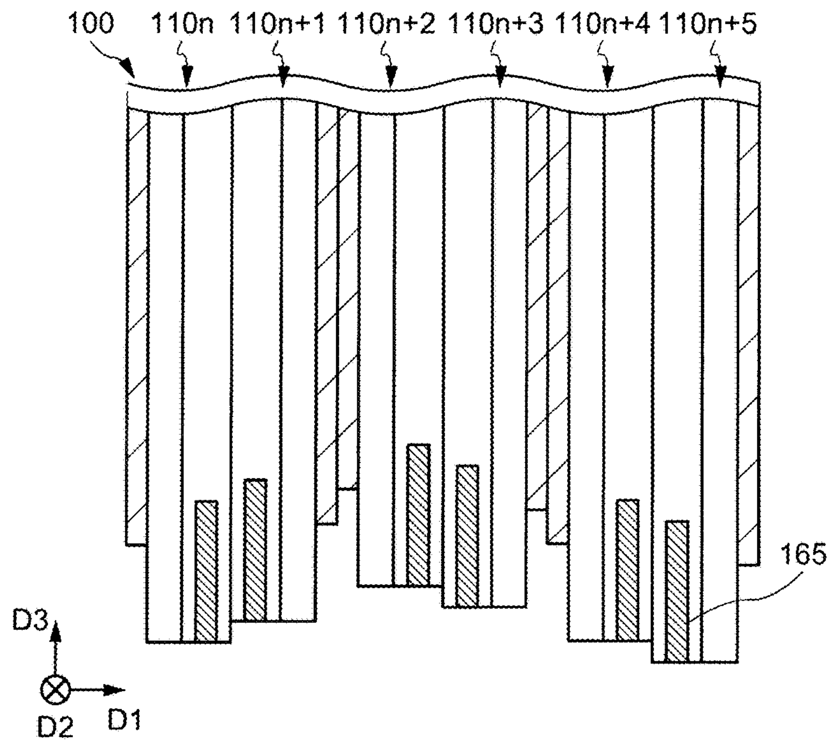

100
110n
110n+1
110n+2
110n+3
110n+4
110n+5
165
D3
D2
D1

FIG. 6B

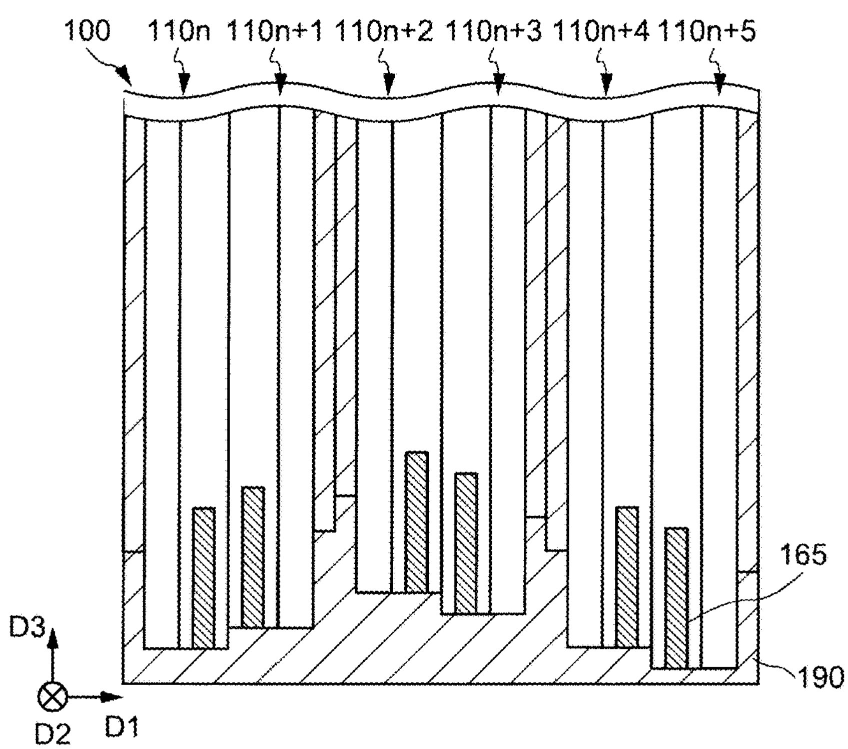

100
110n
110n+1
110n+2
110n+3
110n+4
110n+5
165
190
D3
D2
D1

FIG. 7A

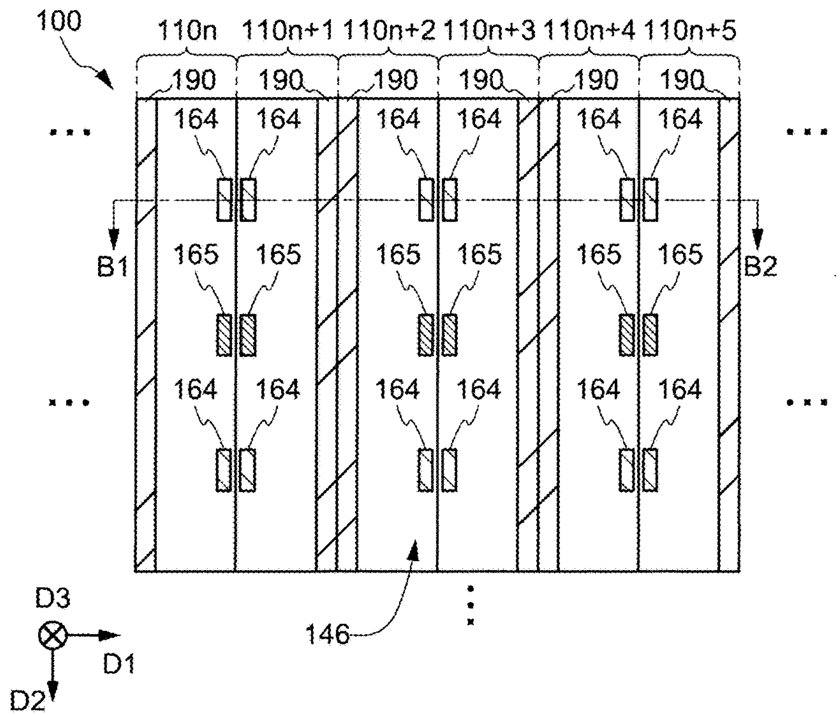

100
110n
110n+1
110n+2
110n+3
110n+4
110n+5
190
164
165
B1
B2
146
D3
D1
D2

FIG. 7B

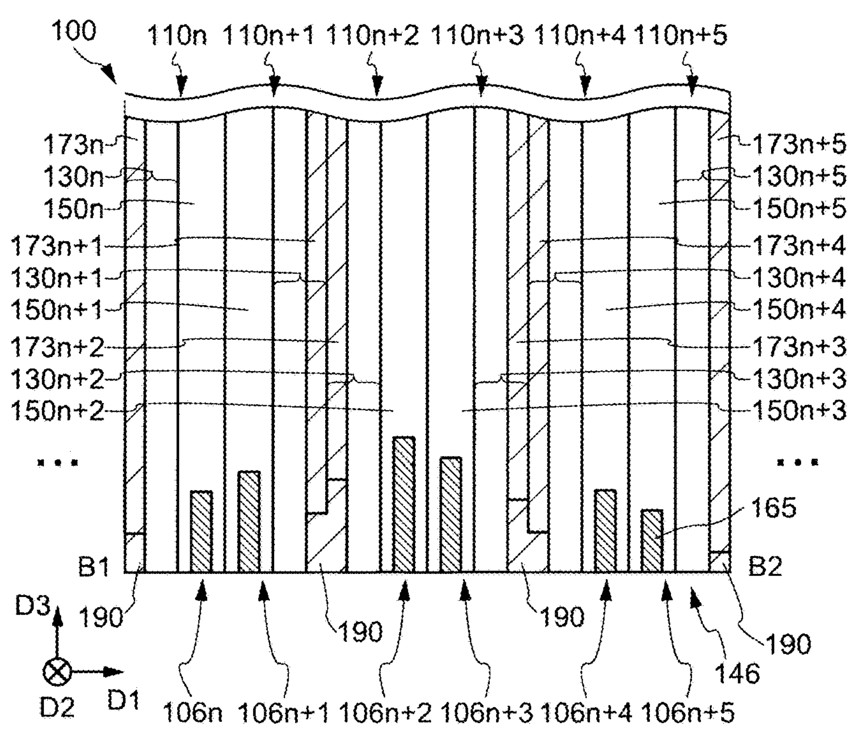

100
110n
110n+1
110n+2
110n+3
110n+4
110n+5
173n
130n
150n
173n+1
130n+1
150n+1
173n+2
130n+2
150n+2
173n+5
130n+5
150n+5
173n+4
130n+4
150n+4
173n+3
130n+3
150n+3
165
B1
B2
190
146
D3
D2
D1
106n
106n+1
106n+2
106n+3
106n+4
106n+5

FIG. 8A

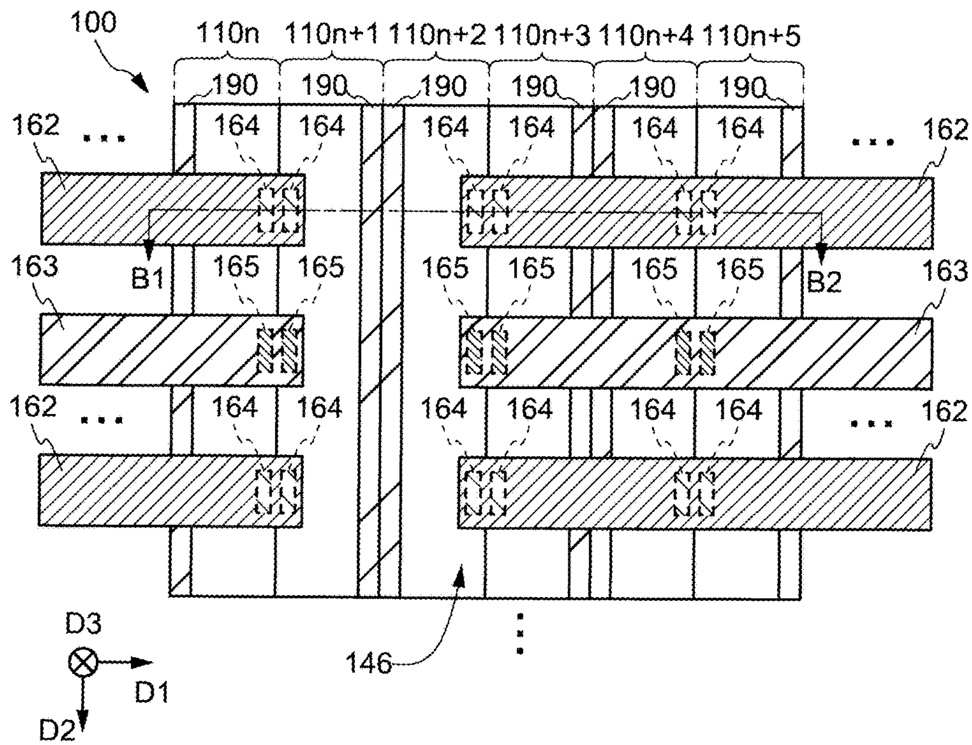

100
110n
110n+1
110n+2
110n+3
110n+4
110n+5
190
162
163
164
165
B1
B2
146
D3
D1
D2

FIG. 8B

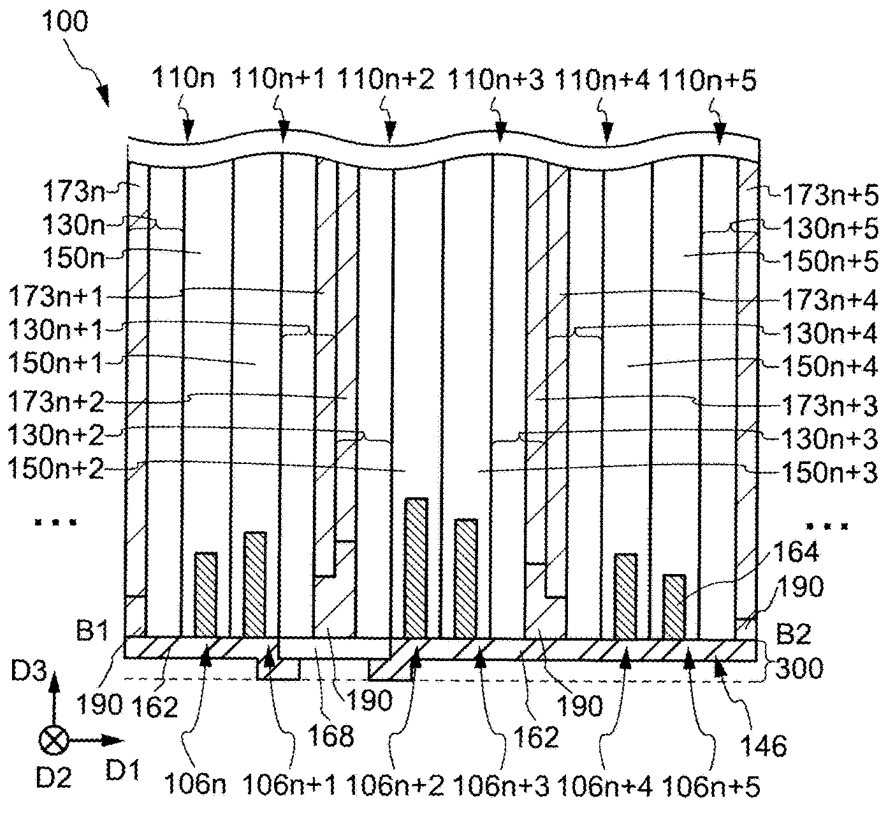

100
110n
110n+1
110n+2
110n+3
110n+4
110n+5
173n
130n
150n
173n+1
130n+1
150n+1
173n+2
130n+2
150n+2
173n+5
130n+5
150n+5
173n+4
130n+4
150n+4
173n+3
130n+3
150n+3
164
190
B1
B2
300
162
168
146
D3
D2
D1
106n
106n+1
106n+2
106n+3
106n+4
106n+5

FIG. 9A

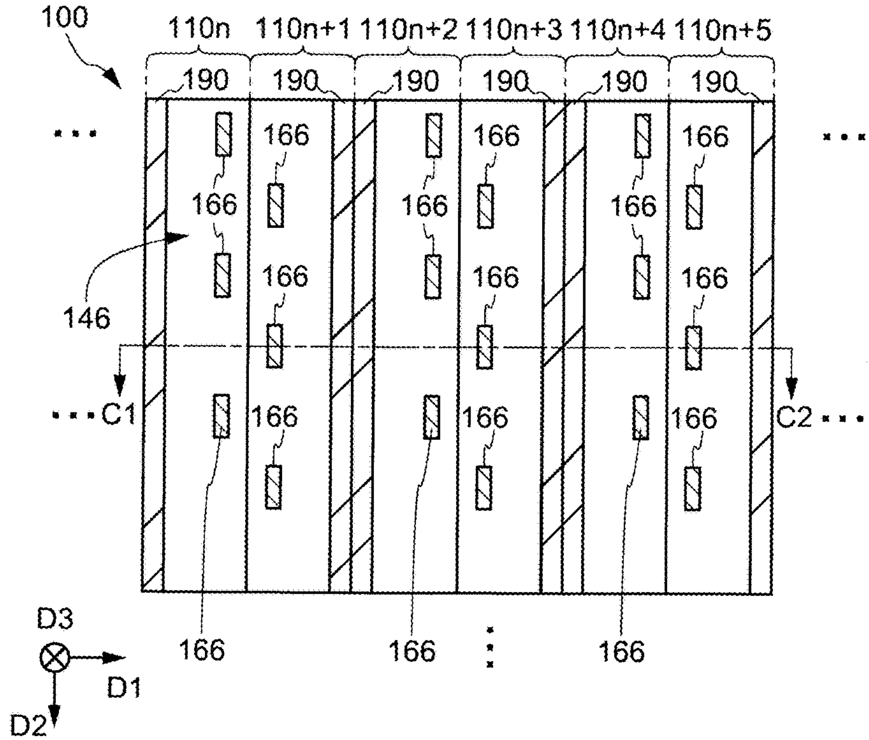

100
110n
110n+1
110n+2
110n+3
110n+4
110n+5
190
166
146
C1
C2
D3
D1
D2

FIG. 9B

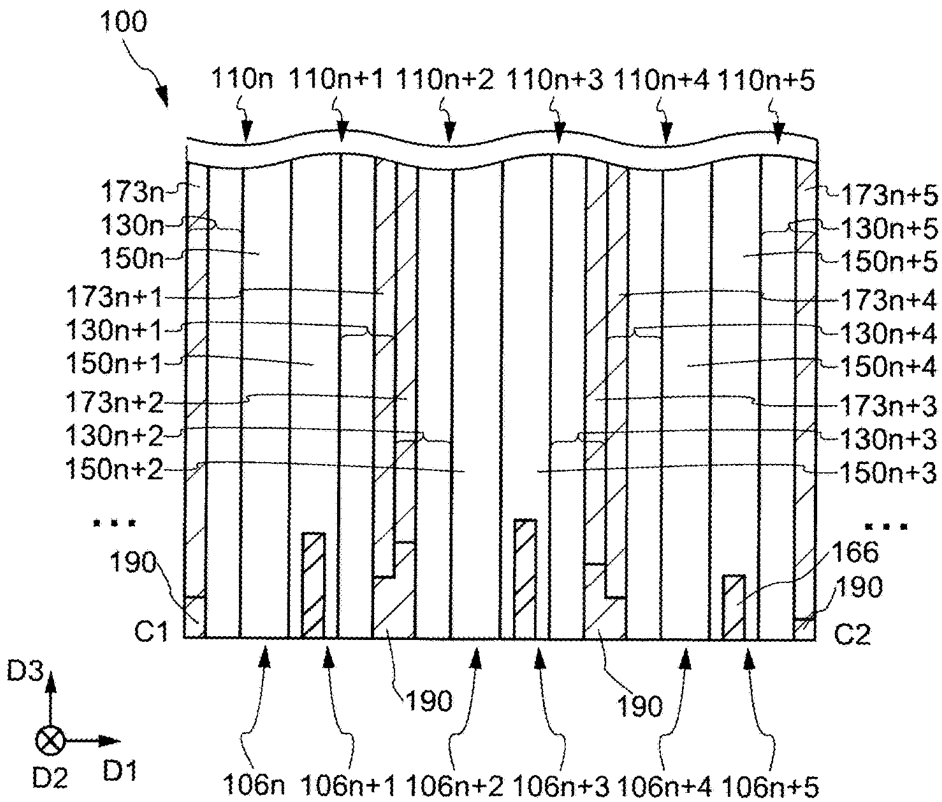

100
110n
110n+1
110n+2
110n+3
110n+4
110n+5
173n
130n
150n
173n+1
130n+1
150n+1
173n+2
130n+2
150n+2
173n+5
130n+5
150n+5
173n+4
130n+4
150n+4
173n+3
130n+3
150n+3
190
166
C1
C2
106n
106n+1
106n+2
106n+3
106n+4
106n+5
D3
D2
D1

FIG. 10A

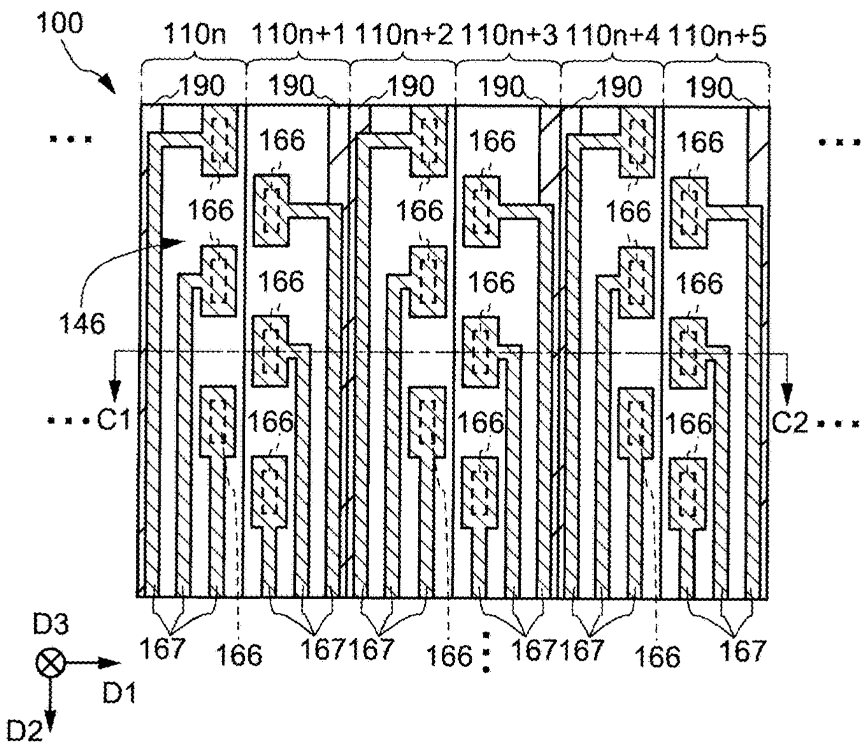


FIG. 10B

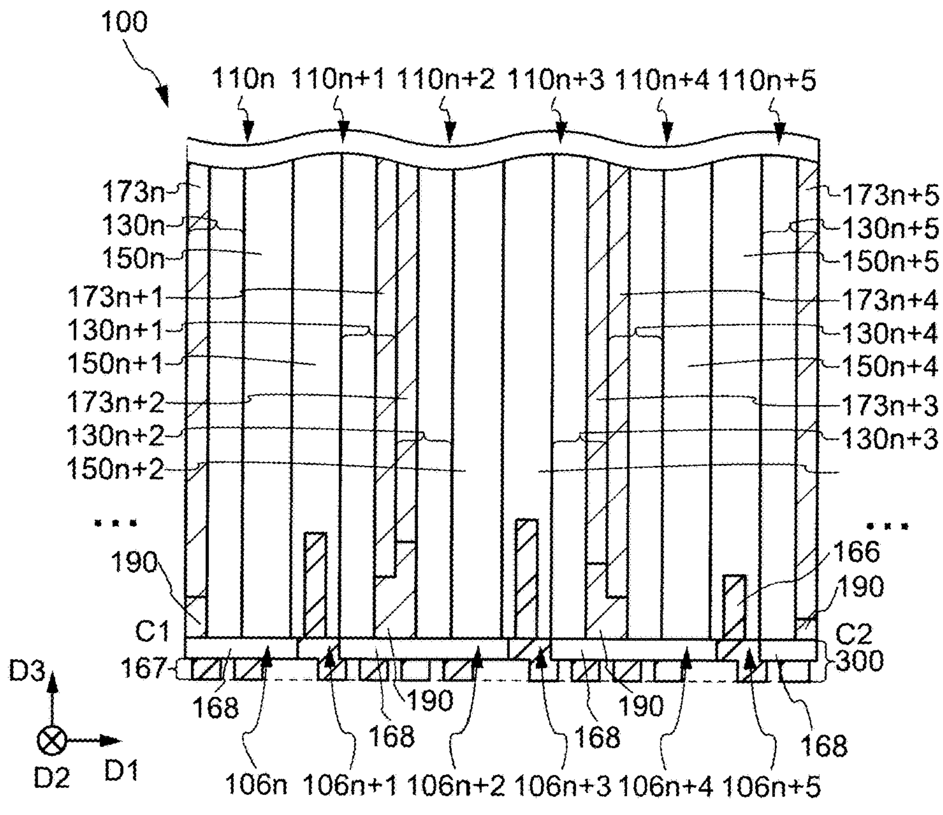


FIG. 11A

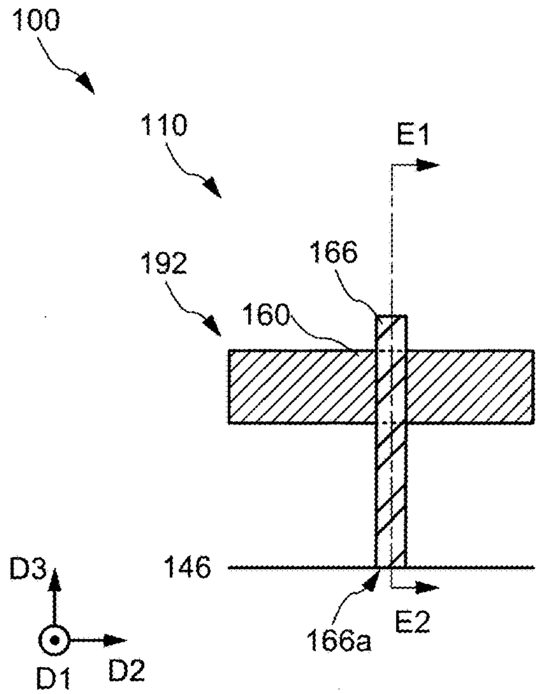


FIG. 11B

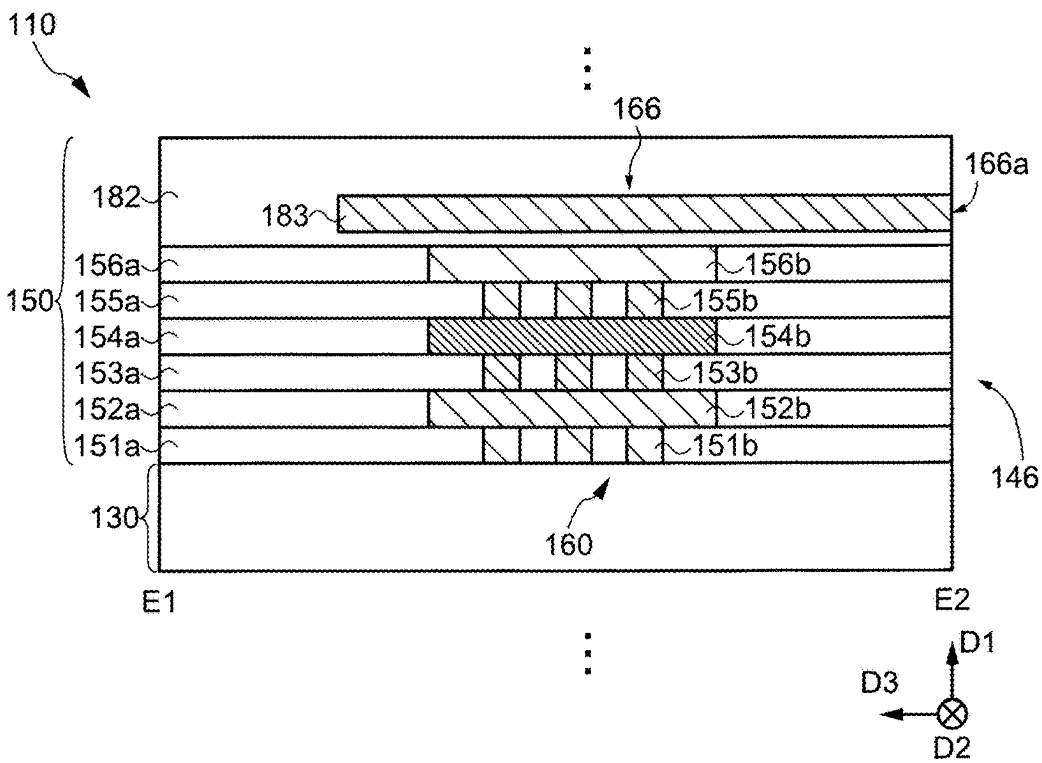


FIG. 12

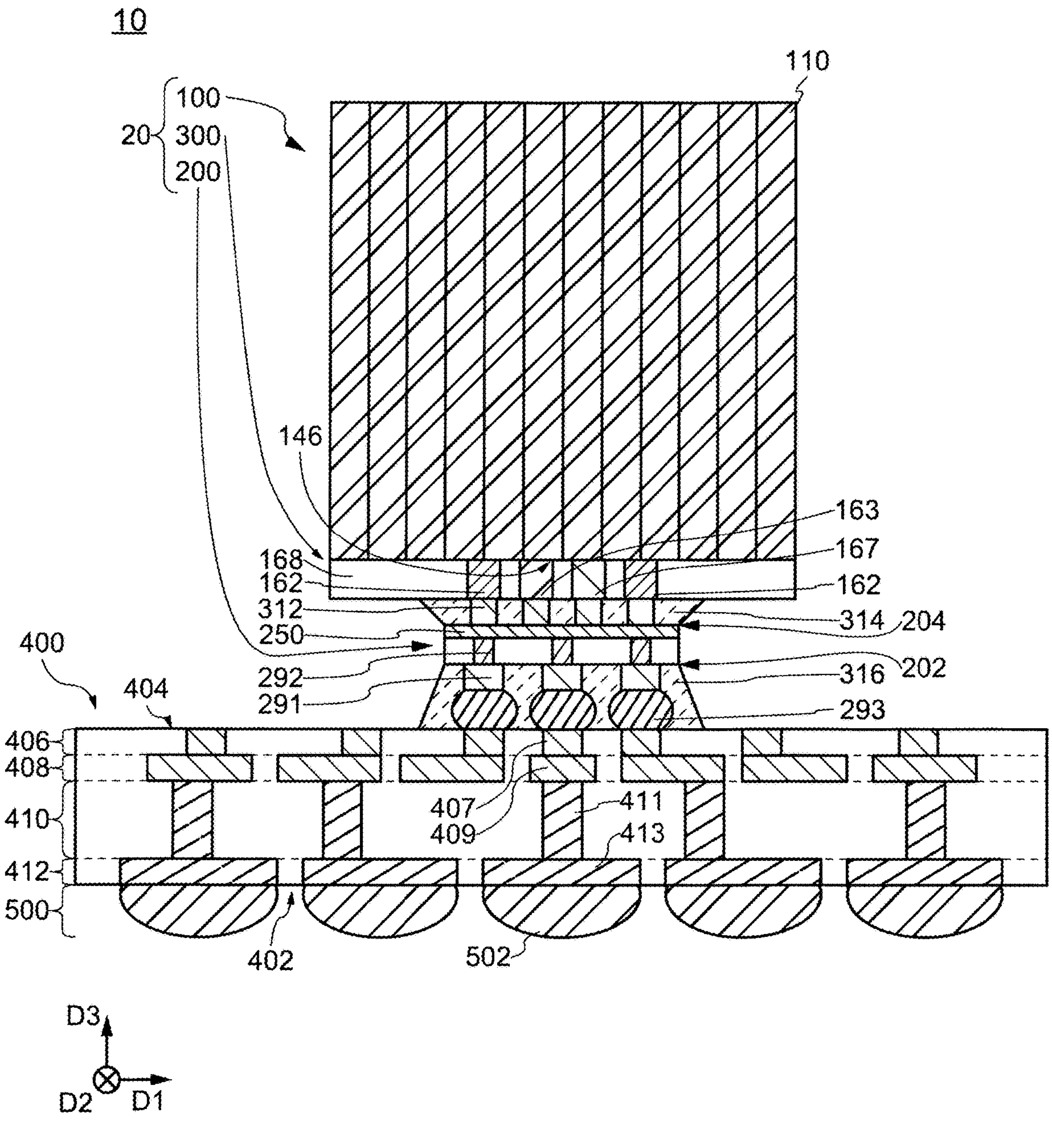

10
20
100
300
200
110
146
163
167
168
162
162
312
314
204
250
202
292
316
291
293
400
404
406
408
410
407
409
411
413
412
500
402
502
D3
D2
D1

FIG. 13

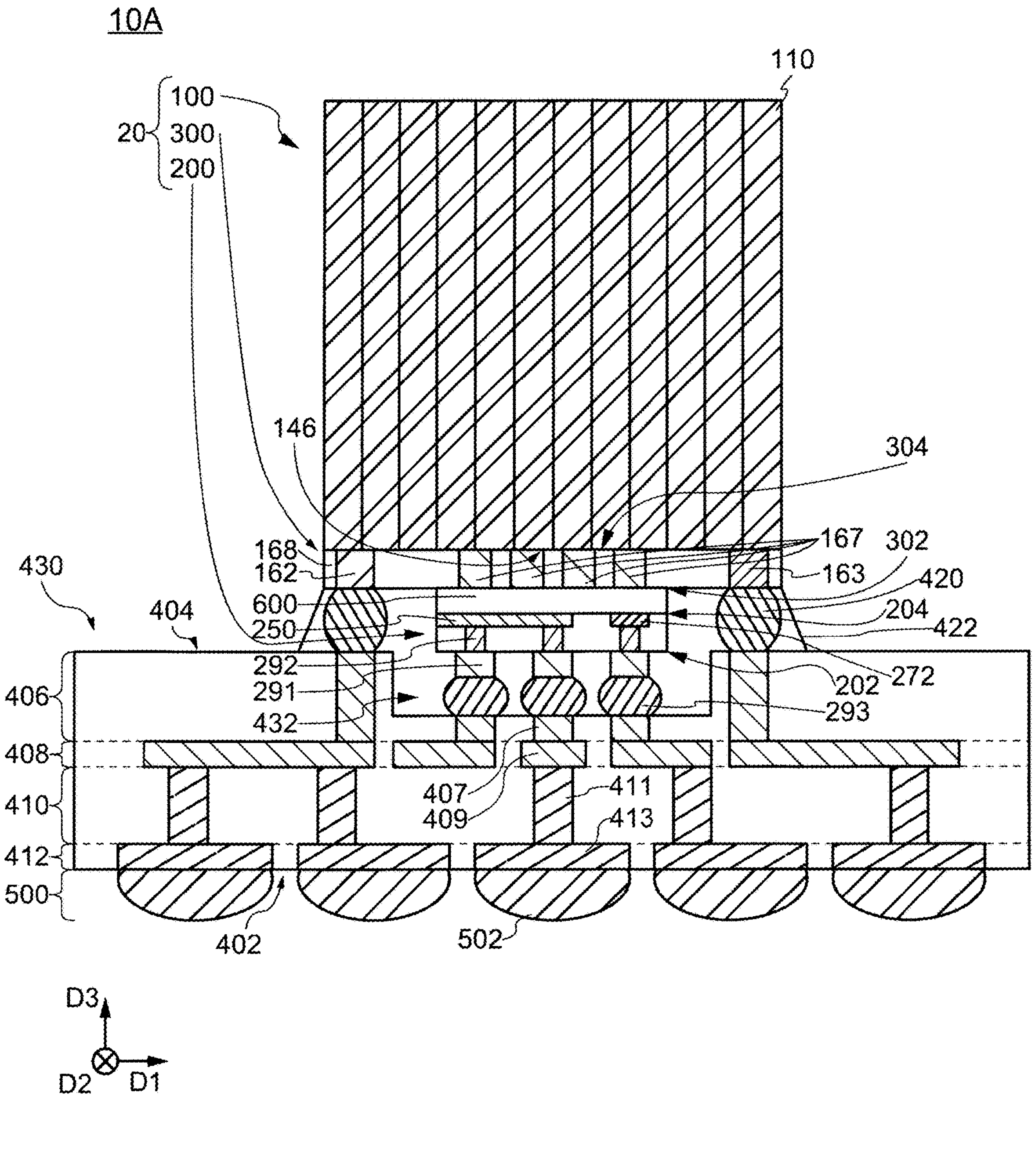

10A
110
100
20
300
200
146
304
168
167
302
162
163
430
404
600
420
250
204
422
292
202
272
406
291
293
432
408
410
407
411
409
413
412
500
402
502
D3
D2
D1

FIG. 14

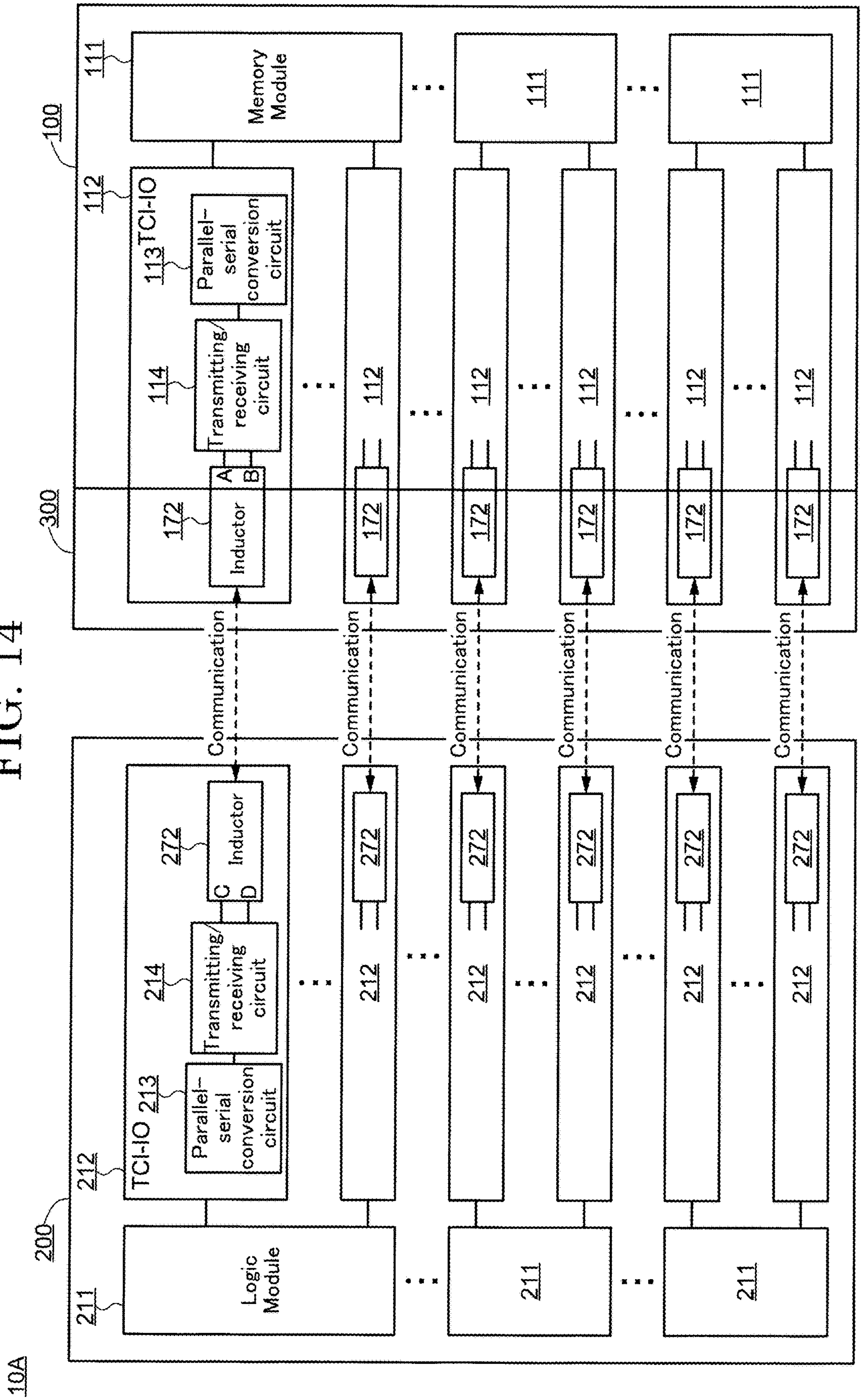

10A
100
300
200
111
Memory Module
111
111
112
113 TCI-IO
Parallel-serial conversion circuit
114
Transmitting/receiving circuit
A
B
172
Inductor
112
172
Communication
211
Logic Module
211
211
212
TCI-IO
213
Parallel-serial conversion circuit
214
Transmitting/receiving circuit
C
D
272
Inductor
212
272

FIG. 15

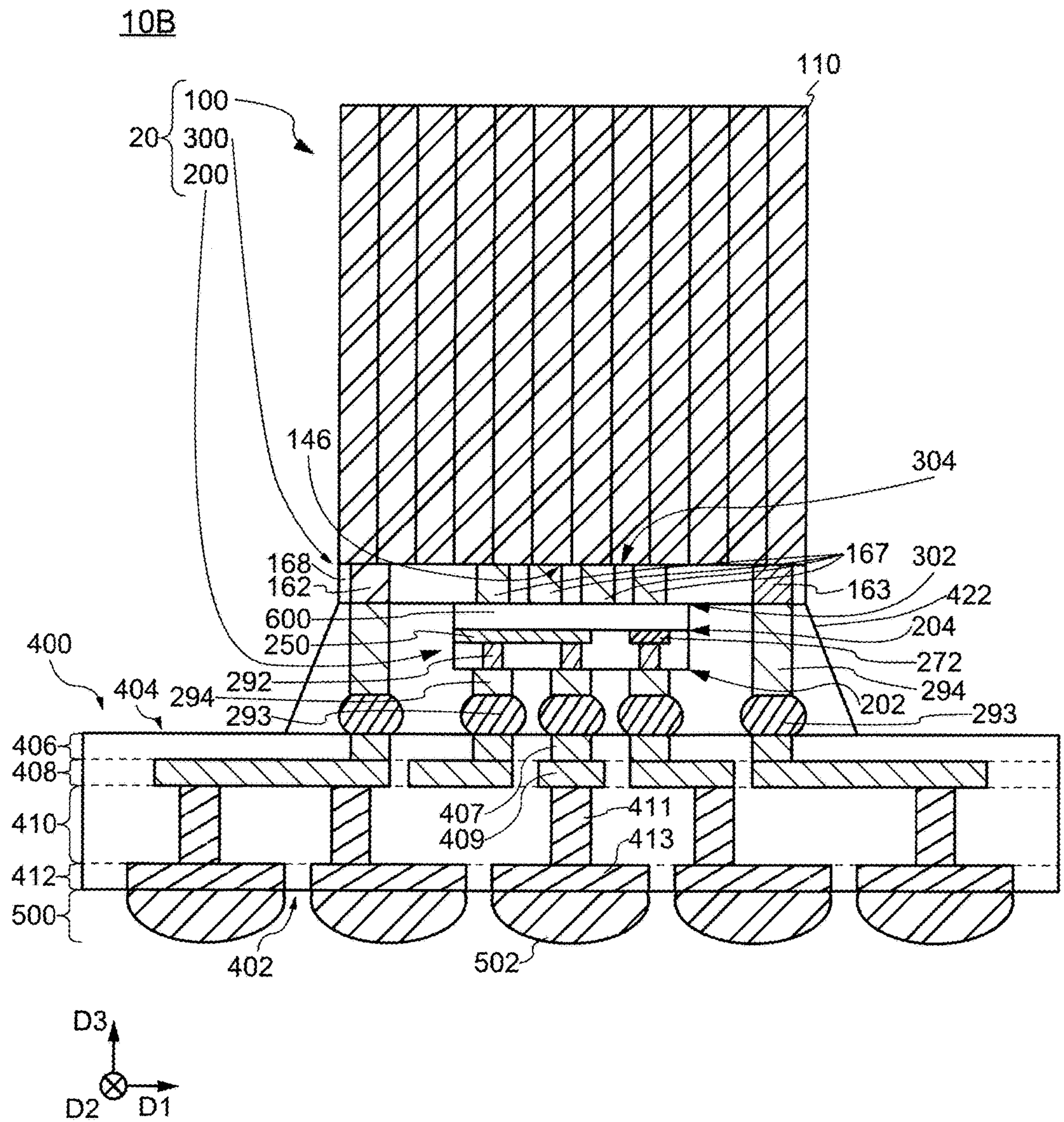

10B
110
20
100
300
200
146
304
168
162
167
302
163
422
600
204
250
272
400
404
292
294
202
294
293
293
406
408
407
411
410
409
413
412
500
402
502
D3
D2
D1

FIG. 16

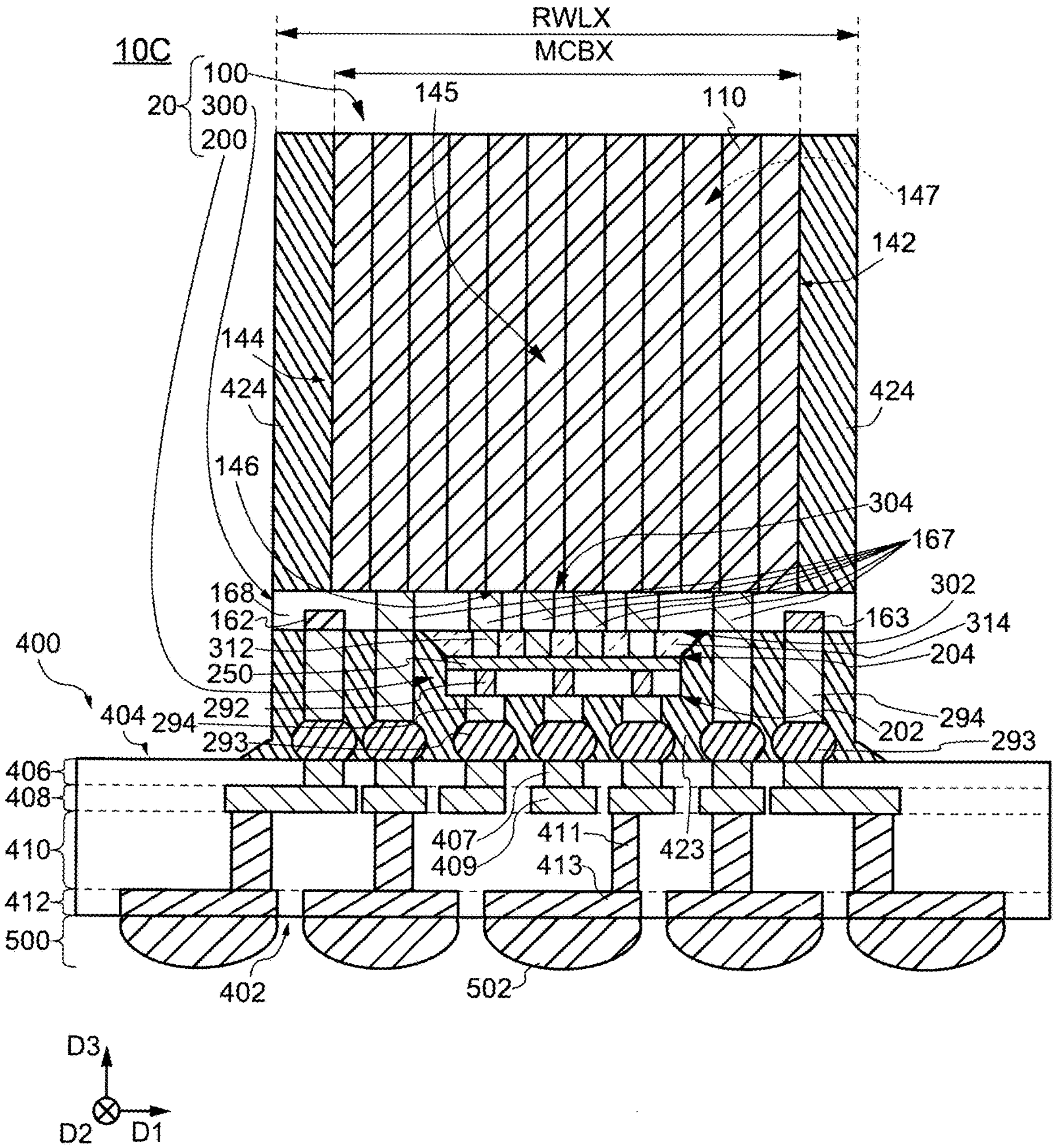

RWLX
MCBX
10C
100
20
300
200
145
110
147
142
144
424
424
146
304
167
168
302
162
163
314
400
312
204
250
292
294
404
294
202
293
293
406
408
410
407
411
423
409
413
412
500
402
502
D3
D2
D1

FIG. 17

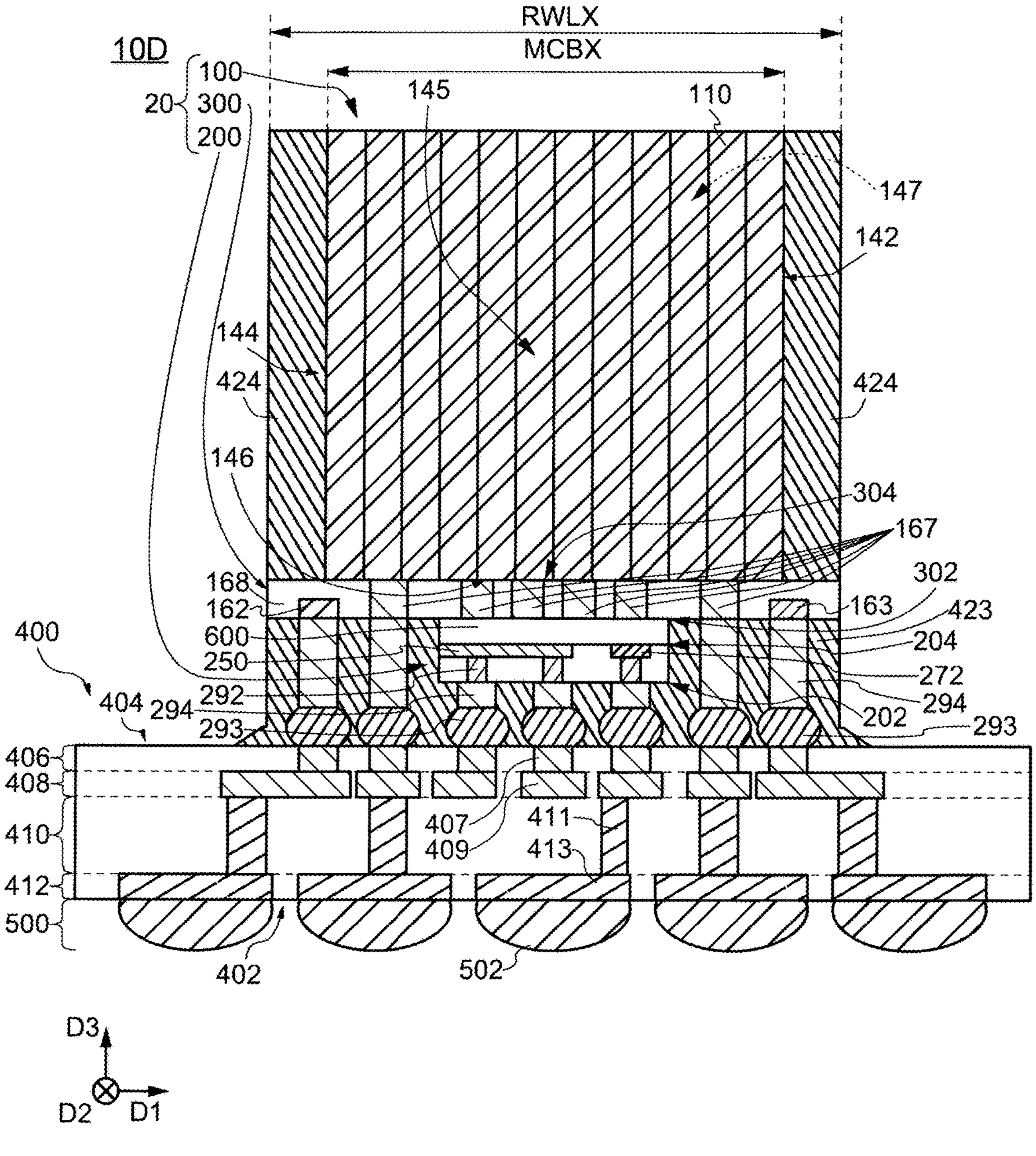
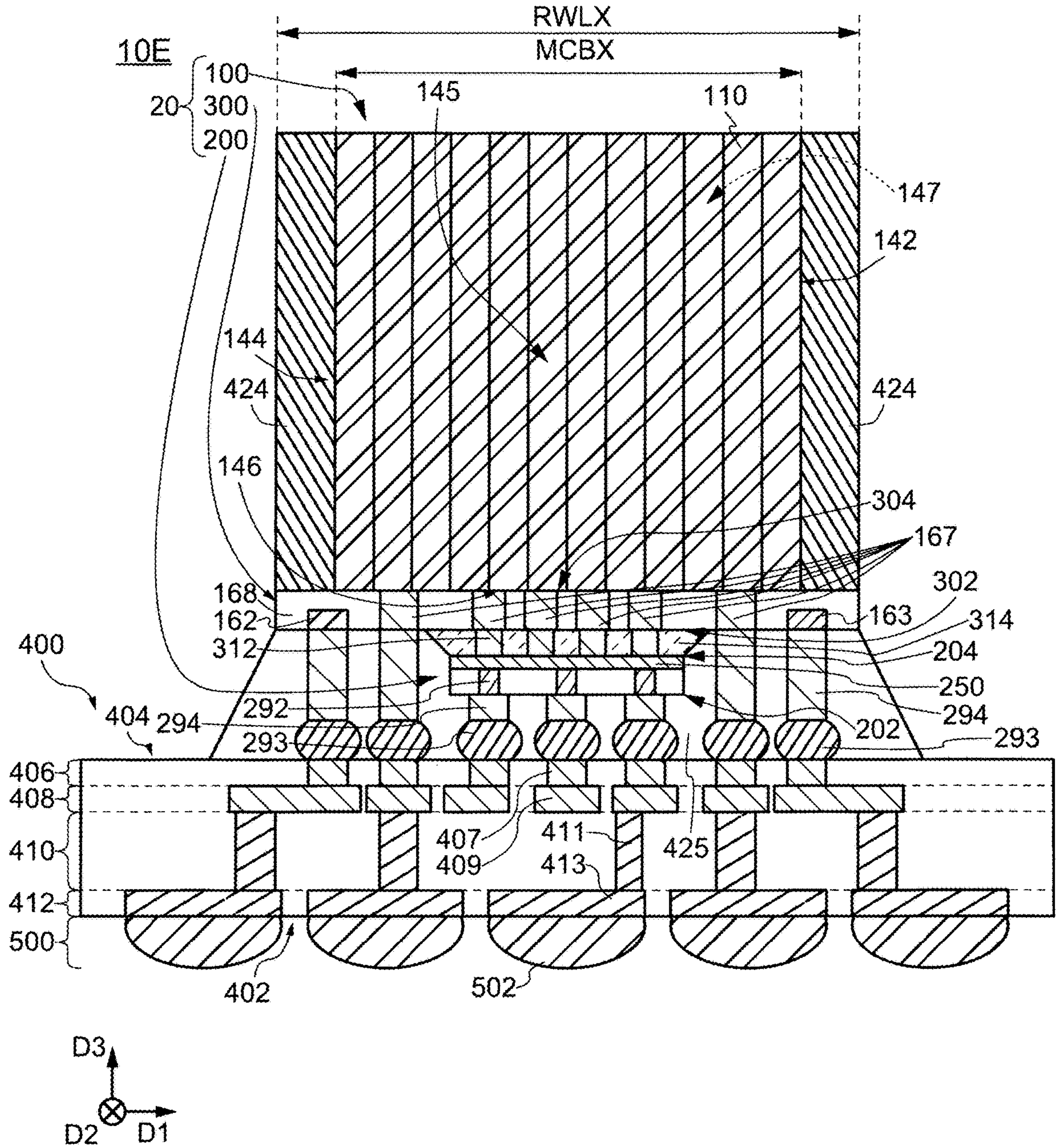

RWLX
MCBX
10D
100
20
300
200
145
110
147
142
144
424
424
146
304
167
168
302
162
163
423
600
204
250
272
292
294
404
294
202
293
293
400
406
408
410
407
409
411
413
412
500
402
502
D3
D2
D1

FIG. 18



10E
RWLX
MCBX
20
100
300
200
145
110
147
142
144
424
424
146
304
167
168
162
163
302
312
314
204
400
250
292
294
404
294
202
293
293
406
408
410
407
409
411
413
425
412
500
402
502
D3
D2
D1

FIG. 19A

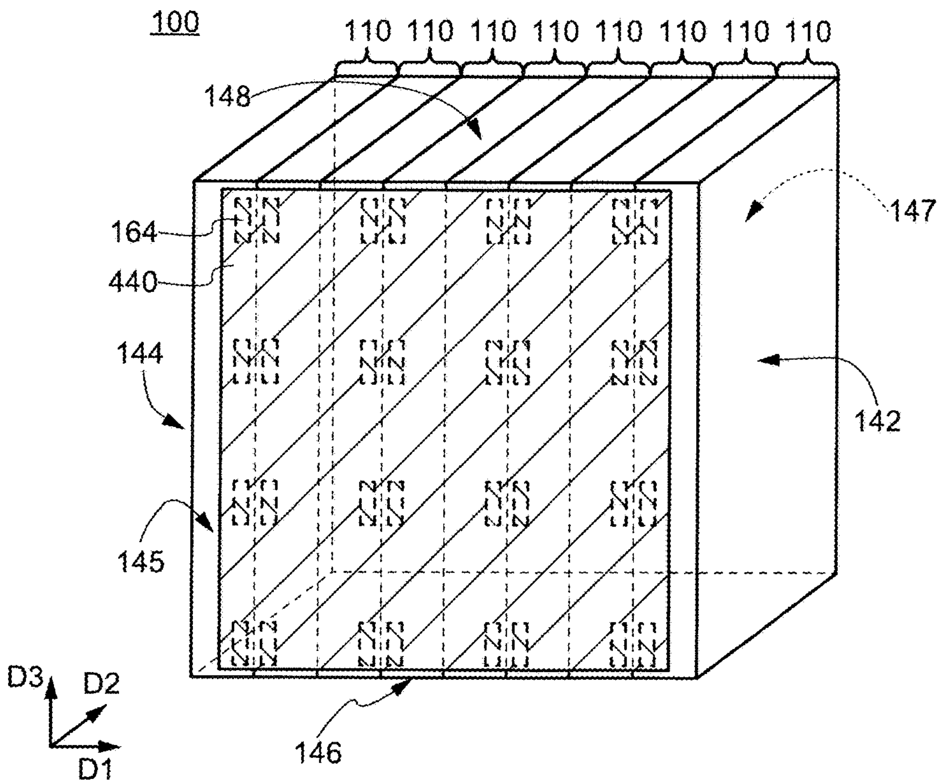


FIG. 19B

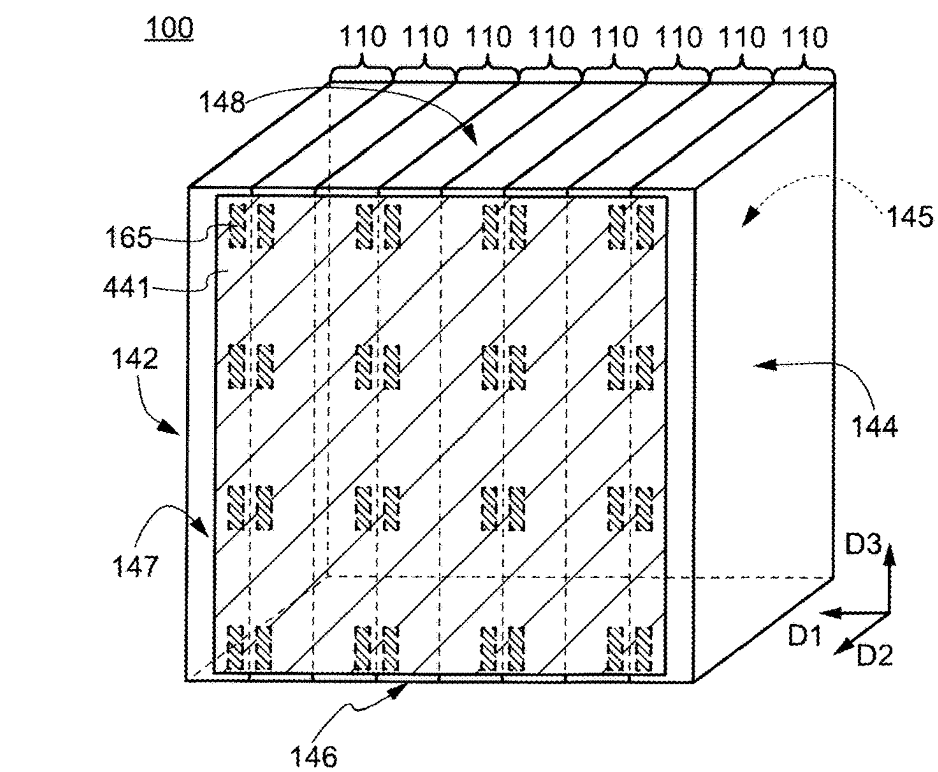


FIG. 20A

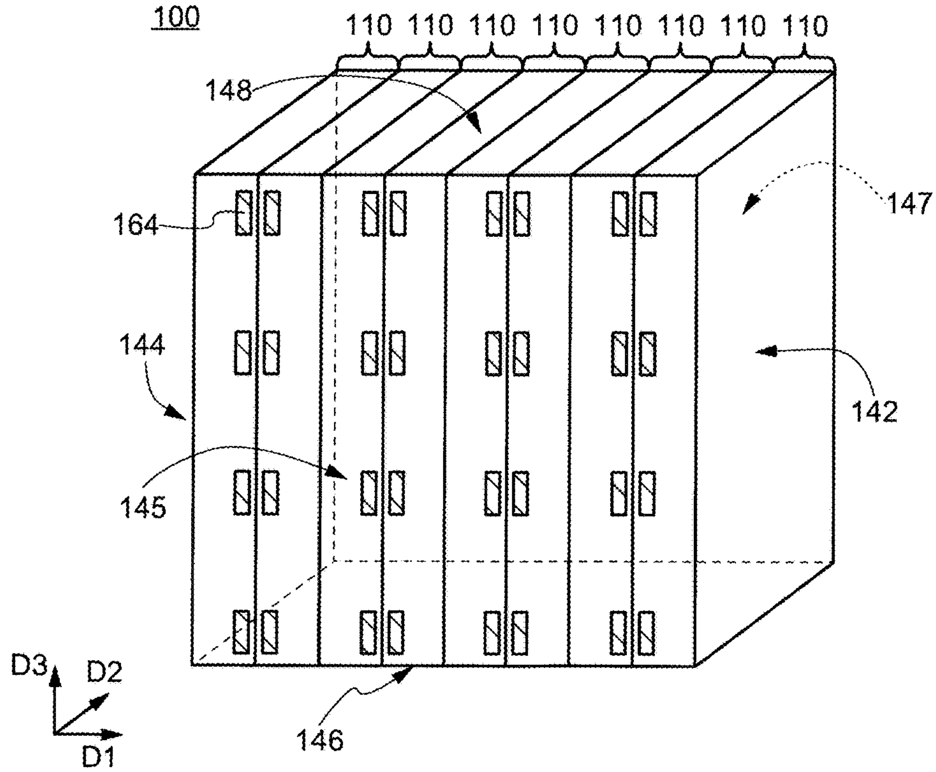


FIG. 20B

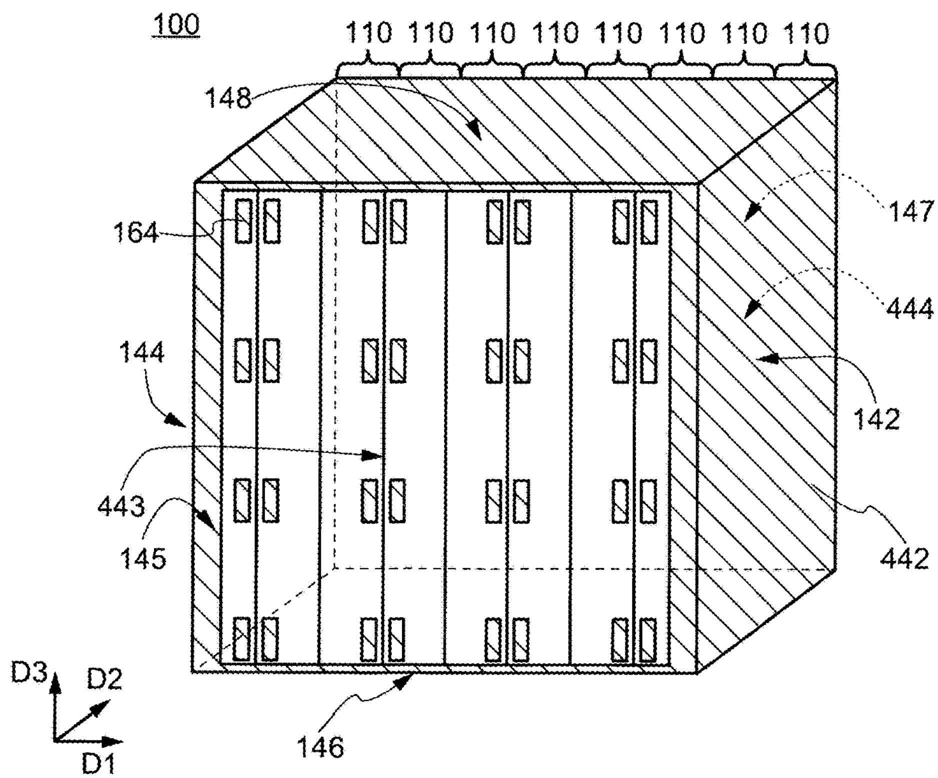


FIG. 21A

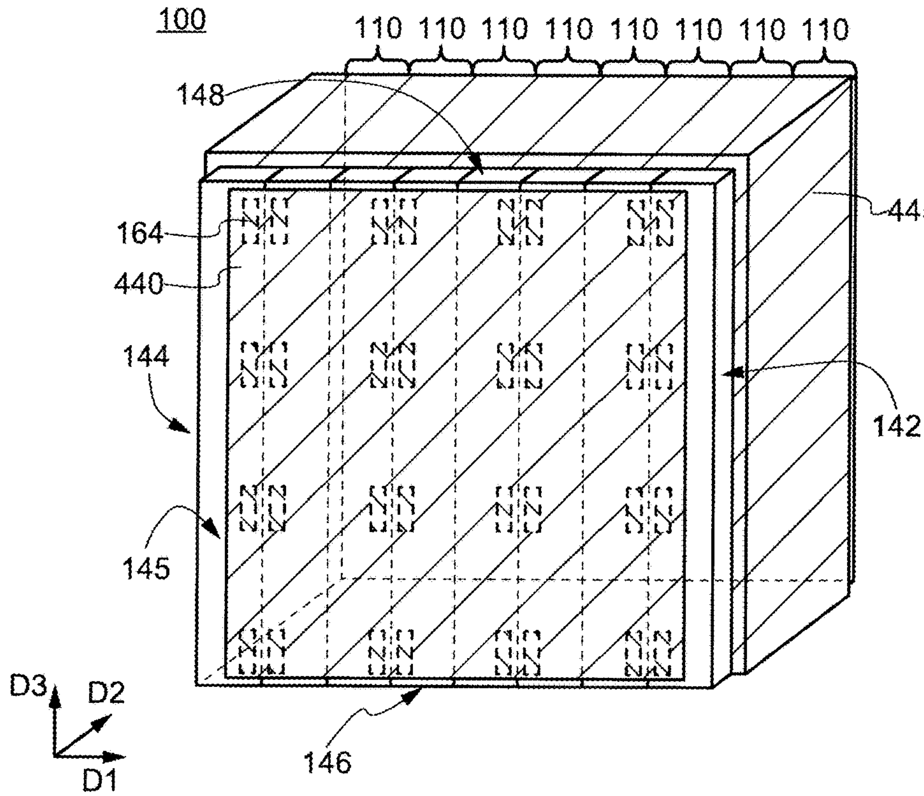


FIG. 21B

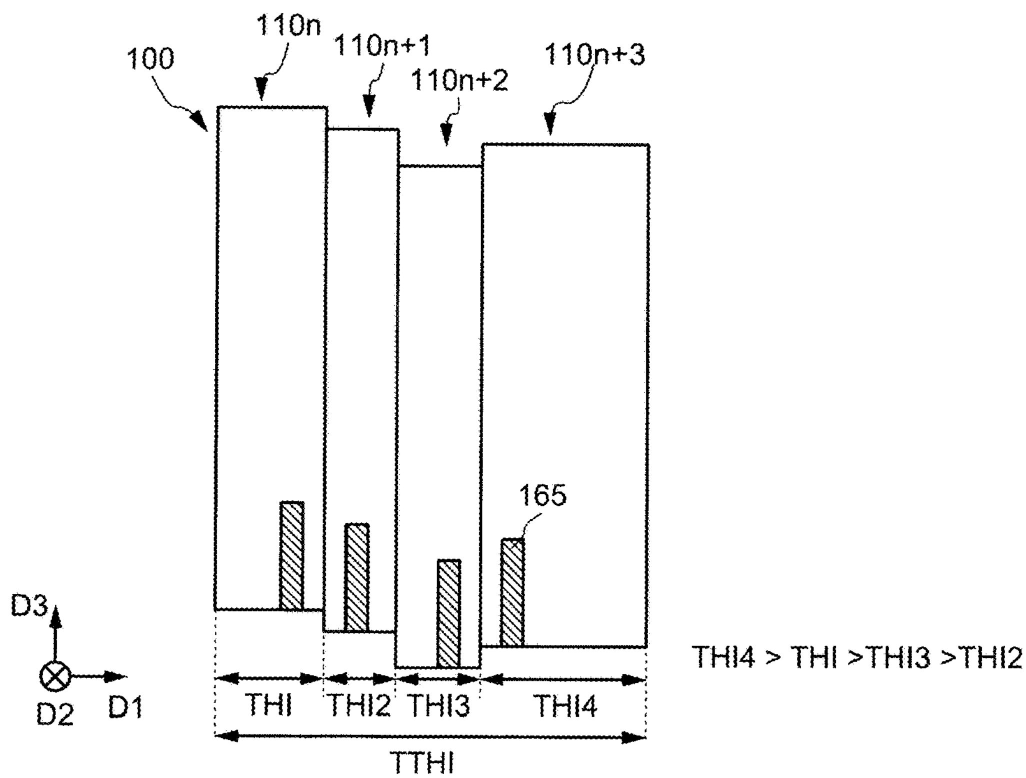


FIG. 22A

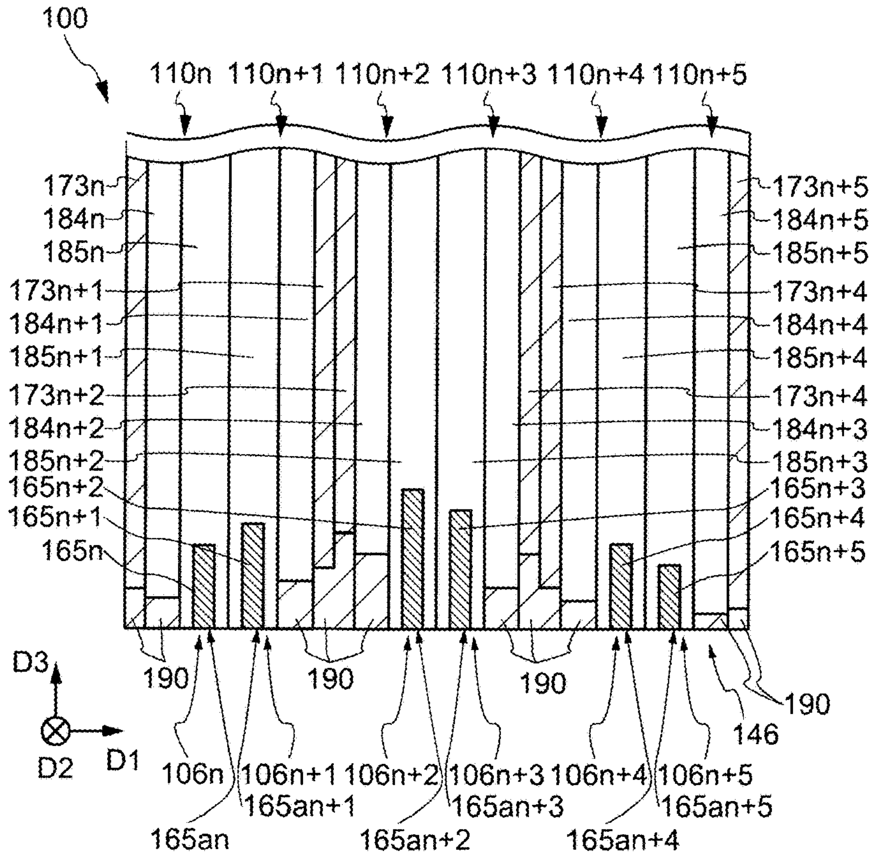


FIG. 22B

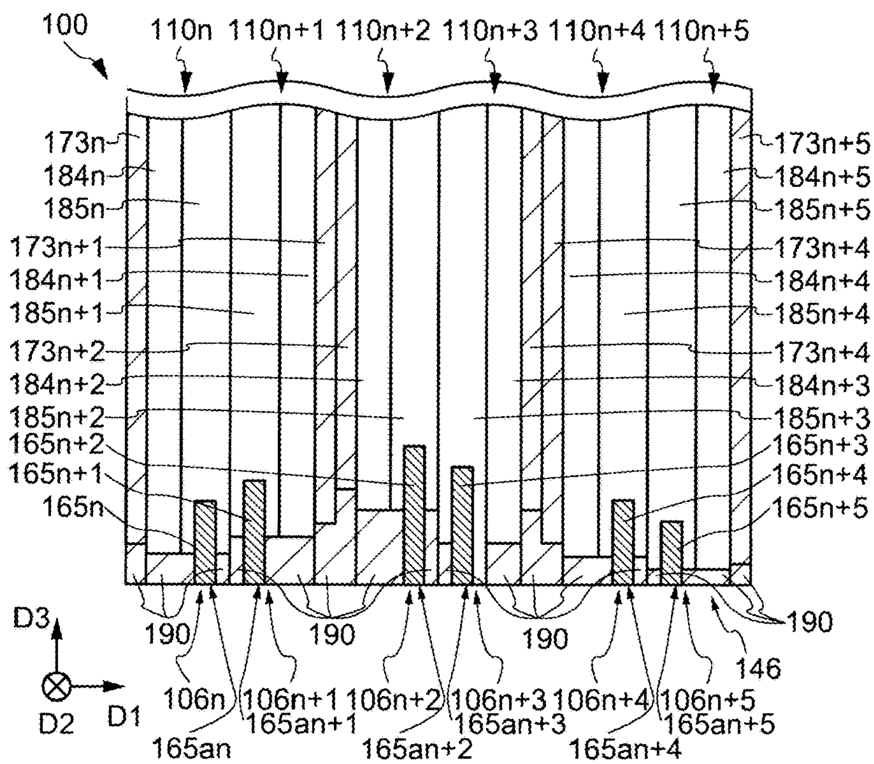


FIG. 23A

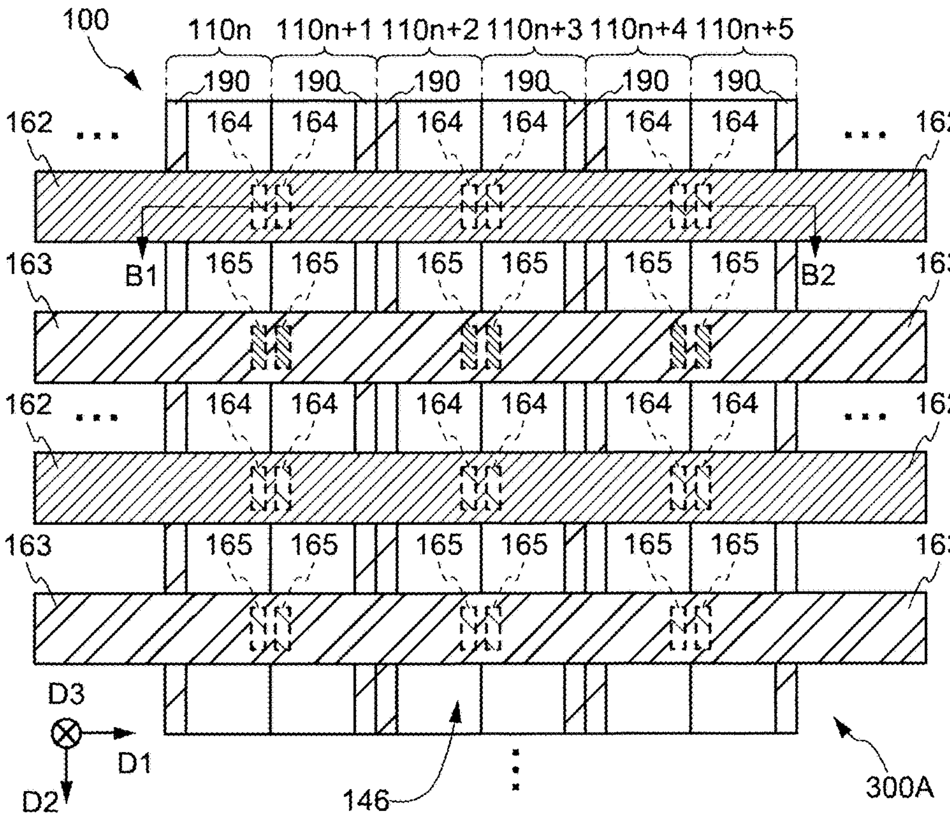


FIG. 23B

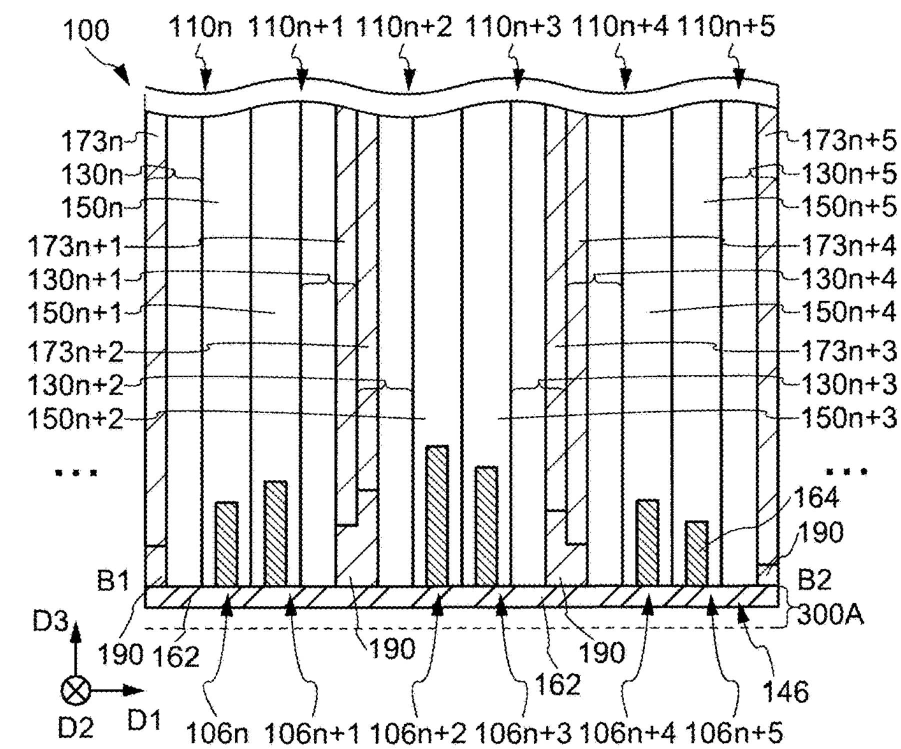


FIG. 24A

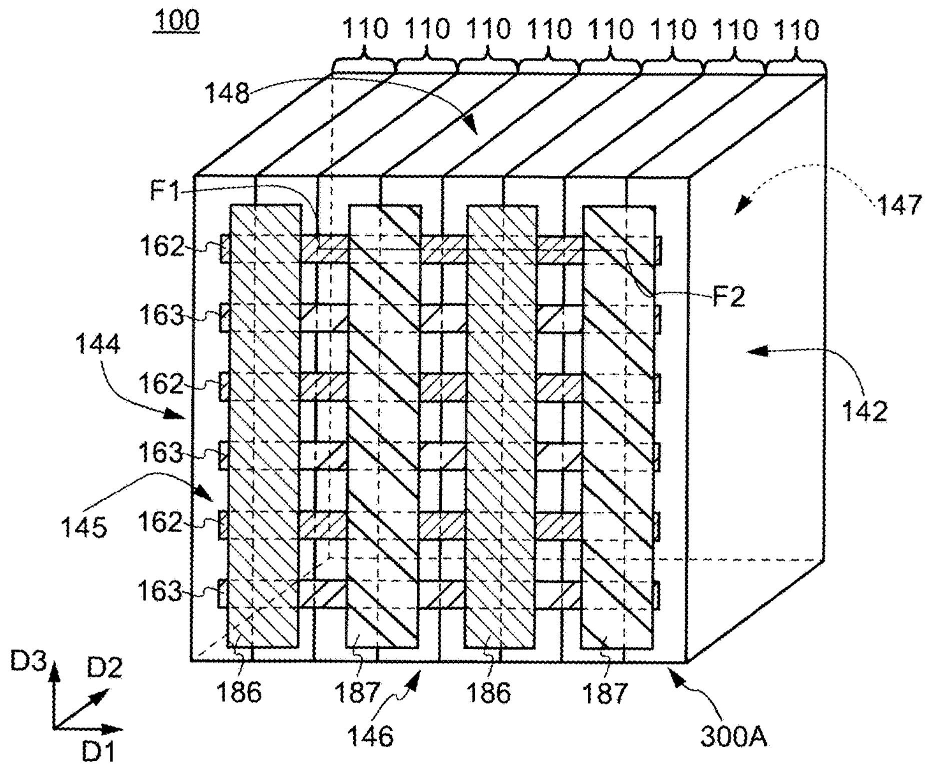


FIG. 24B

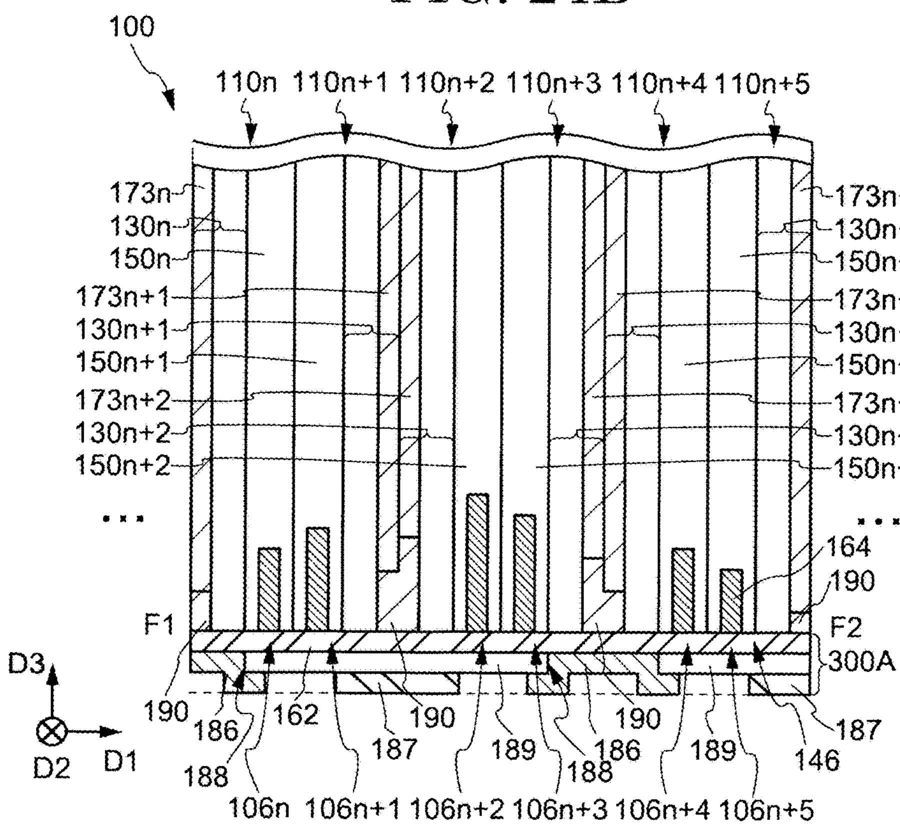
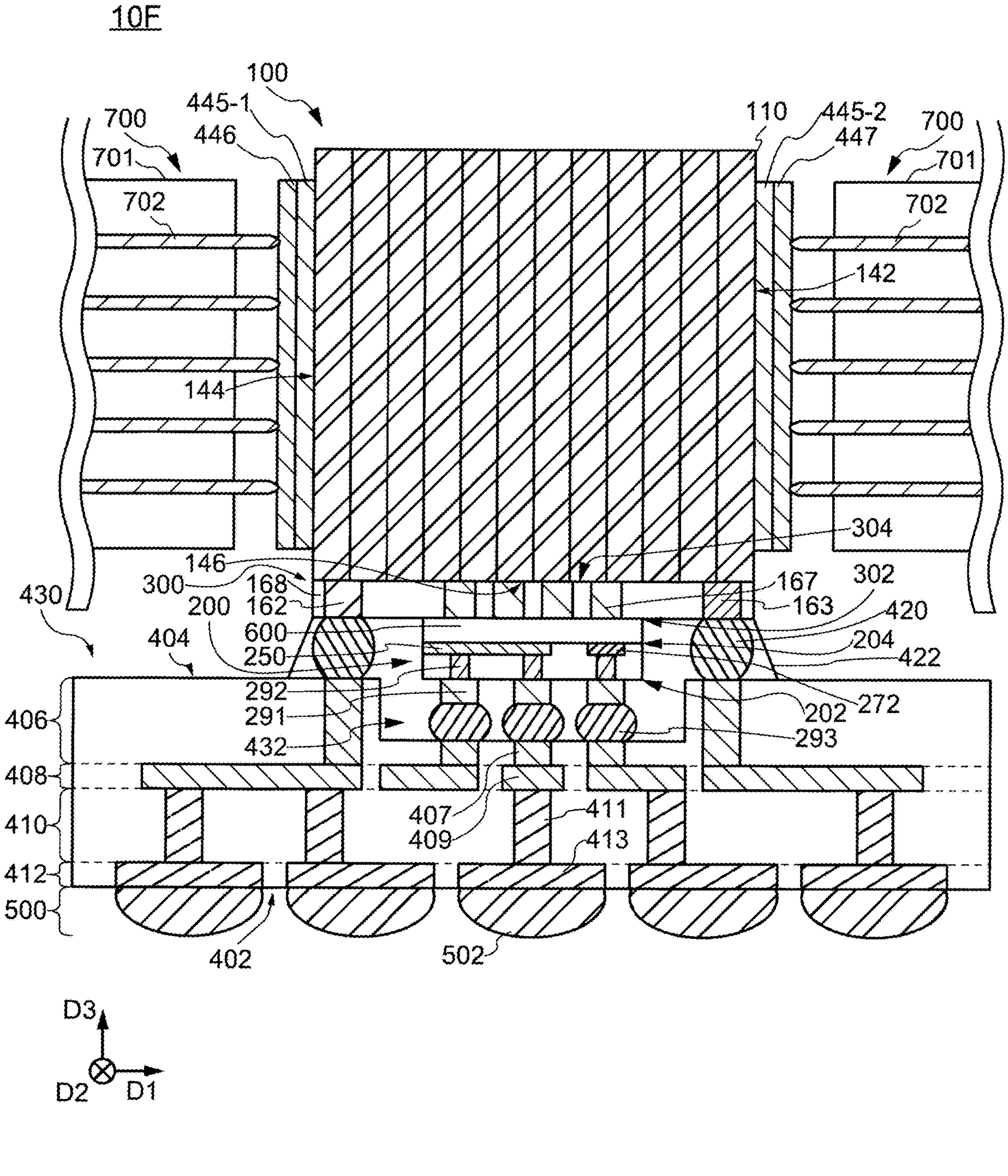


FIG. 25



10F
100
445-1
446
700
701
702
110
445-2
447
700
701
702
142
144
304
146
300
168
162
200
167
163
302
420
430
404
600
250
204
422
292
291
272
202
406
293
432
408
407
409
411
410
413
412
500
402
502
D3
D2
D1

FIG. 26

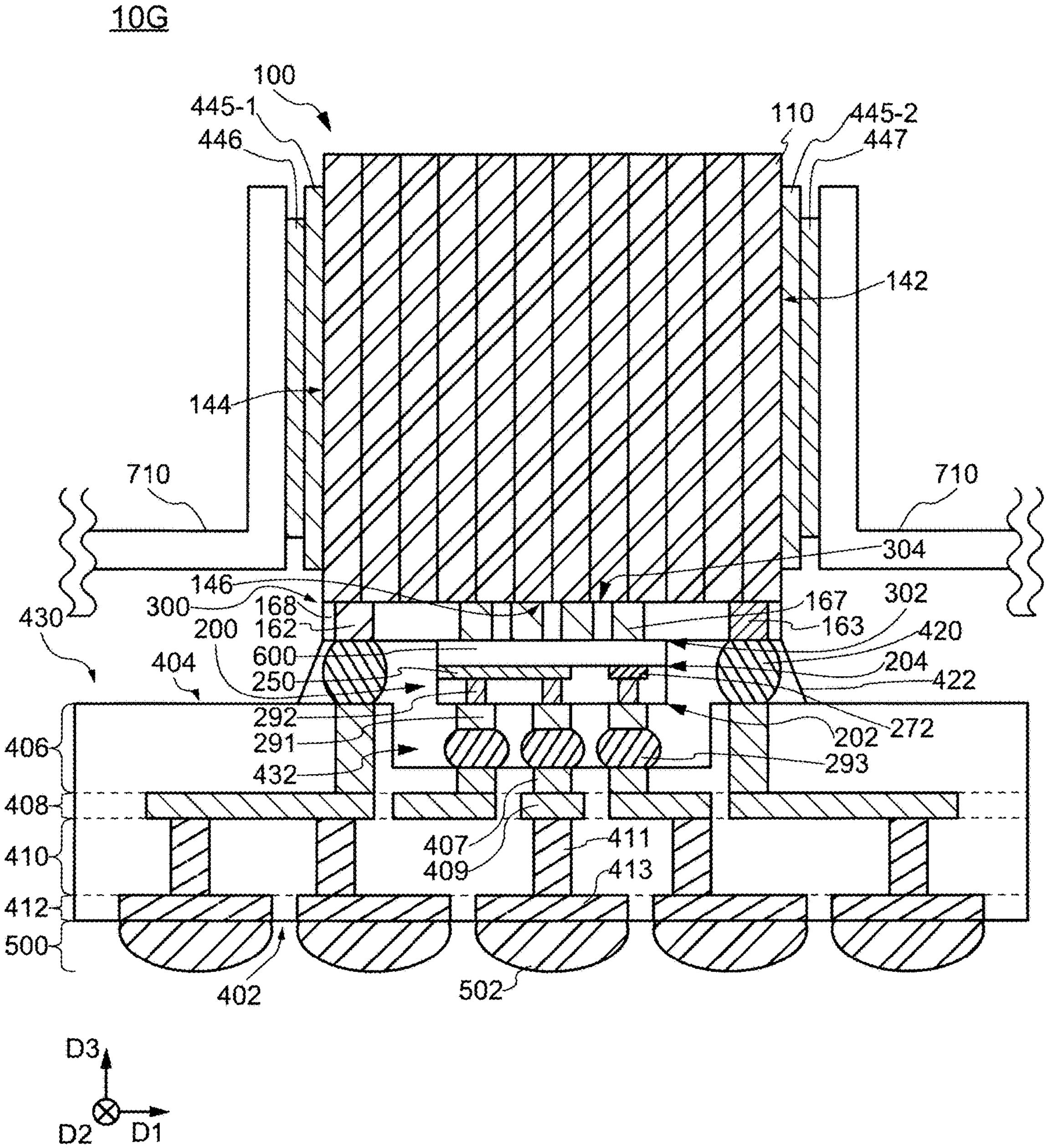

10G
100
445-1
446
110
445-2
447
142
144
710
710
304
146
300
168
162
167
163
302
420
430
200
600
204
404
250
422
272
292
202
291
293
406
432
408
407
411
410
409
413
412
500
402
502
D3
D2
D1

FIG. 27

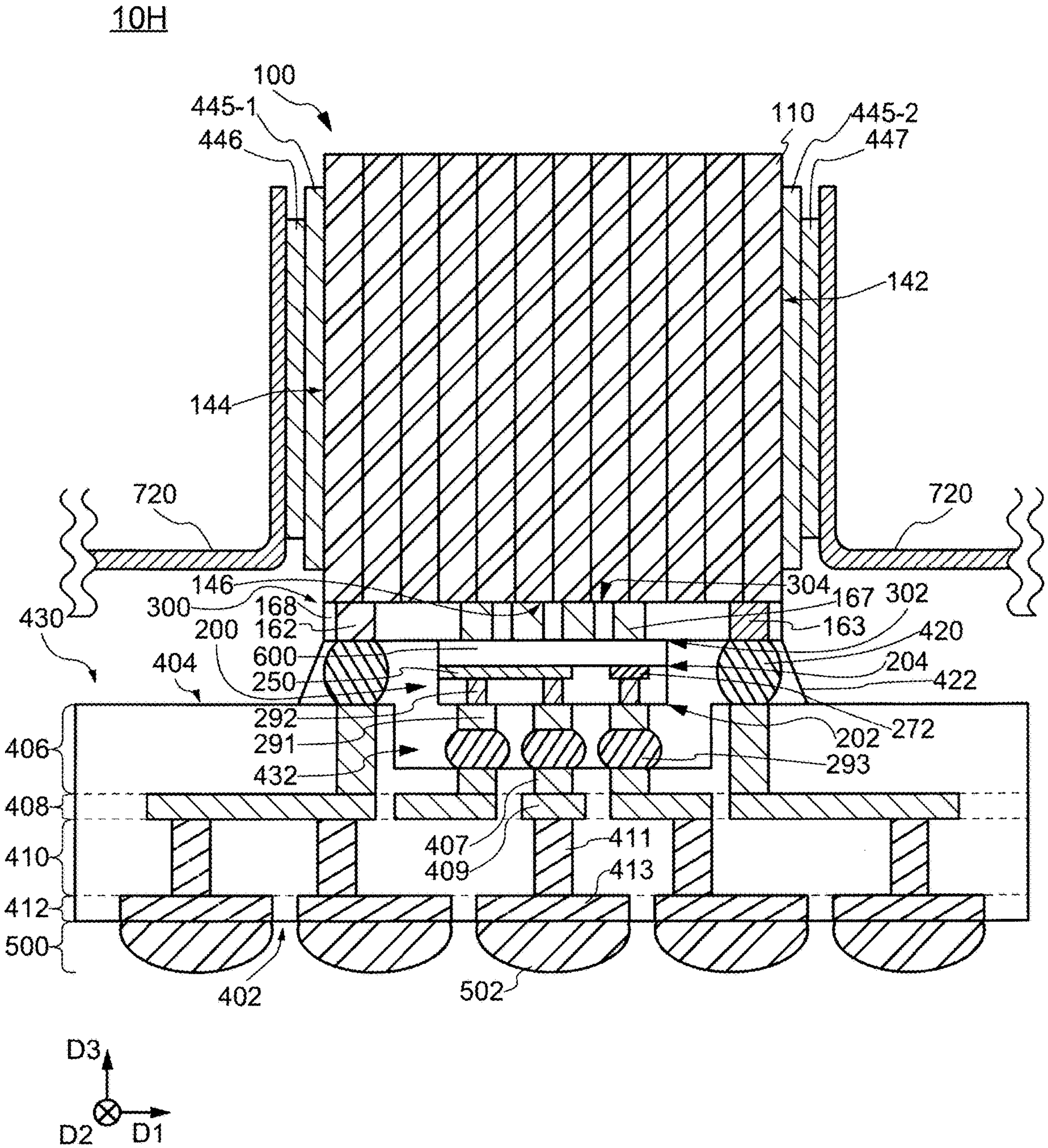
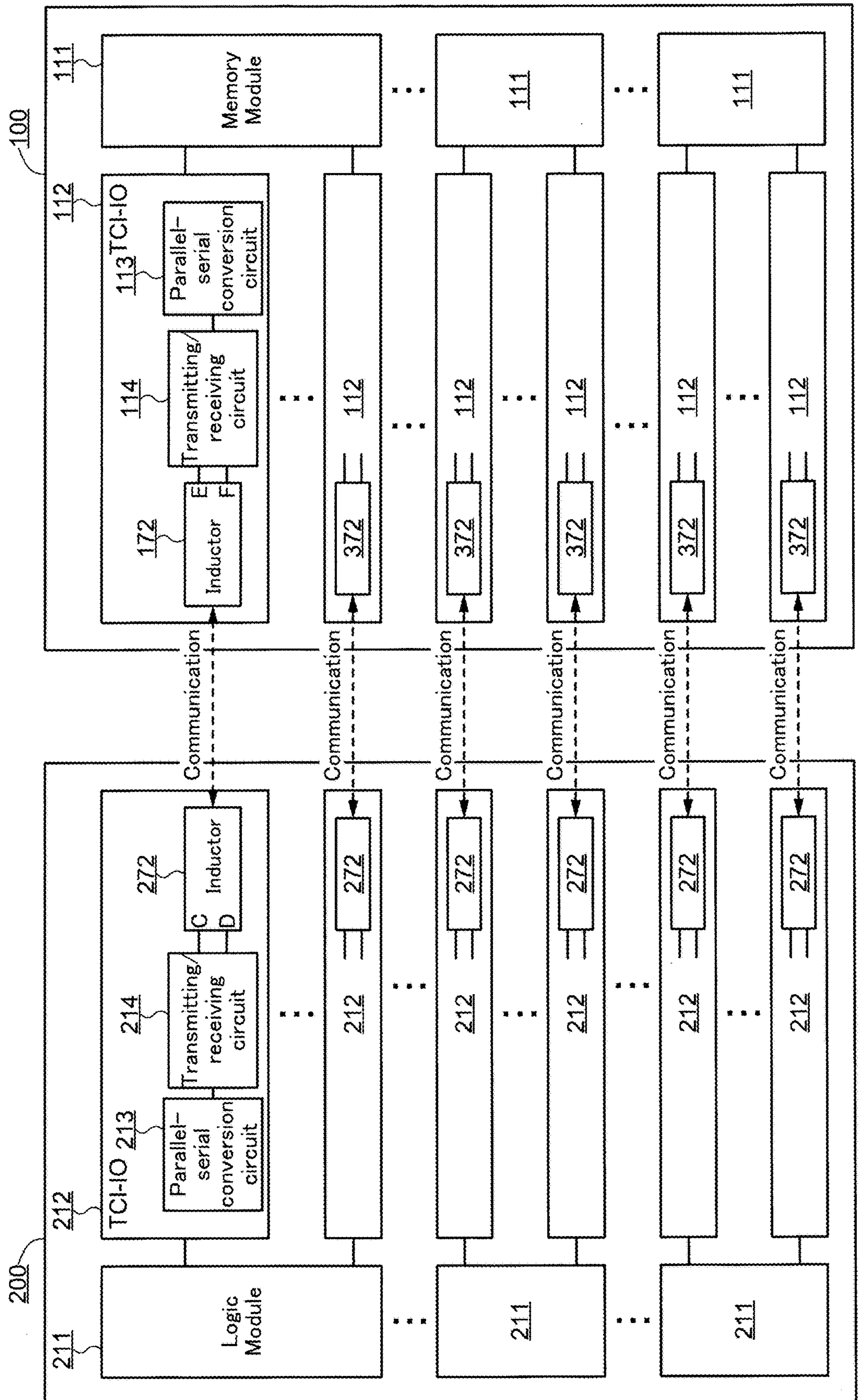

10H
100
445-1
446
110
445-2
447
142
144
720
720
146
304
167
302
300
168
163
430
200
162
420
600
204
404
250
422
292
202
272
406
291
293
432
408
407
411
409
413
410
412
500
402
502
D3
D2
D1

FIG. 28


100
111
Memory Module
112
TCI-IO
113
Parallel-serial conversion circuit
114
Transmitting/receiving circuit
172
Inductor
E
F
372
Communication
200
211
Logic Module
212
TCI-IO
213
Parallel-serial conversion circuit
214
Transmitting/receiving circuit
272
Inductor
C
D

10J

FIG. 29

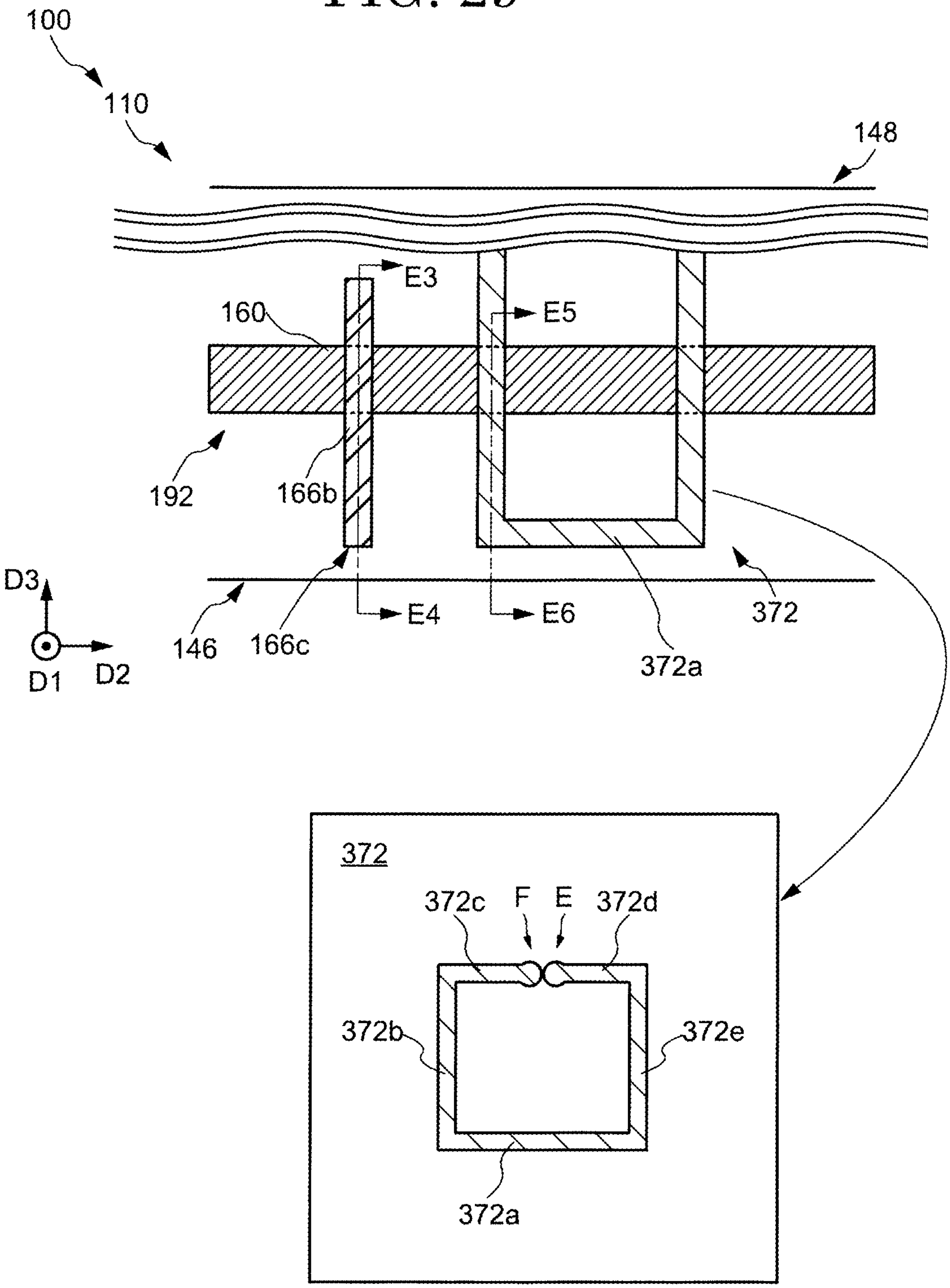
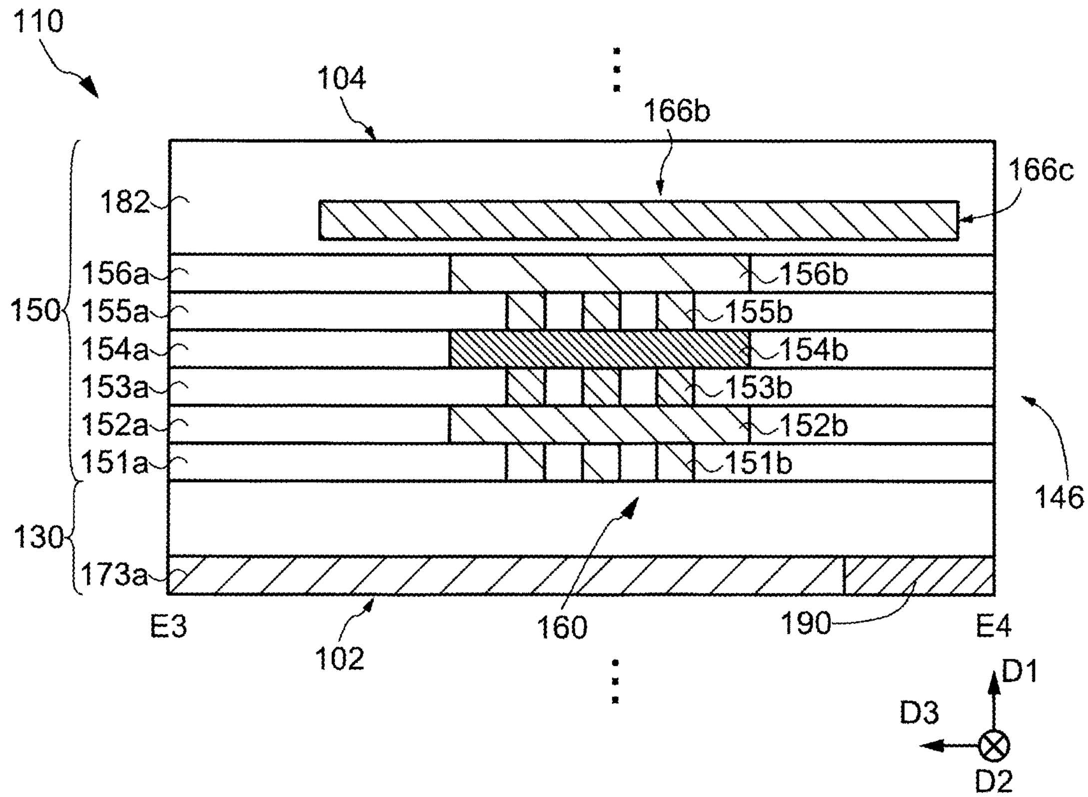

100
110
148
E3
E5
160
192
166b
D3
D1
D2
146
166c
E4
E6
372a
372
372
372c
F
E
372d
372b
372e
372a

FIG. 30



110
104
166b
166c
182
156a
156b
155a
155b
150
154a
154b
153a
153b
152a
152b
151a
151b
146
130
173a
E3
102
160
190
E4
D1
D3
D2

FIG. 31

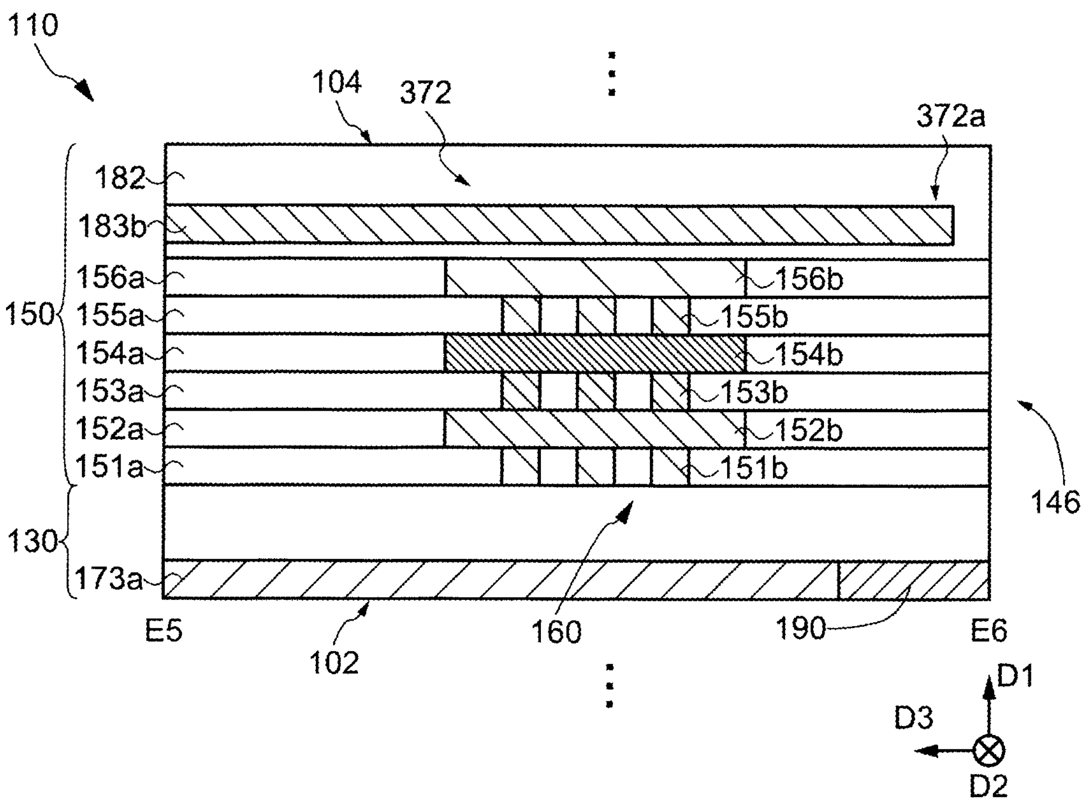

110
104
372
372a
182
183b
156a
156b
150
155a
155b
154a
154b
153a
153b
152a
152b
151a
151b
146
130
173a
E5
102
160
190
E6
D1
D3
D2

FIG. 32

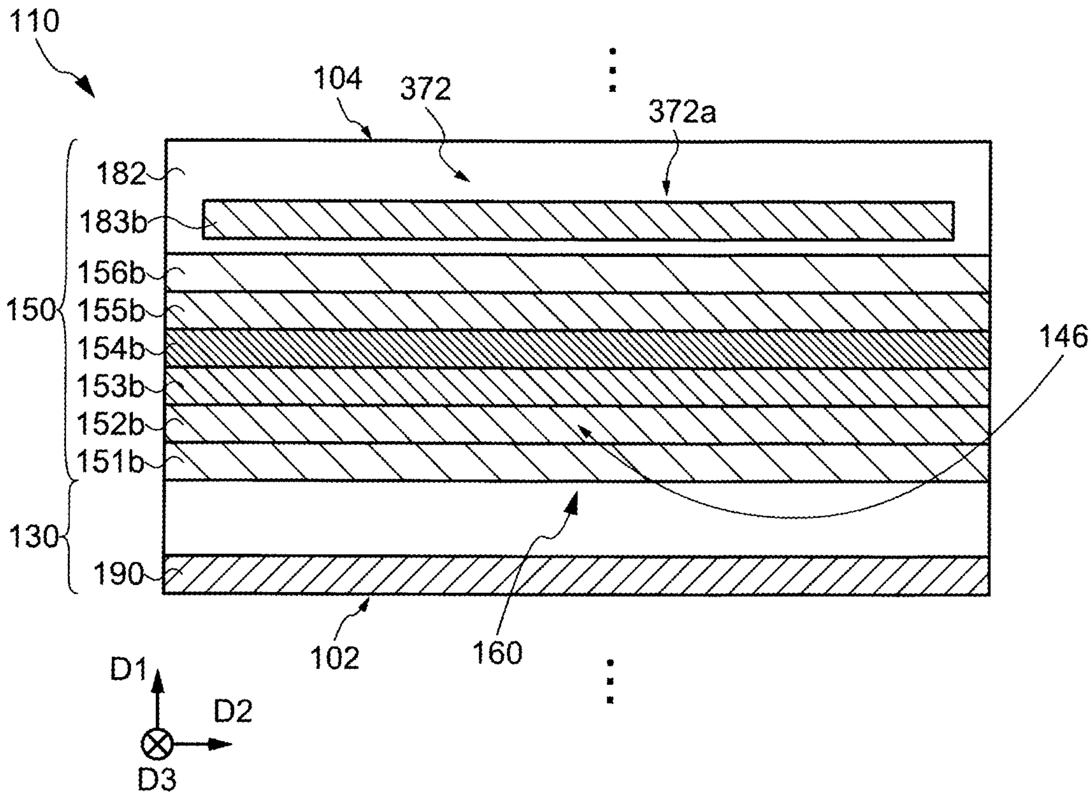

110
104
372
372a
182
183b
156b
155b
150
154b
153b
152b
151b
146
130
190
102
160
D1
D2
D3

FIG. 33

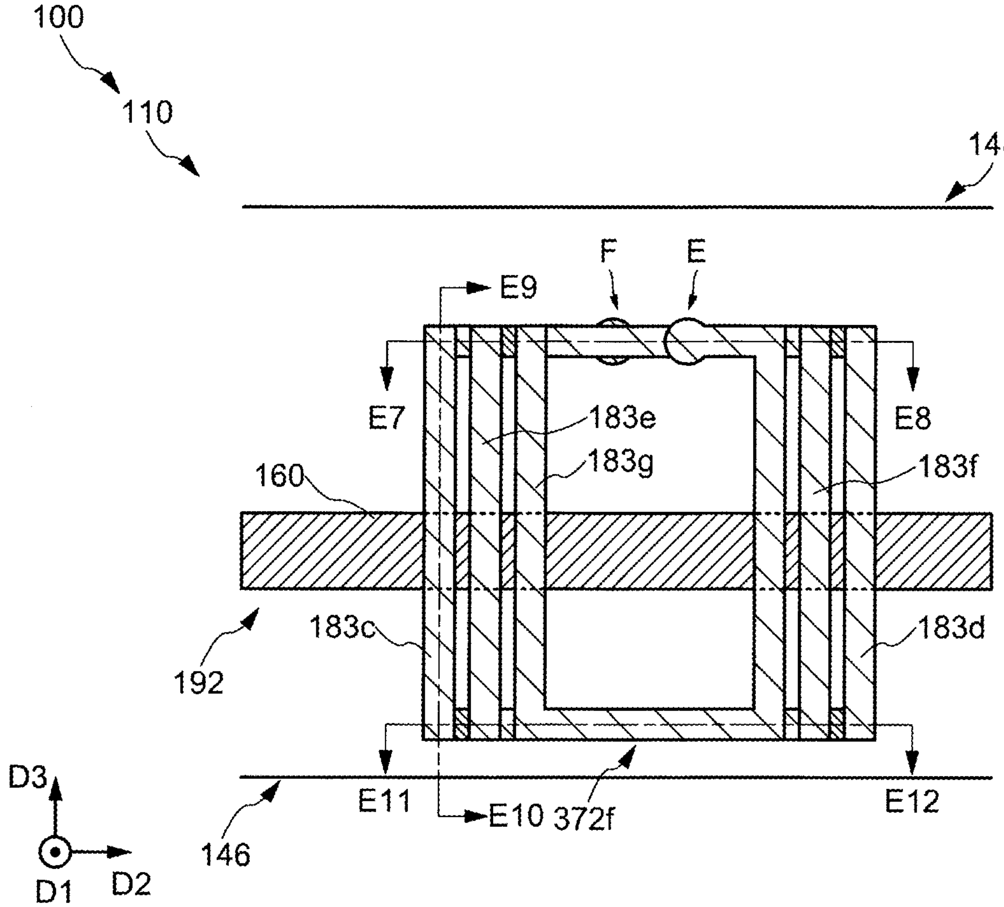

100
110
148
F
E
E9
E7
E8
183e
183g
183f
160
183c
183d
192
D3
D1
D2
146
E11
E10
372f
E12

FIG. 34

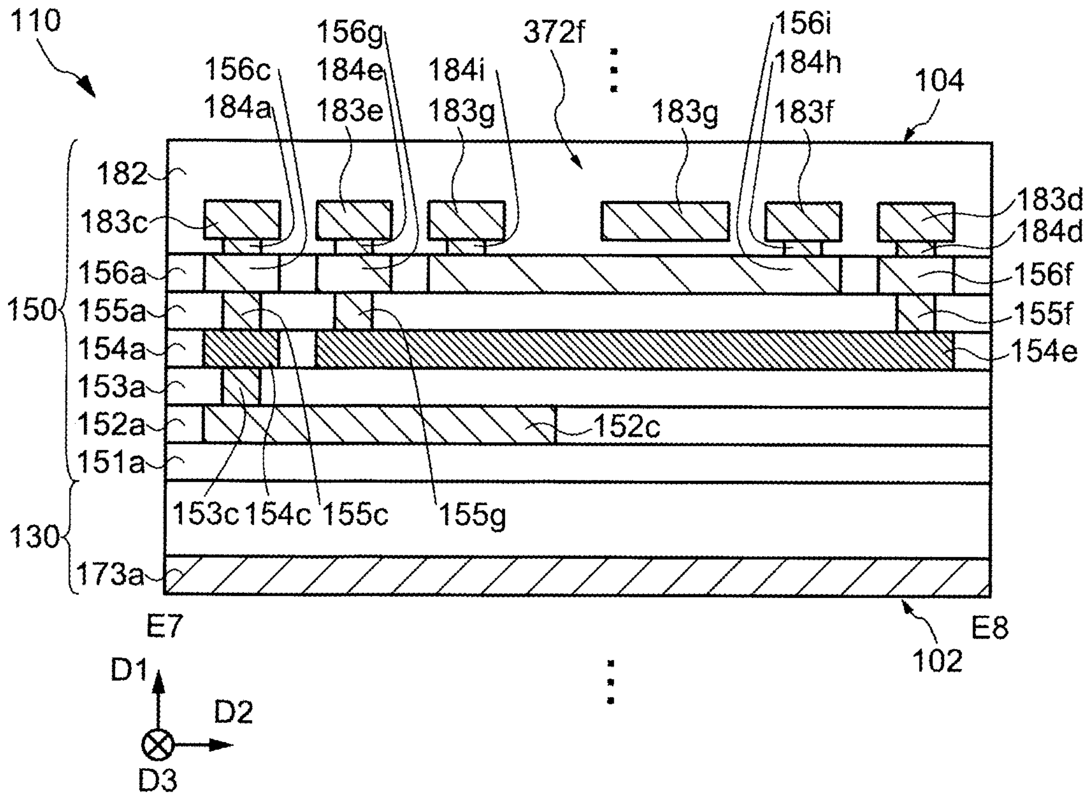

110
156g
372f
156i
156c
184e
184i
184h
104
184a
183e
183g
183g
183f
182
183c
183d
184d
156a
156f
150
155a
155f
154a
154e
153a
152a
152c
151a
153c
154c
155c
155g
130
173a
E7
E8
102
D1
D2
D3

FIG. 35

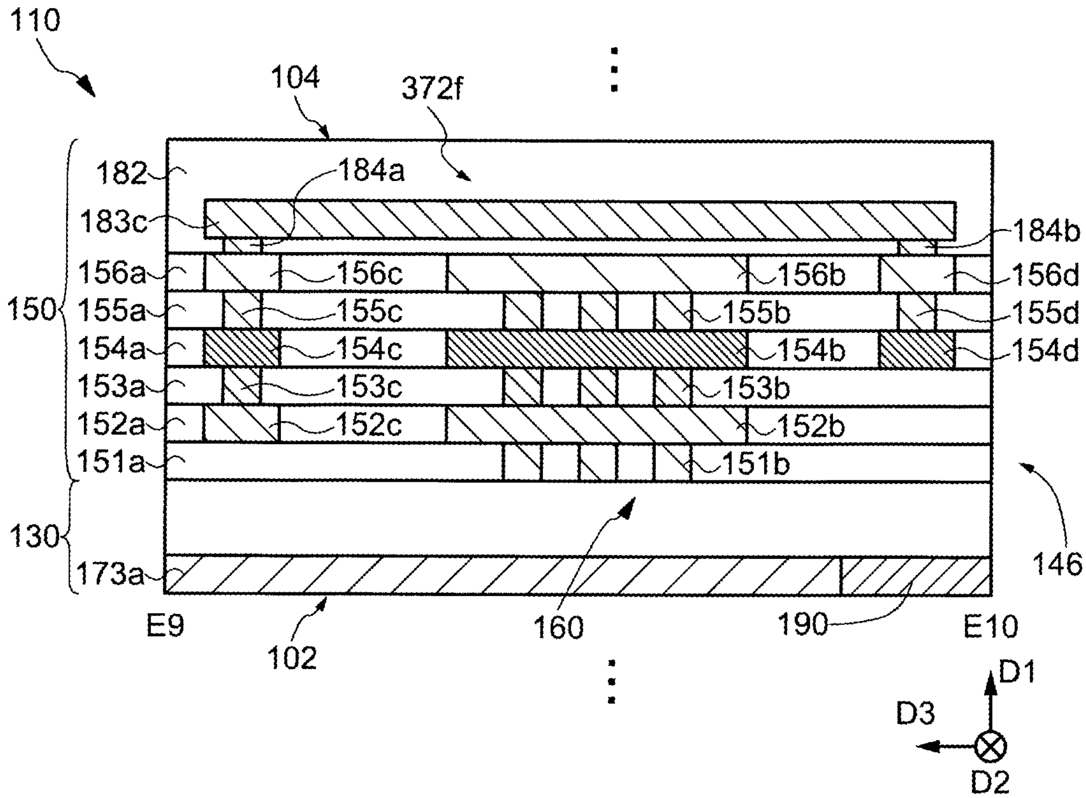

110
104
372f
182
184a
183c
184b
156a
156c
156b
156d
150
155a
155c
155b
155d
154a
154c
154b
154d
153a
153c
153b
152a
152c
152b
151a
151b
130
146
173a
E9
160
190
E10
102
D1
D3
D2

FIG. 36

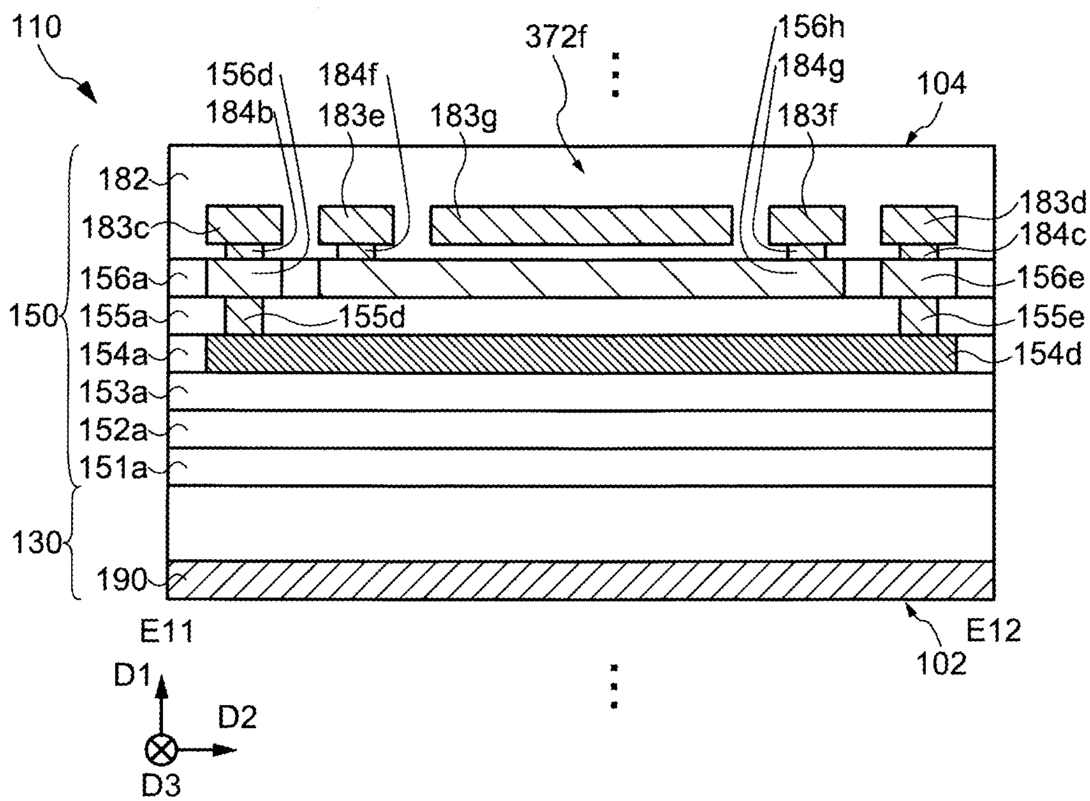

110
372f
156h
184g
156d
184f
184b
183e
183g
183f
104
182
183c
183d
184c
156a
156e
150
155a
155d
155e
154a
154d
153a
152a
151a
130
190
E11
E12
102
D1
D2
D3

FIG. 37

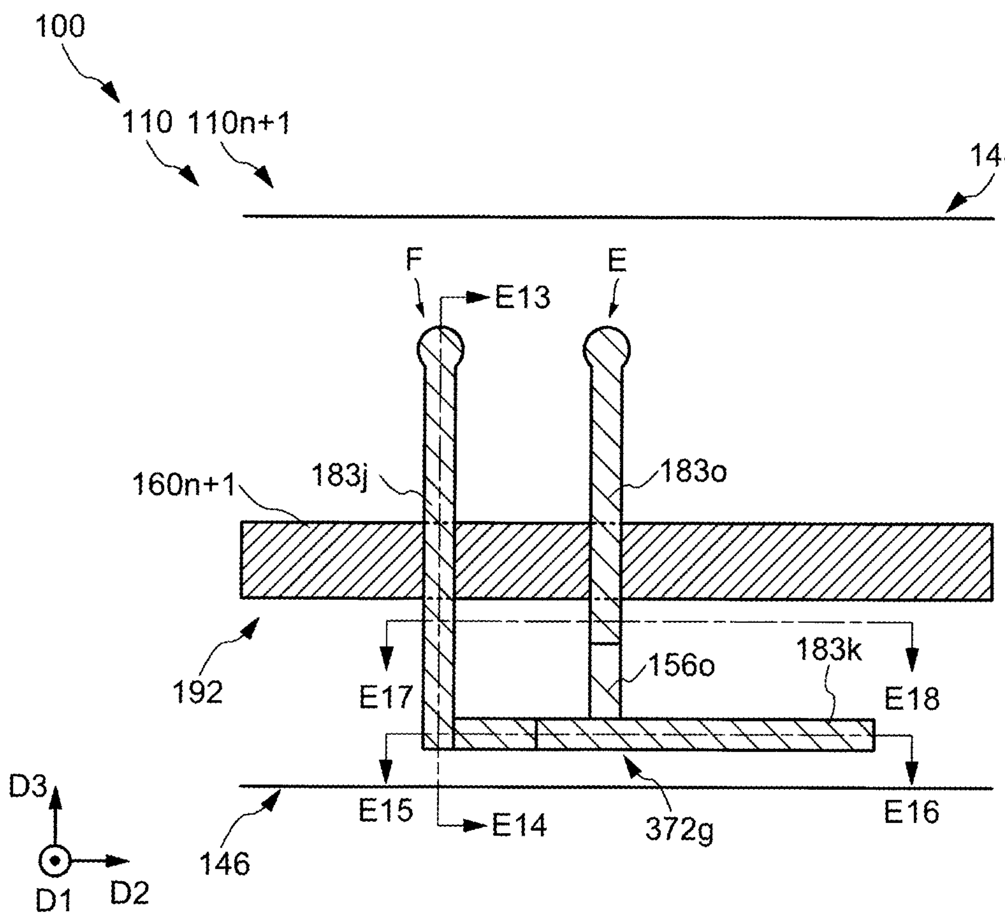

100
110
110n+1
148
F
E
E13
183j
183o
160n+1
192
E17
156o
183k
E18
E15
E14
372g
E16
146
D3
D1
D2

FIG. 38

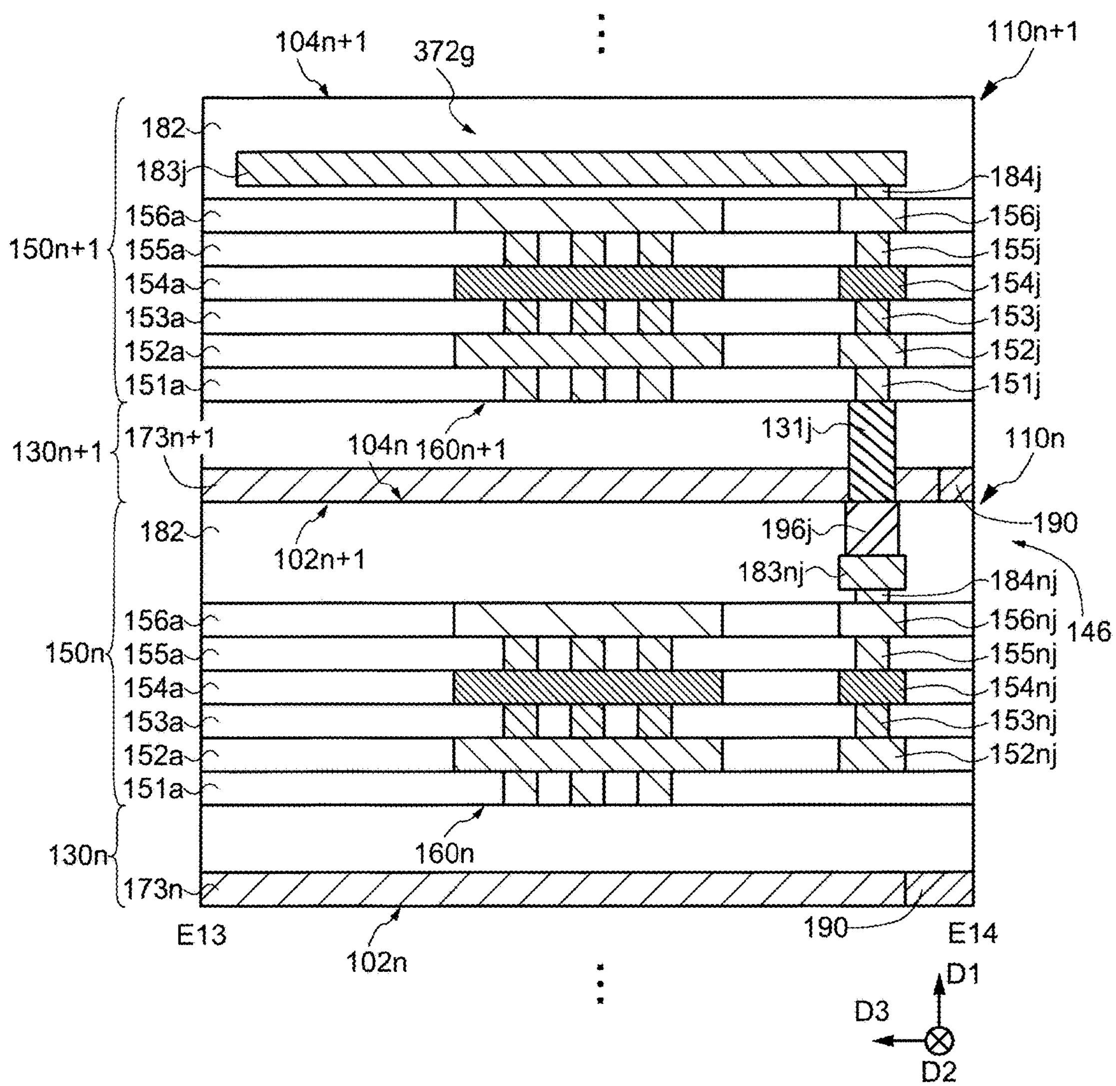

104n+1
372g
110n+1
182
183j
184j
156a
156j
155a
155j
150n+1
154a
154j
153a
153j
152a
152j
151a
151j
131j
130n+1
173n+1
104n
160n+1
110n
190
182
196j
102n+1
183nj
184nj
156a
156nj
146
155a
155nj
150n
154a
154nj
153a
153nj
152a
152nj
151a
130n
160n
173n
E13
190
E14
102n
D1
D3
D2

FIG. 39

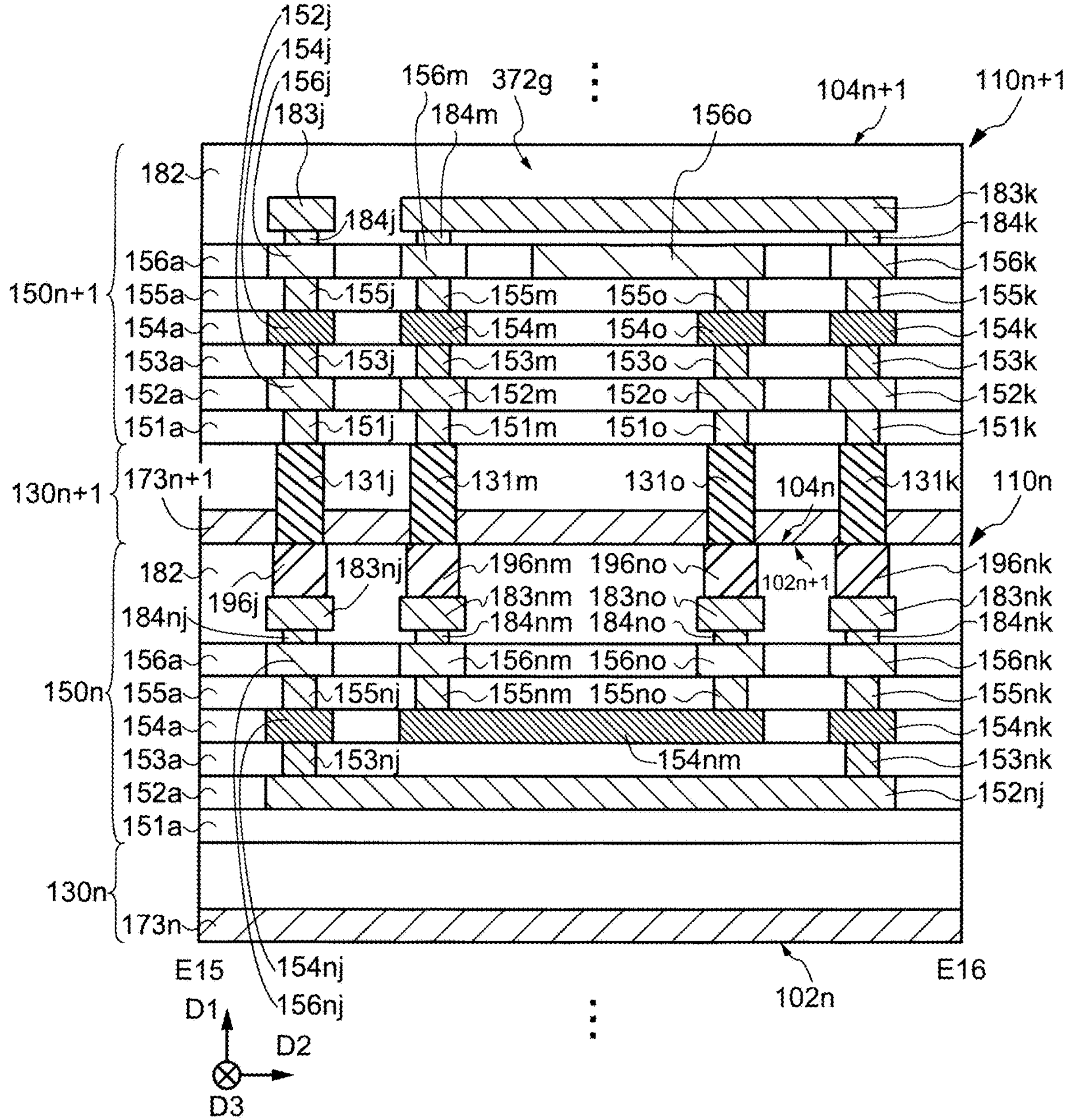

152j
154j
156j
183j
156m
184m
372g
104n+1
110n+1
156o
182
183k
184j
184k
156a
156k
150n+1
155a
155j
155m
155o
155k
154a
154m
154o
154k
153a
153j
153m
153o
153k
152a
152m
152o
152k
151a
151j
151m
151o
151k
173n+1
131j
131m
131o
104n
131k
110n
130n+1
182
196nm
196no
196nk
183nj
102n+1
196j
183nm
183no
183nk
184nj
184nm
184no
184nk
156a
156nm
156no
156nk
150n
155a
155nj
155nm
155no
155nk
154a
154nk
153a
153nj
154nm
153nk
152a
152nj
151a
130n
173n
E15
154nj
E16
156nj
102n
D1
D2
D3

FIG. 40

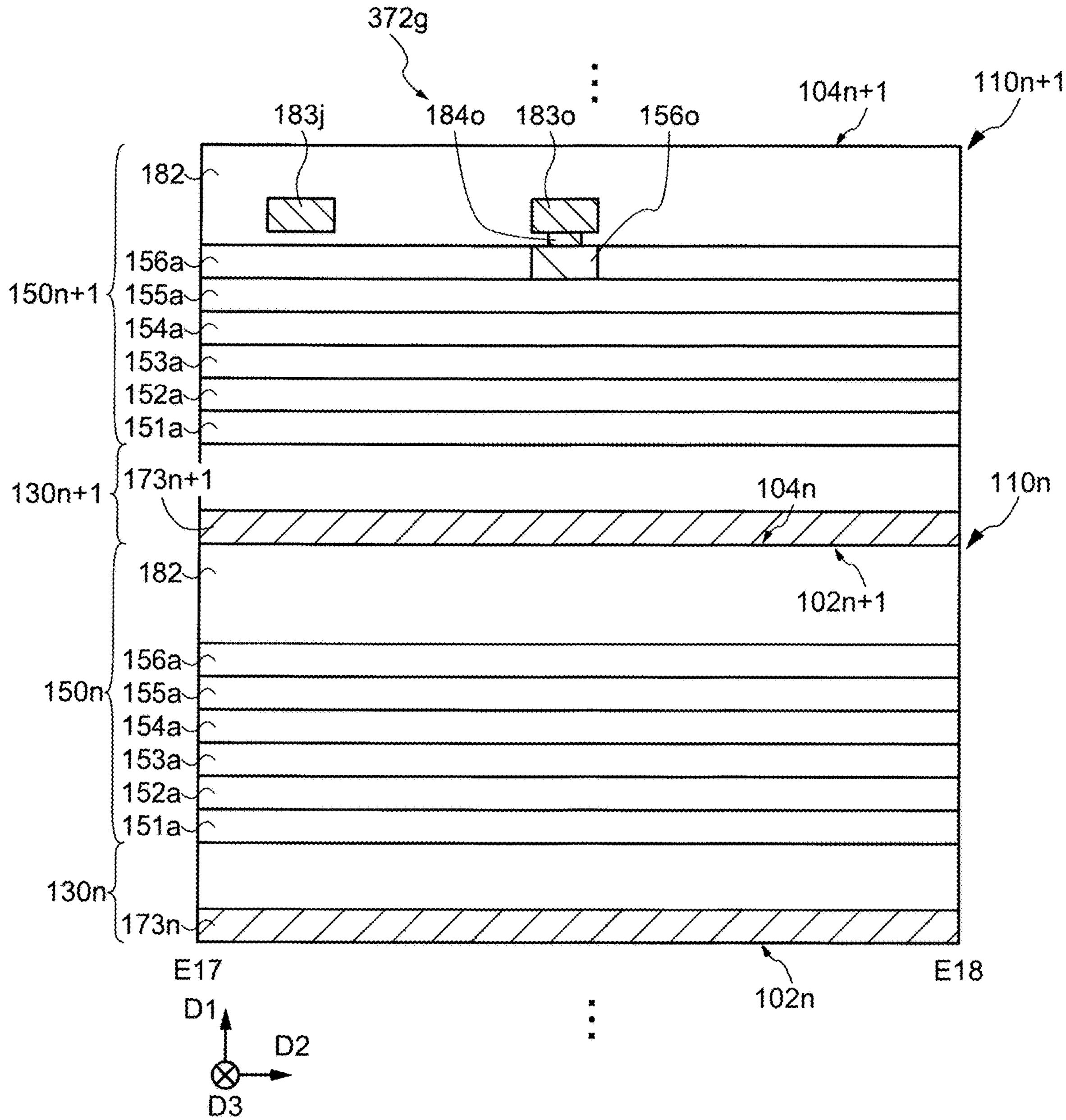
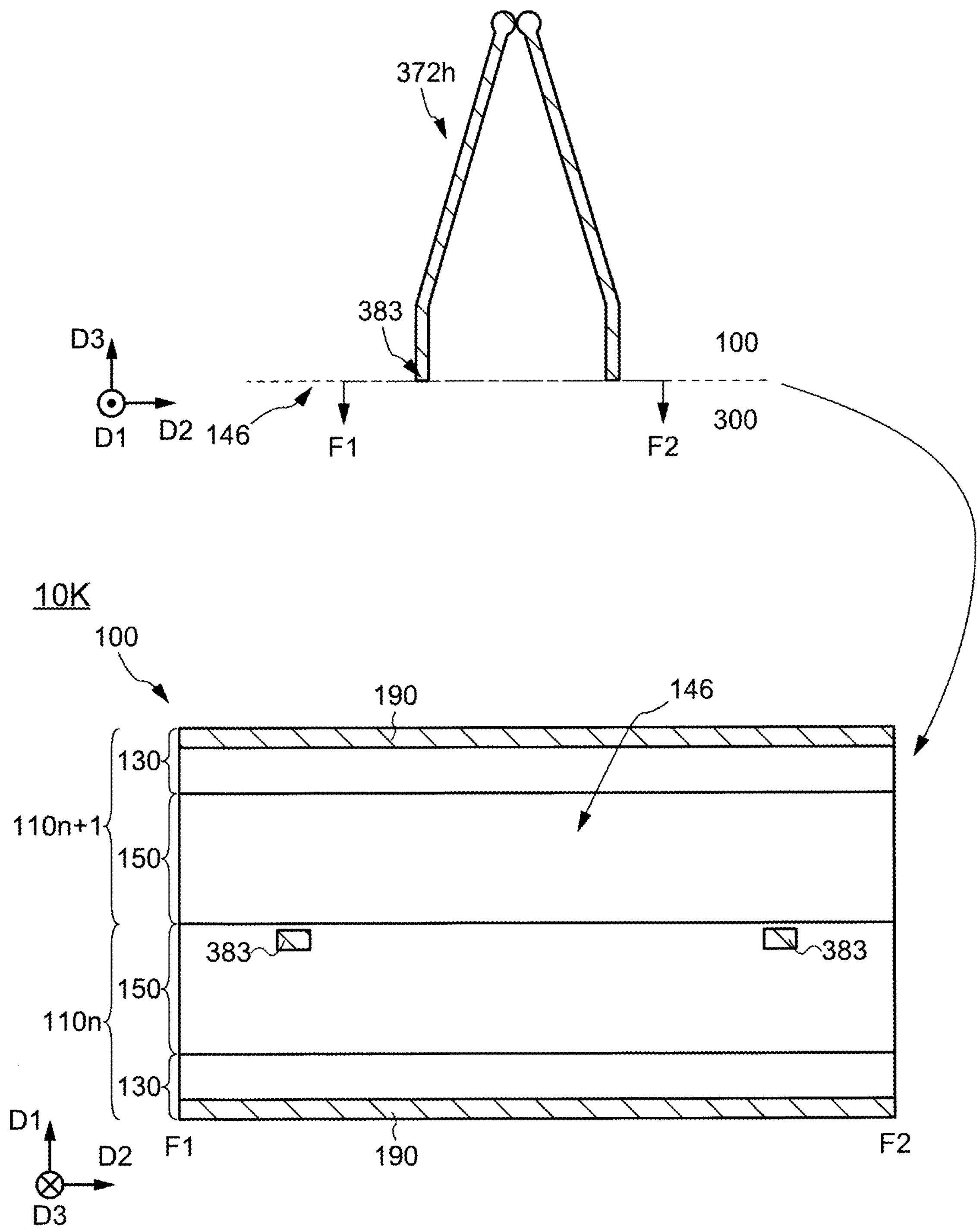
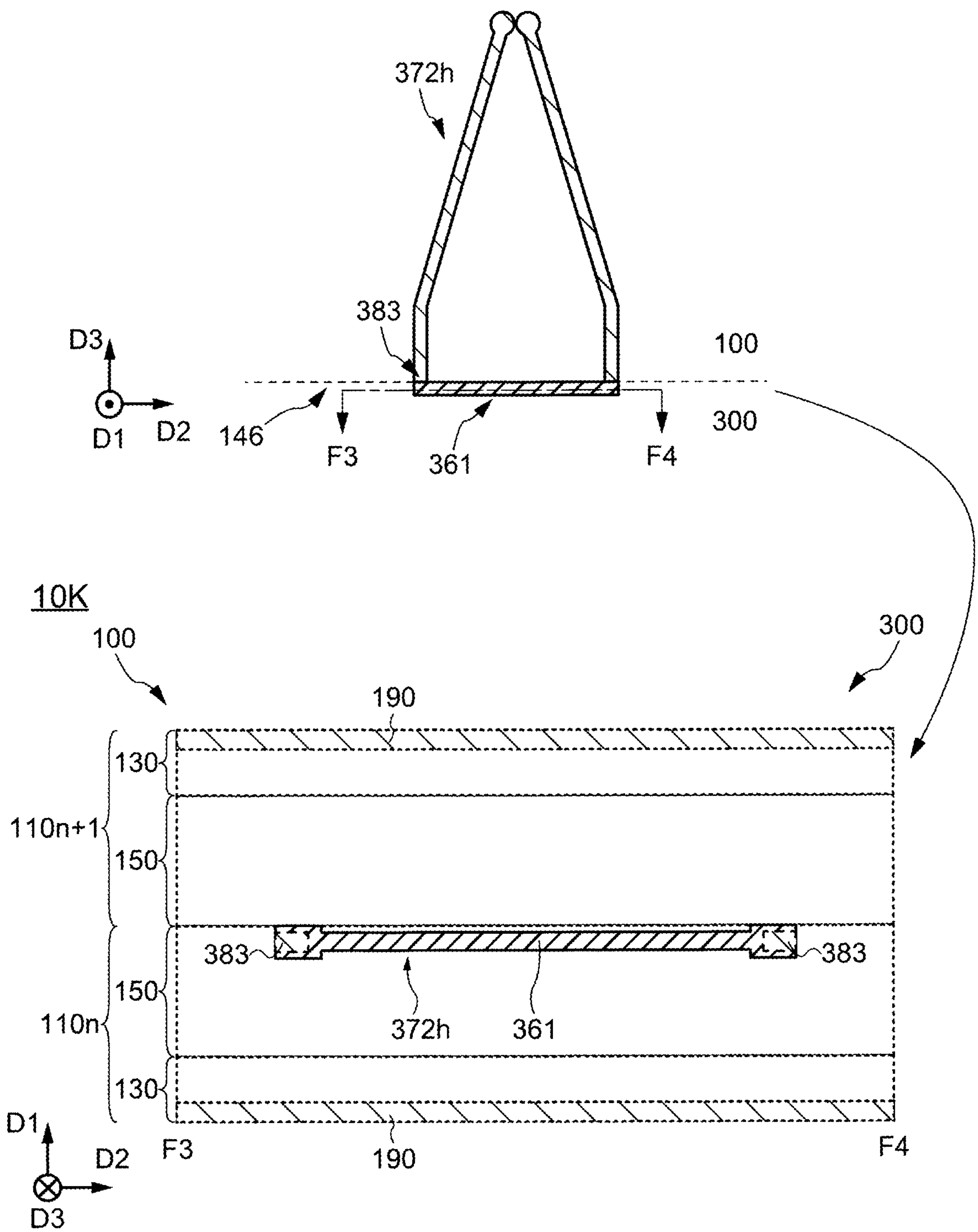

372g
183j
184o
183o
156o
104n+1
110n+1
182
156a
155a
154a
153a
152a
151a
150n+1
130n+1
173n+1
104n
110n
102n+1
182
156a
155a
154a
153a
152a
151a
150n
130n
173n
E17
E18
102n
D1
D2
D3

FIG. 41



372h
383
D3
100
D1
D2
146
F1
F2
300
10K
100
190
146
130
110n+1
150
383
383
150
110n
130
D1
D2
F1
190
F2
D3

FIG. 42



372h
383
100
D3
D1
D2
146
F3
361
F4
300
10K
100
300
190
130
110n+1
150
150
383
383
372h
361
110n
130
190
D1
D2
D3
F3
F4

FIG. 43

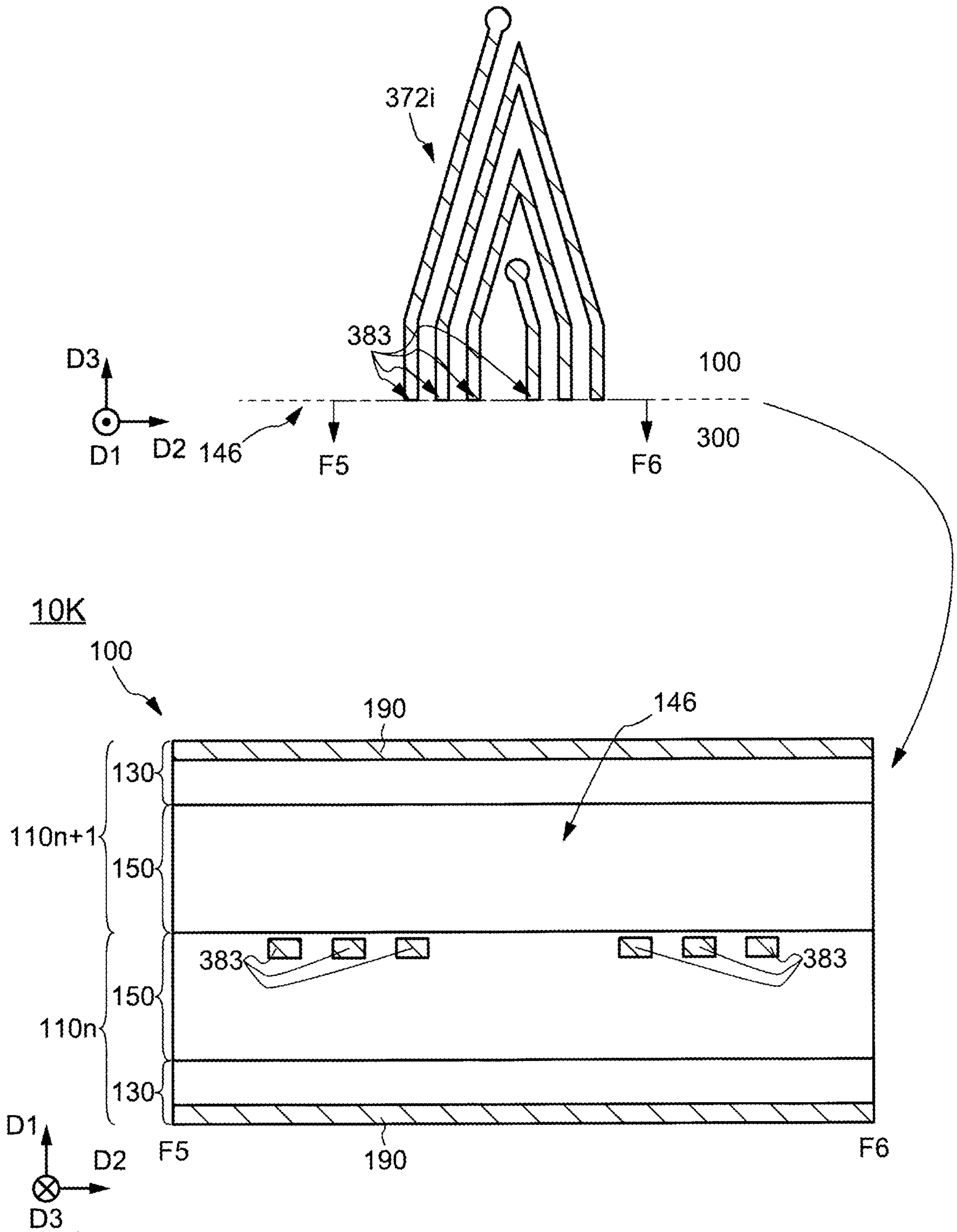
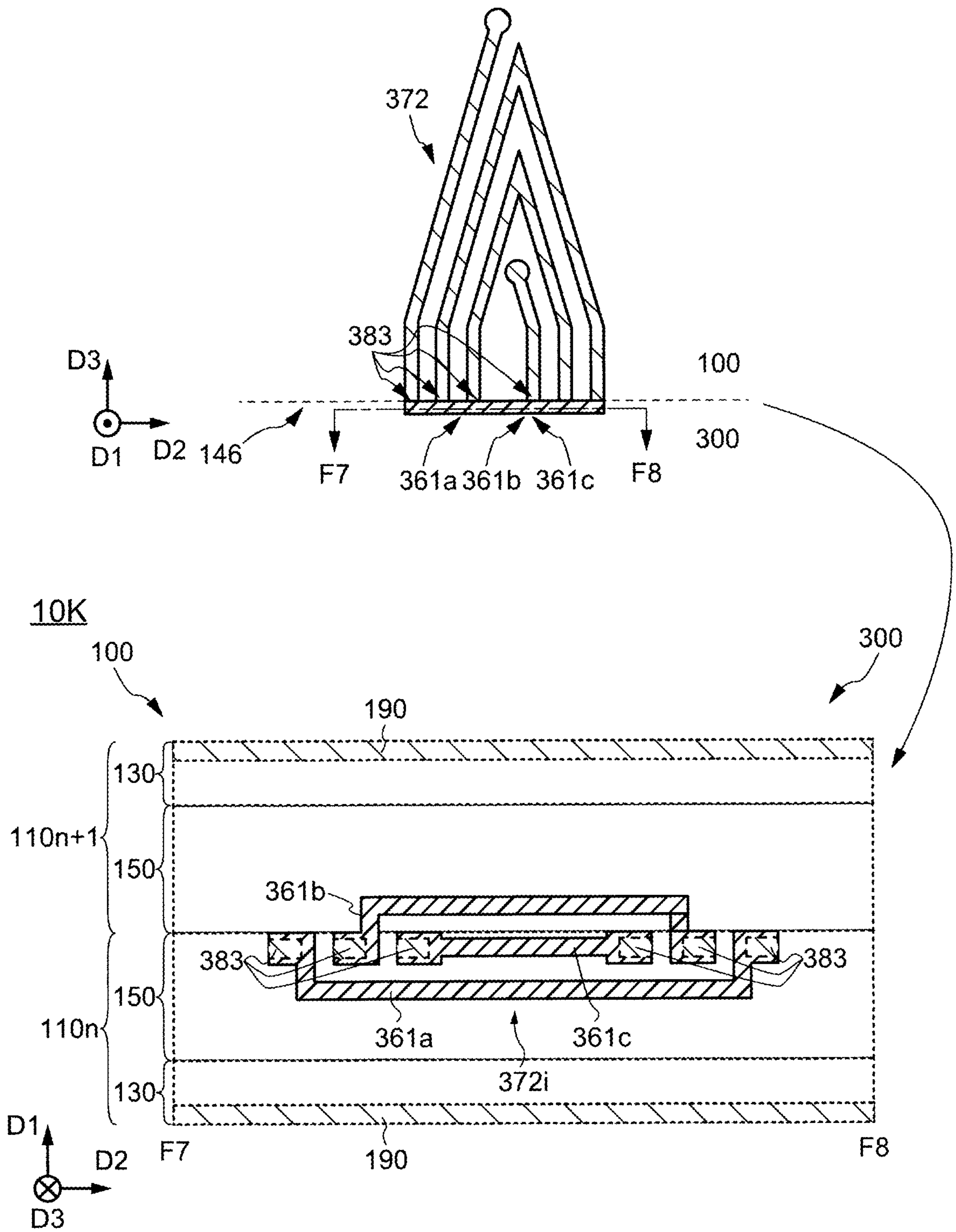

372i
383
D3
D1
D2
146
F5
F6
100
300
10K
100
190
146
130
110n+1
150
383
150
110n
130
383
D1
D2
D3
F5
190
F6

FIG. 44



372
383
100
D3
D1
D2
146
300
F7
361a
361b
361c
F8
10K
100
300
190
130
110n+1
150
361b
383
383
150
110n
361a
361c
372i
130
190
D1
D2
D3
F7
F8

FIG. 45

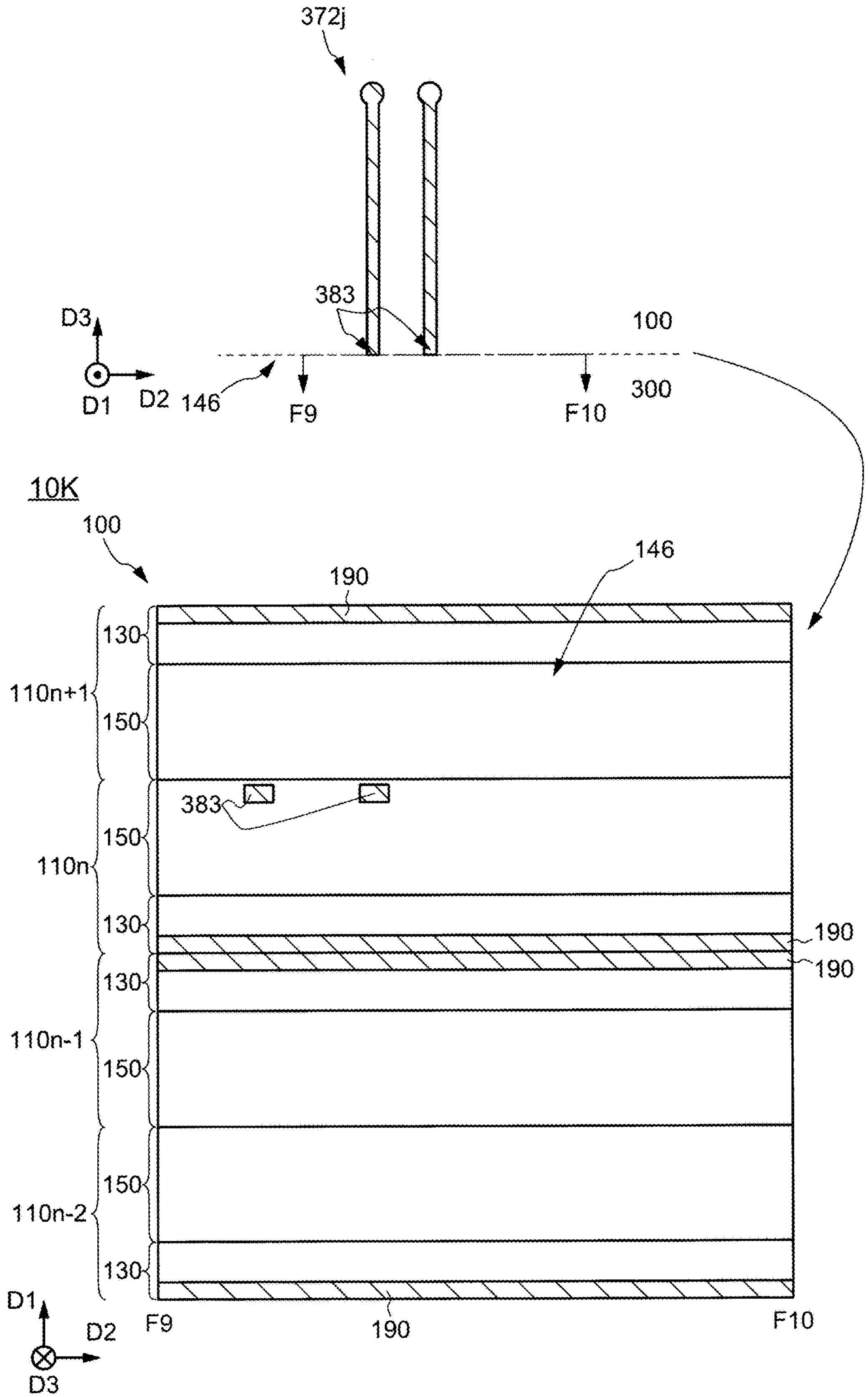

372j
383
100
D3
D1
D2
146
F9
F10
300
10K
100
146
190
130
110n+1
150
383
150
110n
130
190
190
130
110n-1
150
150
110n-2
130
D1
D2
D3
F9
190
F10

FIG. 46

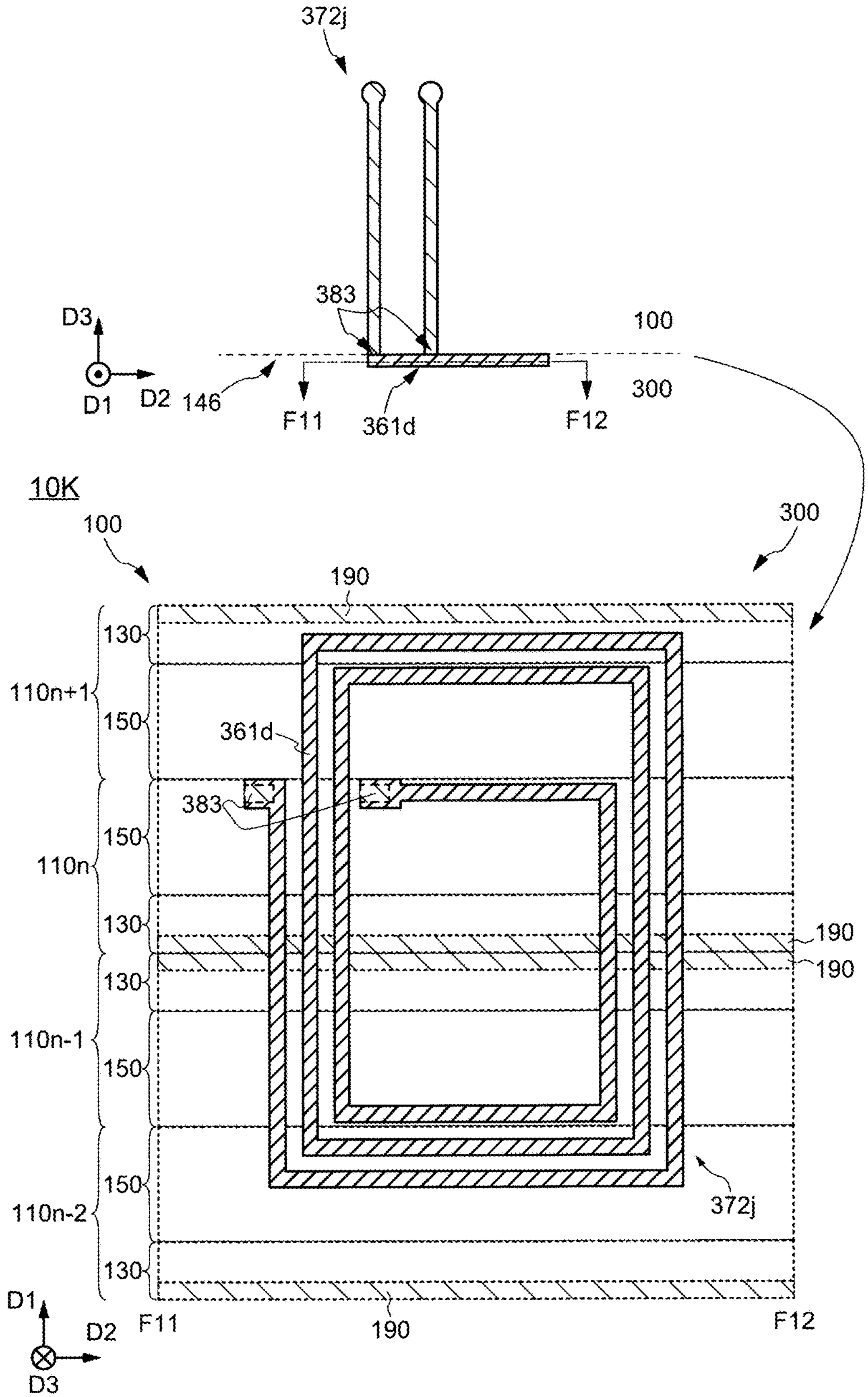

372j
383
100
D3
D1
D2
146
F11
361d
F12
300
10K
100
300
190
130
110n+1
150
361d
383
150
110n
130
190
190
130
110n-1
150
150
110n-2
372j
130
D1
D2
D3
F11
190
F12

FIG. 47

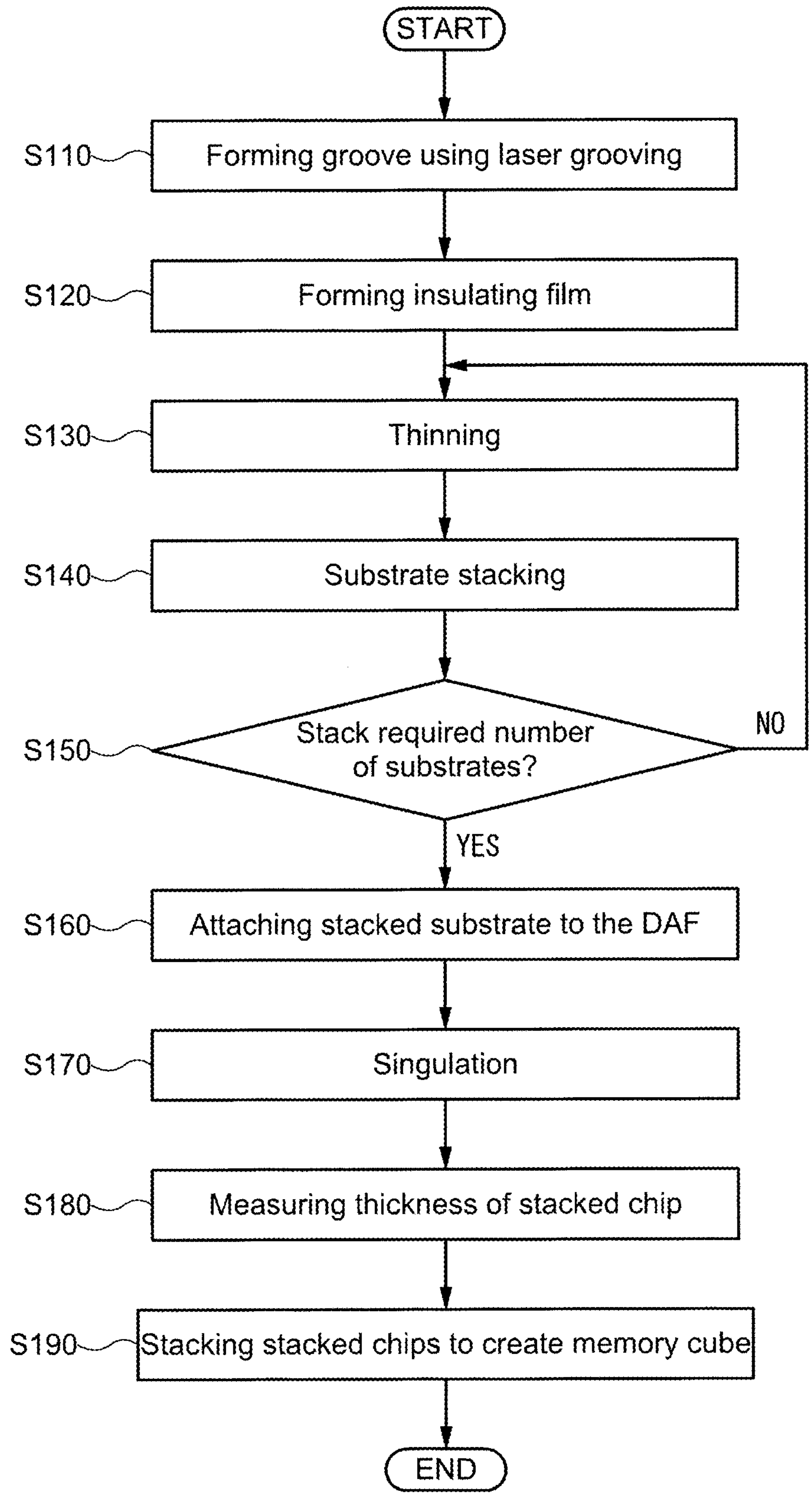

START
S110
Forming groove using laser grooving
S120
Forming insulating film
S130
Thinning
S140
Substrate stacking
S150
Stack required number of substrates?
NO
YES
S160
Attaching stacked substrate to the DAF
S170
Singulation
S180
Measuring thickness of stacked chip
S190
Stacking stacked chips to create memory cube
END

FIG. 48

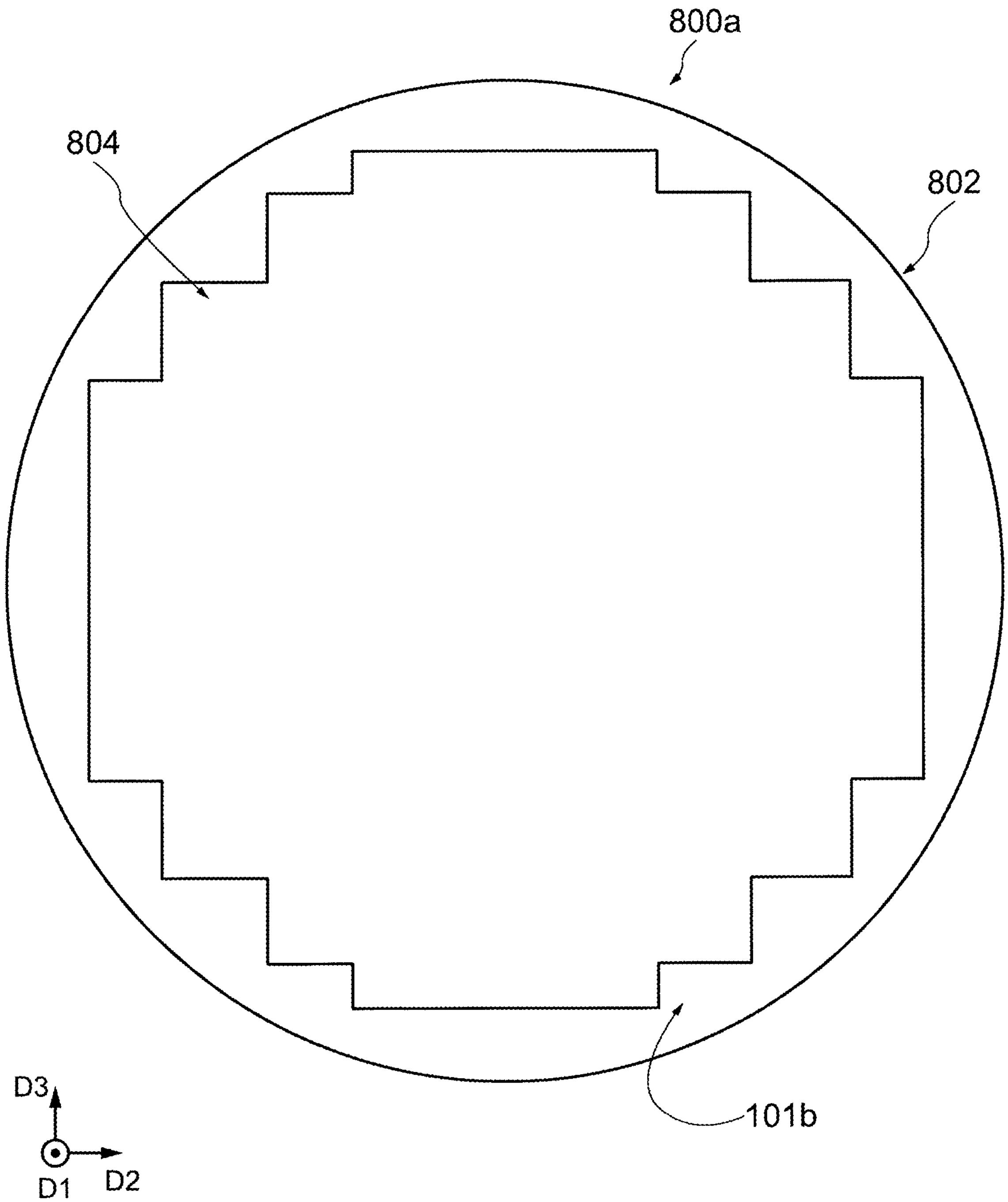


FIG. 49

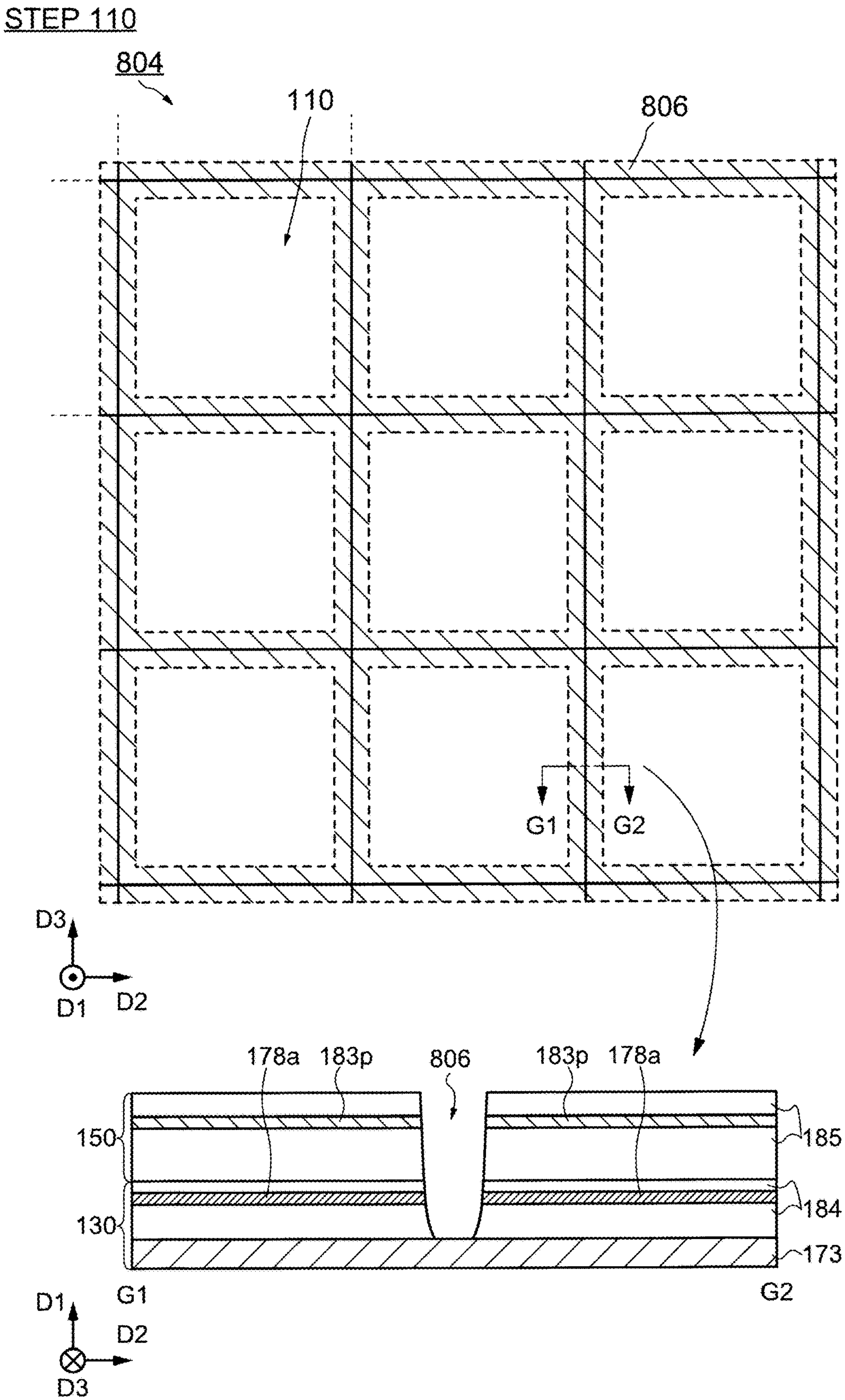

STEP 110
804
110
806
G1
G2
D3
D1
D2
178a
183p
806
183p
178a
150
185
130
184
173
G1
G2
D1
D2
D3

FIG. 50

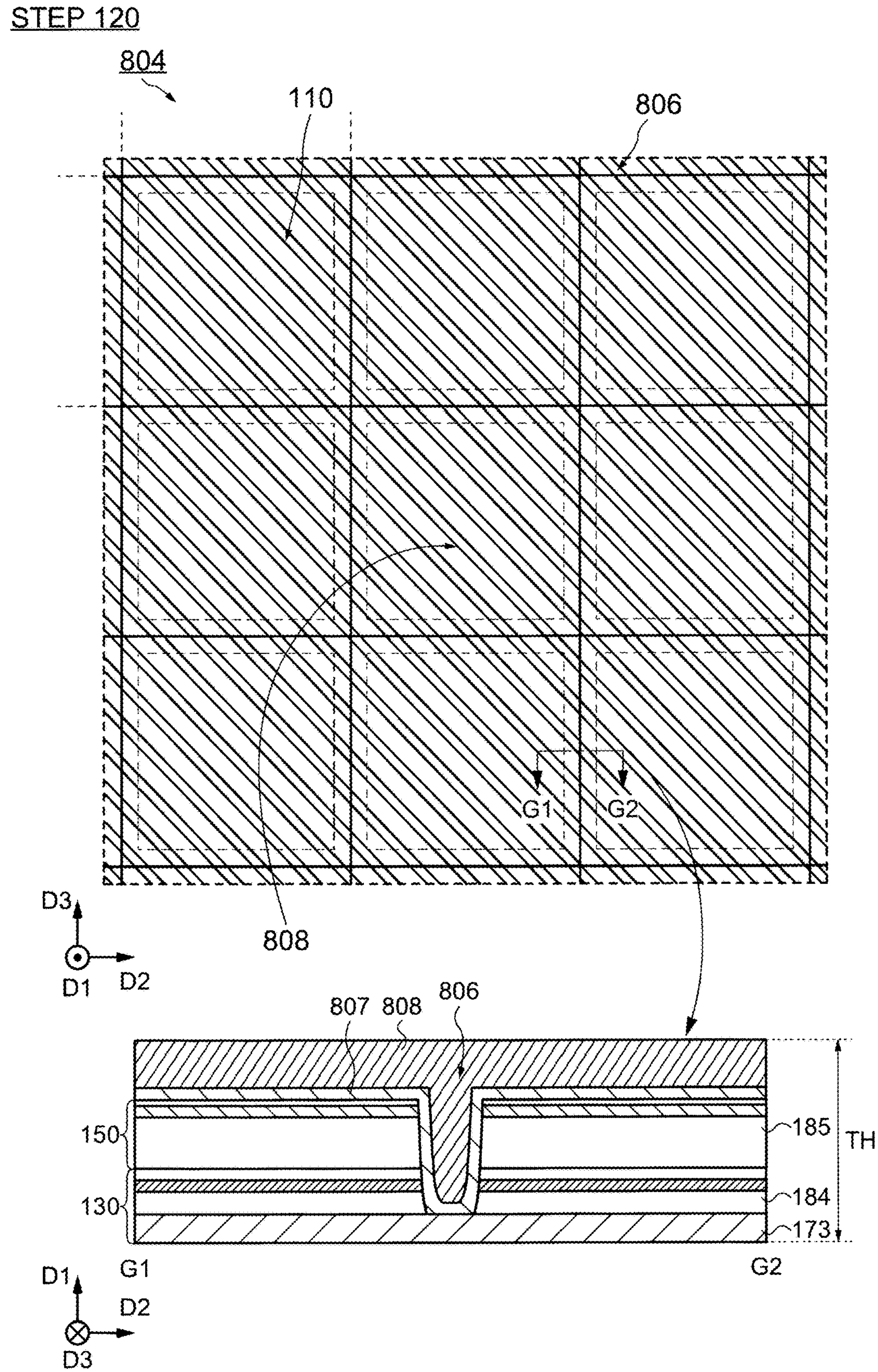
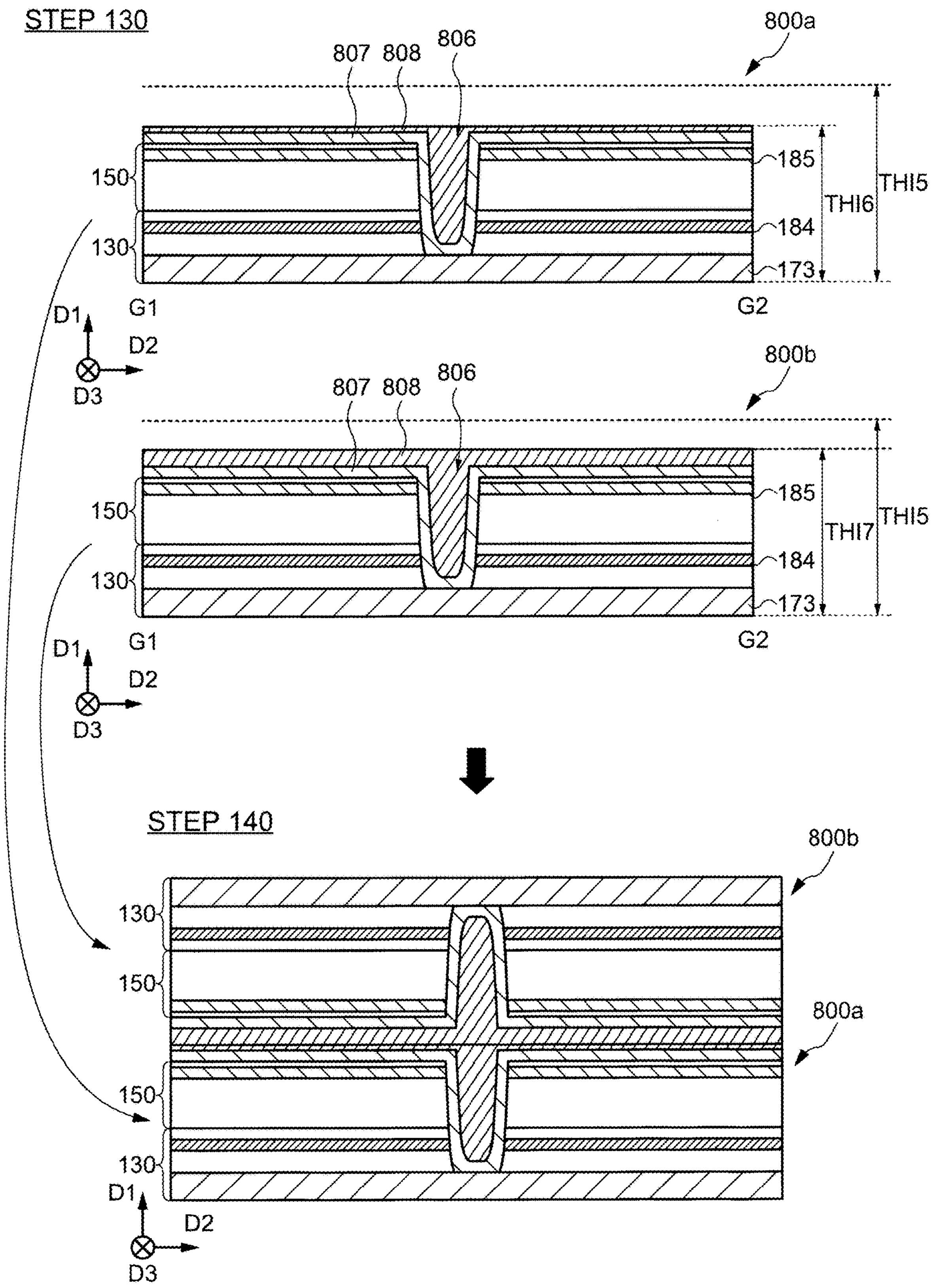

STEP 120
804
110
806
G1
G2
808
D3
D1
D2
807
808
806
150
130
185
184
173
THI5
G1
G2
D1
D2
D3

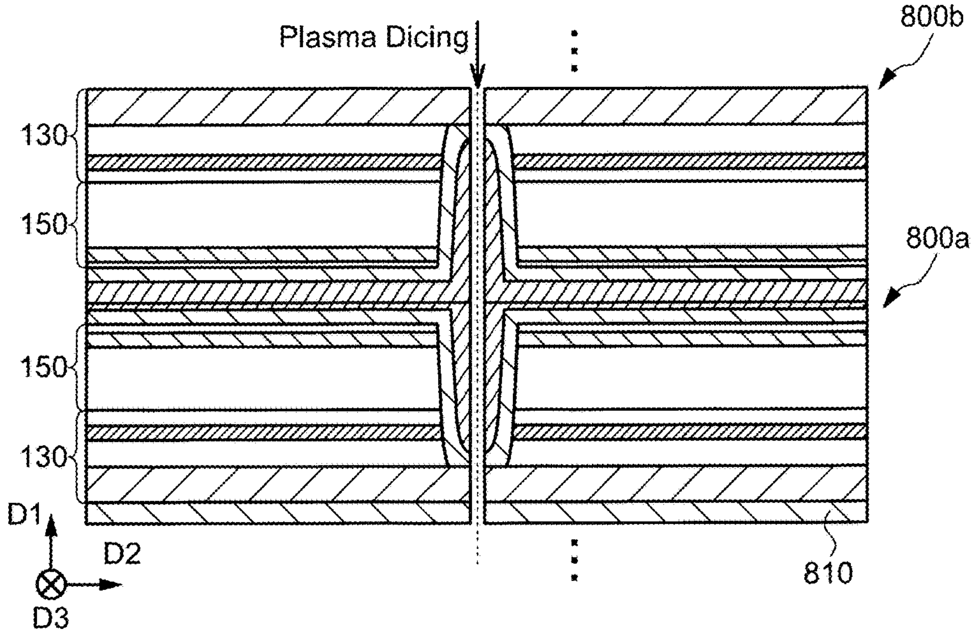
FIG. 51



STEP 130
800a
807 808
806
185
THI5
150
THI6
184
130
173
G1
G2
D1
D2
D3
800b
807 808
806
185
150
THI7
THI5
184
130
173
G1
G2
D1
D2
D3
STEP 140
800b
130
150
800a
150
130
D1
D2
D3

FIG. 52

STEP170



STEP 180

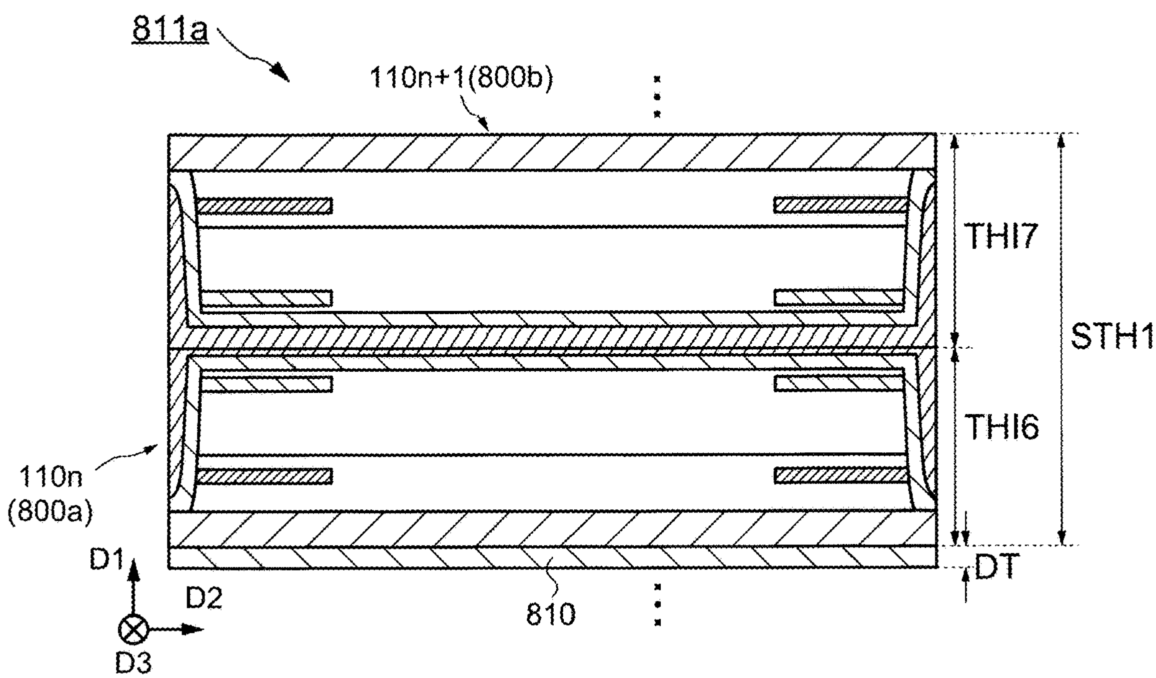
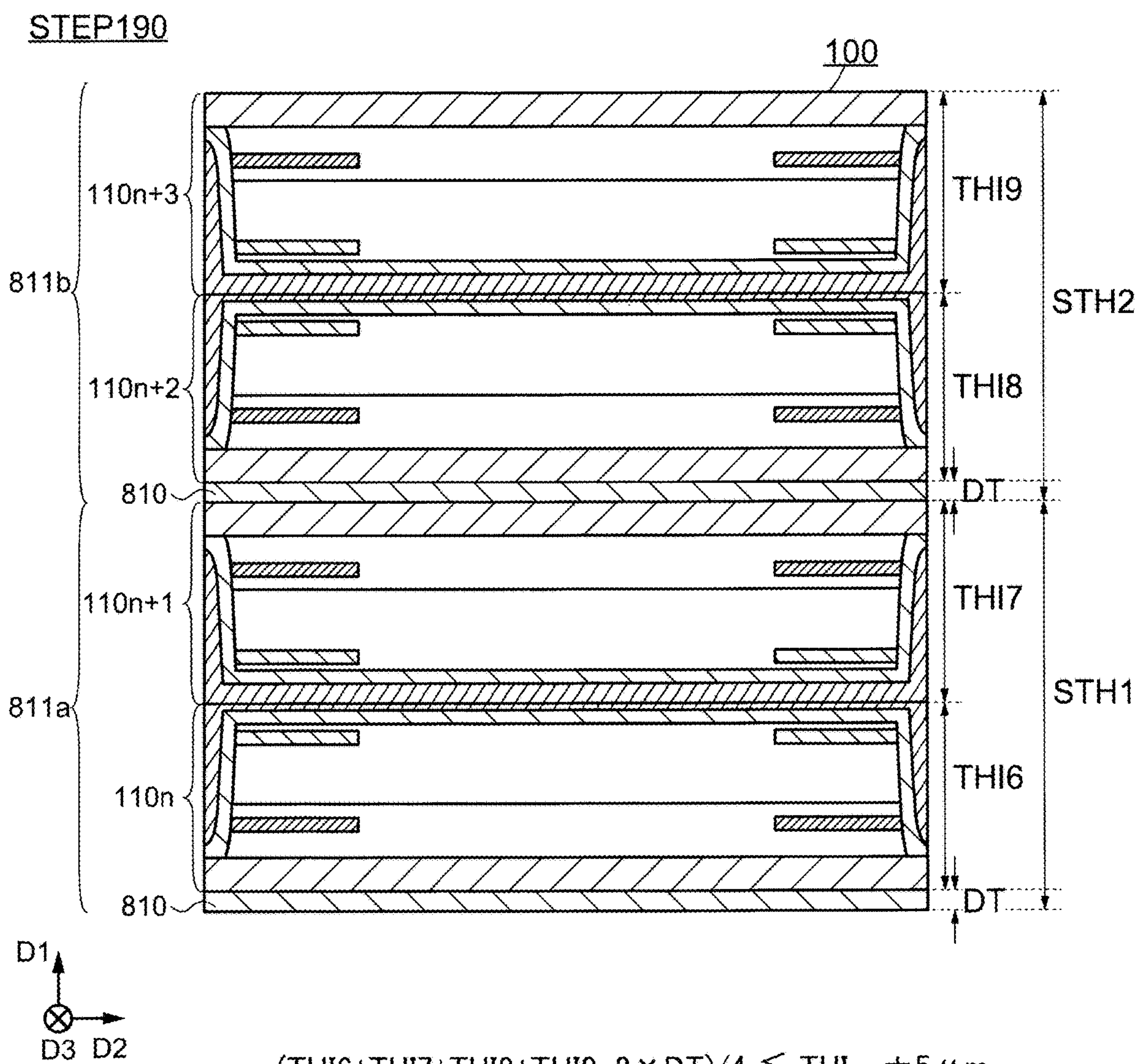


FIG. 53



$(THI6+THI7+THI8+THI9-2\times DT)/4 \leqq THI_{ave} \pm 5\,\mu m$

FIG. 54

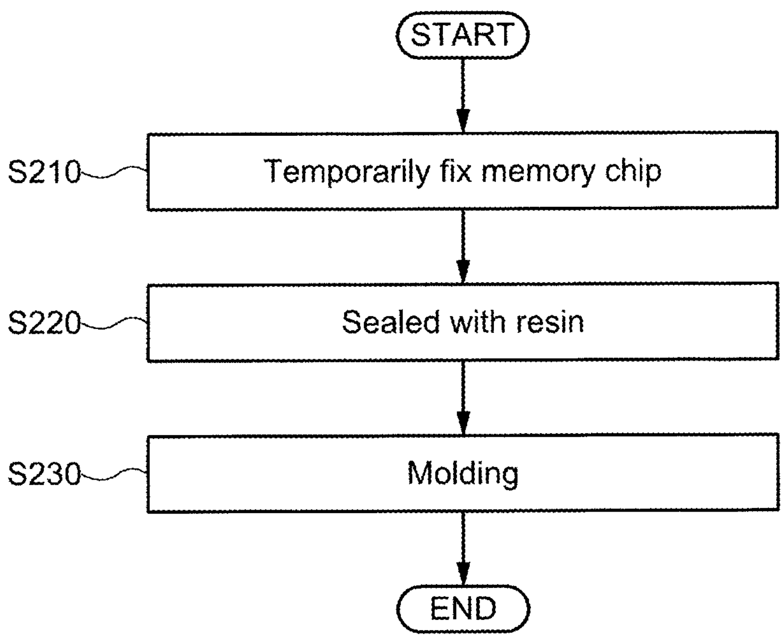

START
S210
Temporarily fix memory chip
S220
Sealed with resin
S230
Molding
END

FIG. 55

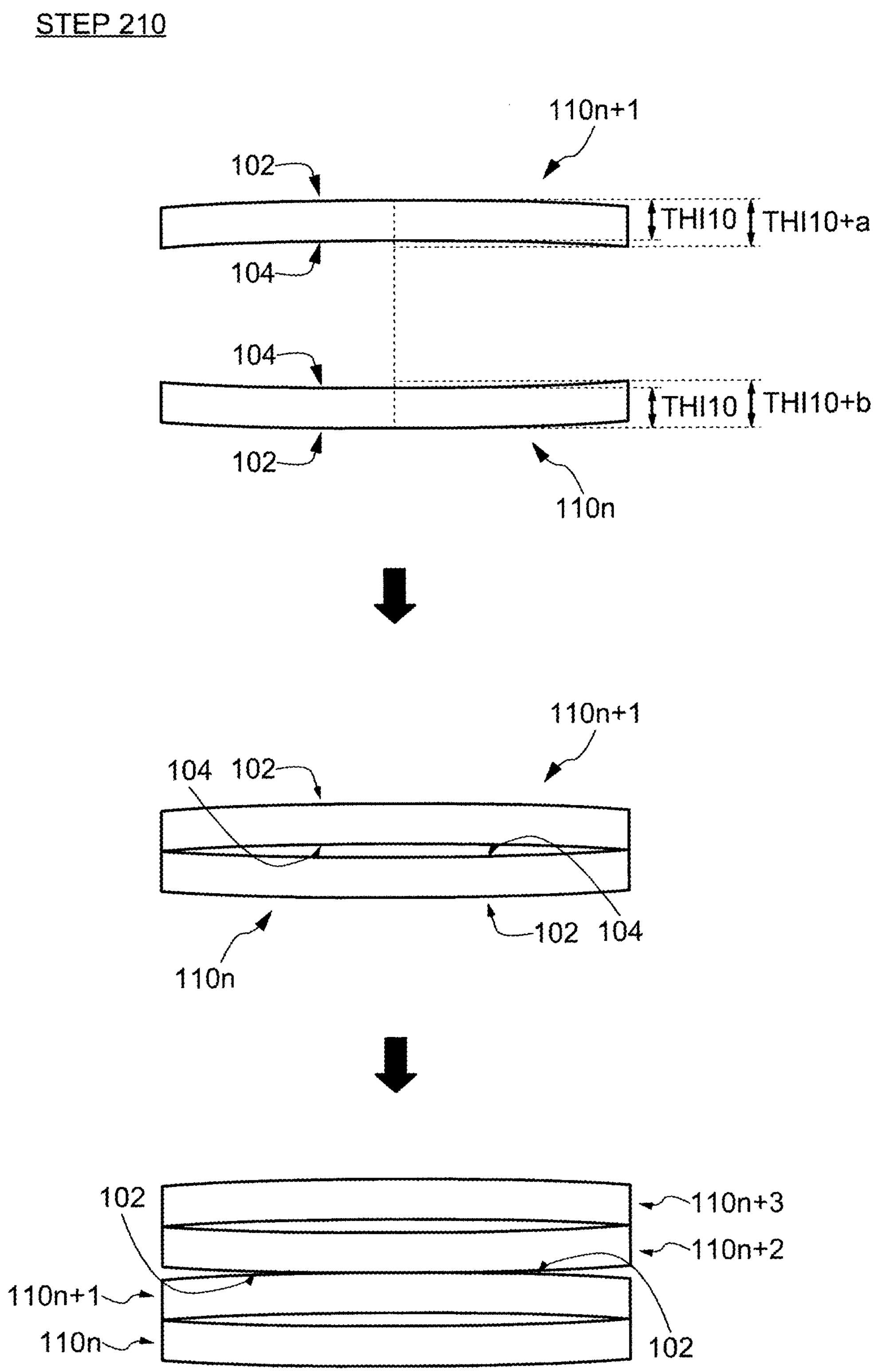
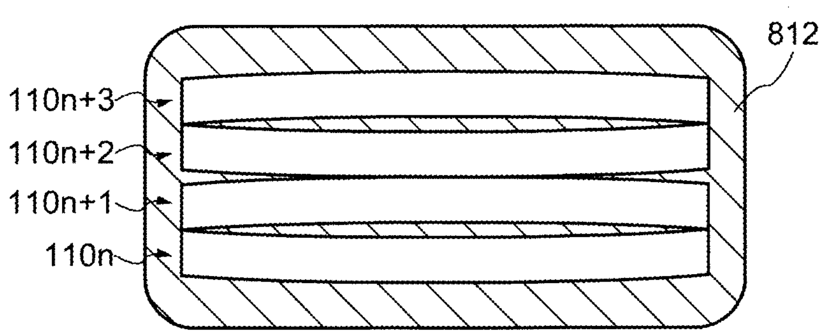

STEP 210
110n+1
102
104
104
102
THI10
THI10+a
THI10
THI10+b
110n
110n+1
104 102
102 104
110n
102
110n+3
110n+2
110n+1
110n
102

FIG. 56

STEP 220



STEP 230

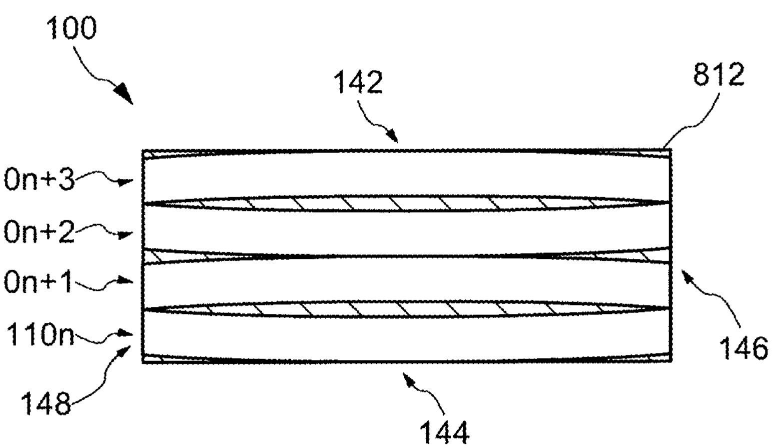


FIG. 57

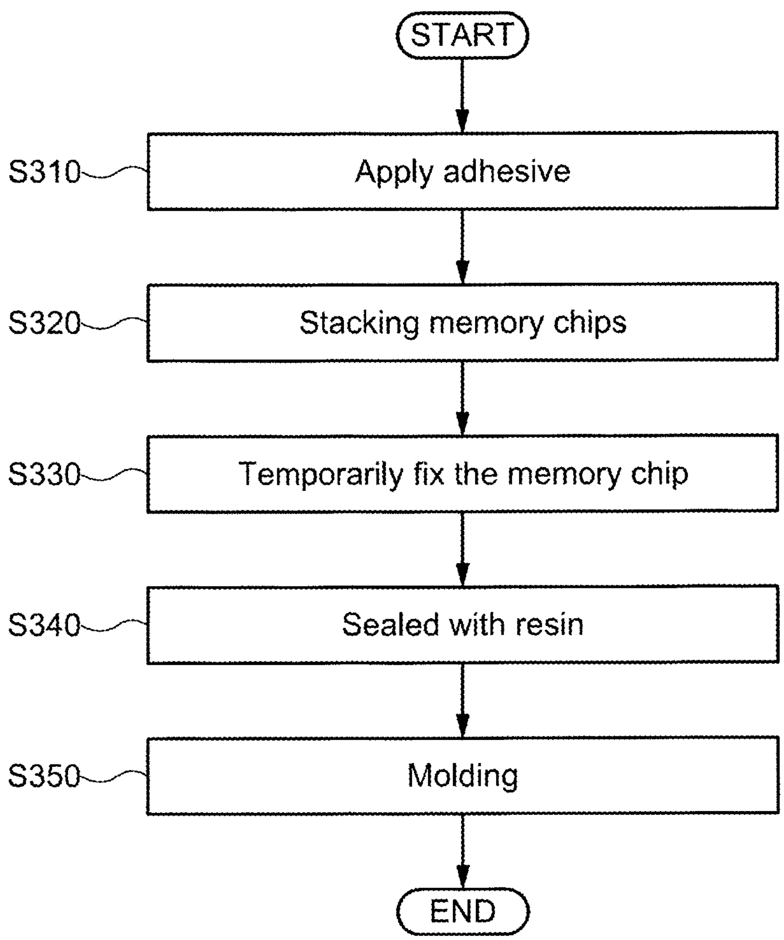

START
S310
Apply adhesive
S320
Stacking memory chips
S330
Temporarily fix the memory chip
S340
Sealed with resin
S350
Molding
END

FIG. 58

STEP 310

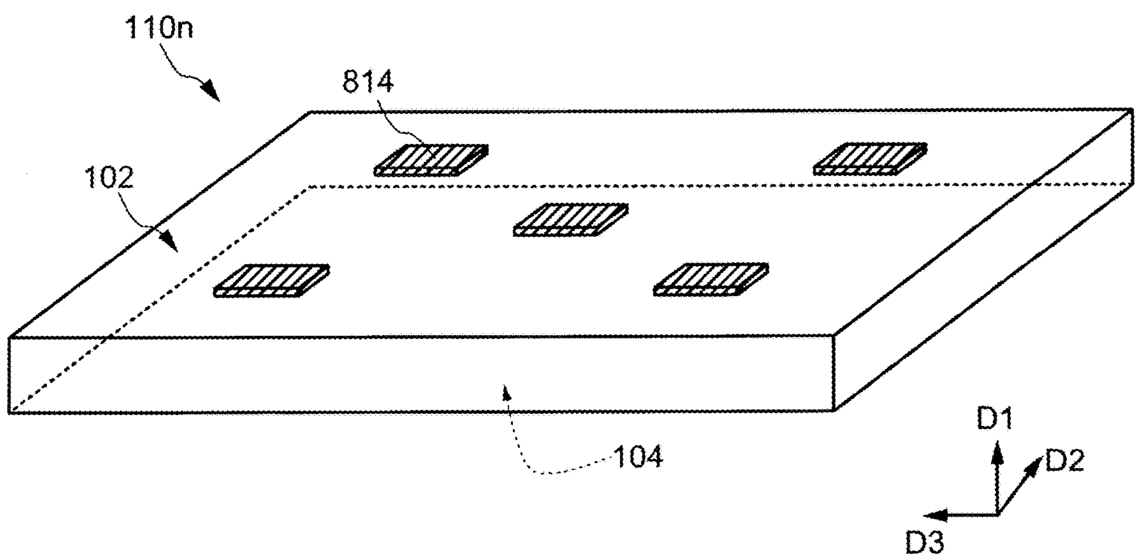


FIG. 59

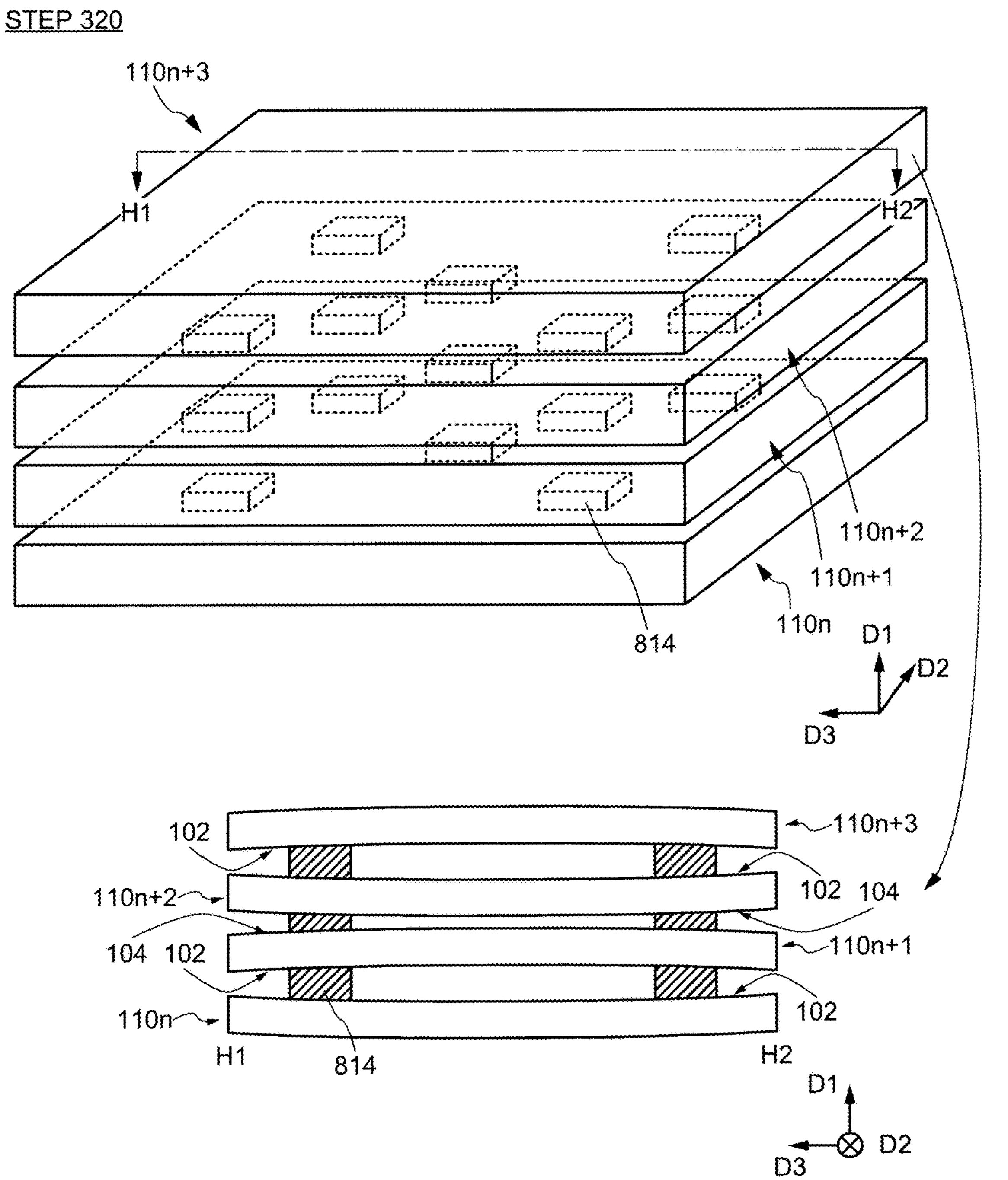
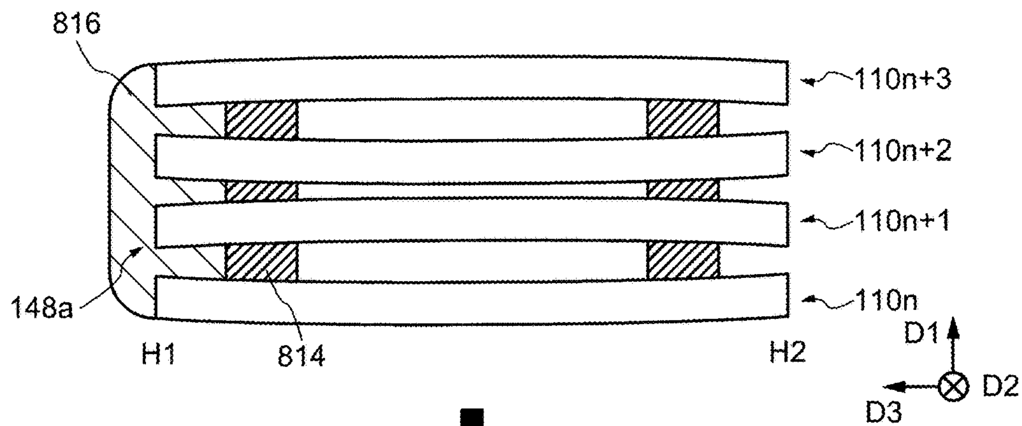
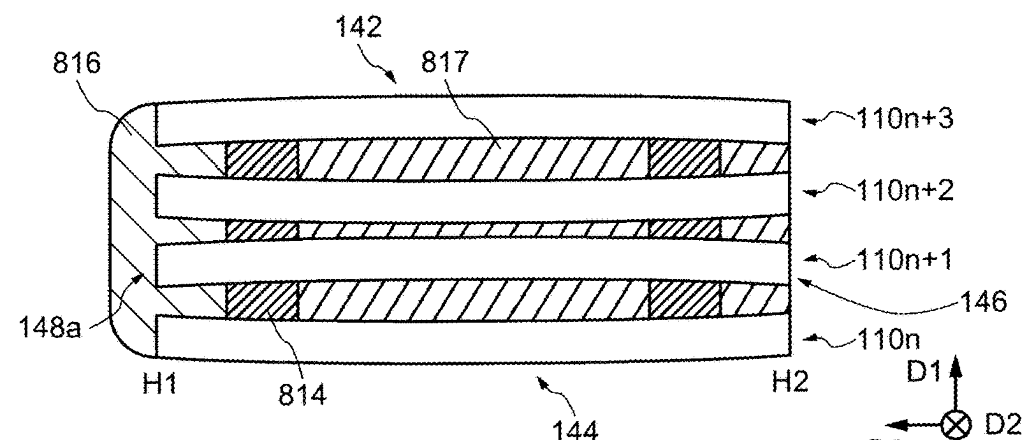

STEP 320
110n+3
H1
H2
110n+2
110n+1
110n
814
D1
D2
D3
102
110n+3
110n+2
102 104
104 102
110n+1
110n
102
H1
814
H2
D1
D3
D2

FIG. 60

STEP 330



STEP 340



STEP 350

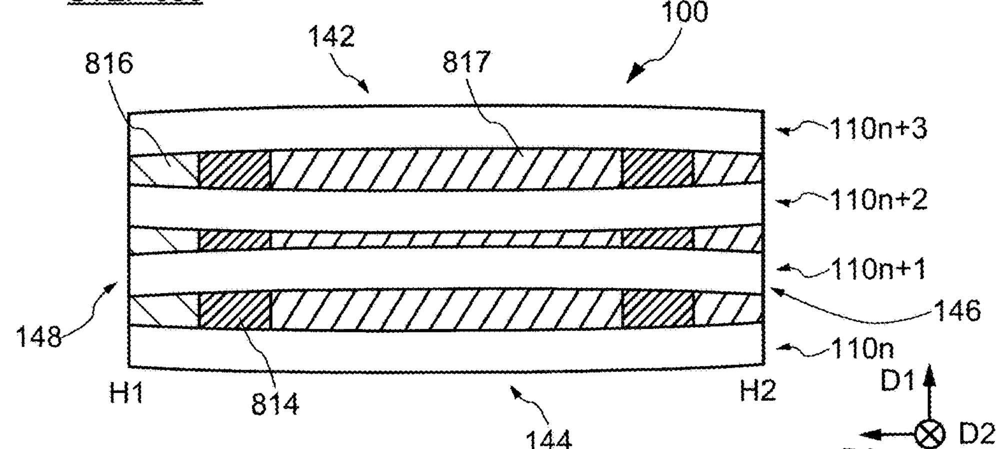


FIG. 61

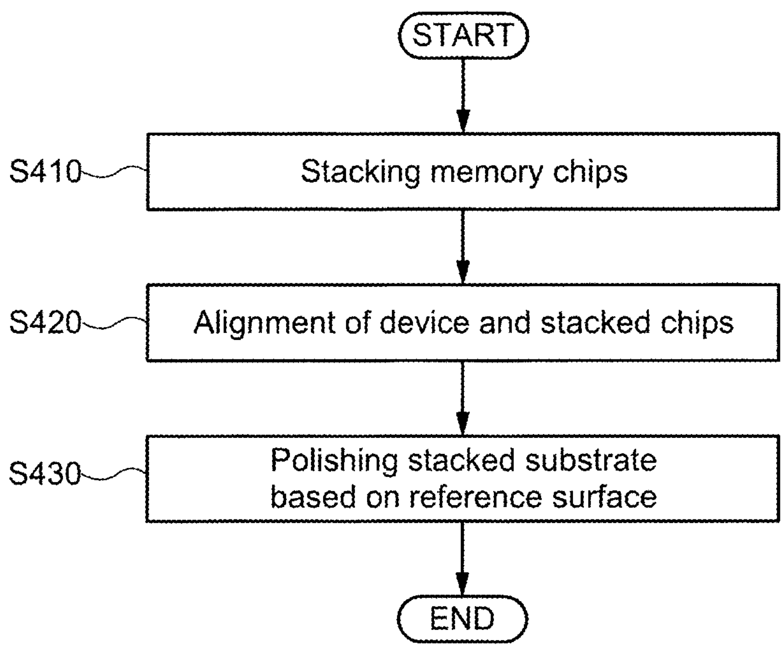

START
S410
Stacking memory chips
S420
Alignment of device and stacked chips
S430
Polishing stacked substrate
based on reference surface
END

FIG. 62

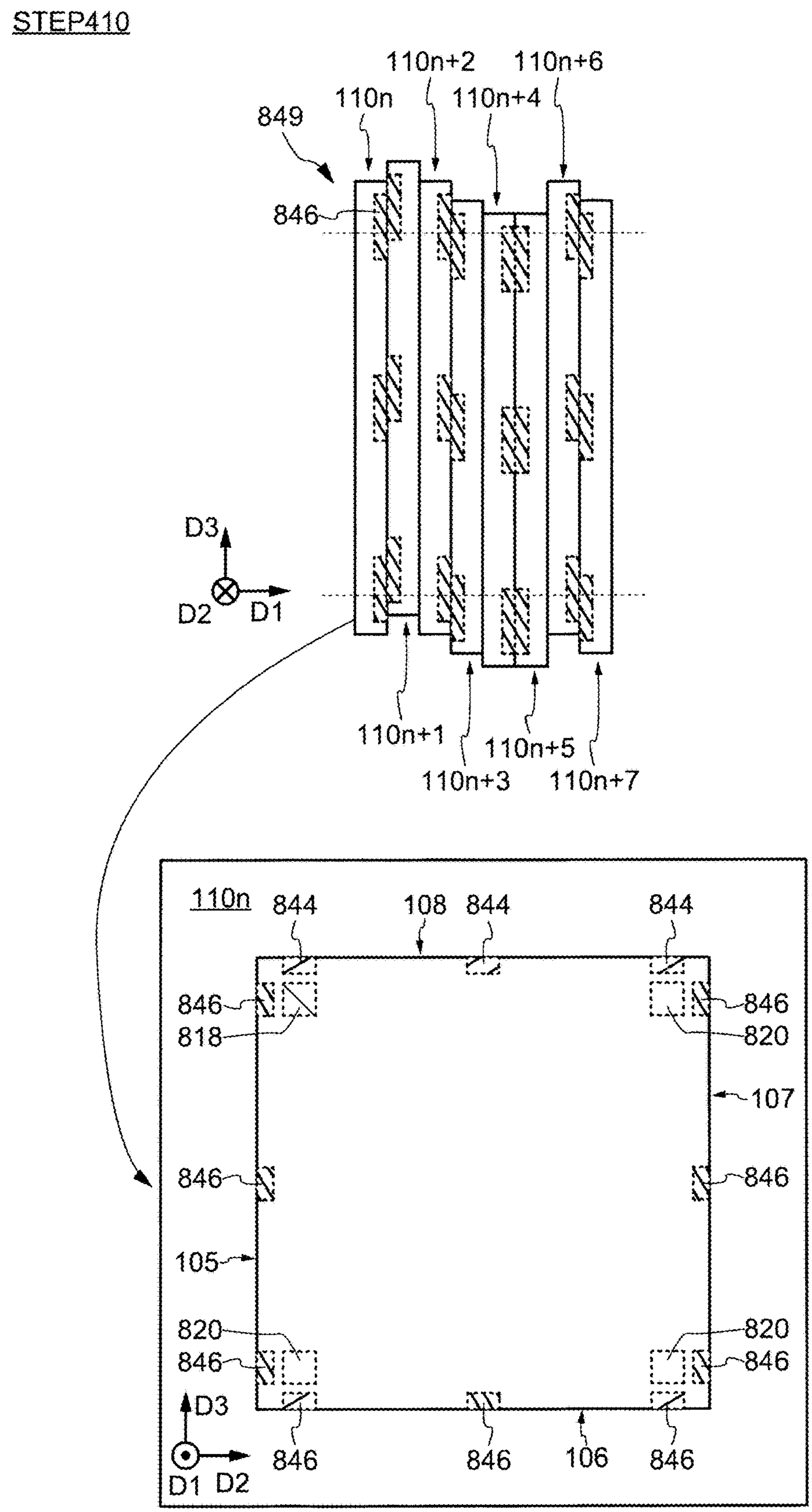

STEP410
849
110n
110n+2
110n+4
110n+6
846
D3
D2
D1
110n+1
110n+3
110n+5
110n+7
110n
844
108
844
844
846
818
846
820
107
846
846
105
820
846
820
846
D3
D1
D2
846
846
106
846

FIG. 63

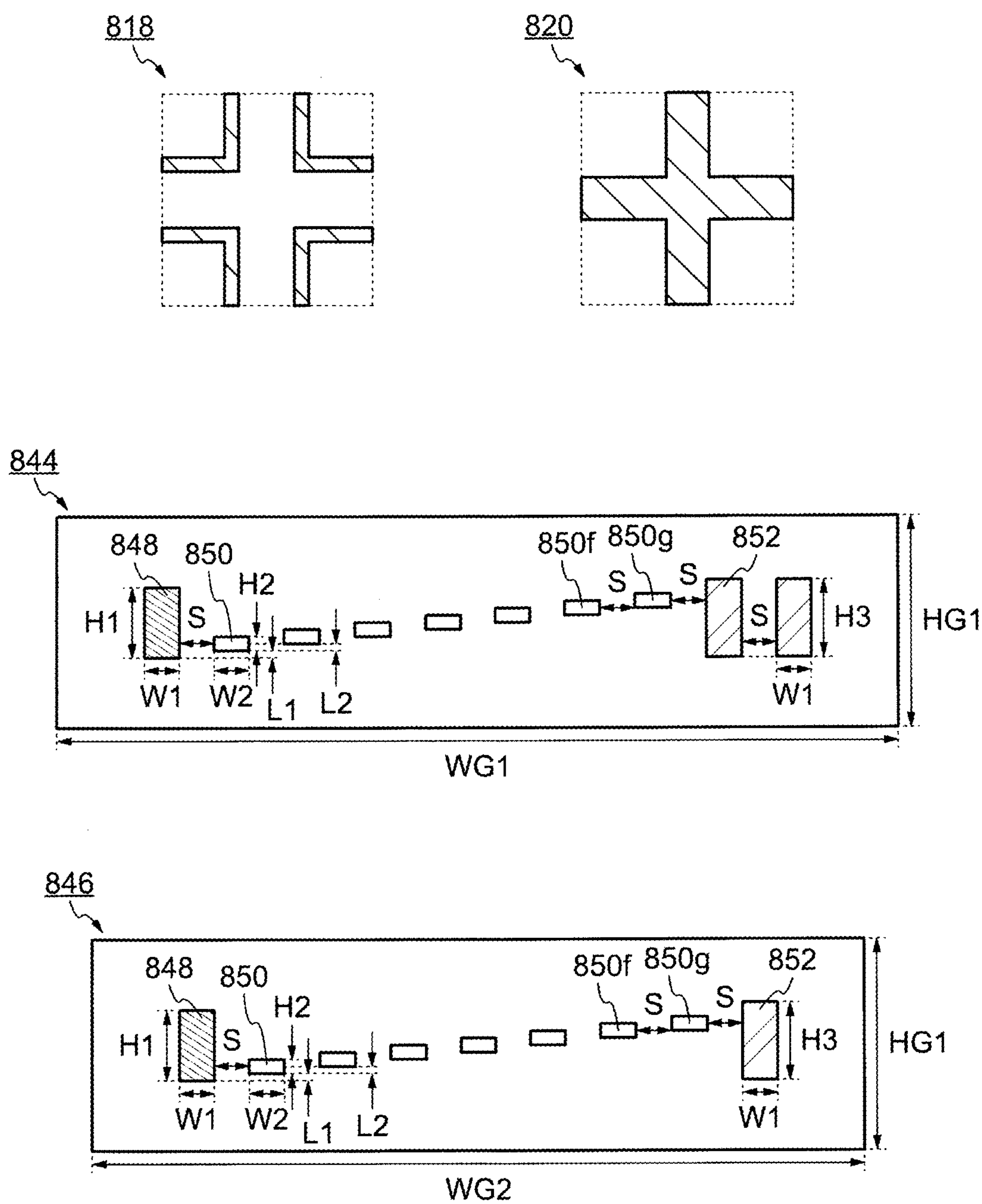

818
820
844
848
850
H2
850f
850g
852
S
H1
H3
HG1
W1
W2
L1
L2
WG1
846
WG2

FIG. 64

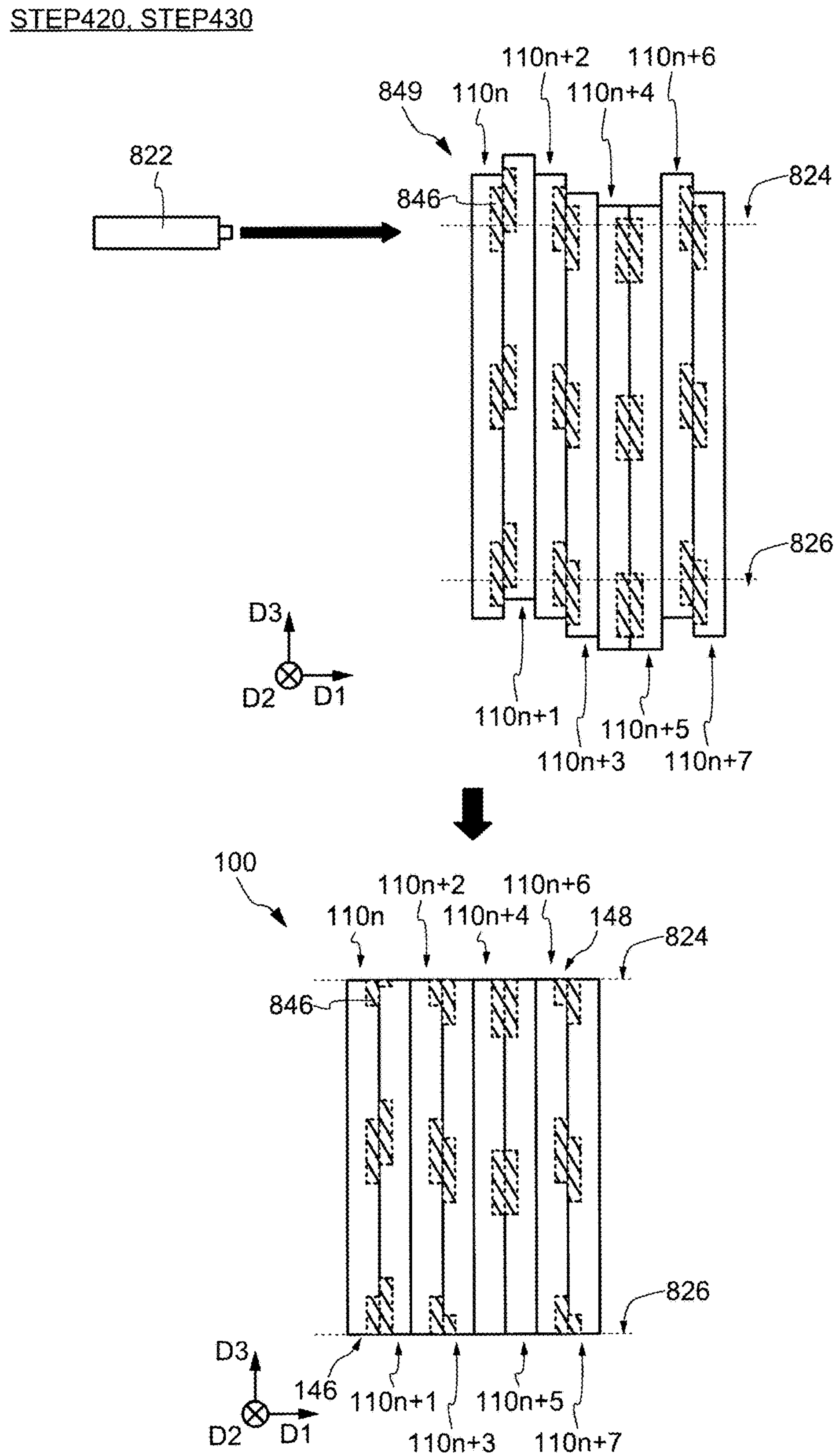

STEP420, STEP430
822
849
110n
110n+2
110n+4
110n+6
846
824
826
110n+1
110n+3
110n+5
110n+7
D3
D2
D1
100
148
146

FIG. 65

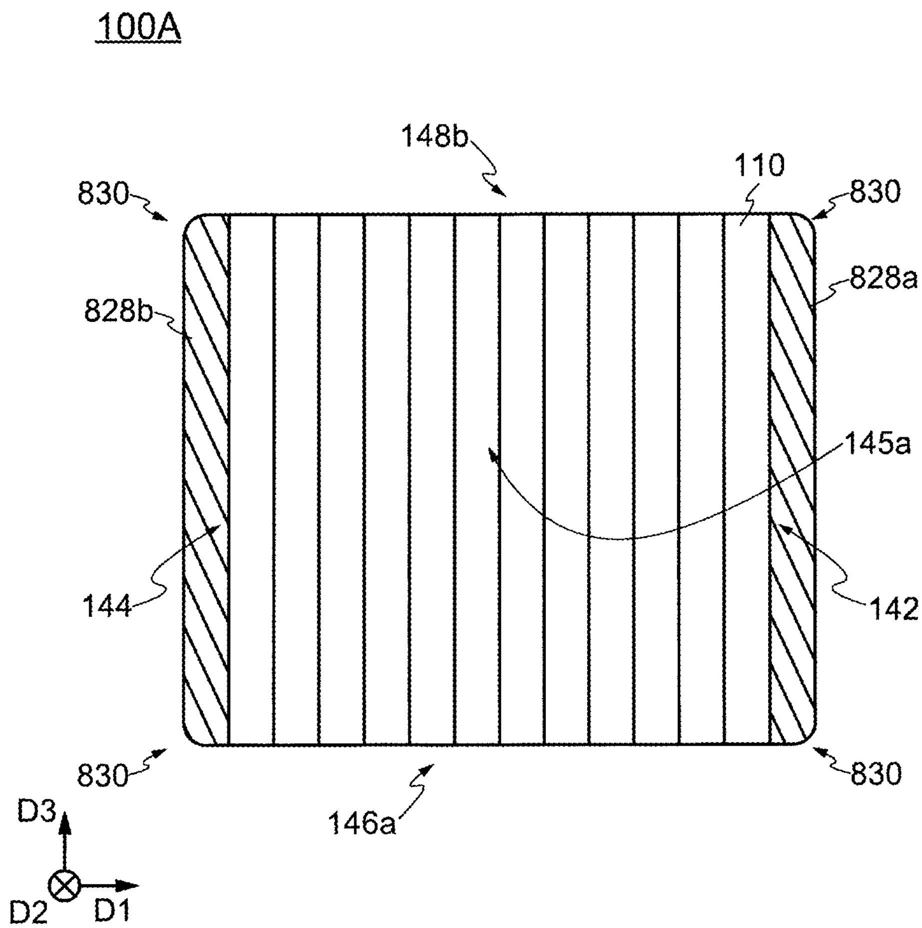

100A
148b
110
830
830
828b
828a
145a
144
142
830
830
146a
D3
D2
D1

FIG. 66

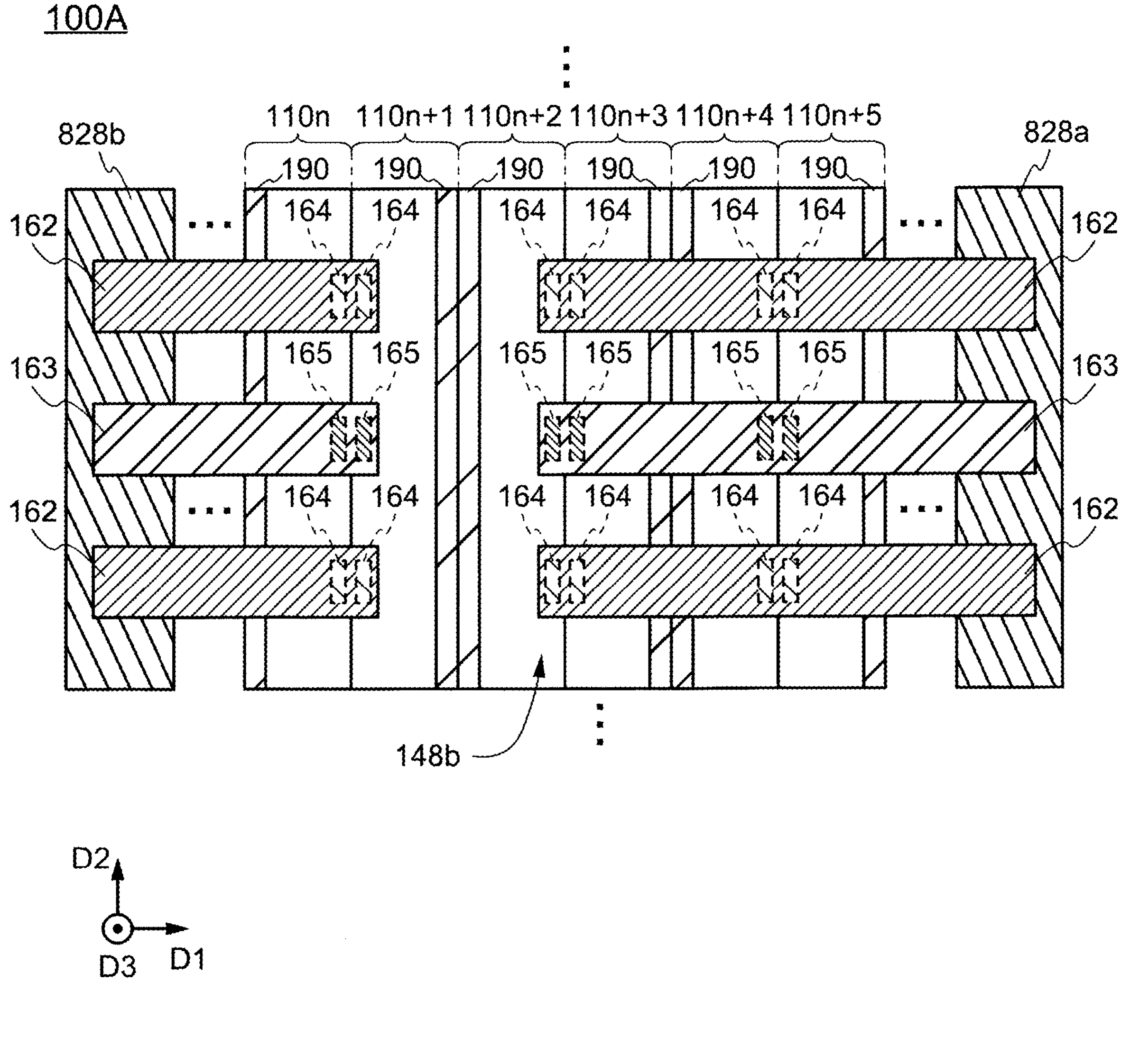

100A
110n
110n+1
110n+2
110n+3
110n+4
110n+5
828b
828a
190
162
163
164
165
148b
D2
D3
D1

FIG. 67

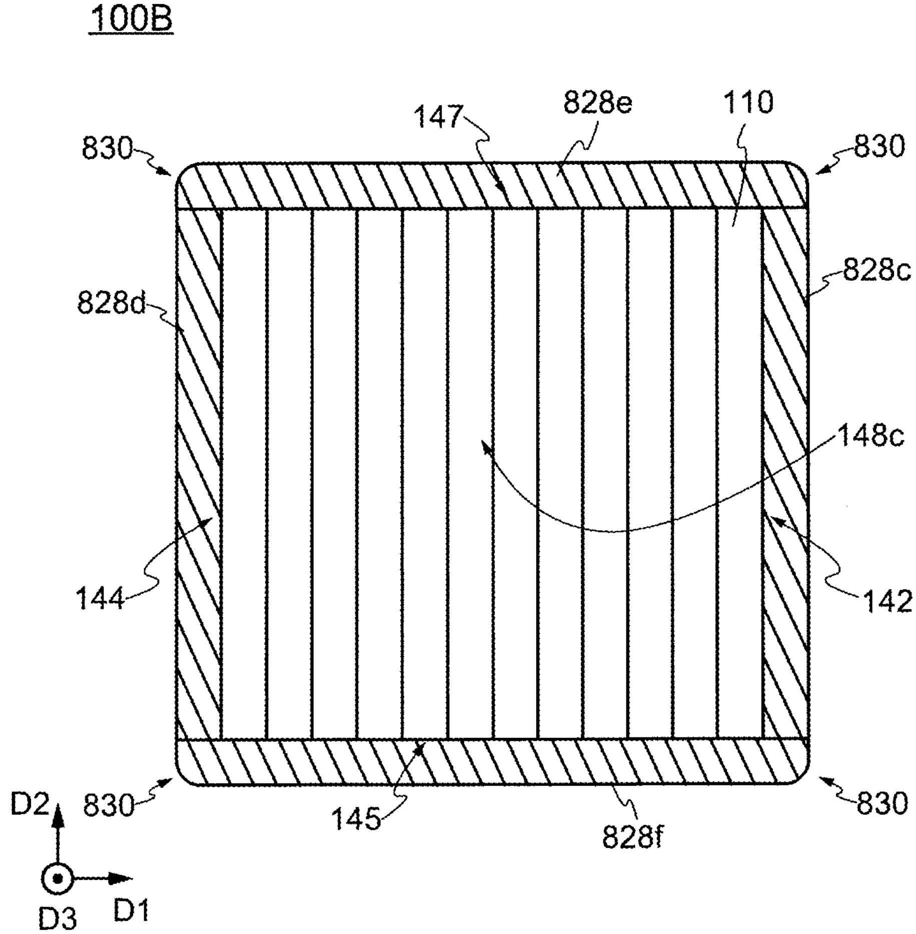
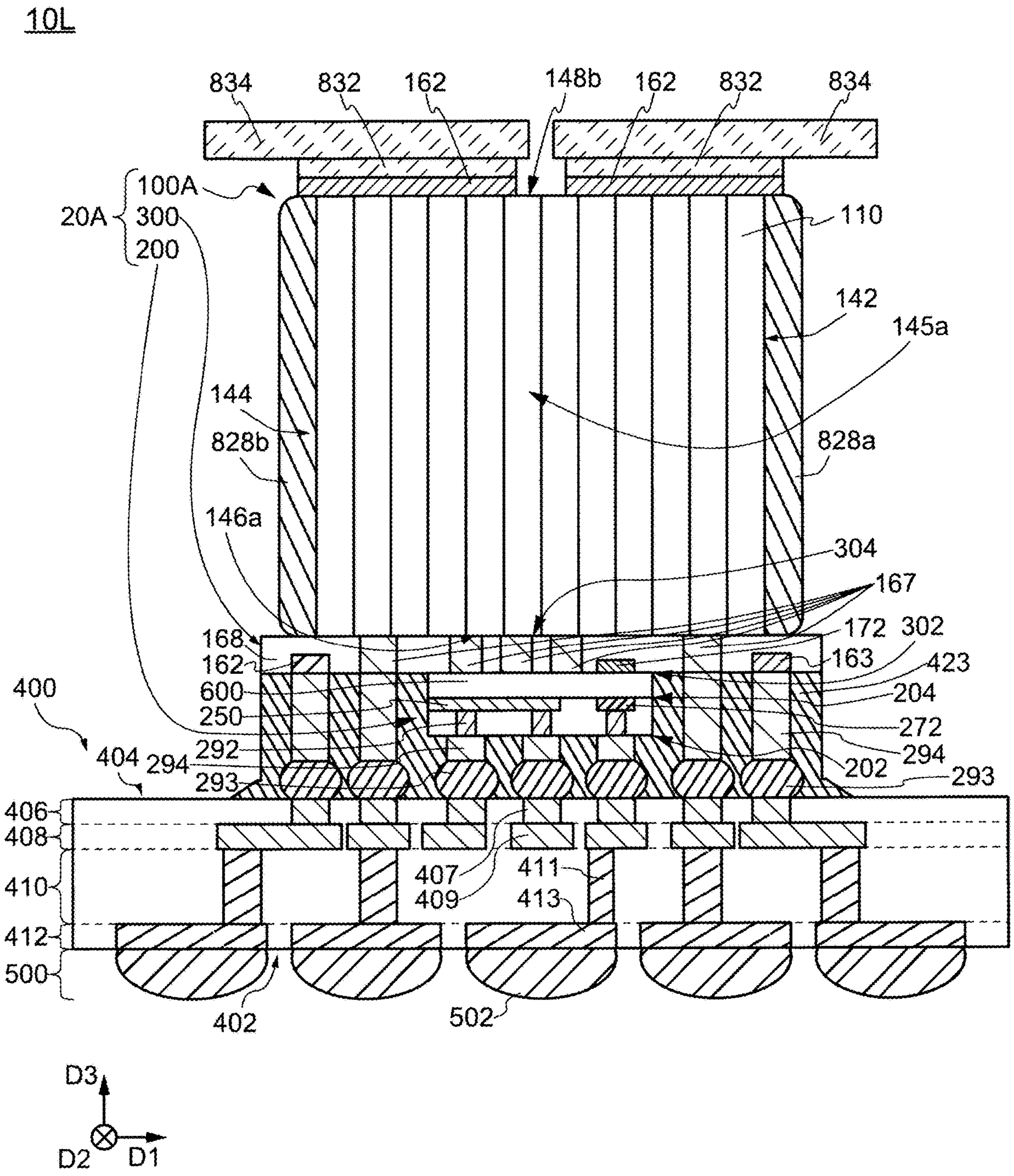

100B
147
828e
110
830
830
828d
828c
148c
144
142
830
145
828f
830
D2
D3
D1

FIG. 68



10L
834
832
162
148b
162
832
834
20A
100A
300
200
110
142
145a
144
828b
828a
146a
304
167
168
162
600
250
292
294
293
172
302
163
423
204
272
294
202
293
400
404
406
408
410
412
500
407
409
411
413
402
502
D3
D2
D1

FIG. 69

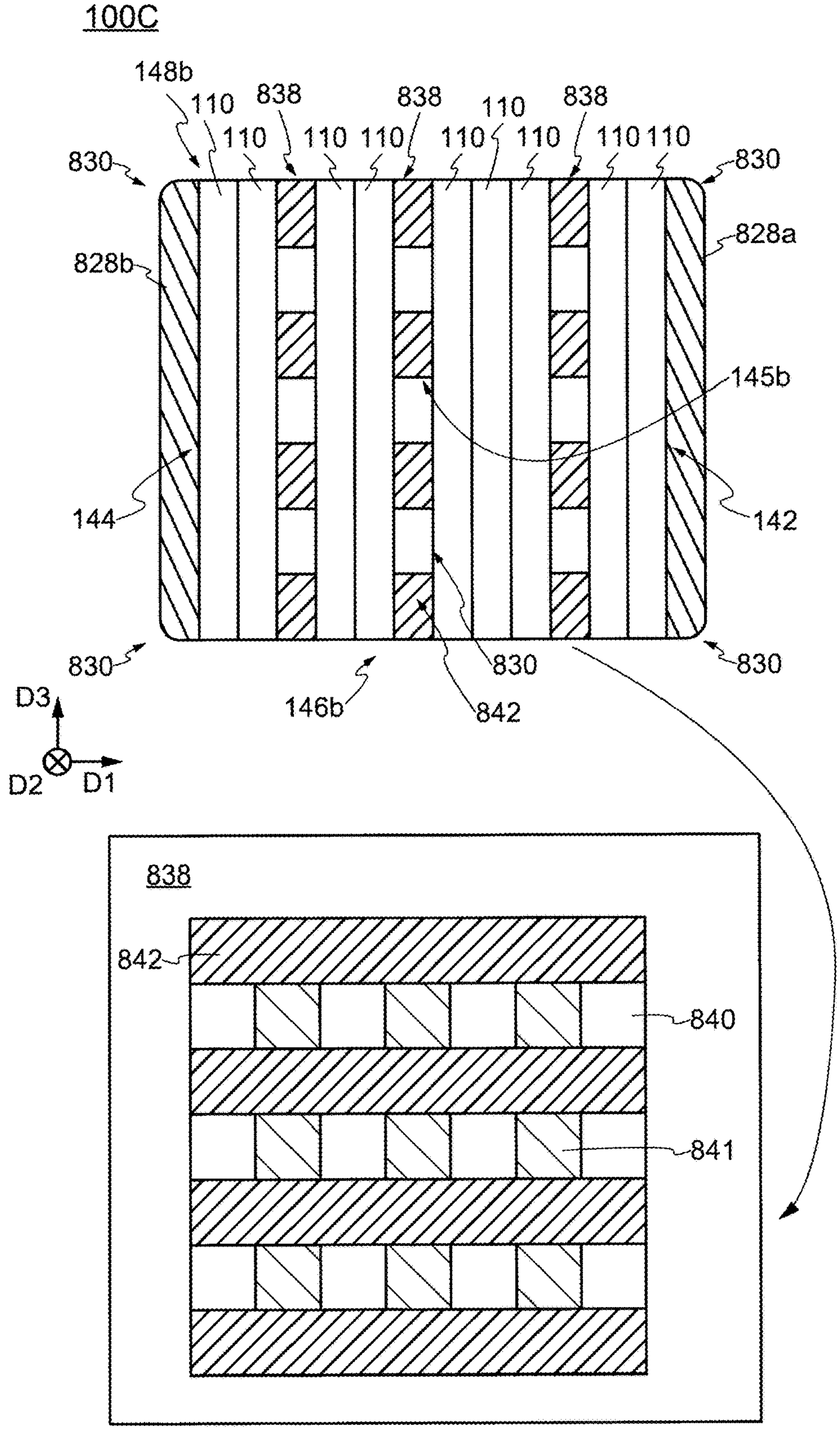

100C
148b
110
838
110
110
110
838
110
110
110
838
110
110
830
830
828b
828a
145b
144
142
830
830
830
146b
842
D3
D2
D1
838
842
840
841

FIG. 70

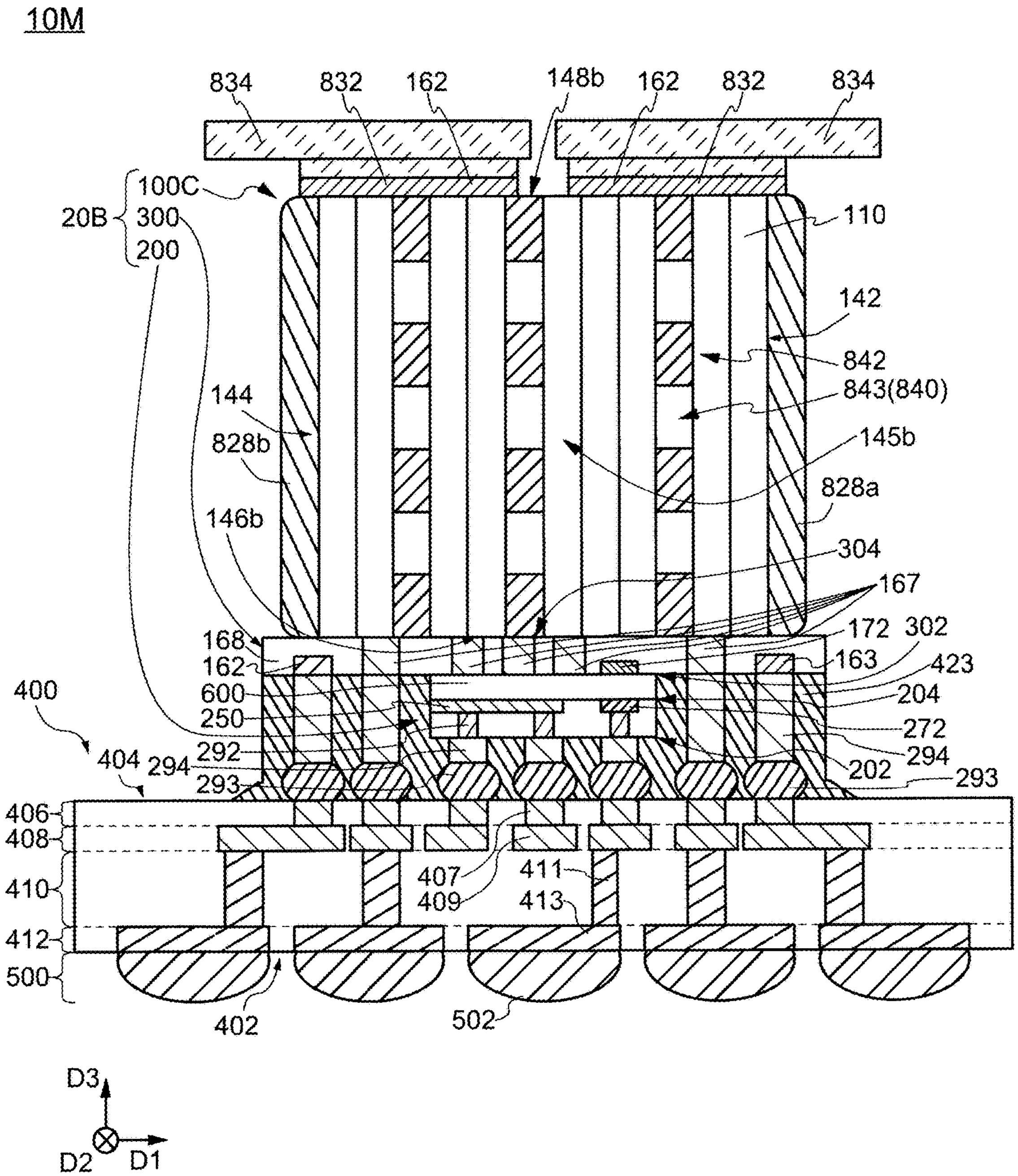

10M
834
832
162
148b
162
832
834
20B
100C
300
200
110
142
842
843(840)
145b
144
828b
828a
146b
304
167
168
172
302
162
163
423
600
204
250
272
292
294
202
400
404
294
293
293
406
408
407
409
411
413
410
412
500
402
502
D3
D2
D1

FIG. 71

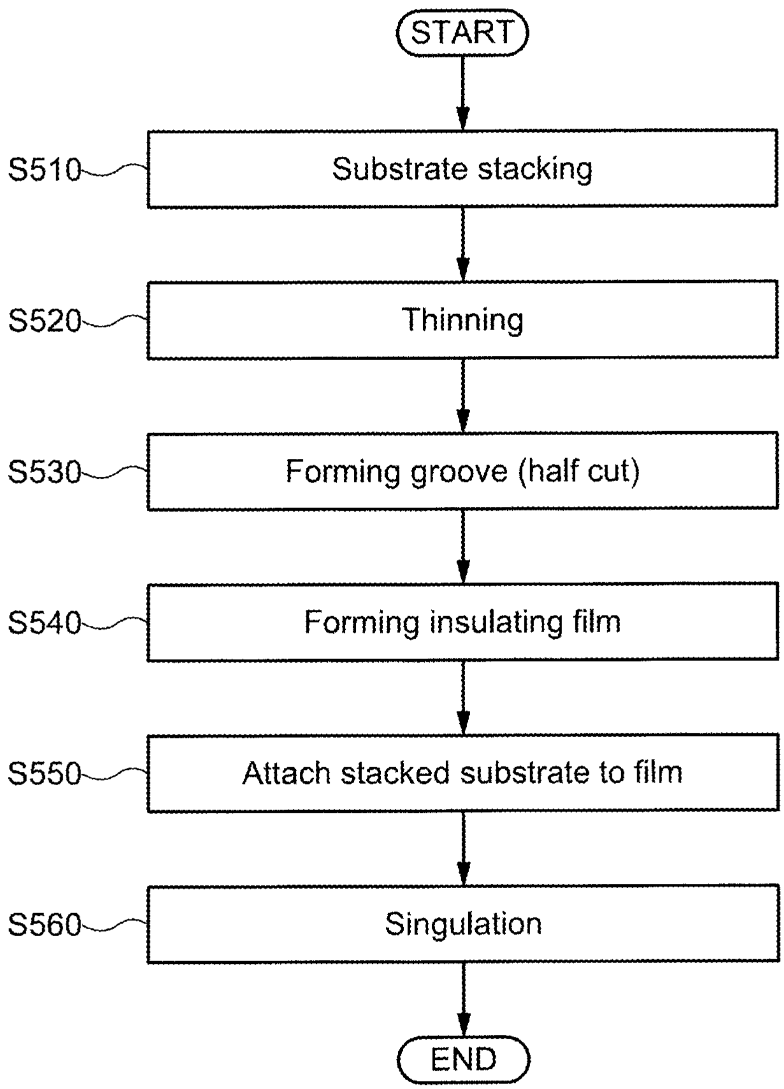
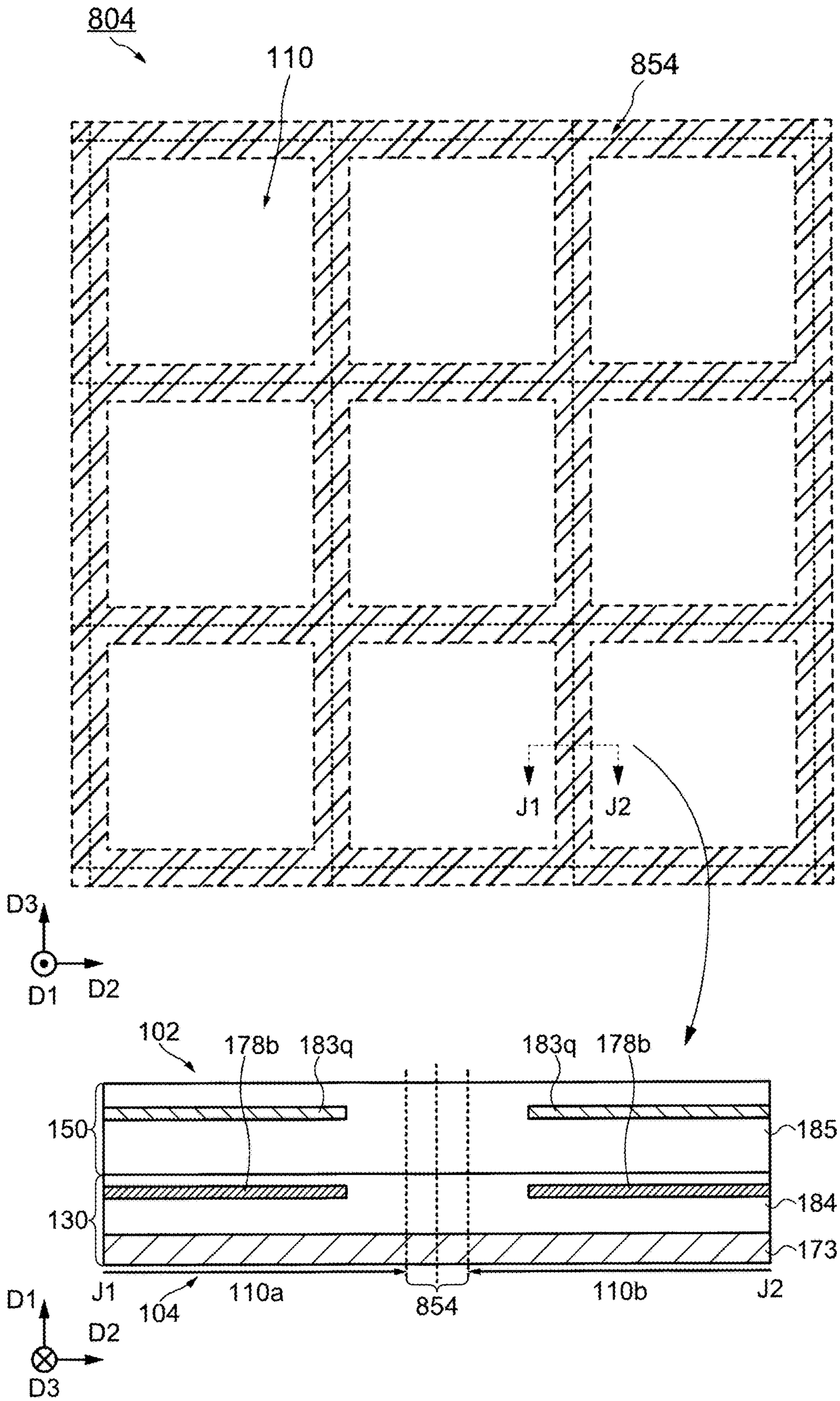

START
S510
Substrate stacking
S520
Thinning
S530
Forming groove (half cut)
S540
Forming insulating film
S550
Attach stacked substrate to film
S560
Singulation
END

FIG. 72



804
110
854
J1
J2
D3
D1
D2
102
178b
183q
183q
178b
150
185
130
184
173
J1
104
110a
854
110b
J2
D1
D2
D3

FIG. 73

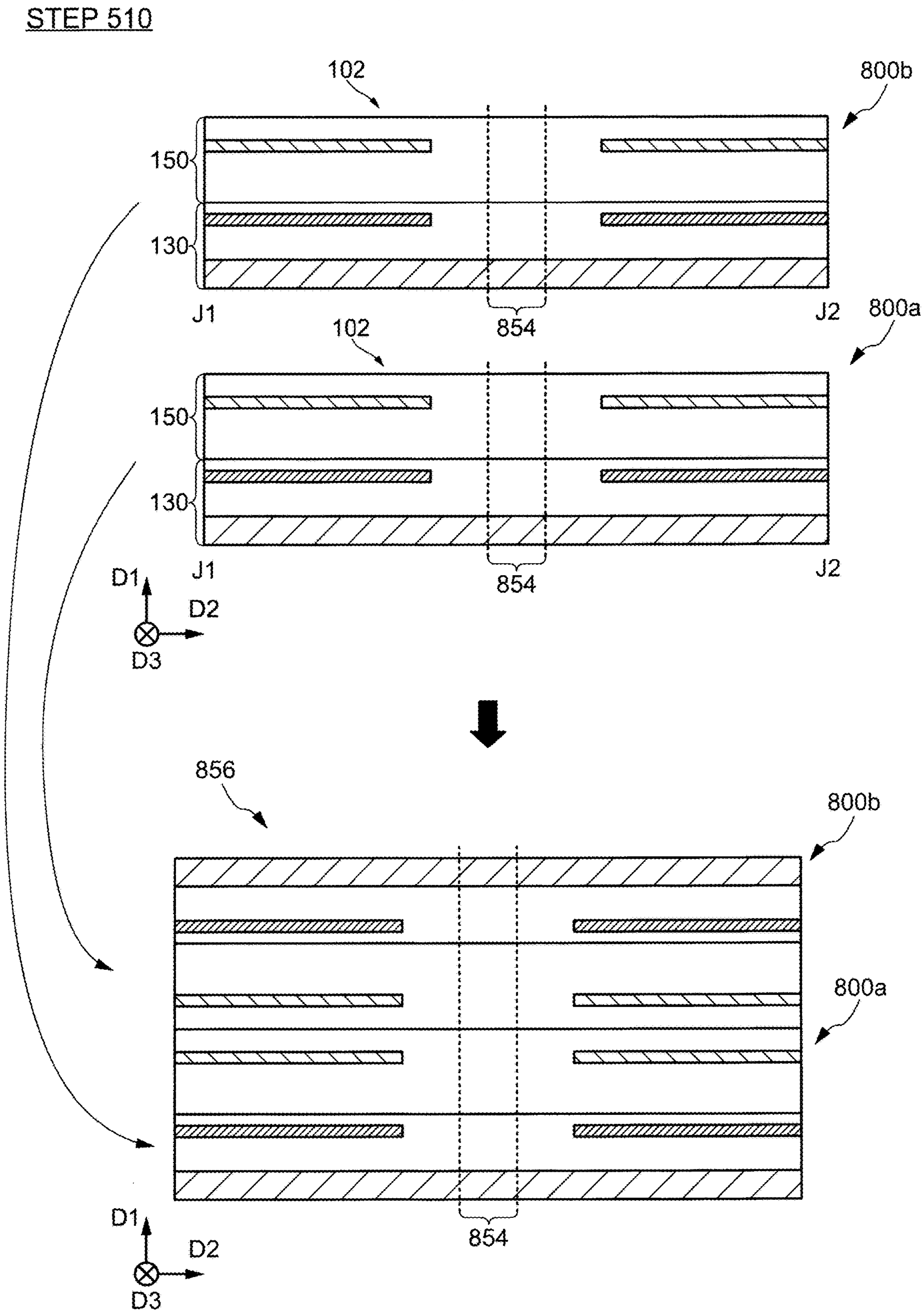
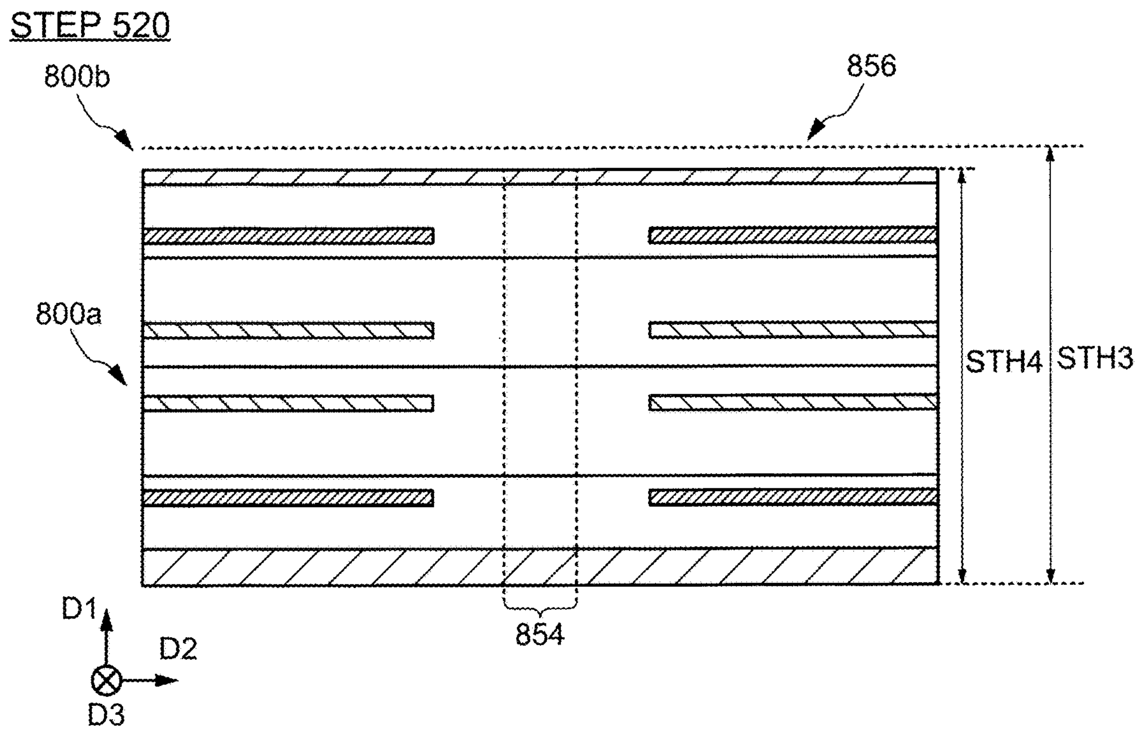

STEP 510
102
800b
150
130
J1
854
J2
102
800a
150
130
J1
854
J2
D1
D2
D3
856
800b
800a
854
D1
D2
D3

FIG. 74


STEP 520
800b
856
800a
STH4
STH3
854
D1
D2
D3

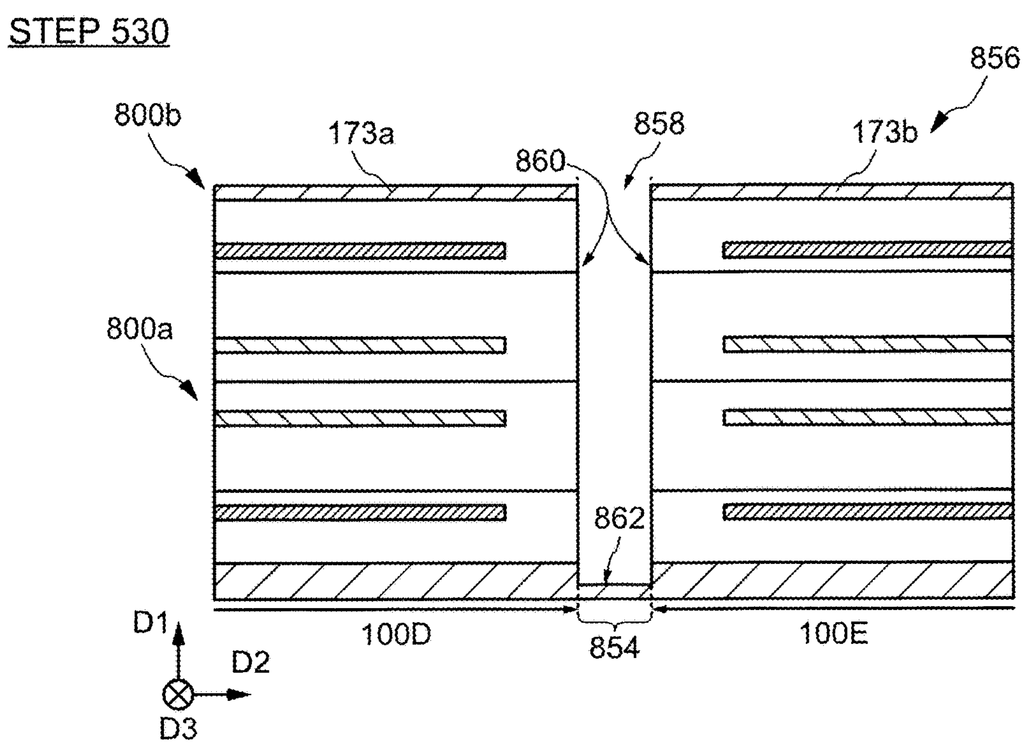
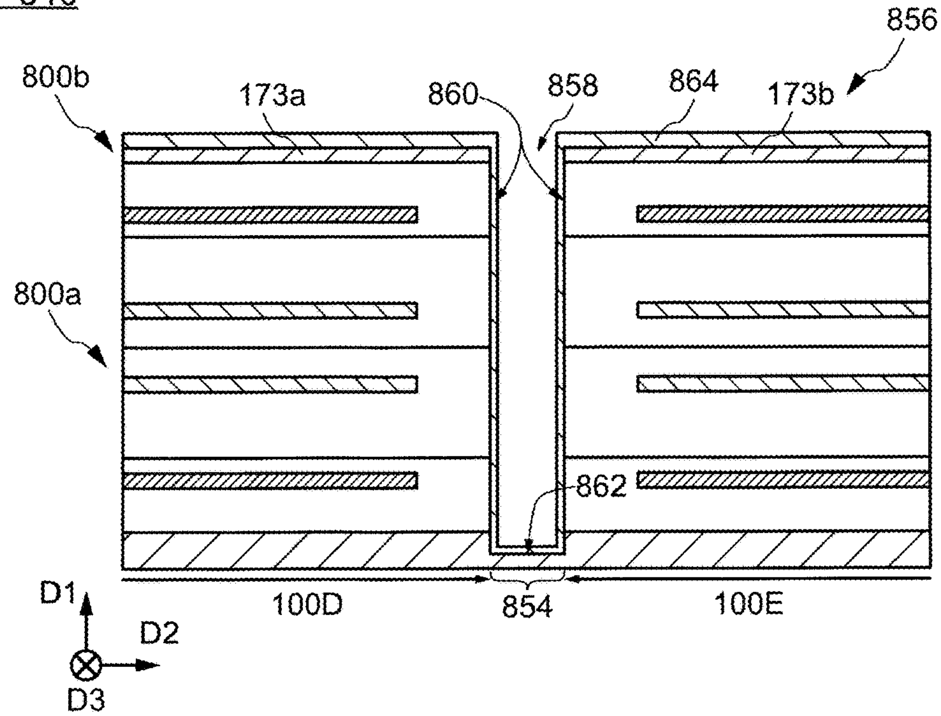

STEP 530
856
800b
173a
858
173b
860
800a
862
D1
100D
854
100E
D2
D3

FIG. 75

STEP 540



STEP 550

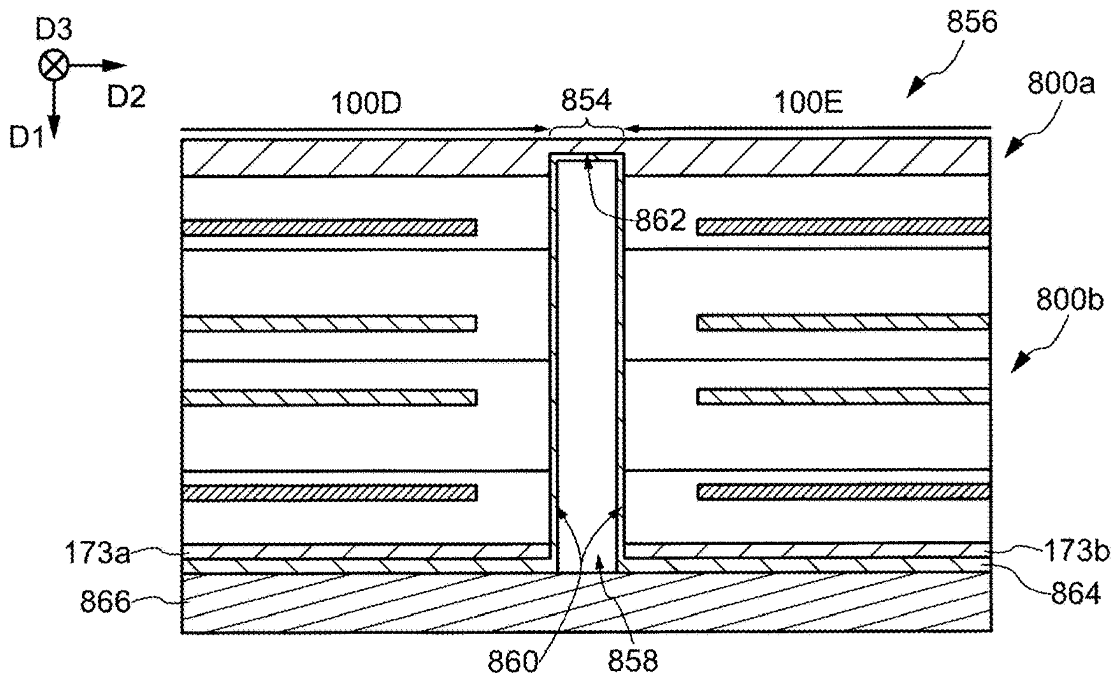
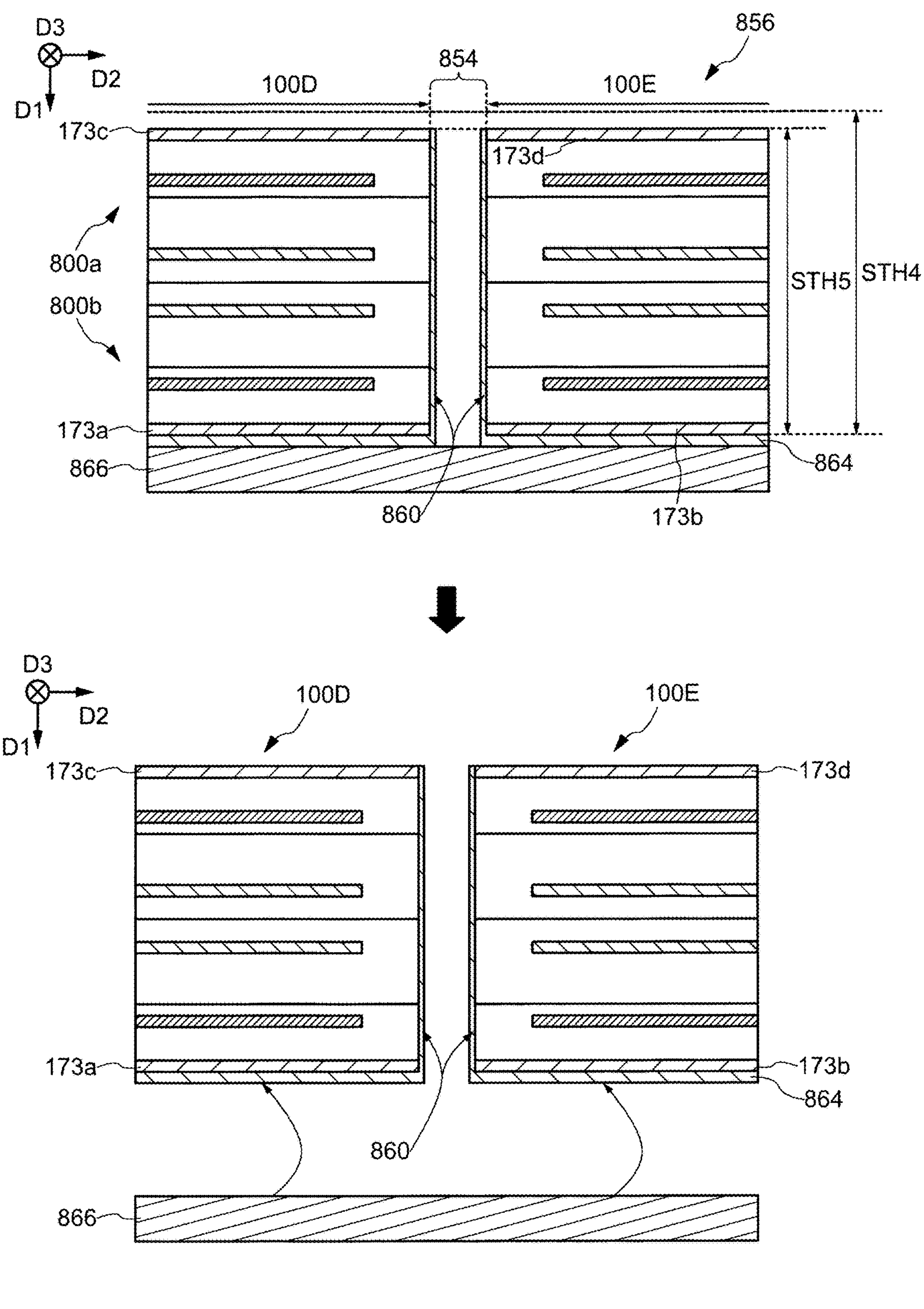


FIG. 76

STEP 560



SEMICONDUCTOR MODULE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a Continuation of International Patent Application No. PCT/JP2023/045466, filed on Dec. 19, 2023, which claims the benefit of priority to Japanese Patent Application No. 2022-203554, filed on Dec. 20, 2022, the entire contents of which are incorporated herein by reference.

FIELD

[0002] An embodiment of the present invention relates to a semiconductor module.

BACKGROUND

[0003] In recent years, the amount of data communication in an electronic computer such as a data center has increased. In addition, with an increase in the amount of data communication, the power consumption, the memory capacity, and the demand for a reduction in power consumption and an increase in capacity of the electronic computer is increasing. For example, the electronic computer includes a plurality of logic chips and a plurality of memory chips electrically connected to the plurality of logic chips. The logic chip is, for example, an IC (Integrated Circuit) chip mounted with a logic circuit, and a memory chip is a semiconductor chip in which a memory circuit is mounted. Data communication in an electronic computer is performed between a logic chip and a memory chip, for example. For example, a three-dimensional implementation method of a stacked logic chip and a memory chip is one effective solution for reducing power consumption of an electronic computer and increasing the capacity of the memory.

[0004] As an example of a three-dimensional implementation method, a semiconductor module including a structure (vertically stacked memory cube) in which a plurality of memory chips are stacked is arranged on a substrate or a logic chip so that the plurality of memory chips is parallel to the substrate or the logic chip, or a semiconductor module including a structure (horizontally stacked memory cube) in which a plurality of memory chips is stacked is placed vertically on a substrate or a logic chip so that the plurality of memory chips is perpendicular to the substrate or the logic chip is known.

[0005] Further, as an example of the three-dimensional implementation method, the memory cube and the logic chip are electrically connected using TSV, micro-bumps, flip-chip bonding, or the like. Further, as an example of a three-dimensional implementation method, a technique of forming a wiring on a sidewall of a vertically stacked memory cube is disclosed.

SUMMARY

[0006] A semiconductor module includes a memory cube in which a plurality of memory chips is stacked along a first direction parallel to a plane formed by a second direction perpendicular to the first direction and a third direction perpendicular to the second direction. The memory cube includes a first outermost surface, a third outermost surface opposite the first outermost surface, and a second outermost surface and a fourth outermost surface opposite the second outermost surface, the second outermost surface and the fourth outermost surface being the two outermost surfaces in the second direction. The first outermost surface is arranged on a side on which a semiconductor chip is arranged in the third direction. Each of the plurality of memory chips includes a substrate and a wiring layer stacked on the substrate. The substrate includes a first side surface along the first outermost surface, a second side surface along the second outermost surface, a third side surface along the third outermost surface and a fourth side surface along the fourth outermost surface. A first insulating film is included between the first outermost surface and the first side surface, between the second outermost surface and the second side surface, between the third outermost surface and the third side surface, and between the fourth outermost surface and the fourth side surface. The wiring layer includes an internal wiring. The internal wiring is connected to an electrode provided on at least one of the second outermost surface, the third outermost surface, and the fourth outermost surface, and the memory cube is configured to transmit signals while being spaced apart from the semiconductor chip.

[0007] A semiconductor module includes a memory cube in which a plurality of memory chips is stacked along a first direction parallel to a plane formed by a second direction perpendicular to the first direction and a third direction perpendicular to the second direction. The memory cube includes a first substrate, a second substrate, a bottom, and an upper surface opposite to the bottom surface. The first substrate and the second substrate sandwiches the plurality of memory chips in the first direction. The bottom surface is an outermost surface on a side where the semiconductor chips are arranged in the third direction. Each of the first substrate and the second substrate does not include a transistor layer including a transistor. Each of the first substrate and the second substrate does not include a transistor layer including a wiring circuit transistor. Each of the plurality of memory chips includes a substrate, and the transistor layer and a wiring layer stacked on the substrate. The wiring layer includes a first internal wiring, a second internal wiring, and a first inductor. The first internal wiring, the second internal wiring, and the first inductor are arranged apart from each other. The semiconductor module includes the first internal wiring, a first conductive layer in contact with the upper surface and an upper surface of the first substrate along the third direction, a second conductive layer electrically connected to the first conductive layer, and a heat conductive member electrically connected to the second conductive layer.

BRIEF DESCRIPTION OF DRAWINGS

[0008] FIG. **1** is a perspective view showing a configuration of a semiconductor module according to a first embodiment of the present invention.

[0009] FIG. **2** is a perspective view showing a configuration of a memory chip according to the first embodiment of the present invention.

[0010] FIG. **3** is a cross-sectional view showing a cross-section structure of the memory chip taken along a line **A1**-**A2** shown in FIG. **2**.

[0011] FIG. **4** is a block diagram showing the configuration of the memory chip according to the first embodiment of the present invention.

[0012] FIG. **5**A is a cross-sectional view showing a cross-section structure of a memory cube according to the first embodiment of the present invention.

[0013] FIG. **5**B is a schematic diagram showing a method for manufacturing the memory cube according to the first embodiment of the present invention.

[0014] FIG. **6**A is a schematic diagram showing a method for manufacturing the memory chip according to the first embodiment of the present invention.

[0015] FIG. **6**B is a schematic diagram showing the method for manufacturing the memory chip according to the first embodiment of the present invention.

[0016] FIG. **7**A is a schematic diagram showing a method for manufacturing the memory cube and a rewiring layer (a side power wiring and a side ground wiring) according to the first embodiment of the present invention.

[0017] FIG. **7**B is a cross-sectional view showing a cross-section structure of the memory cube taken along a line B**1**-B**2** of FIG. **7**A.

[0018] FIG. **8**A is a schematic diagram showing configurations of the memory cube and the rewiring layer (the side power wiring and the side ground wiring) according to the first embodiment of the present invention.

[0019] FIG. **8**B is a cross-sectional view showing cross-section structures of the memory cube and the rewiring layer taken along a line B**1**-B**2** in FIG. **8**A.

[0020] FIG. **9**A is a schematic diagram showing the method for manufacturing the memory cube and the rewiring layer according to the first embodiment of the present invention.

[0021] FIG. **9**B is a cross-sectional view of a cross-section structure of the memory cube taken along a line C**1**-C**2** of FIG. **9**A.

[0022] FIG. **10**A is a schematic diagram showing the configurations of the memory cube and the rewiring layer according to the first embodiment of the present disclosure.

[0023] FIG. **10**B is a cross-sectional view of cross-section structures of the memory cube and the rewiring layer taken along a line C**1**-C**2** in FIG. **10**A.

[0024] FIG. **11**A is a plan view showing configurations of a seal ring and a signal transmission wiring included in the memory chip according to the first embodiment of the present invention.

[0025] FIG. **11**B is a cross-sectional view showing cross sections of the seal ring and the signal transmission wiring included in the memory chip taken along a line E**1**-E**2** of FIG. **11**A.

[0026] FIG. **12** is a cross-sectional view showing a mounting example of the semiconductor module according to the first embodiment of the present invention.

[0027] FIG. **13** is a cross-sectional view showing a mounting example of a semiconductor module according to a second embodiment of the present invention.

[0028] FIG. **14** is a block diagram showing a configuration of the semiconductor module according to the second embodiment of the present invention.

[0029] FIG. **15** is a cross-sectional view showing a mounting example of a semiconductor module according to a third embodiment of the present invention.

[0030] FIG. **16** is a cross-sectional view showing a mounting example of a semiconductor module according to a fourth embodiment of the present invention.

[0031] FIG. **17** is a cross-sectional view showing a mounting example of a semiconductor module according to a fifth embodiment of the present invention.

[0032] FIG. **18** is a cross-sectional view showing a mounting example of a semiconductor module according to a sixth embodiment of the present invention.

[0033] FIG. **19**A is a perspective view showing a conductive film formed on a side surface of a memory cube according to a seventh embodiment of the present invention.

[0034] FIG. **19**B is a perspective view showing the conductive film formed on the side surface of the memory cube according to the seventh embodiment of the present invention.

[0035] FIG. **20**A is a perspective view showing a method for manufacturing the conductive film on the side surface of the memory cube according to the seventh embodiment of the present invention.

[0036] FIG. **20**B is a perspective view showing a method for manufacturing the conductive film on the side surface of the memory cube according to the seventh embodiment of the present invention.

[0037] FIG. **21**A is a perspective view showing the conductive film formed on the side surface of the memory cube according to the seventh embodiment of the present invention.

[0038] FIG. **21**B is a schematic diagram showing a configuration of a memory cube according to an eighth embodiment of the present invention.

[0039] FIG. **22**A is a cross-sectional view showing the cross-section structure of the memory cube according to the first embodiment of the present invention.

[0040] FIG. **22**B is a cross-sectional view showing the cross-section structure of the memory cube according to the first embodiment of the present invention.

[0041] FIG. **23**A is a schematic diagram showing the configurations of the memory cube and the rewiring layer (the side power wiring and the side ground wiring) according to the first embodiment of the present invention.

[0042] FIG. **23**B is a cross-sectional view of the memory cube and the rewiring layer taken along a line B**1**-B**2** in FIG. **23**A.

[0043] FIG. **24**A is a schematic diagram showing the configurations of the memory cube and the rewiring layer (the side power wiring and the side ground wiring) according to the first embodiment of the present invention.

[0044] FIG. **24**B is a cross-sectional view of cross-section structures of the memory cube and the rewiring layer taken along a line F**1**-F**2** in FIG. **24**A.

[0045] FIG. **25** is a cross-sectional view showing a mounting example of a semiconductor module according to a ninth embodiment of the present invention.

[0046] FIG. **26** is a cross-sectional view showing a mounting example of the semiconductor module according to the ninth embodiment of the present invention.

[0047] FIG. **27** is a cross-sectional view showing a mounting example of the semiconductor module according to the ninth embodiment of the present invention.

[0048] FIG. **28** is a block diagram showing a configuration of a semiconductor module according to a tenth embodiment of the present invention.

[0049] FIG. **29** is a plan view showing a configuration of a seal ring and a signal transmission wiring according to the tenth embodiment of the present invention.

[0050] FIG. **30** is a cross-sectional view showing cross sections of the seal ring and the signal transmission wiring taken along a line E**3**-E**4** of FIG. **29**.

[0051] FIG. 31 is a cross-sectional view showing cross sections of the seal ring and an inductor taken along a line E5-E6 of FIG. 29.

[0052] FIG. 32 is a cross-sectional view showing cross sections of the seal ring and the inductor shown in FIG. 29.

[0053] FIG. 33 is a plan view showing configurations of the seal ring and the inductor according to the tenth embodiment of the present invention.

[0054] FIG. 34 is a cross-sectional view showing a cross section of the inductor taken along a line E7-E8 of FIG. 33.

[0055] FIG. 35 is a cross-sectional view showing cross sections of the seal ring and inductor taken along a line E9-E10 of FIG. 33.

[0056] FIG. 36 is a cross-sectional view showing a cross section of the inductor taken along a line E11-E12 of FIG. 33.

[0057] FIG. 37 is a plan view showing the configurations of the seal ring and the inductor according to the tenth embodiment of the present invention.

[0058] FIG. 38 is a cross-sectional view showing cross sections of the seal ring and inductor taken along a line E13-E14 of FIG. 37.

[0059] FIG. 39 is a cross-sectional view showing a cross section of the inductor taken along a line E15-E16 of FIG. 37.

[0060] FIG. 40 is a cross-sectional view showing a cross section of the inductor taken along a line E17-E18 of FIG. 37.

[0061] FIG. 41 is a plan view showing a method for manufacturing an inductor according to an eleventh embodiment and a side view of a semiconductor module taken along a line F1-F2.

[0062] FIG. 42 is a plan view showing the method for manufacturing the inductor according to the eleventh embodiment and a cross-sectional view of the semiconductor module taken along a line F3-F4.

[0063] FIG. 43 is a plan view showing the method for manufacturing the inductor according to the eleventh embodiment and a side view of the semiconductor module taken along a line F5-F6.

[0064] FIG. 44 is a plan view showing the method for manufacturing the inductor according to the eleventh embodiment and a cross-sectional view of the semiconductor module taken along a line F7-F8.

[0065] FIG. 45 is a plan view showing the method for manufacturing the inductor according to the eleventh embodiment and a side view of the semiconductor module taken along a line F9-F10.

[0066] FIG. 46 is a plan view showing the method for manufacturing the inductor according to the eleventh embodiment and a cross-sectional view of the semiconductor module taken along a line F11-F12.

[0067] FIG. 47 is a flowchart showing a method for manufacturing a memory cube according to a twelfth embodiment of the present invention.

[0068] FIG. 48 is a plan view showing the method for manufacturing the memory cube according to the twelfth embodiment of the present invention.

[0069] FIG. 49 is a plan view showing the method for manufacturing the memory cube according to the twelfth embodiment of the present invention.

[0070] FIG. 50 is a plan view showing the method for manufacturing the memory cube according to the twelfth embodiment of the present invention and a cross-sectional view showing a cross section of the memory chip taken along a line G1-G2.

[0071] FIG. 51 is a cross-sectional view showing the method for manufacturing the memory cube according to the twelfth embodiment of the present invention.

[0072] FIG. 52 is a cross-sectional view showing the method for manufacturing the memory cube according to the twelfth embodiment of the present invention.

[0073] FIG. 53 is a cross-sectional view showing the method for manufacturing the memory cube according to the twelfth embodiment of the present invention.

[0074] FIG. 54 is a flowchart showing a method for manufacturing a memory cube according to a thirteenth embodiment of the present invention.

[0075] FIG. 55 is a cross-sectional view showing the method for manufacturing the memory cube according to the thirteenth embodiment of the present invention.

[0076] FIG. 55 is a cross-sectional view showing the method for manufacturing the memory cube according to the thirteenth embodiment of the present invention.

[0077] FIG. 56 is a cross-sectional view showing the method for manufacturing the memory cube according to the thirteenth embodiment of the present invention.

[0078] FIG. 57 is a flowchart showing a method for manufacturing a memory cube according to a fourteenth embodiment of the present invention.

[0079] FIG. 58 is a perspective view showing a method for manufacturing the memory cube according to the fourteenth embodiment of the present invention.

[0080] FIG. 59 is a cross-sectional view showing the method for manufacturing the memory cube according to the fourteenth embodiment and a cross-sectional view showing a cross section taken along a line H1-H2.

[0081] FIG. 60 is a cross-sectional view showing the method for manufacturing the memory cube according to the fourteenth embodiment of the present invention.

[0082] FIG. 61 is flowchart showing a method for manufacturing a memory cube according to a fifteenth embodiment of the present invention.

[0083] FIG. 62 is a cross-sectional view and a plan view showing the method for manufacturing the memory cube according to the fifteenth embodiment of the present invention.

[0084] FIG. 63 is a plan view showing a marker of a memory chip according to the fifteenth embodiment of the present invention.

[0085] FIG. 64 is a cross-sectional view showing the method for manufacturing the memory cube according to the fifteenth embodiment of the present invention.

[0086] FIG. 65 is a side view showing a configuration of a memory cube according to a sixteenth embodiment of the present invention.

[0087] FIG. 66 is a side view (top view) showing the configuration of the memory cube according to the sixteenth embodiment of the present invention.

[0088] FIG. 67 is a side view showing the configuration of the memory cube according to the sixteenth embodiment of the present invention.

[0089] FIG. 68 is a cross-sectional view showing a mounting example of a semiconductor module according to the sixteenth embodiment of the present invention.

[0090] FIG. **69** is a side view showing a configuration of a memory cube according to a seventeenth embodiment of the present invention.

[0091] FIG. **70** is a cross-sectional view showing a mounting example of a semiconductor module according to the seventeenth embodiment of the present invention.

[0092] FIG. **71** is a flowchart showing a method for manufacturing a memory cube according to an eighteenth embodiment of the present invention.

[0093] FIG. **72** is a plan view showing the method for manufacturing the memory cube according to the eighteenth embodiment of the present invention and a cross-sectional view showing a cross section of the memory chip taken along a line **J1**-**J2**.

[0094] FIG. **73** is a cross-sectional view showing the method for manufacturing the memory cube according to the eighteenth embodiment of the present invention.

[0095] FIG. **74** is a cross-sectional view showing the method for manufacturing the memory cube according to the eighteenth embodiment of the present invention.

[0096] FIG. **75** is a cross-sectional view showing the method for manufacturing the memory cube according to the eighteenth embodiment of the present invention.

[0097] FIG. **76** is a cross-sectional view showing the method for manufacturing the memory cube according to the eighteenth embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

[0098] For example, well-known three-dimensional implementation techniques may expose sidewalls (for example, also referred to as sides and end surfaces) of a memory cube. When the sidewalls of the memory cube are exposed, the memory cube may become contaminated with metal or the like, which may cause a defect in electrical characteristics of the memory cube. Further, when the sidewalls of the memory cube are exposed, there is a possibility that a problem arises in insulation between respective memory chips in the memory cube and between respective electrodes even in the same chip.

[0099] That is, if the sidewalls of the memory cube are exposed, a problem may arise in long-term reliability of a semiconductor module or the memory cube. Therefore, in the three-dimensional implementation technology, it is a problem to maintain long-term reliability.

[0100] In view of such problems, an object of an embodiment of the present invention is to provide a semiconductor module capable of maintaining long-term reliability.

[0101] Hereinafter, embodiments of the present invention will be described with reference to the drawings and the like. However, the present invention can be implemented in many different aspects, and should not be construed as being limited to the description of the embodiments exemplified below. Although the drawings may be schematically represented with respect to the width, thickness, shape, and the like of each part as compared with the actual embodiment in order to make the description clearer, the drawings are merely examples, and do not limit the interpretation of the present invention. In addition, in the present specification and the drawings, the same reference signs (or reference signs denoted by a, b, and the like) are given to the same elements as those described above with respect to the previous drawings, and detailed description thereof may be omitted as appropriate. Furthermore, the terms "first" and "second" with respect to the respective elements are convenient signs used to distinguish the respective elements, and do not have any further meaning unless otherwise specified.

[0102] In one embodiment of the invention, in the case where a member or region is "above (or below)" another member or region, this includes not only a case where it is directly above (or directly below) the other member or region, unless otherwise limited, but also a case where it is above (or below) the other member or region, that is, a case where another component is included between above (or below) the other member or region.

[0103] In an embodiment of the present disclosure, a direction D**1** intersects a direction D**2**, and a direction D**3** intersects the direction D**1** and the direction D**2** (a plane D**1**D**2**). For example, the direction D**1** is orthogonal to the direction D**2** and the direction D**3**, and the direction D**2** is orthogonal to the direction D**3**. The direction D**1** is referred to as a first direction, the direction D**2** is referred to as a second direction, and the direction D**3** is referred to as a third direction.

[0104] In one embodiment of the present invention, in the case where the terms "same" and "coincide" are used, the same and coincide may include errors within the scope of the design. In addition, in an embodiment of the present invention, in the case where an error within the scope of design is included, the expression "substantially the same" and "substantially coincide" may be used in some cases.

First Embodiment

[0105] The semiconductor module **10** according to the first embodiment will be described with reference to FIG. **1** to FIG. **12**.

[1-1. Overview of Semiconductor Module **10**]

[0106] An overview of a semiconductor module **10** will be described referring to FIG. **1** to FIG. **5**A. FIG. **1** is a perspective view showing a configuration of the semiconductor module **10**. FIG. **2** is a perspective view showing a configuration of a memory chip **110**. FIG. **3** is a cross-sectional view of the memory chip **110** taken along a line A**1**-A**2** shown in FIG. **2**. FIG. **4** is a block diagram showing the configuration of the memory chip **110**. FIG. **5**A is a cross-sectional view of a memory cube **100**.

[0107] First, the configuration of the semiconductor module **10** will be described with reference to FIG. **1**. The semiconductor module **10** includes the memory cube **100**, a logic chip **200**, and a rewiring layer **300**. For example, the memory cube **100**, the logic chip **200**, and the rewiring layer **300** constitute a structure **20**. The semiconductor module **10** may include a wiring substrate **400** and a bump layer **500**. The logic chip **200** may be referred to as a semiconductor chip. Although details will be described later, the structure **20** may include elements other than the memory cube **100**, the logic chip **200**, and the rewiring layer **300**.

[0108] The memory cube **100** includes, for example, a function of storing received data and a function of transmitting stored data. The memory cube **100** includes a configuration in which a plurality of memory chips **110** are stacked. The memory cube **100** includes a first surface **142** parallel to the directions D**2** and D**3** and a second surface **144** opposite the first surface **142** and parallel to the first surface **142** with respect to the direction D**1**. The memory cube **100** also includes a first side surface **146** that is perpendicular to the first surface **142** and the second surface **144**, a second side surface **145** that is adjacent to the first side surface **146**, a third side surface **148** that is adjacent to the second side surface **145**, and a fourth side surface **147** that is adjacent to the third side surface **148** and the first side surface **146**. The first side surface **146** is in contact with and electrically connected to the rewiring layer **300**, and the memory cube **100** is arranged on a second surface **204** of the logic chip **200**.

[0109] In the memory cube **100**, the first side surface **146** is the outermost surface on a side where the logic chip **200** is arranged in the direction D**3**, and the third side surface **148** is the outermost surface on a side opposite the first side surface **146** in the direction D**3**. The second side surface **145** and the fourth side surface **147** opposite the second side surface **145** are the two outermost surfaces of the memory cube **100** in the direction D**2**. Further, the first surface **142** and the second surface **144** opposite the first surface **142** are the two outermost surfaces of the memory cube **100** in the direction D**1**. The first side surface **146** may be referred to as a bottom surface or a cube bottom surface, the second side surface **145** may be referred to as a second outermost surface or a cube front side surface, the fourth side surface **147** may be referred to as a fourth outermost surface or a cube rear side surface, and the third side surface **148** may be referred to as a third outermost surface or a cube upper surface.

[0110] The logic chip **200** includes a function of driving the memory cube **100**. The logic chip **200** includes, for example, a first surface **202** and the second surface **204** which are parallel to the direction D**1** and the direction D**2** intersecting the direction D**1**, a wiring layer **250**, and a transistor layer **230**. The first surface **202** is a surface on which the wiring substrate **400** is arranged, and the second surface **204** is a surface on which the rewiring layer **300** is arranged. The wiring layer **250** and the transistor layer **230** are stacked parallel to the first surface **202** and the second surface **204** in the direction D**3**. In the logic chip **200** according to the present embodiment, a substrate **273** is arranged downward with respect to the direction D**3** (toward the first surface **202** side), and the wiring layer **250** is stacked above the substrate **273** with respect to the direction D**3**. In the semiconductor module **10**, for example, the substrate **273** (the first surface **202**) of the logic chip **200** is arranged on the wiring substrate **400**, and the logic chip **200** is mounted face-up on the wiring substrate **400**.

[0111] The rewiring layer **300** is arranged between the first side surface **146** of the memory cube **100** and the second side surface **204** of the logic chip **200**, and is directly connected to the memory cube **100**, and includes a function of electrically connecting the memory cube **100** and the logic chip **200**.

[0112] Although details will be described later, the wiring substrate **400** includes a multilayer wiring structure in which wirings and insulating layers are alternately stacked, and includes a function of connecting the memory cube **100**, the logic chip **200**, and the like to an external substrate, an external circuit, and the like. The insulating layer may include a low dielectric constant film that allows moisture absorption. The bump layer **500** has a function of connecting the wiring substrate **400** on which the memory cube **100** and the logic chip **200** are mounted to an external substrate, an external circuit, and the like, as will be described in detail later in the same manner as the wiring substrate **400**.

[0113] The memory cube **100** has a length MCBX, a length MCBY, and a length MCBZ along the direction D**1**, the direction D**2**, and the direction D**3**. The rewiring layer **300** has the length MCBX and the length MCBY along the directions D**1** and D**2**. A length of the rewiring layer **300** in the direction D**3** may be any length. The logic chip **200** has a length LCX and a length LCY along the direction D**1** and the direction D**2**. A length of the logic chip **200** in the direction D**3** may be any length. The length MCBX is longer than the length LCX, and the length MCBY is longer than the length LCY. That is, a size of the memory cube **100** is larger than a size of the logic chip **200**.

[0114] Next, the configuration of the memory chip will be described referring to FIG. **1** to FIG. **5**A. As shown in FIG. **1** or FIG. **5**A, each of the plurality of memory chips **110** has the same configuration and function, and includes, for example, a transistor layer **130** and a wiring layer **150**. Each of the plurality of memory chips **110** includes, for example, a memory chip **110***n*, a memory chip **110***n*+1 adjacent to the memory chip **110***n*, a memory chip **110***n*+2 adjacent to the memory chip **110***n*+1, a memory chip **110***n*+3 adjacent to the memory chip **110***n*+2, a memory chip **110***n*+4 adjacent to the memory chip **110***n*+3, and a memory chip **110***n*+5 adjacent to the memory chip **110***n*+4.

[0115] In the case where each of the plurality of memory chips **110** is not distinguished, the memory chip is represented as the memory chip **110**. In the case where each of the plurality of memory chips **110** is distinguished, the memory chip is represented as the memory chip **110***n*, the memory chip **110***n*+1, the memory chip **110***n*+2, and the like.

[0116] As shown in FIG. **2**, the memory chip **110** includes a first surface **102** and a second surface **104** that are parallel to the directions D**2** and D**3**. The first surface **102** is a surface on which the transistor layer **130** is arranged, and the second surface **104** is a surface on which the wiring layer **150** is arranged. The first surface **102** and the second surface **104** are parallel to the first surface **142** and the second surface **144**.

[0117] In addition, the memory chip **110** also includes a first side surface **106** perpendicular to the first side surface **102** and the second side surface **104**, a second side surface **105** adjacent to the first side surface **106**, a third side surface **108** adjacent to the second side surface **105**, and a fourth side surface **107** adjacent to the third side surface **108** and the first side surface **106**. The first side surface **106** is part of the first side surface **146**, the second side surface **105** is part of the second side surface **145**, the third side surface **108** is part of the third side surface **148**, and the fourth side surface **107** is part of the fourth side surface **147**.

[0118] As shown in FIG. **3**, the wiring layer **150** is stacked on the transistor layer **130** in the direction D**1**. The transistor layer **130** includes, for example, portions of a substrate **173**, an element isolation region **174**, an activation region **175**, a transistor **176**, an insulating layer **177**, and a wiring **178**. The substrate **173** is, for example, a Si substrate or a Si-wafer.

[0119] The wiring layer **150** includes a multilayer wiring structure in which wirings and insulating layers are alternately stacked. The wiring layer **150** includes, for example, portions of the wiring **178**, an insulating layer **179**, a wiring **180**, an insulating layer **181**, an insulating layer **182**, and a plurality of ground wirings **165**. An end surface **165***a* of the ground wiring **165** of a part of the plurality of ground wirings **165** is exposed to the first side surface **106** (the first side surface **146** of the memory cube **100**) of the memory chip **110**, and the end surface **165***a* is flush with the first side surface **106**. A plurality of power wirings **164** (see FIG. **2**) and a plurality of signal transmission wirings **166** (see FIG. **2**) are arranged in the same layer as the plurality of ground wirings **165**. Similar to the plurality of ground wirings **165**, part of the plurality of power wirings **164** and part of a plurality of signal transmission wirings **166** are exposed to the first side surface **106** of the memory chip **110**, and each end surface is flush with the first side surface **106**. That is, inner wirings (the ground wiring **165**, the power wiring **164**, and the signal transmission wiring **166**) of the memory chip **110** extend in the direction D**3** and are exposed to the first side surface **146** of the memory cube **100**, and the first side surface **146** is flattened.

[0120] The number of layers of the multilayer wiring in the wiring layer **150** is not limited to three layers shown in FIG. **3**. The number of layers of the multilayer wiring in the wiring layer **150** may be four or more. The number of layers of the multilayer wiring in the wiring layer **150** can be appropriately changed according to the specifications, applications, and the like of the semiconductor module **10**.

[0121] In addition, the logic chip **200** includes a configuration and a function similar to those of the stacked structure shown in FIG. **3**. Therefore, although detailed explanation is omitted, for example, the substrate **273** is a Si substrate or a Si-wafer, similar to the substrate **173**. The wiring layer **250** includes a multilayer wiring structure in which wirings and insulating layers are alternately stacked in the same manner as the wiring layer **150**. Further, the logic chip **200** includes a plurality of transistors formed in the same manner as the transistor **176**, and includes a plurality of logic circuits formed using the plurality of transistors or the like, and can drive the memory cube **100** using the plurality of logic circuits.

[0122] As shown in FIG. **4**, the memory chip **110** includes a plurality of memory modules **111**. Each of the plurality of memory modules **111** includes a memory cell array **115**.

[0123] Each of the plurality of memory modules **111** includes a function of storing data included in the received signal in the memory cell array **115**, a function of reading data from the memory cell array **115**, and a function of transmitting a signal including data. The plurality of memory modules **111** is electrically connected to the plurality of power wirings **164**, the plurality of ground wirings **165**, and the plurality of signal transmission wirings **166**.

[0124] Although explained in detail later, the plurality of power wirings **164** and the plurality of ground wirings **165** are electrically connected to, for example, the logic chip **200** and an external circuit (not shown), and a power supply voltage VDD, a voltage VSS, and the like are supplied from the external circuit. The power supply voltage VDD is, for example, 3 V or 5 V. The voltage VSS is, for example, a ground voltage of 0 V or the like. Further, the plurality of signal transmission wirings **166** is electrically connected to, for example, the logic chip **200** and an external circuit (not shown), and an address signal, an enable signal, and the like for controlling the memory chip **110** are supplied from the logic chip **200** and the external circuit.

[0125] The memory cell array **115** includes a plurality of memory cells (not shown). Each of the plurality of memory cell arrays **115** is, for example, a SRAM (Static Random Access Memory), and each of the plurality of memory cells is a SRAM cell. In addition, SRAM, SRAM cell, and the memory module **111** for SRAM can employ a technique used in the technical field of SRAM. Therefore, detailed description will be omitted here. In addition, the plurality of memory cell arrays **115** and the plurality of memory cells may be memory cell arrays and memory cells other than SRAM, and may be, for example, DRAM (Dynamic Random Access Memory) and DRAM cells, MRAM (Magnetoresistive Random Access Memory) and MRAM cells, or the like.

1-2. Example of Manufacturing Method of Memory Cube **100**

[0126] An example of a method for manufacturing the memory cube **100** will be mainly described referring to FIG. **5**A to FIG. **10**B. FIG. **5**A is a cross-sectional view of the memory cube **100**, and FIG. **5**B to FIG. **6**B are schematic views showing the method for manufacturing the memory cube **100**. The same or similar configurations as those in FIG. **1** to FIG. **4** will not be described here.

[0127] Stacking (bonding) memory chips **110** so as to face the second surfaces **104** facing each other on the wiring layers **150** is referred to as, for example, F2F bonding (Face to Face Fusion). Stacking (bonding) memory chips **110** such that the first surfaces **102** of the transistor layers **130** are opposite each other on the substrate **173** side, is referred to as, for example, B2B bonding (Back to Back Fusion). Stacking (bonding) memory chips **110** so that the second surface **104** on the wiring layer **150** side faces the first surface **102** on the substrate **173** side included in the transistor layer **130** is referred to as, for example, F2B bonding (Face to Back Fusion). For example, a technique such as welding (Fusion Bonding) or silicon direct bonding (Silicon Direct Bonding (SDB)) can be used for stacking (bonding) the memory chips. Since welding and silicon direct bonding are techniques used in the art, detailed description is omitted here.

[0128] As shown in FIG. **5**A, the memory cube **100** includes, for example, a structure in which the memory chips **110***n* to **110***n*+5 are stacked. For the sake of clarity, in the memory chip **110**, the transistor layer **130** including the substrate **173** and the wiring layer **150** including the ground wiring **165** are shown, and illustration of other components is omitted.

[0129] The memory chip **110***n* includes a transistor layer **130***n* including a substrate **173***n* and a wiring layer **150***n* including a ground wiring **165***n*. Similar to the memory chip **110***n*, the memory chip **110***n*+1 includes a transistor layer **130***n*+1 including the substrate **173***n*+1 and the wiring layer **150***n*+1 including the ground wiring **165***n*+1. As with the memory chip **110***n* and the memory chip **110***n*+1, the transistor layer **130** including the substrate **173** and the wiring layer **150** including the ground wiring **165** corresponding to each of the memory chips **110***n*+2 to 110n+5 are labeled with the subscripts n+2 to n+5 of the memory chips **110**.

[0130] Next, a configuration of the memory chip **110** in a cross-sectional view shown in FIG. **5**A will be described. In the memory chip **110***n*, the ground wiring **165***n* is exposed to a first side surface **106***n*, and an end surface **165**an of the ground wiring **165***n* coincides with the first side surface **106***n*. In addition, a length between an end surface of the substrate **173***n* and the first side surface **106***n* is a length Widn, a space between the end surface of the substrate **173***n* and the first side surface **106***n* is filled with an insulating film **190**, and an end surface of the insulating film **190** coincides with an exposed wiring surface of the first side surface **106***n*.

In addition, the insulating film **190** may cover side surfaces of an insulating layer on the first side surface **106***n*. Although details will be described later, it is important to cover the insulating layer with an insulating film **190** that prevents moisture absorption from side surfaces thereof, particularly in the case where the insulating layer includes a low dielectric constant film. In this case, SiO_2, SIN, SiON, SICN, a composite film thereof, or the like can be used as the insulating film. The first surface **142** and the second surface **144** of the memory cube **100** may be covered. The insulating film **190** may be referred to as a first insulating film. In the memory chip **110** (the memory cube **100**), the end surface of the substrate **173** is the outermost surface on which the logic chip **200** is arranged in the direction D**3**. The end surface of the substrate **173** may be referred to as a substrate bottom surface.

[0131] Similar to the memory chip **110***n*, in the memory chip **165***n*+1, the ground wiring **165***n*+1 is exposed to a first side surface **106***n*+1, and an end surface **165**an +1 of the ground wiring **165***n*+1 coincides with the first side surface **106***n*+1. Further, a length from the end surface of the substrate **173***n*+1 to the first side surface **106***n*+1 is a length Widn+1, the insulating film **190** is filled between the end surface of the substrate **173***n*+1 and the first side surface **106***n*+1, and the end surface of the insulating film **190** coincides with the first side surface **106***n*+1.

[0132] In the memory chip **110***n*+2, a length from an end surface of a substrate **173***n*+2 to the first side surface **106***n*+2 is a length Widn+2. In the memory chip **110***n*+3, a length from an end surface of a substrate **173***n*+3 to a first side surface **106***n*+3 is a length Widn+3. In the memory chip **110***n*+4, a length from an end surface of a substrate **173***n*+4 to a first side surface **106***n*+4 is a length Widn+4. In the memory chip **110***n*+5, a length from an end surface of a substrate **173***n*+5 to a first side surface **106***n*+5 is a length Widn+5. The other configurations of the memory chips **110***n*+2 to 110n+5 are the same as those of the memory chip **110***n* and the memory chip **110***n*+1, and detailed explanation thereof is omitted here.

[0133] Although not shown, similar to ground wirings **165***n* to **165***n*+5 included in the memory chips **110***n* to **110***n*+5, each of the power wirings and the signal transmission wirings is exposed to the first side surfaces **106***n* to **106***n*+5, and end surfaces of the power wirings and the signal transmission wirings coincide with the first side surfaces **106***n* to **106***n*+5. In addition, the memory chips **110***n* to **110***n*+5 include the plurality of ground wirings **165**, the plurality of power wirings **164** and the plurality of signal transmission wirings **166**, part of the ground wiring **165** of the plurality of ground wirings **165**, part of the power wiring **164** of the plurality of power wirings **164**, and part of the signal transmission wiring **166** of the plurality of signal transmission wirings **166** are exposed to the first side surface **106***n* to **106***n*+5.

[0134] In the memory cube **100**, the first side surface **146** is polished and planarized, as described in detail below. That is, the first side surface **146** coincides with the first side surfaces **106***n* to **106***n*+5 of each of the memory chips **110***n* to **110***n*+5. Further, although details will be described later, in the memory cube **100**, a length of each memory chip **110** from the end surface of the substrate **173** to the first side surface **106** is different. That is, the lengths Widn to Widn+5 differ in length.

[0135] Next, the method for manufacturing the memory cube **100** will be described. Step **1** of the method for manufacturing the memory cube **100** is shown in FIG. **5**B. In step **1**, the second surface **104** of the memory chip **110***n* and the second surface **104** of the memory chip **110***n*+1 are stacked (bonded) so as to face each other, the second surface **104** of the memory chip **110***n*+2 and the second surface **104** of the memory chip **110***n*+3 are stacked (bonded) so as to face each other, and the second surface **104** of the memory chip **110***n*+4 and the second surface **104** of the memory chip **110***n*+5 are stacked (bonded) so as to face each other. That is, in step **1**, the two memory chips **110***n* and **110***n*+1, the two memory chips **110***n*+2 and 110n+3, and the two memory chips **110***n*+4 and 110n+5 are bonded by the F2F bonding.

[0136] Further, in step **1**, the memory chip **110***n* and the memory chip **110***n*+1 which are F2F bonded to each other are bonded to the memory chip **110***n*+2 and the memory chip **110***n*+3 which are F2F bonded to each other, and the memory chip **110***n*+2 and the memory chip **110***n*+3 which are F2F bonded to each other are bonded to the memory chip **110***n*+4 and the memory chip **110***n*+5 which are F2F bonded to each other.

[0137] For example, the memory chip **110***n* and the first surface **102** of the memory chip **110***n*+1 on a side of the memory chip **110***n* are bonded to the memory chip **110***n*+2 and the first surface **102** of the memory chip **110***n*+3 on a side of the memory chip **110***n*+2. That is, the four memory chips **110***n* to **110***n*+3 are B2B bonded. Further, although not shown, a memory chip **110***n*+6 and a first surface **102** of a memory chip **110***n*+7 on a side of the memory chip **110***n*+6 are bonded to the memory chip **110***n*+4 and the first surface **102** of the memory chip **110***n*+5 on a side of the memory chip **110***n*+4. That is, the four memory chips **110***n*+4 to 110n+7 are B2B bonded. The four B2B bonded memory chips **110***n* to **110***n*+3 are B2B bonded to the four memory chips **110***n*+4 to 110n+7.

[0138] For example, memory chips **110***n* to **110***n*+63 may be stacked (bonded) to form the memory cube **100** in which the **64**-layer memory chips **110** are stacked. The number of stacked memory chips **110** is not limited to 64 layers, and may be 128 layers or 256 layers. Further, the method for stacking the memory chips **110** is not limited to the method described in the present embodiment. F2B bonding may be repeated as the method for stacking the memory chips **110**. The number of stacked memory chips **110** and the method of stacking can be appropriately selected based on the specifications, applications, and the like of the semiconductor module **10** without departing from the scope of the present invention.

[0139] As shown in FIG. **5**B, in a state where the plurality of memory chips **110***n* to **110***n*+5 is stacked in step **1**, a step-like step occurs at a joint surface between the memory chips. That is, in a state where the plurality of memory chips **110***n* to **110***n*+5 is stacked in step **1**, the first side surfaces **106***n* to **106***n*+5 of the respective memory chips are not in the same plane.

[0140] Next, in step **2**, the substrate **173** is selectively etched by anisotropic dry etching (see FIG. **6**A). Since the first side surfaces **106***n* to **106***n*+5 of each memory chip are not flush with each other, a step-like step occurs at the joint surface between the substrates **173***n* to **173***n*+5 of each memory chip.

[0141] Next, in step **3**, the insulating film **190** is provided in a step-like step at the joint surface between the memory chips and in a step-like step at the joint surface between the substrates **173***n* to **173***n*+5 of the memory chips (see FIG. **6**B). Since the insulating film **190** is embedded in the step-like step at the joint surface between the memory chips and the step-like step at the joint surface between the substrates **173***n* to **173***n*+5 of the memory chips, the step-like step is relaxed on the surface of the memory cube **100** provided with the insulating film **190**.

[0142] Next, in step **4**, a surface of the memory cube **100** is polished and the first side surface **146** of the memory cube **100** is planarized (see FIG. **5**A). Consequently, as shown in FIG. **5**A, for example, the end surface **165***a* of the ground wiring **165** is exposed to the first side surface **146**. Polishing can be performed using, for example, a chemical mechanical polishing (Chemical Mechanical Polishing (CMP)) device.

[0143] Here, a case in which the insulating layer in the semiconductor module **10** includes a low dielectric constant film will be described with reference to FIG. **22**A and FIG. **22**B. FIG. **22**A and FIG. **22**B are cross-sectional views showing a cross section configuration of the memory cube **100**. In the semiconductor module **10**, the insulating layer (the element isolation region **174**) and the insulating layer **177** are sometimes referred to as insulating layers **184**. Insulating layers **184** (**184***n*, **184***n*+1, 184n+2, 184n+3, and 184n+4) are formed using, for example, a low dielectric constant film having a dielectric constant lower than that of the insulating film **190**. In this case, after FIG. **6**A, the insulating layers **184***n* to **184***n*+4 are selectively etched. Further, the insulating film **190** is provided in the step-like step at the joint surface between the memory chips and in the step-like step at the joint surface between the substrates **173***n* to **173***n*+5 of the memory chips. Subsequently, the surface of the memory cube **100** is polished, and the first side surface **146** of the memory cube **100** is planarized (see FIG. **22**A).

[0144] In addition, in the semiconductor module **10**, the insulating layer **179**, the insulating layer **181**, and the insulating layer **182** may be referred to as insulating layers **185**. The insulating layers **185** (**185***n*, **185***n*+1, 185n+2, 185n+3, and 185n+4) are formed using, for example, a low dielectric constant film having a dielectric constant lower than that of the insulating film **190**. In the case where the insulating layers **184** to **184***n*+4 and the insulating layers **185***n* to **185***n*+4 are formed using a low dielectric constant film, the insulating layers **184** and the insulating layers **185** are selectively etched after FIG. **6**A. Further, the insulating film **190** is provided in the step-like step at the joint surface between the memory chips and in the step-like step at the joint surface between the substrates **173***n* to **173***n*+5 of the memory chips. Subsequently, the surface of the memory cube **100** is polished, and the first side surface **146** of the memory cube **100** is planarized (see FIG. **22**B).

[0145] In FIG. **5**A to FIG. **6**B, a method for manufacturing the first side **146** of the memory cube **100** is shown. Similar to the first side surface **146** of the memory cube **100**, the second side surface **145**, the third side surface **148**, and the fourth side surface **147** of the memory cube **100** are planarized by steps **1** to **4**, and the end surface **165***a* of the ground wiring **165**, an end surface of the power wiring **164**, or an end surface of the signal transmission wiring **166** is exposed to the respective side surfaces. Although an example is shown in which the method for manufacturing the memory cube **100** includes steps **1** to **4**, the method for manufacturing the memory cube **100** is not limited to steps **1** to **4**. For example, if the first side surface **146**, the second side surface **145**, the third side surface **148**, and the fourth side surface **147** of the memory cube **100** are flattened using the same steps as those of step **1** to step **4**, and the end surface of the ground wiring **165**, the end surface of the power wiring **164**, or the end surface of the signal transmission wiring **166** is exposed to each side surface, there is no restriction on the method for manufacturing the memory cube **100**.

[0146] The memory cube **100** can be manufactured as described above. In the semiconductor module **10**, capacity of the memory cube **100** can be increased by stacking the plurality of memory chips **110**.

[0147] In the method for manufacturing the memory cube **100**, the substrate **173** is etched, and the step-like step at the joint surface between the memory chips and the step-like step at the joint surface between the substrates of the memory chips can be filled with the insulating film **190**. As a result, since the first side surface **146**, the second side surface **145**, the third side surface **148**, and the fourth side surface **147** of the memory cube **100** are covered with the insulating film **190**, it is possible to prevent the first side surface **106** of the substrate **173** from being contaminated by metal, and prevent corrosion and deterioration of each device in the memory cube **100** due to moisture absorption and intrusion of impurities. Therefore, in the semiconductor module **10**, contamination from the side surface of the memory cube **100** is suppressed, and reliability can be maintained without impairing long-term reliability of the semiconductor module **10**.

[0148] In addition, in the method for manufacturing the memory cube **100**, since the first side surface **146**, the second side surface **145**, the third side surface **148**, and the fourth side surface **147** of the memory cube **100** are covered with the insulating film **190**, for example, it is possible to suppress short circuits between adjacent electrodes and memory chips caused by metal contamination of the first side surface **146**, the second side surface **145**, the third side surface **148**, and the fourth side surface **147**. Therefore, in the semiconductor module **10**, it is possible to ensure insulation between adjacent electrodes and memory chips, and it is possible to maintain the reliability without impairing the long-term reliability of the semiconductor module **10**.

[0149] Further, in the method for manufacturing the memory cube **100**, for example, by polishing the first side surface **146**, the second side surface **145**, the third side surface **148**, and the fourth side surface **147**, the ground wiring **165**, the power wiring **164**, or the signal transmission wiring **166** can be exposed on the first side surface **146**, the second side surface **145**, the third side surface **148**, and the fourth side surface **147** in a self-aligned manner. Therefore, in the method for manufacturing the memory cube **100**, a side surface wiring can be formed on the wiring exposed on the side surface by a simple process, rather than the manufacturing method of forming each wiring using a complicated process while aligning. In addition, in the method for manufacturing the memory cube **100**, since the side surface wiring can be formed on the exposed wiring, the exposed wiring is capped with the side surface wiring to prevent oxidation of the exposed wiring, and the side surface wiring can be formed on the side surface of the memory cube **100**, so that corrosion, deterioration, and the like of each device in the memory cube **100** caused by moisture absorption, impurities, and the like can be suppressed. That is, the reliability can be maintained without impairing the long-term reliability of the semiconductor module **10**.

[0150] In addition, since the method for manufacturing the memory cube **100** can form the side surface wiring on the side surface of the memory cube **100**, it is possible to expand choices of electrical connection between the memory cube **100** and various sizes of logic chips, wiring substrates, external circuits, and the like. That is, the method for manufacturing the memory cube **100** can improve degree of freedom in design.

[0151] In addition, in the method for manufacturing the memory cube **100**, the first side surface **146**, the second side surface **145**, the third side surface **148**, and the fourth side surface **147** of the memory cube **100** can be made flat by using the insulating film **190**. Therefore, in the method for manufacturing the memory cube **100**, it is possible to allow positional deviation immediately after stacking of the plurality of memory chips **110**, and it is possible to insulate the memory chips **110** adjacent to each other by using the insulating film **190**. Therefore, in the semiconductor module **10**, it is possible to ensure insulation between adjacent electrodes and memory chips, and it is possible to maintain the reliability without impairing the long-term reliability of the semiconductor module **10**. In addition, since each side surface of the memory cube **100** is flattened, irregularities on the memory cube **100** are relaxed, for example, effective heat conduction from the memory cube **100** to the logic chip **200**, the wiring substrate **400**, and the like can be improved.

[0152] Further, in the method for manufacturing the memory cube **100**, a length between the first side surface **106** of the memory chip **110** and the end surface of the insulating film **190** is different for each memory chip. Therefore, since a difference in contrast between adjacent memory chips can be easily recognized, for example, in the side formation of the side surface wiring after step **4**, each layer of the plurality of memory chips **110** can be easily recognized.

1-3. Example of Method for Manufacturing Rewiring Layer **300**

[0153] An example of a method for manufacturing the rewiring layer **300** will be mainly described referring to FIG. **7**A to FIG. **10**B. FIG. **7**A is a schematic diagram showing a method for manufacturing the memory cube **100** and the rewiring layer **300** (a side power wiring **162** and a side ground wiring **163**), and FIG. **7**B is a cross-sectional view of the memory cube **100** taken along a line B**1**-B**2** in FIG. **7**A. Further, FIG. **7**A and FIG. **7**B are schematic diagrams showing a state similar to the state of FIG. **5**A described in "1-2. Example of Manufacturing Method of Memory Cube **100**". Further, FIG. **7**A is an enlarged view of the first side surface **146** of the memory cube **100** as viewed from the direction D**3**. FIG. **8**A is a schematic diagram showing the configuration of the memory cube **100** and the rewiring layer **300** (the side power wiring **162** and the side ground wiring **163**), and FIG. **8**B is a cross-sectional view showing cross section configurations of the memory cube **100** and the rewiring layer **300** taken along a line B**1**-B**2** of FIG. **8**A. FIG. **9**A is a schematic diagram showing the method for manufacturing the memory cube **100** and the rewiring layer **300** (a side signal transmission wiring **167**), and FIG. **9**B is a cross-sectional view of the memory cube **100** taken along a line C**1**-C**2** of FIG. **9**A. Further, FIG. **9**A and FIG. **9**B are schematic diagrams showing a state similar to the state of FIG. **5**A described in "1-2. Example of Manufacturing Method of Memory Cube **100**". FIG. **9**A is an enlarged view of the first side surface **146** of the memory **100** as viewed from the direction D**3**. FIG. **10**A is a schematic diagram showing the configurations of the memory cube **100** and the rewiring layer **300** (the side signal transmission wiring **167**), and FIG. **10**B is a cross-sectional view showing the cross section structure of the memory cube **100** taken along a line C**1**-C**2** of FIG. **10**A. The same or similar configurations as those in FIG. **1** to FIG. **6**B will not be described here.

[0154] First, referring to FIG. **7**A and FIG. **8**B, a method for forming the plurality of side power wirings **162** and the plurality of side surface ground wirings **163** included in the rewiring layers **300** will be described.

[0155] As shown in FIG. **7**A, the insulating film **190**, the plurality of power wirings **164**, and the plurality of ground wirings **165** included in the plurality of memory chips **110*n*** to **110*n***+5 are exposed to the first side surface **146** of the memory cube **100**. As described above, FIG. **7**B is a schematic diagram showing a state similar to the state of FIG. **5**A described in "1-2. Example of Manufacturing Method of Memory Cube **100**" (that is, a state in which the manufacturing method of step **4** is shown), and is a diagram in which the ground wiring **165** in FIG. **5**A is replaced with the power wiring **164**. Since other configurations are the same as those in FIG. **5**A, detailed explanation is omitted here.

[0156] The method for manufacturing the rewiring layers **300** is shown in FIG. **8**A and FIG. **8**B. For the sake of clarity, an insulating film **168** is not shown in FIG. **8**A. For example, in the manufacturing method of the rewiring layer **300**, step **4** of the manufacturing method of the memory cube **100** is followed by step **5**. In step **5**, for example, the insulating film **168** is formed so that the insulating film **168** remains between the two side power wirings **162** and between the two ground wirings **165**. For example, as shown in FIG. **8**B, the insulating film **168** is formed on a part of the first side surfaces **106*n***+1 and 106n+2 of the memory chips **110*n***+1 and 110n+2 located between the two side power wirings **162** and between the two ground wirings **165**.

[0157] In step **6**, the side power wirings **162** are formed on the plurality of power wirings **164**, and the side ground wirings **163** are formed on the plurality of ground wirings **165** as shown in FIG. **8**A and FIG. **8**B. For example, the side power wiring **162** is arranged so as to be in contact with portions of a side surface and an upper surface of the insulating film **168**, the four power wirings **164** exposed to the first side surfaces **106*n***+2 to 106n+5 of the memory chips **110*n***+2 to 110n+5, and the first side surfaces **106*n***+2 to 106n+5 of the memory chips **110*n***+2 to 110n+5. Further, another side power wiring **162** is arranged so as to be in contact with portions of the side surface and the upper surface of the insulating film **168**, the two power wirings **164** exposed to the first side surfaces **106*n*** to **106*n***+1 of the memory chips **110*n*** to **110*n***+1, and the first side surfaces **106*n*** to **106*n***+1 of the memory chips **110*n*** to **110*n***+1.

[0158] Further, for example, similar to the side power wiring **162**, the side ground wiring **163** is arranged so as to be in contact with a part of the side surface and the upper surface of the insulating film **168**, the four ground wirings **165** exposed to the first side surfaces **106*n***+2 to 106n+5 of the memory chips **110*n***+2 to 110n+5, and the first side surfaces **106*n***+2 to 106n+5 of the memory chips **110*n***+2 to 110n+5. Further, for example, similar to the other side power wiring **162**, the side ground wiring **163** is arranged so as to be in contact with portions of the side surface and the upper surface of the insulating film **168**, the two ground wirings **165** exposed to the first side surfaces **106***n* to **106***n*+1 of the memory chips **110***n* to **110***n*+1, and the first side surfaces **106***n* to **106***n*+1 of the memory chips **110***n* to **110***n*+1. That is, the memory chips that are F2F bonded to each other are electrically connected to each other by the same side power wiring **162** and the same side ground wiring **163**.

[0159] The side power wirings **162** and the side ground wirings **163** extend in the direction D**1**, for example, and are alternately arranged along the direction D**2** in parallel to the direction D**1**.

[0160] Next, referring to FIG. **9**A to FIG. **10**B, a method for forming a plurality of side signal transmission wirings **167** included in the rewiring layers **300** will be described. The method for forming the plurality of side signal transmission wirings **167** explained with reference to FIG. **9**A to FIG. **10**B is similar to the method for forming the plurality of side power wirings **162** and the plurality of side ground wirings **163** explained with reference to FIG. **7**A to FIG. **8**B, in which the configuration of the plurality of side power wirings **162** and the plurality of side ground wirings **163** are replaced with the configuration of the side signal transmission wirings **167**. Therefore, in the method for forming the plurality of side signal transmission wirings **167** explained referring to FIG. **9**A to FIG. **10**B, the configurations of the plurality of side grounded wirings **163** differing from those described with reference to FIG. **7**A to FIG. **8**B will be mainly described.

[0161] The plurality of memory chips **110***n* to **110***n*+5 shown in FIG. **9**A to FIG. **10**B are enlarged views of regions of the first side surfaces **146** of the plurality of memory chips **110***n* to **110***n*+5 that differ from the plurality of memory chips **110***n* to **110***n*+5 shown in FIG. **7**A to FIG. **8**B.

[0162] The plurality of side surface signal transmission wirings **167** included in the rewiring layer **300** is formed in steps **5** and **6** in the same manner as the plurality of side power wirings **162** and the plurality of side ground wirings **163** included in the rewiring layer **300**.

[0163] As shown in FIG. **9**A, the insulating film **190** and the plurality of signal transmission wirings **166** included in the plurality of memory chips **110***n* to **110***n*+5 are exposed on the first side surface **146** of the memory cube **100**. For the sake of clarity, the insulating film **168** is not shown in FIG. **9**A. The plurality of signal transmission wirings **166** are arranged in a staggered manner. For example, the plurality of signal transmission wirings **166** exposed on the first side surfaces **106***n* and **106***n*+1 of the memory chips **110***n* and **110***n*+1 are arranged in a staggered manner, the plurality of signal transmission wirings **166** exposed on the first side surfaces **106***n*+2 and 106n+3 of the memory chips **110***n*+2 and 110n+3 are arranged in a staggered manner, and the plurality of signal transmission wirings **166** exposed on the first side surfaces **106***n*+4 and 106n+5 of the memory chips **110***n*+4 and 110n+5 are arranged in a staggered manner. Also, in the case where the plurality of signal transmission wirings **166** exposed on the first side surface **146** are viewed from the direction D**3**, the plurality of signal transmission wirings **166** are arranged in a staggered manner.

[0164] Therefore, as shown in FIG. **9**B, in the cross-sectional view taken along a line C**1**-C**2**, the plurality of signal transmission wirings **166** are exposed on the first side surfaces (**106***n*+1, 106n+3, and 106n+5) of the memory chips **110***n*+1, 110n+3, and 110n+5.

[0165] In step **5**, for example, the insulating film **168** is formed so that the insulating film **168** remains between the two side surface signal transmission wirings **167**. For example, as shown in FIG. **10**B, the insulating film **168** is formed between the two side signal transmission wirings **167** on part of the first side surfaces **106***n*+1 to 106n+3 of the memory chips **110***n*+1 to 110n+3 and part of the first side surfaces **106***n*+3 to 106n+5 of the memory chips **110***n*+3 to 110n+5.

[0166] In step **6**, as shown in FIG. **10**A and FIG. **10**B, the plurality of side signal transmission wirings **167** is formed on each of the plurality of signal transmission wirings **166**. For example, the side signal transmission wiring **167** is provided so as to be in contact with portions of the side surface and the upper surface of the insulating film **168** and the signal transmission wiring **166** exposed on the first side surface **106***n*+1 of the memory chip **110***n*+1. Similar to the signal transmission wiring **166** exposed on the first side surface **106***n*+1 of the memory chip **110***n*+1, the signal transmission wirings **166** exposed on the first side surfaces **106***n*+1 to 106n+3 of the memory chips **110***n*, **110***n*+2 to 110n+5 are in contact with the side signal transmission wirings **167** corresponding to the respective signal transmission wirings **166**.

[0167] Each of the plurality of side signal transmission wirings **167** is provided so as to cover, for example, the signal transmission wiring **166**, and extends parallel to the direction D1, and then extends parallel to the direction D**2**.

[0168] The insulating film **168**, the plurality of side power wirings **162**, the plurality of side ground wirings **163**, and the plurality of side signal transmission wirings **167** are formed, for example, by photolithography.

[0169] As described above, the plurality of power wirings **164**, the plurality of ground wirings **165**, and the plurality of signal transmission wirings **166** exposed in the memory cube **100** are electrically connected to the side power wirings **162**, the plurality of side ground wirings **163**, and the plurality of side signal transmission wirings **167** included in the rewiring layer **300**. As a result, the memory cube **100** is electrically connected to the rewiring layer **300**.

[0170] In addition, although an example is shown in which the method for manufacturing the memory cube **100** and the rewiring layer **300** includes steps **1** to **6**, the method for manufacturing the memory cube **100** and the rewiring layer **300** is not limited to steps **1** to **6**. For example, if the method in which the memory cube **100** and the rewiring layer **300** are formed using the same steps as steps **1** to **6**, there is no restriction on the method for manufacturing the memory cube **100** and the rewiring layer **300**.

[0171] In the method for manufacturing the memory cube **100** and the rewiring layer **300**, wiring can be formed on a side surface by a simpler process than a manufacturing method for forming each wiring using a complicated process while aligning. In addition, in the method for manufacturing the memory cube **100** and the rewiring layer **300**, since wiring can be formed on the side surface of the memory cube **100**, it is possible to expand choices of electrical connection between the memory cube **100** and various sizes of logic chips, wiring substrates, external circuits, and the like. That is, the manufacturing method of the semiconductor module **10** can improve the degree of freedom in design.

[0172] Here, a rewiring layer **300**A having a configuration differing from the rewiring layer **300** shown in FIG. **8**A to FIG. **10**B will be described referring to FIG. **23**A, FIG. **23**B, FIG. **24**A, and FIG. **24**B. The rewiring layer **300**A is formed using two wiring layers.

[0173] FIG. **23**A is a schematic diagram showing configurations of the memory cube **100** and the rewiring layer **300**A (side power wiring and side ground wiring), FIG. **23**B is a cross-sectional view showing cross-sectional configurations of the memory cube **100** and the rewiring layer **300**A taken along a line B**1**-B**2** of FIG. **23**A. FIG. **24**A is a schematic diagram showing the configurations of the memory cube **100** and the rewiring layer **300**A (side power wiring and side ground wiring), FIG. **24**B is a cross-sectional view showing the cross-sectional configuration of the memory cube **100** and the rewiring layer **300**A taken along a line F1-F2 of FIG. **24**A. In the explanation of the rewiring layer **300**A with reference to FIG. **23**A to FIG. **24**B, differences from the rewiring layer **300** with reference to FIG. **8**A to FIG. **10**B are mainly explained.

[0174] The rewiring layer **300**A includes at least an insulating layer **189** including the side power wiring **162**, the side ground wiring **163**, a second side power wiring **186**, a second side ground wiring **187**, and an opening **188**.

[0175] As shown in FIG. **23**A and FIG. **23**B, the side power wirings **162** and the side ground wirings **163** of the rewiring layer **300**A extend in the direction D**1**, for example, and are alternately arranged along the direction D**2** parallel to the direction D**1**. The side power wiring **162** and the side ground wiring **163** are formed in the same layer. For example, the side power wiring **162** of the rewiring layer **300**A is arranged so as to be in contact with portions of the side surface and the upper surface of the insulating film **168** and the four power wirings **164** exposed to the first side surfaces **106***n* to **106***n*+5 of the memory chips **110***n* to **110***n*+5. Similar to the side power wiring **162** of the rewiring layer **300**, for example, the side ground wiring **163** of the rewiring layer **300**A is arranged so as to be in contact with the four ground wirings **165** exposed on portions of the side surface and the upper surface of the insulating film **168** and the first side surfaces **106***n* to **106***n*+5 of the memory chips **110***n* to **110***n*+5. The side power wiring **162** and the side ground wiring **163** include a conductor made of a metal material, and are, for example, a conductor containing copper, gold, silver, or the like. In the method for manufacturing the rewiring layer **300**A, after the same configuration as that of the method for manufacturing the rewiring layer **300** is formed up to step **4**, the side power wiring **162** and the side ground wiring **163** are formed (for example, a step **7**).

[0176] After step **7**, in step **8**, the insulating layer **189** is formed between the side power wiring **162** and the side ground wiring **163**, on the upper surfaces and the side surfaces of the side power wiring **162** and the side ground wirings **163**, and on portions of the first side surfaces **106***n* to **106***n*+5 where the side power wiring **162** and the side ground wiring **163** are not formed.

[0177] After step **8**, in step **9**, the opening **188** is formed in the insulating layer **189**, and a portion of the side power wiring **162** and a portion of the side ground wiring **163** are exposed. After step **9**, in step **10**, the second side power wiring **186** and the second side ground wiring **187** extend in the direction D3 for example, and are alternately arranged in the direction D**1** parallel to the direction D**3**. The second side power wiring **186** and the second side ground wiring **187** are formed in the same layer. The second side power wiring **186** and the second side ground wiring **187** include a conductor made of metal, and are, for example, a conductor containing copper, gold, silver, tin, nickel, or the like.

[0178] In addition, the opening **188** is formed on the side power wiring **162** connected to the second side power wiring **186**, and is formed on the side ground wiring **163** connected to the second side ground wiring **187**. The opening **188** is not formed on the side ground wiring **163** intersecting the second side power wiring **186** and is not formed on the side power wiring **162** intersecting the second side ground wiring **187**, and the insulating layer **189** remains formed on the side ground wiring **163** intersecting the second side power wiring **186** and on the side power wiring **162** intersecting the second side ground wiring **187**.

[0179] In the semiconductor module **10** using the rewiring layer **300**A, for example, a signal can be input/output from the first side surface **146** (the bottom surface of the memory cube **100**), and a voltage such as a power supply voltage or a ground voltage can be supplied from, for example, the second side surface **145** and the fourth side surface **147** (the side surface of the memory cube **100**). Therefore, the semiconductor module **10** can use the bottom surface only for signal transmission to other chips such as the logic chip **200**. As a result, the semiconductor module **10** can suppress the number of input/output signals related to signal transmission, and can increase a plurality of free input/output terminals. In addition, a power supply or the like from the outside can be supplied to a plurality of input/output terminals conventionally used as input/output of signals related to signal transmission.

[0180] In addition, in the semiconductor module **10** using the rewiring layer **300**A, a voltage such as the power supply voltage or the ground voltage can be directly supplied to the second side surface **145** and the fourth side surface **147** (the side surface of the memory cube **100**) without passing through a chip other than the memory chip. That is, the semiconductor module **10** can stabilize the power supply voltage by performing the low-impedance power supply connection from both side surfaces (the second side surface **145** and the fourth side surface **147**) of the memory cube **100**.

1-4. Example of Seal Ring **160**

[0181] A seal ring **160** of the semiconductor module **10** will be described with respect to FIG. **11**A and FIG. **11**B. FIG. **11**A is a plan view showing the configuration of the signal transmission wiring **166** included in the seal ring **160** and the memory chip **110**, and FIG. **11**B is a cross-sectional view showing a cross section of the signal transmission wiring **166** included in the seal ring **160** and the memory chip **110** taken along a line D**1**-D**2** of FIG. **11**A. The same or similar configurations as those in FIG. **1** to FIG. **10**B will not be described here.

[0182] As shown in FIG. **11**A and FIG. **11**B, the memory cube **100** includes a seal ring **160**. The seal ring **160** is provided in an outer periphery **192** and is formed in the wiring layer **150**. The signal transmission wiring **166** is formed so as to overlap the seal ring **160**. The outer periphery **192** is a peripheral portion of the memory chip **110** and is a portion close to each side surface of the memory chip **110** (for example, the first side surface **106**, the second side surface **105**, the third side surface **108**, and the fourth side surface **107** shown in FIG. **2**). A portion of the signal transmission wiring **166** is arranged outside the outer periphery **192**. As described in "1-1. Overview of Semiconductor Module **10**", the wiring layer **150** includes a multilayer wiring structure. In addition, similar to the signal transmission wiring **166**, the power wiring **164** (for example, FIG. **8**A) and the ground wiring **165** (for example, FIG. **8**A) are formed so as to overlap the seal ring **160**, and a portion of the power wiring **164** and a portion of the ground wiring **165** are arranged on the outer side of the outer periphery **192**.

[0183] As shown in the cross-sectional view of FIG. **11**B, the wiring layer **150** includes, for example, a multilayer wiring structure of seven layers (first to seventh layers). The seven-layer multilayer wiring structure includes an insulating layer **151***a*, a via **151***b*, an insulating layer **152***a*, a wiring **152***b*, an insulating layer **153***a*, a via **153***b*, an insulating layer **154***a*, a wiring **154***b*, an insulating layer **155***a*, a via **155***b*, an insulating layer **156***a*, a wiring **156***b*, the insulating layer **182**, and a wiring **183**. The first insulating layer **151***a* is formed on the transistor layer **130**, and the first via **151***b* is formed on the transistor layer **130** through the insulating layer **151***a*. The second insulating layer **152***a* is formed on the insulating layer **151***a* and the via **151***b*, and the second wiring **152***b* is formed on the via **151***b* through the insulating layer **152***a*. In the same manner as in the first and second layers of the multilayer wiring structure, the third insulating layer **153***a* and the via **153***b*, the fourth insulating layer **154***a* and the wiring **154***b*, the fifth insulating layer **155***a* and the via **155***b*, the sixth insulating layer **156***a* and the wiring **156***b*, and the seventh insulating layer **182** and the wiring **183** are formed in this order from a lower side to an upper side in the direction D**1**. For example, the signal transmission wiring **166**, the power wiring **164**, and the ground wiring **165** are formed using the wiring **183**.

[0184] The insulating layers **182**, **156***a*, **155***a*, and **154***a* are formed using, for example, an insulating material that differs from a material having a lower dielectric constant (low-k material). The insulating layers **182**, **156***a*, **155***a*, and **154***a* are formed of, for example, SiO_2, SiCN, SIN, SION, or the like.

[0185] The seal ring **160** has a function of suppressing moisture absorption, impurities, and the like from, for example, the second side surface **145**, the third side surface **148**, and the fourth side surface **147** of the memory cube **100**. The semiconductor module **10** can suppress corrosion, deterioration, and the like of the wiring caused by moisture absorption and intrusion of impurities and the like by the seal ring **160**.

1-5. Mounting Example of Semiconductor Module 10

[0186] A mounting example of the semiconductor module **10** will be described with reference to FIG. **12**. FIG. **12** is a cross-sectional view showing a mounting example of the semiconductor module **10** according to the first embodiment of the present invention. The same or similar configurations as those in FIG. **1** to FIG. **11**B will not be described here.

[0187] As shown in FIG. **12**, the semiconductor module **10** includes the structure **20**, an electrode pad **291**, a bump **293**, a UF agent **316**, the circuit substrate **400**, and the bump layers **500**. The structure **20** is configured to include the memory cube **100**, the rewiring layer **300**, the logic chip **200**, the plurality of pillars **312**, and a UF agent **314**. As described in "1-1. Overview of Semiconductor Module **10**", a size of the memory cube **100** is larger than a size of the logic chip **200**.

[0188] In the semiconductor module **10**, the memory cube **100**, the rewiring layer **300**, and the logic chip **200** are electrically connected by the plurality of pillars **312** using, for example, a flip chip (Flip-Chip (FC)) mounting technique, and the memory cube **100**, the rewiring layer **300**, and the logic chip **200** are fixed to each other by the UF agent **314**. In the semiconductor module **10**, for example, the structure **20** and the wiring substrate **400** are electrically connected to each other by the electrode pad **291** and the bump **293**, and the structure **20** and the wiring substrate **400** are fixed to each other by the UF agent **316** using the flip-chip (Flip-Chip (FC)) mounting technique.

[0189] More specifically, the first side surface **146** of the memory cube **100** is electrically connected to the side power wiring **162**, the side ground wiring **163**, and the side signal transmission wiring **167** of the rewiring layer **300**. The side power wiring **162**, the side ground wiring **163**, and the side signal transmission wiring **167** of the rewiring layer **300** are electrically connected to the wiring layer **250** (a plurality of wirings) on the second surface **204** side of the logic chip **200** by using the pillar **312**, and the rewiring layer **300** and the logic chip **200** are fixed to each other by using the UF agent **314**. The wiring layer **250** includes a plurality of wirings. The logic chip **200** includes the wiring layer **250**, a through-hole electrode **292**, and the electrode pad **291**. The wiring layer **250** is electrically connected to the electrode pad **291** formed on the first surface **202** via the through-hole electrode **292**. The logic chip **200** (the electrode pad **291**) is electrically connected to the wiring substrate **400** using the bump **293**, and the logic chip **200** and the wiring substrate **400** are fixed to each other using the UF agent **316**.

[0190] Further, for example, the power supply voltage VDD, the voltage VSS, various types of signals, and the like are supplied from an external circuit (not shown) to the logic chip **200** via the wiring substrate **400**, the bump **293**, and the electrode pad **291**. Further, the power supply voltage VDD, the voltage VSS, various types of signals, and the like are supplied from an external circuit (not shown) to the memory cube **100** via the wiring substrate **400**, the bump **293**, the electrode pad **291**, the through-hole electrode **292**, the wiring layer **250**, the pillar **312**, and the rewiring layer **300**. The plurality of power wirings **164** and the plurality of ground wirings **165** are electrically connected to the logic chip **200** and the external circuit using, for example, the wiring substrate **400**, the bump **293**, the electrode pad **291**, the through-hole electrode **292**, the wiring layer **250**, the pillar **312**, and the rewiring layer **300**, and the power supply voltage VDD, the voltage VSS, and the like are supplied from the external circuit. Further, the plurality of signal transmission wirings **166** are electrically connected to the logic chip **200** and an external circuit (not shown) using, for example, the wiring substrate **400**, the bump **293**, the electrode pad **291**, the through-hole electrode **292**, the wiring layer **250**, the pillar **312**, and the rewiring layer **300**, and an address signal, an enable signal, and the like for controlling the memory chip **110** are supplied from the logic chip **200** and the external circuit.

[0191] The flip-chip (Flip-Chip (FC)) mounting technique is one technique for mounting a semiconductor chip and a substrate via bumps, pillars, and the like. The substrate is, for example, an interposer, a printed circuit substrate capable of high-density interconnection (High-density interconnect (HDI)), or the like. By using flip-chip (Flip-Chip (FC)) mounting techniques, the semiconductor chip and the substrate can be electrically connected, for example, using pillars or bumps, and secured together using adhesives such as under fill (Under Fill (UF)) agents, and anisotropic conductive adhesives (Anisotropic Conductive Adhesives (ACA)). The UF agent is, for example, an insulating adhesive.

[0192] The through-hole electrode **292**, the electrode pad **291**, and the pillar **312** are formed using, for example, a conductor made of metal. The conductor made of metal is, for example, a conductor containing copper, tin, or the like.

[0193] The rewiring layer **300** of the semiconductor module **10** is, for example, a fan-in type rewiring layer. The fan-in type is a mounting type in which, in a chip including a plurality of input/output terminals arranged in an array on a surface of the chip, a wiring group electrically connected to the chip is routed from the outside of the chip toward the plurality of input/output terminals inside the chip. Although not shown in the drawings, for example, the wiring group is electrically connected to the input/output terminal so as to collapse inward from the periphery (outer side) of the chip in a plan view.

[0194] The wiring substrate **400** includes a multilayer wiring structure in which wirings and insulating layers are alternately stacked. The wiring substrate **400** includes, for example, a second surface **404**, a first surface **402**, and a plurality of wiring layers **406**, **408**, **410**, and **412**, and includes a function of connecting the memory cube **100**, the logic chip **200**, and the like to an external substrate, an external circuit, and the like. In FIG. **12**, the insulating layers alternately stacked with the wiring are not shown. The circuit substrate **400** is, for example, a printed circuit substrate capable of high-density interconnection (High-density interconnect (HDI)). For example, the wiring layers **406**, **408**, **410**, and **412** are arranged parallel to the direction D**1** and the direction D**2** and stacked in this order in the direction D**3**. The plurality of wiring layers **406**, **408**, **410**, and **412** includes a plurality of wirings **407**, a plurality of wirings **409**, a plurality of wirings **411**, and a plurality of wirings **413**. For example, the wiring **407** is electrically connected to the wiring **409**, the wiring **409** is electrically connected to the wiring **411**, and the wiring **411** is electrically connected to the wiring **413**. In addition, the number of stacked layers of the multilayer wiring structure of the wiring substrate **400** is not limited to the number of stacked layers (four layers) shown in FIG. **12**. The number of layers of the multilayer wiring structure of the wiring substrate **400** can be appropriately changed based on the application or specification of the semiconductor module **10**.

[0195] The bump layer **500** includes a plurality of bumps **502**, and has a function of connecting the wiring substrate **400** to an external substrate, an external circuit, or the like.

[0196] The semiconductor module **10** may be implemented as described above. In the method for manufacturing the memory cube **100** and the rewiring layer **300**, a wiring (side wiring) can be formed on a side surface by a simpler process than a manufacturing method for forming each wiring using a complicated process while aligning. In addition, the memory cube **100** and the fan-in type rewiring layer **300** can be electrically connected to each other by using side wiring of each other. Further, in the semiconductor module **10**, the memory cube **100** in which the plurality of memory chips **110** are stacked can be vertically arranged (arranged in the direction D**3**) on the logic chip **200** by using the wirings (side wirings) formed in the memory cube **100** and the rewiring layer **300**. In the semiconductor module **10**, by vertically mounting the memory cube **100** on the logic chip **200**, it is possible to keep the distance between the respective memory chips **110** and the logic chip **200** close and constant, rather than stacking the plurality of memory chips **110** and the logic chip **200** in parallel. By using the semiconductor module **10**, a memory with low power consumption and large capacity can be realized.

[0197] Further, in the semiconductor module **10**, the memory cube **100**, the fan-in type rewiring layer **300**, the logic chip **200** having a size smaller than that of the memory cube **100**, and the wiring substrate **400** can be electrically connected using the wiring (side wiring) formed in the memory cube **100** and the rewiring layer **300** and the flip-chip technology. That is, in the semiconductor module **10**, the side surface wiring and the flip-chip technology can be used in combination based on the wiring (side surface wiring) formed in the memory cube **100** and the rewiring layer **300**, and thus the degree of freedom in design of the memory cube **100** is improved.

2. Second Embodiment

[0198] An exemplary implementation of a semiconductor module **10**A will be described referring to FIG. **13** and FIG. **14**. FIG. **13** is a cross-sectional view of the semiconductor module **10**A according to a second embodiment of the present disclosure. FIG. **14** is a diagram showing a configuration of the semiconductor module **10**A according to the second embodiment of the present disclosure. The same or similar configurations as those in FIG. **1** to FIG. **12** will not be described here.

[0199] First, the configuration of the semiconductor module **10**A will be described. As shown in FIG. **13**, the semiconductor module **10**A includes the structure **20** and a cavity substrate **430**. The structure **20** includes the memory cube **100**, the rewiring layer **300**, the logic chip **200**, and an adhesive layer **600**.

[0200] The memory cube **100** has the structure and function described in the first embodiment. A detailed description of the memory cube **100** is omitted here.

[0201] The rewiring layer **300** is a fan-in type rewiring layer including the same functions and configurations as those of the first embodiment. The side signal transmission wiring **167** included in the rewiring layer **300** includes, for example, an inductor **172** (see FIG. **14**). The rewiring layer **300** according to the second embodiment includes the plurality of side signal transmission wirings **167** and the plurality of inductors **172**. For example, each of the side power wiring **162** and the side ground wiring **163** included in the rewiring layer **300** is electrically connected to a bump **420**. In addition, the rewiring layer **300** according to the second embodiment has the same structure and function as those described in the first embodiment. A detailed description of the rewiring layer **300** will be omitted here.

[0202] The adhesive layer **600** is arranged between a first surface **302** of the rewiring layer **300** and the second surface **204** of the logic chip **200**, and adheres the memory cube **100**, the rewiring layer **300**, and the logic chip **200** to each other. The adhesive layer **600** may be, for example, an adhesive containing an epoxy resin, an acrylic polymer, or the like, a die bonding film (Die Bonding Film (DBF)) containing an epoxy resin or an acrylic polymer, or an adhesive film such as a die attach film (Die Attached Film (DAF)).

[0203] The logic chip **200** includes, for example, a plurality of wirings (not shown) included in the wiring layer **250** and a plurality of inductors **272**. Further, the logic chip **200** includes a plurality of through-hole electrodes **292** and a plurality of electrode pads **291**. The plurality of wirings and the plurality of inductors **272** are provided in the same process, for example. Each of the plurality of wirings and the plurality of inductors **272** is electrically connected to the electrode pad **291** using the through-hole electrode **292**. Each of the plurality of electrode pads **291** is electrically connected to the corresponding bump **293**. In the semiconductor module **10**A, similar to the semiconductor module **10**, for example, the substrate **273** (the first surface **202**) of the logic chip **200** is arranged on the cavity substrate **430**, and the logic chip **200** is mounted face-up on the cavity substrate **430**. In addition, for example, a mounting configuration in which the wiring layers **250** (second surfaces **204**) of the logic chip **200** are arranged on the cavity substrate **430** and the stacking direction is downward in the direction D3 is referred to as face-down mounting.

[0204] The cavity substrate **430** includes a cavity part **432**. The cavity part **432** may be referred to as a recess. The cavity substrate **430** can absorb a height DH of a portion including the adhesive layer **600** and the logic chip **200** which are convexly arranged in the memory cube **100** in the cavity part **432**. In addition, the cavity substrate **430** includes a multilayer wiring structure in which wirings and insulating layers are alternately stacked, similar to the wiring substrate **400** according to the first embodiment. In the description of the cavity substrate **430**, description which are the same as or similar to that of the wiring substrate **400** will be omitted here.

[0205] The bump layer **500** includes the plurality of bumps **502**, is electrically connected to the plurality of wirings **413** included in the cavity substrate **430**, and has a function of connecting the cavity substrate **430** to an external substrate, an external circuit, or the like.

[0206] The structure **20** is electrically connected to the cavity substrate **430** using the plurality of bumps **420** and the plurality of bumps **293**, and the structure **20** and the cavity substrate **430** are fixed to each other using a UF agent **422**. Specifically, the structure **20** is electrically connected to the wiring **407** exposed to the cavity part **432** by using the plurality of bumps **420**, and is electrically connected to the wiring **407** exposed to the second surface **404** by using the plurality of bumps **293**. In addition, a portion of the rewiring layer **300**, the adhesive layer **600**, the logic chip **200**, the plurality of bumps **293**, and the plurality of bumps **420** are fixed to the cavity part **432** and a portion of the second surface **404** of the cavity substrate **430** using the UF agent **422**.

[0207] Further, for example, the power supply voltage VDD, the voltage VSS, various types of signals, and the like are supplied from an external circuit (not shown) to the logic chip **200** via the cavity substrate **430**, the bumps **293**, and the electrode pads **291**. Further, the power supply voltage VDD, the voltage VSS, various types of signals, and the like are supplied from an external circuit (not shown) to the memory cube **100** via the cavity substrate **430**, the bumps **420**, and the rewiring layers **300**. The plurality of power wirings **164** and the plurality of ground wirings **165** are electrically connected to the external circuit using, for example, the cavity substrate **430**, the bumps **420**, and the rewiring layers **300**, and the power supply voltage VDD, the voltage VSS, and the like are supplied from the external circuit. Further, the plurality of signal transmission wirings **166** may be electrically connected to an external circuit (not shown) using, for example, the cavity substrate **430**, the bump **420**, and the rewiring layer **300**, and a part of the signals for controlling the memory chip **110** may be supplied from the external circuit.

[0208] Next, FIG. **14** is a schematic diagram showing the configuration of the semiconductor module **10**A. As shown in FIG. **14**, the memory cube **100** includes a plurality of magnetic field coupled chip-to-chip interfaces (Through Chip Interface-IO (TCI-IO)) **112** and the plurality of memory modules **111**. The plurality of TCI-IO **112** is electrically connected to the memory module **111**.

[0209] The TCI-IO **112** includes the inductor **172**, a transmitting/receiving circuit **114**, and a parallel-serial conversion circuit **113**. The inductor **172** is electrically connected to the transmitting/receiving circuit **114** using a terminal A and a terminal B. The transmitting/receiving circuit **114** is electrically connected to the parallel-serial conversion circuit **113**. The parallel-serial conversion circuit **113** is electrically connected to the memory module **111**.

[0210] The inductor **172** has a function of performing inductor communication with the inductor **272** of the logic chip **200** in a non-contact manner.

[0211] The transmitting/receiving circuit **114** has, for example, a function of amplifying a signal (data) received by the inductor **172** and a function of removing noise from the received signal (data). The transmitting/receiving circuit **114**, for example, has a function of placing a desired signal (data) converted by using the parallel-serial conversion circuit **113** on a radio wave. The signal received by inductor **172** includes a number of parallel signals from the logic chip **200**. The desired signal includes a number of parallel signals from the memory module **111**.

[0212] For example, in step **10**, the parallel-serial conversion circuit **113** performs parallel-serial conversion on a large number of parallel signals from the logic chip **200** to convert them into serial signals (serial signals). The serial signal is transferred at high speed using one signal path (wiring). In step **11**, the parallel-serial conversion circuit **113** performs parallel-serial conversion on a serial signal described above immediately before the memory module **111**, returns the serial signal to a number of parallel signals, and then transmits the plurality of parallel signals to the memory module **111**. In the case where a signal (data) is transmitted from the memory module **111** to the logic chip **200**, the parallel-serial conversion circuit **113** executes, for example, step **10** following step **11**. The parallel-serial conversion circuit **113** is referred to as, for example, a SerDes circuit (Serialize and Deserialise Circuit).

[0213] The memory module **111** includes, for example, a function of generating a plurality of parallel signals to be transmitted and a function of controlling a plurality of received parallel signals and storing them in the memory cell array **115** (see FIG. **4**).

[0214] The logic chip **200** includes a plurality of field coupled chip-to-chip interfaces (Through Chip Interface-IO (TCI-IO)) **212** and a plurality of logic modules **211**. The plurality of TCI-IO **212** is electrically connected to the logical module **211**.

[0215] The TCI-IO **212** includes the inductor **272**, a transmitting/receiving circuit **214**, and a parallel-serial conversion circuit **213**. The inductor **272** is electrically connected to the transmitting/receiving circuit **214** using a terminal C and a terminal D. The transmitting/receiving circuit **214** is electrically connected to the parallel-serial conversion circuit **213**. The parallel-serial conversion circuit **213** is electrically connected to the logic module **211**.

[0216] The configurations and functions of the inductor **272**, the transmitting/receiving circuit **214**, the parallel-serial conversion circuit **213**, and the logic module **211** are the same as those of the inductor **172**, the transmitting/receiving circuit **114**, the parallel-serial conversion circuit **113**, and the memory module **111**. Therefore, the configurations and functions of the inductor **272**, the transmitting/receiving circuit **214**, the parallel-serial conversion circuit **213**, and the logic module **211** will not be described here.

[0217] The plurality of inductors **172**, the circuits included in the plurality of TCI-IO **112**, and the circuits included in the plurality of memory modules **111** are formed using the transistor layer **130** and the wiring layer **150**. Further, the plurality of inductors **272**, the respective circuits included in the plurality of TCI-IO **212**, and the respective circuits included in the plurality of logic modules **211** are formed using the transistor layer **230** and the wiring layer **250**.

[0218] Next, an outline of inductor communication between the inductor **172** and the inductor **272** of the semiconductor module **10**A will be described. As described above, the inductor **172** is included in the rewiring layer **300**, and the inductor **272** is provided together with a plurality of wirings (not shown) included in the wiring layer **250**. Generally, only the adhesive layer **600** is arranged between the inductor **172** and the inductor **272**. Therefore, since the inductor **172** and the inductor **272** are arranged close to each other, good communication at a short distance is possible. A shape of the inductor **172** and a shape of the inductor **272** may be, for example, a triangular shape, a square shape, a trapezoidal shape, or a pentagonal shape. There is no limitation on the shape of the inductor **172** and the inductor **272** as long as the inductor **172** and the inductor **272** can communicate with each other in a one-to-one manner by magnetic field coupling. In addition, the communication between the inductors associated with the magnetic field coupling is referred to as, for example, inductor communication, signal communication, data communication, and the like.

[0219] The semiconductor module **10**A is implemented as described above. In the method for manufacturing the memory cube **100** and the rewiring layer **300** included in the semiconductor module **10**A, the wiring can be formed on the side surface by a simpler process than the manufacturing method for forming the wiring using a complicated process while aligning. Further, in the semiconductor module **10**A, the memory cube **100**, the fan-in type rewiring layer **300**, and the cavity substrate **430** can be electrically connected by using the wiring (side wiring) formed on the side surface of the memory cube **100**. In the semiconductor module **10**A, by vertically mounting the memory cube **100** on the logic chip **200**, a distance between the respective memory chips **110** and the logic chip **200** can be kept close and constant, rather than the plurality of memory chips **110** and the logic chip **200** being stacked in parallel. By using the semiconductor module **10**A, a low power consumption and large capacity memory can be realized.

[0220] Further, the semiconductor module **10**A includes the functions and configurations described above, and the memory cube **100** connected to the fan-in type rewiring layer **300** using the wiring (side wiring) formed on the side of the memory cube **100**, and the logic chip **200**, which is smaller than the memory cube **100**, can communicate with each other in a non-contact manner using inductor communication without being directly electrically connected. In the semiconductor module **10**A, by using the contactless inductor communication, a load caused by wiring resistance and parasitic capacitance between the memory chip **110** and the logic chip **200** is suppressed, so that the power consumption of the semiconductor module **10**A can be further suppressed.

[0221] Further, in the semiconductor module **10**A, the structure **20** and the cavity substrate **430** can be electrically connected to each other and fixed to each other by using the wiring (side wiring) formed in the memory cube **100** and the rewiring layer **300**, the electrode pads **291**, the bumps **420** and the bumps **293**, and UF agent **422**. That is, in the semiconductor module **10**A, the degree of freedom in design of the memory cube **100** is improved based on the wirings (side wirings) formed in the memory cube **100** and the rewiring layer **300**, and unlike the semiconductor module **10** according to the first embodiment, the memory module can be mounted on the cavity substrate **430**.

[0222] In addition, in the semiconductor module **10**A, although the logic chip **200** is mounted face-up on the cavity substrate **430**, the method for mounting the logic chip **200** is not limited to face-up mounting. For example, even if the rewiring layer **300** (inductor **172**) and the inductor **272** are separated from each other, inductor communication between the inductor **172** and the inductor **272** can be performed if a distance between the inductor **172** and the inductor **272** along the direction D3 is a constant distance. That is, the constant distance may be any distance that enables inductor communication between the inductor **172** and the inductor **272**. In this case, the implementation of the logic chip **200** may be a face-down implementation.

3. Third Embodiment

[0223] FIG. **15** is a block diagram showing an exemplary embodiment of a semiconductor module **10**B. FIG. **15** is a cross-sectional view showing the semiconductor module **10**B according to a third embodiment of the present disclosure. Descriptions of the same or similar configurations as those in FIG. **1** to FIG. **14** will be omitted.

[0224] As shown in FIG. **15**, the semiconductor module **10**B includes the structure **20**, the wiring substrate **400**, pillars **294**, the bumps **293**, the UF agent **422**, and bump layer **500**. The structure **20** includes the memory cube **100**, the rewiring layer **300**, the logic chip **200**, and the adhesive layer **600**.

[0225] In the semiconductor module **10**B, the structure **20** and the wiring substrate **400** are directly electrically connected to each other by using the pillars **294** and the bumps **293**, and the structure **20** and the wiring substrate **400** are fixed to each other by using the UF agent **422**. More specifically, the through-hole electrode **292** exposed on the first surface **202** of the logic chip **200** included in the structure **20** is directly electrically connected to the wiring layer **406** (wiring **407**) exposed on the first surface **402** of the wiring substrate **400** by using the pillar **294** and the bump **293**. Further, the side power wiring **162**, the side ground wiring **163**, and the side signal transmission wiring **167**

exposed on the first surface **302** of the rewiring layer **300** included in the structure **20** are directly electrically connected to the wiring layer **406** (the wiring **407**) exposed on the first surface **402** of the wiring substrate **400** by using the pillar **294** and the bump **293**. On the other hand, in the semiconductor module **10**A, the structure **20** is electrically connected to the cavity substrate **430** by using the plurality of bumps **420** and the plurality of bumps **293**, and the structure **20** and the cavity substrate **430** are fixed to each other by using the UF agent **422**. The semiconductor module **10**B has the same configuration and function as those of the semiconductor module **10**A except for the electric coupling described above and the fixing using the UF agent **422**. Therefore, in the explanation of the semiconductor module **10**B, differences from the semiconductor module **10**A are mainly explained. In addition, the wiring substrate **400** may be referred to as an external substrate.

[0226] The memory cube **100** has the structure and function described in the first embodiment. The memory cube **100** is electrically connected to the rewiring layer **300** using the side power wiring **162**, the side ground wiring **163**, and the side surface signal transmission wiring **167**. A detailed description of the memory cube **100** is omitted here.

[0227] The rewiring layer **300** is a fan-in type rewiring layer including the same functions and configurations as those of the first embodiment. The side signal transmission wiring **167** included in the rewiring layer **300** includes, for example, an inductor **172**. The rewiring layer **300** according to the third embodiment includes the plurality of side signal transmission wirings **167** and the plurality of inductors **172**. For example, each of the side power wiring **162** and the side ground wiring **163** included in the rewiring layer **300** is electrically connected to the bump **420**. In addition, the rewiring layer **300** has the same structure and function as those described in the first embodiment. A detailed description of the rewiring layer **300** will be omitted here.

[0228] The logic chip **200** and the adhesive layers **600** have the same configuration and function as those of the semiconductor module **10**A described in the second embodiment. The wiring substrate **400** and the bump layer **500** have the same configuration and function as those of the first embodiment. Detailed description of the logic chip **200**, the wiring substrate **400**, and the bump layer **500** will be omitted here. In the semiconductor module **10**B, similar to the semiconductor module **10**A, although an example is shown in which the logic chip **200** is mounted face-up on the wiring substrate **400**, the logic chip **200** may be mounted face-down as long as the inductor **172** and the inductor **272** can communicate with each other.

[0229] Further, for example, the power supply voltage VDD, the voltage VSS, various types of signals, and the like are supplied from an external circuit (not shown) to the logic chip **200** via the wiring substrate **400**, the bumps **293**, and the pillars **294**. In addition, the power supply voltage VDD, the voltage VSS, various types of signals, and the like are supplied from an external circuit (not shown) to the memory cube **100** via the wiring substrate **400**, the bumps **293**, the pillars **294**, and the rewiring layer **300**. The plurality of power wirings **164** and the plurality of ground wirings **165** are electrically connected to the external circuit using, for example, the wiring substrate **400**, the bumps **293**, the pillars **294**, and the rewiring layers **300**, and the power supply voltage VDD, the voltage VSS, and the like are supplied from the external circuit. Further, the plurality of signal transmission wirings **166** may be electrically connected to an external circuit (not shown) by using the wiring substrate **400**, the bumps **293**, the pillars **294**, and the rewiring layer **300**, and a part of the signals for controlling the memory chip **110** may be supplied from the external circuit.

[0230] The semiconductor module **10**B is implemented as described above. In the semiconductor module **10**B, similar to the semiconductor module **10**, a wiring (side wiring) can be formed on the side surface in a simple process, and a distance between each memory chip **110** and the logic chip **200** can be kept close and constant. Therefore, a low power consumption and large capacity memory can be realized by using the semiconductor module **10**B.

[0231] In addition, the semiconductor module **10**B, the memory cube **100** and the logic chip **200** can communicate (electrically connected) using inductor communication, similar to the semiconductor module **10**A. Therefore, in the semiconductor module **10**B, by using the contactless inductor communication, the load caused by wiring resistance and parasitic capacitance between the memory chip **110** and the logic chip **200** is suppressed, so that the power consumption of the semiconductor module **10**B can be further suppressed.

[0232] Further, in the semiconductor module **10**B, the structure **20** and the wiring substrate **400** can be electrically connected to each other and fixed to each other by using the wiring (side wiring) formed in the memory cube **100** and the rewiring layer **300**, the pillar **294** and the bump **293**, and the UF agent **422**. That is, in the semiconductor module **10**B, the degree of freedom in design of the memory cube **100** is improved based on the wiring (side wiring) formed in the memory cube **100** and the rewiring layer **300**, and can be implemented differently from the first embodiment and the second embodiment. In addition, for example, the pillar **294** may be constructed using similar materials as the pillar **312** and may be referred to as copper pillars, solder balls, solder bumps, gold stud bumps, copper stud bumps, and the like.

4. Fourth Embodiment

[0233] FIG. **16** is a diagram showing a mounting example of a semiconductor module **10**C. FIG. **16** is a cross-sectional view of the semiconductor module **10**C according to a fourth embodiment of the present disclosure. The same or similar configurations as those in FIG. **1** to FIG. **15** will not be described here.

[0234] The semiconductor module **10**C includes the structure **20**, the wiring substrate **400**, the pillars **294**, the bumps **293**, a molding material **423**, a molding material **424**, and the bump layer **500**. The structure **20** is configured to include the memory cube **100**, the rewiring layer **300**, and the logic chip **200**. A length RWLX of the rewiring layer **300** in the direction D**1** is longer than a length MCBX of the memory cube **100** in the direction D**1**. The molding material is an insulating resin material, and includes, for example, a resin material such as epoxy, a curing agent, a filler, an additive, and the like. In addition, the molding material **423** or the molding material **424** may be referred to as a second insulating film.

[0235] In the semiconductor module **10**C, the memory cube **100**, the rewiring layer **300**, and the logic chip **200** are electrically connected to each other, and the memory cube **100**, the rewiring layer **300**, and the logic chip **200** are fixed to each other by using a FC mounting technique. For example, the plurality of side surface signal transmission wirings **167** included in the rewiring layer **300** is electrically connected to the plurality of wirings (not shown) included in the wiring layer **250** of the logic chip **200** by using the plurality of pillars **312**, and the rewiring layer **300** and the logic chip **200** are fixed to each other by using the UF agent **314**.

[0236] In addition, the semiconductor module **10**C, the structure **20** and the wiring substrate **400** are electrically connected to each other using the pillars **294** and the bumps **293**, and the structure **20** and the wiring substrate **400** are fixed to each other using the molding material **423** and the molding material **424**. More specifically, the through-hole electrode **292** exposed on the first surface **202** of the logic chip **200** included in the structure **20** is directly electrically connected to the wiring layer **406** (wiring **407**) exposed on the first surface **402** of the wiring substrate **400** by using the pillar **294** and the bump **293**. Further, the side power wiring **162**, the side ground wiring **163**, and the side signal transmission wiring **167** exposed on the first surface **302** of the rewiring layer **300** included in the structure **20** are directly electrically connected to the wiring layer **406** (the wiring **407**) exposed on the first surface **402** of the wiring substrate **400** by using the pillar **294** and the bump **293**.

[0237] In addition, in the memory cube **100**, the first surface **142**, the second surface **144**, the second side surface **145**, and the fourth side surface **147** are fixed to the rewiring layer **300** and are protected by using the molding material **424**. Further, the logic chip **200** electrically connected to and fixed to the rewiring layer **300** using the FC mounting technique, and the plurality of pillars **294** and the plurality of bumps **293** electrically connecting the rewiring layer **300** and the logic chip **200** to the wiring substrate **400** are fixed to the rewiring layer **300** and the wiring substrate **400** and are protected by using the molding material **423**.

[0238] The molding materials **423** and **424** can suppress vibrations and impacts that the semiconductor module **10** receives from the outside and can suppress the intrusion of water from the outside into the semiconductor module **10**C. Further, the semiconductor module **10**C can efficiently discharge the heat generated in the semiconductor module to the outside by the molding materials **423** and **424**.

[0239] On the other hand, in the semiconductor module **10**C, the structure **20** and the wiring substrate **400** are electrically connected by using the pillars **294** and the bumps **293**, and the structure **20** and the wiring substrate **400** are fixed to each other by using the UF agent **422**. The semiconductor module **10**C has the same configuration and functions as those of the semiconductor module **10** except for the electric connection described above and the fixing using the molding materials **423** and **424**. Therefore, in the explanation of the semiconductor module **10**C, differences from the semiconductor module **10** are mainly explained.

[0240] The memory cube **100** has the structure and function described in the first embodiment. A detailed description of the memory cube **100** is omitted here.

[0241] The rewiring layer **300** of the semiconductor module **10**C is a fan-out type rewiring layer. The fan-out type is a mounting type in which, in a chip including a plurality of input/output terminals arranged in an array on a surface of the chip, a wiring group electrically connected to the chip is routed from the plurality of input/output terminals toward the outside (peripheral side) of the chip. Although not shown in the drawings, for example, in a plan view, the plurality of input/output terminals is electrically connected to the wiring group so as to spread around (outside) the chip from the plurality of input/output terminals arranged in an array on the inside of the chip. The wiring group is, for example, the side power wiring **162**, the side ground wiring **163**, the side surface signal transmission wiring **167**, or the like included in the rewiring layer **300**.

[0242] The logic chip **200**, the wiring substrate **400**, and the bump layer **500** have the same configurations and functions as those of the first embodiment. Detailed description of the logic chip **200**, the wiring substrate **400**, and the bump layer **500** will be omitted here. In the semiconductor module **10**C, similar to the semiconductor module **10**, for example, the semiconductor module is mounted face-up on the circuit substrate **400**.

[0243] Further, for example, the power supply voltage VDD, the voltage VSS, various types of signals, and the like are supplied from an external circuit (not shown) to the logic chip **200** via the wiring substrate **400**, the bumps **293**, and the pillars **294**. In addition, the power supply voltage VDD, the voltage VSS, various types of signals, and the like are supplied from an external circuit (not shown) to the memory cube **100** via the wiring substrate **400**, the bumps **293**, the pillars **294**, and the rewiring layer **300**. The plurality of power wirings **164** and the plurality of ground wirings **165** are electrically connected to the external circuit using, for example, the wiring substrate **400**, the bumps **293**, the pillars **294**, and the rewiring layer **300** (the side power wiring **162** and the side ground wiring **163**), and the power supply voltage VDD, the voltage VSS, and the like are supplied from the external circuit. Further, the plurality of signal transmission wirings **166** may be electrically connected to an external circuit (not shown) by using the wiring substrate **400**, the bump **293**, the pillar **294**, and the rewiring layer **300** (side signal transmission wiring **167**), and a part of the signals for controlling the memory chip **110** may be supplied from the external circuit.

[0244] The semiconductor module **10**C is implemented as described above. In the semiconductor module **10**C, similar to the semiconductor module **10**, the wiring (side wiring) can be formed on the side surface in a simple process, and a distance between each memory chip **110** and the logic chip **200** can be kept close and constant. Therefore, by using the semiconductor module **10**C, a low power consumption and large capacity memory can be realized.

[0245] Further, in the semiconductor module **10**C, the structure **20** and the wiring substrate **400** can be electrically connected to each other and fixed to each other using the wiring (side wiring) formed in the memory cube **100** and the fan-out type rewiring layer **300**, the pillars **294** and the bumps **293**, and the molding materials **423** and **424**. That is, in the semiconductor module **10**C, the degree of freedom in design of the memory cube **100** is improved based on the wiring (side wiring) formed in the memory cube **100** and the rewiring layer **300**, and implementations different from those of the first to third embodiments are possible.

5. Fifth Embodiment

[0246] FIG. **17** is a diagram showing an exemplary embodiment of a semiconductor module **10**D. FIG. **17** is a cross-sectional view showing a mounting example of the semiconductor module **10**D according to a fifth embodiment of the present disclosure. Descriptions of the same or similar configurations as those in FIG. **1** to FIG. **16** will be omitted.

[0247] As shown in FIG. **17**, the semiconductor module **10**D includes the structure **20**, the wiring substrate **400**, the pillars **294**, the bumps **293**, the molding material **423**, the molding material **424**, and the bump layer **500**. The structure **20** includes the memory cube **100**, the rewiring layer **300**, the logic chip **200**, and the adhesive layer **600**. The length RWLX of the rewiring layer **300** in the direction D**1** is longer than the length MCBX of the memory cube **100** in the direction D**1**.

[0248] In the semiconductor module **10**D, the second surface **204** of the logic chip **200** and the first surface **302** of the fan-out type rewiring layer **300** are bonded using the adhesive layer **600**. In addition, the logic chip **200** includes a configuration and a function similar to those of the semiconductor module **10**A. The semiconductor module **10**D has the same configuration and function as the semiconductor module **10**C except for the configuration using the adhesive layers **600** and the configuration and function of the logic chip **200**. Therefore, in the description of the semiconductor module **10**D, the description of the same configuration and function as those of the semiconductor modules **10**B and **10**C will be omitted.

[0249] The semiconductor module **10**D may be implemented as described above. In the semiconductor module **10**D, as in the case of the semiconductor module **10**C, the wiring (side wiring) can be formed on the side surface in a simple process, and a distance between each memory chip **110** and the logic chip **200** can be kept close and constant. Therefore, by using the semiconductor module **10**D, a low power consumption and large capacity memory can be realized.

[0250] In addition, in the semiconductor module **10**D, the memory cube **100** and the logic chip **200** can communicate (electrically connected) using inductor communication, similar to the semiconductor module **10**A. Therefore, in the semiconductor module **10**D, by using the contactless inductor communication, the load caused by the wiring resistance and the parasitic capacitance between the memory chip **110** and the logic chip **200** is suppressed, so that the power consumed by the semiconductor module **10**D can be further suppressed.

[0251] Further, in the semiconductor module **10**D, the memory cube **100**, the rewiring layer **300**, and the logic chip **200** are bonded to each other using the adhesive layer **600**, and the structure **20** and the wiring substrate **400** can be electrically connected to each other and fixed to each other using the wiring (side wiring) formed in the memory cube **100** and the fan-out type rewiring layer **300**, the pillars **294** and the bumps **293**, and the molding materials **423** and **424**. That is, in the semiconductor module **10**C, the degree of freedom in design of the memory cube **100** is improved based on the wiring (side wiring) formed in the memory cube **100** and the rewiring layer **300**, and can be implemented differently from the first embodiment to the fourth embodiment.

6. Sixth Embodiment

[0252] FIG. **18** is a diagram showing a mounting example of a semiconductor module **10**E. FIG. **18** is a cross-sectional view of the semiconductor module **10**E according to a sixth embodiment of the present disclosure. The same or similar configurations as those in FIG. **1** to FIG. **17** will not be described here.

[0253] The semiconductor module **10**E includes the structure **20**, the wiring substrate **400**, the pillars **294**, the bumps **293**, a UF agent **425**, and the bump layer **500**. The structure **20** is configured to include the memory cube **100**, the rewiring layer **300**, the logic chip **200**, the plurality of pillars **312**, and the UF agent **314**. The length RWLX of the rewiring layer **300** in the direction D**1** is longer than the length MCBX of the memory cube **100** in the direction D**1**.

[0254] In the semiconductor module **10**E, the structure **20** and the wiring substrate **400** are electrically connected by using the pillars **294** and the bumps **293**, and the structure **20** and the wiring substrate **400** are fixed to each other by using the UF agent **425**. More specifically, the logic chip **200** electrically connected to and fixed to the rewiring layer **300** using the FC mounting technique, and the plurality of pillars **294** and the plurality of bumps **293** electrically connecting the rewiring layer **300** and the logic chip **200** to the wiring substrate **400** are fixed to the rewiring layer **300** and the wiring substrate **400** and are protected by using the UF agent **425**. The semiconductor module **10**E has the same configuration and function as those of the semiconductor module **10**C except for the electric coupling described above and fixing using the UF agent **425**. Therefore, in the description of the semiconductor module **10**E, the description of the same configuration and function as those of the semiconductor module **10**C will be omitted.

[0255] The semiconductor module **10**E is implemented as described above. In the semiconductor module **10**E, as in the case of the semiconductor module **10**C, the wiring (side wiring) can be formed on the side surface in a simple process, and a distance between each memory chip **110** and the logic chip **200** can be kept close and constant. Therefore, by using the semiconductor module **10**E, a low power consumption and large capacity memory can be realized.

[0256] Further, in the semiconductor module **10**E, the structure **20** and the wiring substrate **400** can be electrically connected to each other and fixed to each other by using the wiring (side wiring) formed in the memory cube **100** and the fan-out type rewiring layer **300**, the pillar **294**, the bump **293**, the molding material **424**, and the UF agent **425**. That is, in the semiconductor module **10**E, the degree of freedom in design of the memory cube **100** is improved based on the memory cube **100** and the wiring (side wiring) formed in the rewiring layer **300**, and can be implemented differently from the first embodiment to the fifth embodiment.

7. Seventh Embodiment

[0257] A method for manufacturing a conductive film on a side surface of the semiconductor module **10** will be described referring to FIG. **19**A and FIG. **21**A. FIG. **19**A is a perspective view showing a conductive film **440** formed on the second side surface **145** of the memory cube **100** according to a seventh embodiment of the present invention. FIG. **19**B is a perspective view showing a conductive film **441** formed on the fourth side surface **147** of the memory cube **100**. FIG. **20**A to FIG. **21**A are perspective views showing a method for manufacturing the conductive film **440** on the second side surface **145** of the memory cube **100** according to the seventh embodiment of the present disclosure. The same or similar configurations as those in FIG. **1** to FIG. **18** will not be described here.

[0258] The memory cube **100** includes a configuration similar to the configuration described in "1-1. Overview of Semiconductor Module **10**" and "1-2. Example of Manufacturing Method of Memory Cube **100**". As shown in FIG. **19**A or FIG. **19**B, the memory cube **100** includes the conductive film **440** arranged on the second side surface **145**

and the conductive film **441** arranged on the fourth side surface **147**. The conductive film **440** arranged on the second side surface **145** is electrically connected to the plurality of power wirings **164** exposed on the second side surfaces **145** of the plurality of memory chips **110**, and is in contact with the second side surfaces **145** of the plurality of memory chips **110**. The conductive film **441** arranged on the fourth side surface **147** is electrically connected to the plurality of ground wirings **165** exposed on the fourth side surfaces **147** of the plurality of memory chips **110**, and is in contact with the fourth side surfaces **147** of the plurality of memory chips **110**. The conductive film **440** and the conductive film **441** include a conductor made of a metal material, and are, for example, a conductor containing copper, tin, or the like.

[0259] The conductive film **440** and the conductive film **441** are formed on the second side surface **145** and the fourth side surface **147** using a similar manufacturing method. Therefore, in the following description, a method for manufacturing the conductive film **440** is mainly described.

[0260] As described in "1-2. Example of Manufacturing Method of Memory Cube **100**", the first side surface **146**, the second side surface **145**, the third side surface **148**, and the fourth side surface **147** of the memory cube **100** are polished. In the memory cube **100** according to the seventh embodiment, the end surfaces of the plurality of power wirings **164** are exposed on the second side surface **145** (see FIG. **20**A), and the end surfaces of the plurality of ground wirings **165** are exposed on the fourth side surface **147**.

[0261] Next, a resist **442** is applied to the first surface **142**, the second surface **144**, the first side surface **146**, the second side surface **145**, the third side surface **148**, and the fourth side surface **147**. Further, resists **442** of opening regions **443** and **444** are removed, the end surfaces of the plurality of power wirings **164** are exposed on the second side surface **145**, and the end surfaces of the plurality of ground wirings **165** are exposed on the fourth side surface **147** (see FIG. **20**B).

[0262] Next, for example, the conductive film **440** is formed on the end surfaces of the second side surface **145** and the plurality of power wirings **164** exposed in the opening region **443** by electric field plating (plating method), the conductive film **441** is formed on the end surfaces of the fourth side surface **147** and the plurality of ground wirings **165** exposed in the opening region **444**, and the resist **442** is removed (see FIG. **19**A or FIG. **19**B).

[0263] Next, a conductive film **445** may be formed so as to be in contact with the first surface **142**, the second surface **144**, and the third side surface **148** except for the first side surface **146**, the second side surface **145**, and the fourth side surface **147** by using the same manufacturing methods as the conductive film **440** and the conductive film **441** (see FIG. **21**A). Further, although not shown, the conductive film **445** may be formed on the first surface **142** and the second surface **144**.

[0264] Although not shown, the conductive films **440**, **441**, or **445** are electrically connected by using the plurality of side power wirings **162** (see, for example, FIG. **8**B and FIG. **16** to FIG. **18**) or the plurality of side ground wirings **163** (see, for example, FIG. **8**B and FIG. **16** to FIG. **18**) formed in the rewiring layers **300** (see, for example, FIG. **8**B and FIG. **16** to FIG. **18**). The conductive films **440**, **441**, or **445** are electrically connected to, for example, an external circuit (not shown) and are supplied with the power supply voltage VDD, the voltage VSS, and the like.

[0265] By using the method for manufacturing the conductive film on the side surface of the semiconductor module **10** described above, it is possible to form the wiring on the side surface in a simpler process than the manufacturing method for forming each wiring using a complicated process Further, by using the method for manufacturing the while aligning. conductive film on the side surface of the semiconductor module **10**, the conductive film can be formed on substantially the entire side surface of the memory cube **100**. Consequently, a volume of the supply path such as the power supply voltage VDD and the voltage VSS to the memory cube **100** can be increased, and thus the power supply capability to the memory cube **100** is improved. In addition, since the volume of the conductive film on the surface of the semiconductor module **10** can be increased, a heat dissipation path of the semiconductor module **10** can be increased, and improvement of thermal resistance can be suppressed. That is, heat from the semiconductor module **10** is easily dissipated.

8. Eighth Embodiment

[0266] An example of the memory cube **100** including a plurality of memory chips of differing thicknesses will be described with respect to FIG. **21**B. FIG. **21**B is a schematic diagram showing the memory cube **100** according to an eighth embodiment of the present disclosure. The same or similar configurations as those in FIG. **1** to FIG. **21**A will not be described here. The memory cube **100** includes, for example, the memory chips **110***n* to **110***n*+3, and has a configuration similar to that described in "1-2. Example of Manufacturing Method of Memory Cube **100**". That is, in the memory cube **100**, the memory chip **110***n* and the memory chip **110***n*+1 are F2F bonded, the memory chip **110***n*+2 and the memory chip **110***n*+3 are F2F bonded, and the memory chip **110***n*+1 and the memory chip **110***n*+2 are B2B bonded (see FIG. **21**B).

[0267] As shown in FIG. **21**B, a thickness of each chip of the memory chips **110***n* to **110***n*+3 may be different when the memory chips **110***n* to **110***n*+3 are stacked. For example, a thickness THI**4** of the memory chip **110***n*+3 is larger than a thickness THI of the memory chip **110***n*, the thickness THI of the memory chip **110***n* is larger than a thickness THI**3** of the memory chip **110***n*+2, and the thickness THI**3** of the memory chip **110***n*+2 is larger than a thickness THI**2** of the memory chip **110***n*+1. In the example shown in FIG. **21**B, although the thickness of each chip of the memory chips **110***n* to **110***n*+3 is different, it is sufficient that the thickness of at least one of the plurality of memory chips is different from the thickness of the other memory chips.

[0268] In the semiconductor module **10**, the memory cube **100** can be manufactured by stacking the plurality of memory chips **110** having different thicknesses. Consequently, a total thickness TTHI of the memory cubes **100** can be aligned between the different memory cubes **100**. That is, in the semiconductor module **10**, variations in the total thickness TTHI between the memory cubes **100** can be suppressed.

9. Ninth Embodiment

[0269] A mounting example of a semiconductor module will be described with reference to FIG. **25** to FIG. **27**. FIG. **25** to FIG. **27** are cross-sectional views showing a mounting example of semiconductor modules **10**F to **10**H according to a ninth embodiment of the present disclosure. The diagram of the semiconductor modules **10**F to **10**H shown in FIG. **25** to FIG. **27** is a diagram showing, by way of example, an implementation format of power supply to the conductive film **445** arranged on the first surface **142** and the conductive film **445** arranged on the second surface **144** of the memory cube **100** described with reference to FIG. **21**A of the semiconductor module **10**A described with reference to FIG. **13**. Descriptions of the same or similar configurations as those in FIG. **1** to FIG. **24**B will be omitted. Except for the implementation format of the power supply of the semiconductor modules **10**F to **10**H, the configuration is the same as the configuration of the semiconductor module **10**A and the like described with reference to FIG. **13**, FIG. **19**A and FIG. **19**B, and therefore, an implementation format of the power supply of the semiconductor modules **10**F to **10**H is mainly described here.

[0270] As shown in FIG. **25**, the semiconductor module **10**F includes a conductive film **445-1**, a conductive film **445-2**, a conductive film **446**, a conductive film **447**, and a pogo pin pressure bonding mechanism **700**. The semiconductor module **10**F supplies power to the memory cube **100**, for example, using the pogo pin pressure bonding mechanism **700**.

[0271] As an example, the conductive film **445-1** is formed on the first surface **142**, and the conductive film **445-2** is formed on the second surface **144**. The conductive film **446** is formed on the conductive film **445-1**, and the conductive film **447** is formed on the conductive film **445-2**. The conductive films **445-1** and **445-2** and the conductive films **446** and **447** are formed by, for example, electric field plating (plating method). The conductive films **445-1** and **445-2** include a conductor made of a metal material, and include, for example, a conductor containing copper, silver, nickel, or the like. The conductive films **446** and **447** include a conductor made of a metal material, and include, for example, a conductor containing gold, tin, nickel, or the like.

[0272] As an example, the conductive film **445** is formed on the first surface **142** and the second surface **144** by electric field plating (plating method). The conductive film **445** includes a conductor made of a metal material, and includes, for example, a conductor containing copper, silver, nickel, or the like.

[0273] The pogo pin pressure bonding mechanism **700** includes a pogo pin support part **701** and a plurality of pogo pins **702**. The pogo pin support part **701** includes the plurality of pogo pins **702** and has a function of supporting the plurality of pogo pins **702**. Each of the plurality of pogo pins **702** is a connector composed of a terminal part, a cylindrical member connected to the terminal part, and an elastic member inserted into the cylindrical member, and the like.

[0274] The pogo pin pressure bonding mechanism **700** crimps each of the plurality of pogo pins **702** to the conductive film **446** and the conductive film **447**, and connects each of the plurality of pogo pins **702** to the conductive film **446** and the conductive film **447**. Consequently, the pogo pin pressure bonding mechanism **700** can externally supply voltages (for example, the power supply voltage VDD and the voltage VSS) to the plurality of pogo pins **702** and supply voltages to the memory cube **100** through the plurality of pogo pins **702**.

[0275] As shown in FIG. **26**, the semiconductor module **10**G includes the conductive film **445-1**, the conductive film **445-2**, the conductive film **446**, the conductive film **447**, and an L-shaped member **710**. For example, in the semiconductor module **10**G, the conductive film **445-1**, the conductive film **445-2**, the conductive film **447**, and the conductive film **446** are formed in the same manner as in the semiconductor module **10**F. The L-shaped member **710** includes a conductor made of a metal material, and includes, for example, a conductor including copper or the like.

[0276] The L-shaped member **710** is connected to the conductive film **446** and the conductive film **447**. Consequently, in the semiconductor module **10**G, a voltage (for example, the power supply voltage VDD and the voltage VSS) is supplied from the outside to a plurality of L-shaped members **710**, and the memory cube **100** is supplied with the voltage through the plurality of L-shaped members **710**.

[0277] As shown in FIG. **27**, the semiconductor module **10**H includes the conductive film **445-1**, the conductive film **445-2**, the conductive film **446**, the conductive film **447**, and a flexible substrate **720**. For example, in the semiconductor module **10**H, the conductive film **445-1**, the conductive film **445-2**, the conductive film **447**, and the conductive film **446** are formed in the same manner as in the semiconductor modules **10**F and **10**G.

[0278] The flexible substrate **720** is connected to the conductive film **446** and the conductive film **447**. Consequently, in the semiconductor module **10**H, the voltage (for example, the power supply voltage VDD and the voltage VSS) is supplied from the outside to the flexible substrate **720**, and the memory cube **100** is supplied with the voltage through the flexible substrate **720**.

10. Tenth Embodiment

[0279] A mounting example of a semiconductor module **10**J according to a tenth embodiment, an example of a configuration of signal transmission wirings **166*b*** included in the seal ring **160** and the memory chip **110** of the semiconductor module **10**J, and an example of the configuration of inductors **372**, **372*f***, and **372*g*** included in the seal ring **160** and the memory chip **110** of the semiconductor module **10**J will be described referring to FIG. **28** to FIG. **40**.

10-1. Mounting Example of Semiconductor Module **10**J

[0280] A mounting example of the semiconductor module **10**J will be described referring to FIG. **28**. FIG. **28** is a diagram showing a configuration of the semiconductor module **10**J according to the tenth embodiment of the present disclosure. The explanation of the mounting example of the semiconductor module **10**J with reference to FIG. **28** mainly describes differences from the semiconductor module **10**B described with reference to FIG. **14**. Configurations that are the same as or similar to those in FIG. **1** to FIG. **27** will be described as necessary.

[0281] The inductor **172** of the semiconductor module **10**B is mainly included in the rewiring layer **300**, and the inductor **172** is electrically connected to the transmitting/receiving circuit **114** using the terminal A and the terminal B, whereas the inductor **372** of the semiconductor module **10**J is mainly included in the memory cube **100**, and the inductor **372** is electrically connected to the transmitting/receiving circuit **114** using a terminal E and a terminal F. Similar to the inductor **172**, the inductor **372** has a function of performing inductor communication with the inductor

272 of the logic chip 200 in a non-contact manner. Configurations of the semiconductor module 10J other than the inductor 372 with reference to FIG. 28 are the same as the configurations of the semiconductor module 10B described with reference to FIG. 14, and will not be described here.

10-2. Example of Signal Transmission Wiring 166*b* and Inductor 372

[0282] Examples of the signal transmitting wiring 166*b* and an inductor 372 of the semiconductor module 10J will be described referring to FIG. 29 to FIG. 32. FIG. 29 is a plan view showing configurations of the seal ring 160, the signal transmission wiring 166*b*, and the inductor 372. FIG. 30 is a cross-sectional view of the seal ring 160 and the signaling wiring 166*b* taken along a line E3-E4 of FIG. 29. FIG. 31 is a cross-sectional view of the seal ring 160 and the inductor 372 taken along a line E5-E6 of FIG. 29. FIG. 32 is a cross-sectional view of the seal ring 160 and the inductor 372 shown in FIG. 29 when viewed from a depth direction from the first side surface 146 toward the third side surface 148. Configurations that are the same as or similar to those in FIG. 1 to FIG. 28 will be described as necessary.

[0283] First, a signal transmission wiring 166*b* of the semiconductor module 10J will be described. The signal transmission wiring 166*b* is parallel to the first surface 102 and the second surface 104. An end part 166*a* of the signal transmission wiring 166 may include a portion exposed to the first side surface 146 (the insulating layer 182), and may include a portion not exposed to the first side surface 146 (the insulating layer 182). For example, although the end part 166*a* of the signal transmission wiring 166 described in FIG. 11A and FIG. 11B is shown to include a portion exposed to the first side surface 146 (insulating layer 182), an end part 166*c* of the signal transmission wiring 166*b* shown in FIG. 29 and FIG. 30 includes a portion not exposed to the first side surface 146 (insulating layer 182). Further, the signal transmission wiring 166*b* is formed by using a wiring 183*b* so as to overlap the seal ring 160 and straddle the seal ring 160. The wiring 183*b* is formed on the same surface as the wiring 183 (see FIG. 11A).

[0284] The configuration shown in FIG. 29 and FIG. 30 is the same as the configuration shown in FIG. 11A and FIG. 11B except that the end part 166*c* of the signal transmission wiring 166*b* is not exposed to the first side surface 146, and therefore, the explanation thereof will be omitted. Similar to the signal transmission wiring 166*b*, a portion of the ground wiring 165 (see FIG. 8A or FIG. 8B) of the memory chip 110 and a portion of the power wiring 164 (see FIG. 8A or FIG. 8B) may include a portion exposed to the first side surface 146, and may include a portion not exposed to the first side surface 146. For example, the ground wiring 165 and the power wiring 164 that are not connected to the wiring layer (wiring) formed in the rewiring layer 300 are not exposed to the first side surface 146.

[0285] Next, the inductor 372 of the semiconductor module 10J will be described. The inductor 372 shown in FIG. 29 includes the terminal E, the terminal F, a first part 372*a*, a second part 372*b*, a third part 372*c*, a fourth part 372*e*, and a fifth part 372*d*. The inductor 372 is an example of a one-turn inductor when viewed from the second surface 104 of the memory chip 110 through the memory chip 110 along the direction D1. The fifth part 372*d* extends in the direction D2, one end of the fifth part 372*d* is electrically connected to the terminal E, and the other end of the fifth part 372*d* is electrically connected to one end of the fourth part 372*e*. The fourth part 372*e* extends in the direction D3 and the other end of the fourth part 372*e* is electrically connected to one end of the first part 372*a*. The first part 372*a* extends in the direction D2 and the other end of the first part 372*a* is electrically connected to one end of the second part 372*b*. The second part 372*b* extends in the direction D3 and the other end of the second part 372*b* is electrically connected to one end of the third part 372*c*. The third part 372*c* extends in the direction D2 and the other end of the third part 372*c* is electrically connected to the terminal F.

[0286] As shown in FIG. 29 and FIG. 31, the inductor 372 is parallel to the first surface 102 and the second surface 104. The first part 372*a* of the inductor 372 corresponds to an end portion of the inductor 372 closer to the first side surface 146. The first part 372*a* is not exposed to the first side surface 146 as is the case with the end part 166*c* of the signal transmission wiring 166*b*. In addition, the inductor 372 is formed so as to overlap the seal ring 160 and straddle the seal ring 160, similar to the signal transmission wiring 166*b*. That is, a length of the seal ring 160 extending along the direction D3 is shorter than a length of the inductor 372 extending along the direction D3. The inductor 372 is formed using a wiring 183*c*. The wiring 183*c* is formed on the same surface as the wiring 183*b* and the wiring 183.

[0287] Further, as shown in FIG. 32, the inductor 372 and the seal ring 160 viewed through the depth direction from the first side surface 146 toward the third side surface 148 are stacked upward from the transistor layer 130 side along the direction D1. The vias 151*b*, 153*b*, and 155*b* forming the seal ring 160 and the wirings 152*b*, 154*b*, and 156*b* forming the seal ring 160 extend along the direction D2, and lengths of the vias 151*b*, 153*b*, and 155*b* and the wirings 152*b*, 154*b*, and 156*b* along the direction D2 is longer than a length of the first part 372*a* of the inductor 372 along the direction D2. In addition, similar to the length of the first part 372*a* of the inductor 372 along the direction D2, the length of the vias 151*b*, 153*b*, and 155*b* and the wirings 152*b*, 154*b*, and 156*b* forming the seal ring 160 along the direction D2 is longer than the length of the end part 166*c* of the signal transmission wiring 166*b* along the direction D2. In addition, in a cross-sectional view of the semiconductor module 10J, the inductor 372 and the seal ring 160 may be stacked upward from the transistor layer 130 along the direction D1 as shown in FIG. 32, and the lengths of the vias 151*b*, 153*b*, and 155*b* and the wirings 152*b*, 154*b*, and 156*b* forming the seal ring 160 along the direction D2 may be longer than the length of the first part 372*a* of the inductor 372 along the direction D2.

[0288] In addition, although one inductor 372 is shown in FIG. 29 and FIG. 31, the memory chip 110 includes a plurality of inductors 372. For example, the inductors 372 are arranged side by side along the direction D2.

[0289] The semiconductor module 10J includes the configuration described above. As a consequence, the insulating layer 182 and the seal ring 160 prevent the plurality of power wirings 164, the plurality of ground wirings 165, and the plurality of signal transmission wirings 166*b* from absorbing moisture from the first side surface 146 and from corroding or deteriorating the wiring due to the intrusion of impurities or the like.

[0290] Also, the inductor 372 is covered by the insulating layer 182 and is not exposed to the side surface of the memory cube 100. Further, the inductor 372 is formed to straddle the seal ring **160** and be shorter than the seal ring **160** when viewed from the first side surface **146**. As a consequence, in the inductor **372**, as in the case of the signal transmission wiring **166***b*, the insulating layer **182** and the seal ring **160** suppress moisture absorption, corrosion, deterioration, and the like of the wiring due to the entry of impurities and the like.

[0291] In addition, by suppressing moisture absorption, corrosion, deterioration, and the like of the wiring due to the intrusion of impurities and the like, insulation between adjacent inductors **372** is ensured, and short circuits between adjacent inductors **372** are suppressed. As a result, a distance between the adjacent inductors **372** and the distance between the inductors **372** and **272** can be shortened. Therefore, the semiconductor module **10**J can increase the number of inductors **372** in the memory chip **110** and improve the communication between the inductor **372** and the inductor **272**.

[0292] Further, since the inductor **372** is not exposed to the respective side surfaces of the memory cube **100**, breakage of the inductor **372** can be suppressed by ESD. Consequently, for example, since an ESD protection circuit of the inductor **372** is not required, power consumption caused by the ESD protection circuit is suppressed, and power consumption of the semiconductor module **10**J is reduced. In addition, a delay caused by the ESD protection circuit is suppressed, and the semiconductor module **10**J can operate at high speed.

[0293] Therefore, reliability of the semiconductor module **10**J can be maintained without impairing long-term reliability.

10-3. Another Example of Inductor 372 (Inductor 372*f*)

[0294] The inductor **372***f* of the semiconductor module **10**J will be described referring to FIG. **33** to FIG. **36**. FIG. **33** is a plan view showing an example of a configuration of the seal ring **160** and the inductor **372***f*. FIG. **34** is a cross-sectional view of the inductor **372***f* taken along a line E**7**-E**8** of FIG. **33**. FIG. **35** is a cross-sectional view of the seal ring **160** and the inductor **372***f* taken along a line E**9**-E**10** of FIG. **33**. FIG. **36** is a cross-sectional view of the inductor **372***f* taken along a line E**11**-E**12** of FIG. **33**. Configurations that are the same as or similar to those in FIG. **1** to FIG. **32** will be described as necessary.

[0295] The inductor **372***f* shown in FIG. **33** to FIG. **36** includes the terminal E, the terminal F, a wiring **152***c*, a via **153***c*, a wiring **154***c*, a via **155***c*, a wiring **156***c*, a via **184***a*, the wiring **183***c*, a via **184***b*, a wiring **156***d*, a via **155***d*, a wiring **154***d*, a via **155***e*, a wiring **156***e*, a via **184***c*, a wiring **183***d*, a via **184***d*, a wiring **156***f*, a via **155***f*, a wiring **154***e*, a via **155***g*, a wiring **156***g*, a via **184***e*, a wiring **183***e*, a via **184***f*, a wiring **156***h*, a via **184***g*, a wiring **183***f*, a via **184***h*, a wiring **156***i*, a via **184***i*, and a wiring **183***g*. The inductor **372***f* is an example of a three-turn inductor when the memory chip **110** is viewed through the second surface **104** of the memory chip **110** along the direction D**1**. The inductor **372***f* is parallel to the first surface **102** and the second surface **104**.

[0296] The wiring **152***c* is formed in the same layer as the wiring **152***b*, and the via **153***c* is formed in the same layer as the via **153***b*. The wiring **154***c*, the wiring **154***d*, and the wiring **154***e* are formed in the same layer as the wiring **154***b*. The via **155***c*, the via **155***d*, the via **155***e*, the via **155***f*, and the via **155***g* are formed in the same layer as the via **155***b*. The wiring **156***c*, the wiring **156***d*, the wiring **156***e*, the wiring **156***f*, the wiring **156***g*, the wiring **156***h* and the wiring **156***i* are formed in the same layer as the wiring **156***b*. The wiring **183***c*, the wiring **183***d*, the wiring **183***e*, the wiring **183***f*, and the wiring **183***g* are formed on the same layer as the wiring **183***b*.

[0297] The via **184***a*, the via **184***b*, the via **184***c*, the via **184***d*, the via **184***e*, the via **184***f*, the via **184***g*, the via **184***h*, and the via **184***i* are formed between the layer where the insulated layer **156***a* and wirings **156***b* to **156***i* are formed and the layer where the wiring **183***c*, wiring **183***d*, wiring **183***e*, wiring **183***f*, and wiring **183***g* are formed in the direction D**1**.

[0298] The wiring **152***c* extends in the direction D**2**, one end of the wiring **152***c* is electrically connected to the terminal F, and the other end of the wiring **152***c* is electrically connected to the via **153***c*. The via **153***c*, the wiring **154***c*, the via **155***c*, the wiring **156***c*, and the via **184***a* extend in the direction D**1** and are electrically connected to each other. The via **184***a* is electrically connected to one end of the wiring **183***c*. The wiring **183***c* extends in the direction D**3**, and the other end of the wiring **183***c* is electrically connected to the via **184***b*. The wiring **183***c* is configured to overlap the seal ring **160** and straddle the seal ring **160**.

[0299] The via **184***b*, the wiring **156***d*, and the via **155***d* extend in the direction D**1** and are electrically connected to each other. The via **155***d* is electrically connected to one end of the wiring **154***d*. The wiring **154***d* extends in the direction D**2**, and the other end of the wiring **154***d* is electrically connected to the via **155***e*. The via **155***e*, the wiring **156***e*, and the via **184***c* extend in the direction D**1** and are electrically connected to each other. The via **184***c* is electrically connected to one end of the wiring **183***d*. The wiring **183***d* extends in the direction D**3**, and the other end of the wiring **183***d* is electrically connected to the via **184***d*. The wiring **183***d* is configured to overlap the seal ring **160** and straddle the seal ring **160**.

[0300] The via **184***d*, the wiring **156***f*, and the via **155***f* extend in the direction D**1** and are electrically connected to each other. The via **155***f* is electrically connected to one end of the wiring **154***e*. The wiring **154***e* extends in the direction D**2**, and the other end of the wiring **154***e* is electrically connected to the via **155***g*. The via **155***g*, the wiring **156***g*, and the via **184***e* extend in the direction D**1** and are electrically connected to each other. The via **184***e* is electrically connected to one end of the wiring **183***e*. The wiring **183***e* extends in the direction D**3**, and the other end of the wiring **183***e* is electrically connected to the via **184***f*. The wiring **183***e* is configured to overlap the seal ring **160** and straddle the seal ring **160**.

[0301] The via **184***f* extends in the direction D**1** and is electrically connected to one end of the wiring **156***h*. The wiring **156***h* extends in the direction D**2**, and the other end of the wiring **156***h* is electrically connected to the via **184***g*. The via **184***g* extends in the direction D**1** and is electrically connected to one end of the wiring **183***f*. The wiring **183***f* extends in the direction D**3**, and the other end of the wiring **183***f* is electrically connected to the via **184***h*. The wiring **183***f* is configured to overlap the seal ring **160** and straddle the seal ring **160**.

[0302] The via **184***h* extends in the direction D**1** and is electrically connected to one end of the wiring **156***i*. The wiring **156***i* extends in the direction D**2**, and the other end of the wiring **156***i* is electrically connected to the via **184***i*. The via **184***i* extends in the direction D**1** and is electrically connected to one end of the wiring **183***g*.

[0303] The wiring **183***g* extends in the direction D**3** and is configured to overlap the seal ring **160** and straddle the seal ring **160**. Further, the wiring **183***g* is bent in the direction D**2** and extends in the direction D**2** after straddling the seal ring **160**. Further, the wiring **183***g* extends in the direction D**2**, and then bends in the direction D**3** and extends in the direction D**3**. Further, the wiring **183***g* extends in the direction D**3** and is configured to overlap the seal ring **160** and straddle the seal ring **160**. Further, the wiring **183***g* is bent in the direction D**2** and extends in the direction D**2** after straddling the seal ring **160**. The other end of the wiring **183***g* extending in the direction D**2** is electrically connected to the terminal E.

[0304] As shown in FIG. **33**, the wiring **183***c*, the wiring **183***d*, the wiring **183***e*, the wiring **183***f*, and the wiring **183***g* overlapping the seal ring **160** are formed in the same plane, and parts of the wiring **183***c*, the wiring **183***e*, the wiring **183***d*, the wiring **183***e*, the wiring **183***f*, and the wiring **183***g* are formed in parallel along the direction D**3**.

[0305] As shown in FIG. **33** or FIG. **34**, parts of the wiring **152***c*, the wiring **154***e*, wiring **156***i*, and the wiring **183***g* overlap each other along the direction D**1** and are formed in parallel along the direction D**2**. The wiring **152***c* is formed in the same layer as the wiring **152***b* constituting the seal ring **160**, the wiring **154***e* is formed in the same layer as the wiring **154***b* constituting the seal ring **160**, and the wiring **156***i* is formed in the same layer as the wiring **156***b* forming the seal ring **160**.

[0306] As shown in FIG. **33** or FIG. **36**, parts of the wiring **154***d*, the wiring **156***h*, and the wiring **183***g* are stacked along the direction D**1** and overlap each other, and are formed in parallel along the direction D**2**. The wiring **154***d* is formed in the same layer as the wiring **154***b* constituting the seal ring **160**, and the wiring **156***h* is formed in the same layer as the wiring **156***b* constituting the seal ring **160**.

[0307] As shown in FIG. **33**, the wiring **183***c* and the wiring **183***e* face the wiring **183***d* and the wiring **183***f*. As shown in FIG. **33**, FIG. **34**, or FIG. **36**, the wiring **154***d* faces the wiring **154***e*, and the wiring **156***h* faces the wiring **156***i*.

[0308] That is, the inductor **372***f* is composed of the wiring **183***c* and the wiring **183***e*, and the wiring **183***d* and the wiring **183***f*, which are formed in the same layer and overlap the seal ring **160** and face each other, the wiring **154***d* and the wiring **154***e* formed in the same layer as the wiring **154***b* constituting the seal ring **160** and opposed to each other, and the wiring **156***h* and the wiring **156***i* formed in the same layer as the wiring **156***b* constituting the seal ring **160**.

[0309] In addition, as shown in FIG. **33** and FIG. **36**, the wiring **183***g*, the wiring **156***h*, and the wiring **154***d* overlap each other and correspond to an end portion of the inductor **372***f* on the first side surface **146** side. The wiring **183***g*, the wiring **156***h*, and the wiring **154***d* are not exposed to the first side surface **146**, similar to the end part **166***c* of the signal transmission wiring **166***b*. The wiring **183***g*, the wiring **156***i*, the wiring **154***e*, and the wiring **152***c* overlap each other and correspond to an end portion of the inductor **372***f* opposite to the first side surface **146**.

[0310] By forming the end portion of the inductor **372***f* on the side of the first side surface **146** and the end of the inductor **372***f* on the side opposite to the side of the first side surface **146** as shown in FIG. **33** and FIG. **36**, the wirings **183***c* to 183 g of the inductor **372***f* can overlap the seal ring **160** and straddle the seal ring **160** as shown in the cross section shown in FIG. **35**.

[0311] Consequently, the three-turn inductor **372***f* configured as described above is insulated from the seal ring **160** by the insulating layers **151***a* to **182**, and is not short-circuited with the seal ring **160**.

[0312] In addition, similar to the inductor **372**, the memory chip **110** includes a plurality of inductors **372***f* arranged along the direction D**2**.

[0313] The semiconductor module **10**J includes the inductor **372***f* as described above, and has the same advantages as the semiconductor module **10**J including the inductor **372**.

10-4. Other Example of Inductor 372 (Inductor 372*g*)

[0314] The inductor 372 g of the semiconductor module **10**J will be described referring to FIG. **37** to FIG. **40**. FIG. **37** is a plan view showing configurations of the seal ring **160** and the inductor **372***g*. FIG. **38** is a cross-sectional view of the seal ring **160** and the inductor **372***g* taken along a line E**13**-E**14** of FIG. **37**. FIG. **39** is a cross-sectional view of the inductor **372***g* taken along a line E**15**-E**16** of FIG. **37**. FIG. **40** is a cross-sectional view of the inductor **372***g* taken along a line E**17**-E**18** of FIG. **37**. The same or similar configurations as those in FIG. **1** to FIG. **36** will be described as necessary.

[0315] The inductor **372***g* shown in FIG. **37** to FIG. **40** is, unlike the inductor **372***f*, an inductor having a plurality of turns formed over a plurality of memory chips in a cross-sectional view. Since other configurations of the inductor **372***g* are the same as the configuration of the inductor **372***f*, explanation thereof will be omitted.

[0316] As shown in FIG. **38** to FIG. **40**, a first surface **102***n* of the memory chip **110***n*+1 is bonded to a second surface **104***n*+1 of the memory chip **110***n*. The inductor **372***g* is arranged over two memory chips (memory chip **110***n* and **110***n*+1).

[0317] FIG. **39** shows a cross section (structure) of the inductor **372***g* when viewed from the first side surface **146** toward the third side surface **148** in a depth direction. When viewed from the first side surface **146** to the third side surface **148**, the inductor **372***g* shown in FIG. **39** is an example of a two-turn inductor provided over two memory chips (memory chip **110***n* and **110***n*+1). Further, the inductor **372***g* is parallel to the first side surface **146** and the third side surface **148**.

[0318] The inductor **372***g* shown in FIG. **37** to FIG. **40** includes the terminal E, the terminal F, a wiring **183***j*, a via **184***j*, a wiring **156***j*, a via **155***j*, a wiring **154***j*, a via **153***j*, a wiring **152***j*, a via **151***j*, a through-hole electrode **131***j*, a wiring **196***j*, a wiring **183***nj*, a via **184***nj*, a wiring **156***nj*, a via **155***nj*, a wiring **154***nj*, a via **153***nj*, a wiring **152***nj*, a via **153***nk*, a wiring **154***nk*, a via **155***nk*, a wiring **156***nk*, a via **184***nk*, a wiring **183***nk*, a wiring **196***nk*, a through-hole electrode **131***k*, a via **151***k*, a wiring **152***k*, a via **153***k*, a wiring **154***k*, a via **155***k*, a wiring **156***k*, a via **184***k*, a wiring **183***k*, a via **184***m*, a wiring **156***m*, a via **155***m*, a wiring **154***m*, a via **153***m*, a wiring **152***m*, a via **151***m*, a through-hole electrode **131***m*, a wiring 196 nm, a wiring 183 nm, a via 184 nm, a wiring 156 nm, a via 155 nm, a wiring 154 nm, a via **155***no*, a wiring **156***no*, a via **184***no*, a wiring **183***no*, a wiring **196***no*, a through-hole electrode **131***o*, a via **151***o*, a wiring **152***o*, a via **153***o*, a wiring **154***o*, a via **155***o*, a wiring **156***o*, a via **184***o*, and a wiring **183***o*.

[0319] The memory chip **110***n* includes the wiring **196***j*, the wiring **183***nj*, the via **184***nj*, the wiring **156***nj*, the via **155***nj*, the wiring **154***nj*, the via **153***nj*, a wiring **152***nj*, the wiring **153***nk*, the wiring **154***nk*, the via **155***nk*, the wiring **156***nk*, the via **184***nk*, the wiring **183***nk*, the wiring **196***nk*, the wiring 196 nm, the wiring 183 nm, the via 184 nm, the wiring 156 nm, the via 155 nm, the wiring 154 nm, the via **155***no*, the wiring **156***no*, the via **184***no*, the wiring **183***no*, the wiring **196***no*, and a seal ring **160***n*. The function and structure of the seal ring **160***n* is the same as the function and structure of the seal ring **160**. The function and structure of the seal ring **160***n* will be described as required.

[0320] The wiring **152***nj* is formed in the same layer as the wiring **152***b* of the memory chip **110***n*, and the vias **153***nj* and **153***nk* are formed in the same layer as the via **153***b* of the memory chip **110***n*. The wiring **154***nj*, the wiring **154***nk*, and the wiring 154 nm are formed in the same layer as the wiring **154***b* of the memory chip **110***n*. The via **155***nj*, the via **155***nk*, the via 155 nm, and the via **155***no* are formed in the same layer as the via **155***b* of the memory chip **110***n*. The wiring **156***nj*, the wiring **156***nk*, the wiring 156 nm, and the wiring **156***no* are formed in the same layer as the wiring **156***b* of the memory chip **110***n*. The wiring **183***nj*, the wiring **183***nk*, the wiring 183 nm, and the wiring **183***no* are formed in the same layer.

[0321] The via **184***nj*, the via **184***nk*, the via 184 nm, and the via **184***no* are formed between the layer on which the insulation layer **156***a*, the wiring **156***nj*, the wiring **156***nk*, the wiring 156 nm, and the wiring **156***no* are formed and the layer on which the wiring **183***nj*, the wiring **183***nk*, the wiring 183 nm, and the wiring **183***no* are formed in the memory chip **110***n* in the direction D**1**.

[0322] The wiring **196***j*, the wiring **196***nk*, the wiring 196 nm, and the wiring **196***no* are formed on the wiring **183***nj*, the wiring **183***nk*, the wiring 183 nm, and the wiring **183***no*.

[0323] The memory chip **110***n*+1 includes the terminal E, the terminal F, the wiring **183***j*, the via **184***j*, the wiring **156***j*, the via **155***j*, the wiring **154***j*, the via **153***j*, the wiring **152***j*, the via **151***j*, the through-hole electrode **131***j*, the through-hole electrode **131***k*, the via **151***k*, the wiring **152***k*, the via **153***k*, the wiring **154***k*, the via **155***k*, the wiring **156***k*, the via **184***k*, the wiring **183***k*, the via **184***m*, the wiring **156***m*, the via **155***m*, the wiring **154***m*, the via **153***m*, the wiring **152***m*, the via **151***m*, the through-hole electrode **131***m*, the through-hole electrode **131***o*, the via **151***o*, the wiring **152***o*, the via **153***o*, the wiring **1540**, the via **155***o*, the wiring **156***o*, the via **184***o*, the wiring **183***o*, and a seal ring **160***n*+1. In addition, the function and structure of the seal ring **160***n*+1 are the same as the function and structure of the seal ring **160**. The function and structure of the seal ring **160***n*+1 will be described as required.

[0324] The via **151***j*, the via **151***k*, the via **151***m*, and the via **151***o* are formed in the same layer as the via **151***b* of the memory chip **110***n*+1. The wiring **152***j*, the wiring **152***k*, the wiring **152***m*, and the wiring **152***o* are formed in the same layer as the wiring **152***b* of the memory chip **110***n*+1. The via **153***j*, the via **153***k*, the via **153***m*, and the via **153***o* are formed in the same layer as the via **153***b* of the memory chip **110***n*+1. The wiring **154***j*, the wiring **154***k*, the wiring **154***m*, and the wiring **154***o* are formed in the same layer as the wiring **154***b* of the memory chip **110***n*+1. The via **155***j*, the via **155***k*, the via **155***m*, and the via **155***o* are formed in the same layer as the via **155***b* of the memory chip **110***n*+1. The wiring **156***j*, the wiring **156***k*, the wiring **156***m*, and the wiring **156***o* are formed in the same layer as the wiring **156***b* of the memory chip **110***n*+1. The wiring **183***j*, the wiring **183***k*, and the wiring **183***o* are formed in the same layer. The via **184***j*, the via **184***k*, the via **184***m*, and the via **1840** are formed in the same layer.

[0325] The through-hole electrode **131***j*, the through-hole electrode **131***k*, the through-hole electrode **131***m* and the through-hole electrode **131***o* are formed so as to penetrate the transistor layer **130***n*+1, and are connected to the via **151***j*, the via **151***k*, the via **151***m*, and the via **151***o*. Further, when the memory chip **110***n*+1 is bonded to the memory chip **110***n*, the through-hole electrode **131***j*, the through-hole electrode **131***k*, the through-hole electrode **131***m*, and the through-hole electrode **131***o* are connected to the wiring **196***j*, the wiring **196***nk*, the wiring 196 nm, and the wiring **196***no* of the memory chip **110***n*.

[0326] Here, a connection of the inductor **372***g* is explained.

[0327] The wiring **183***j* extends in the direction D**3**, one end of the wiring **183***j* is electrically connected to the terminal F, and the other end of the wiring **183***j* is electrically connected to the via **184***j*. The wiring **183***j* overlaps the seal rings **160***n* and **160***n*+1 and is configured to straddle the seal rings **160***n* and **160***n*+1. The via **184***j*, the wiring **156***j*, the via **155***j*, the wiring **154***j*, the via **153***j*, the wiring **152***j*, the via **151***j*, the through-hole electrode **131***j*, the wiring **196***j*, the wiring **183***nj*, the via **184***nj*, the wiring **156***nj*, the via **155***nj*, the wiring **154***nj*, and the via **153***nj* extend in the direction D**1** and are electrically connected to each other. The wiring **154***nj* is electrically connected to one end of the wiring **152***nj*. The wiring **152***nj* extends in the direction D**2**, and the other end of the wiring **152***nj* is electrically connected to the via **153***nk*.

[0328] The via **153***nk*, the wiring **154***nk*, the via **155***nk*, the wiring **156***nk*, the via **184***nk*, the wiring **183***nk*, the wiring **196***nk*, the through-hole electrode **131***k*, the via **151***k*, the wiring **152***k*, the via **153***k*, the wiring **154***k*, the via **155***k*, the wiring **156***k*, and the via **184***k* extend in the direction D**1** and are electrically connected to each other. The via **184***k* is electrically connected to one end of the wiring **183***k*. The wiring **183***k* extends in the direction D**2**, and the other end of the wiring **183***k* is electrically connected to the via **184***m*. The via **184***m*, the wiring **156***m*, the via **155***m*, the wiring **154***m*, the via **153***m*, the wiring **152***m*, the via **151***m*, the through-hole electrode **131***m*, the wiring 196 nm, the wiring 183 nm, the via 184 nm, the wiring 156 nm, and the via 155 nm extend in the direction D**1** and are electrically connected to each other. The via 155 nm is electrically connected to one end of the wiring 154 nm. The wiring 154 nm extends in the direction D**2**, and the other end of the wiring 154 nm is electrically connected to the via **155***no*.

[0329] The via **155***no*, the wiring **156***no*, the via **184***no*, the wiring **183***no*, the wiring **196***no*, the through-hole electrode **131***o*, the via **151***o*, the wiring **152***o*, the via **153***o*, the wiring **154***o*, and the via **155***o* extend in the direction D**1** and are electrically connected to each other. The via **155***o* is electrically connected to one end of the wiring **156***o*. The wiring **156***o* extends in the direction D**2** and then bends in the direction D**3** and extends in the direction D**3**.

[0330] The other end of the wiring **156***o* extending in the direction D**3** is electrically connected to the via **184***o*. The via **184***o* extends in the direction D**1** and is electrically connected to one end of the wiring **183***o*. The wiring **183***o* extends in the direction D**3** and the other end of the wiring **183***o* is electrically connected to the terminal E. The wiring **183***o* is configured to overlap the seal ring **160** and straddle the seal ring **160**.

[0331] The wiring **183***g* extends in the direction D**3** and is configured to overlap the seal ring **160** and straddle the seal ring **160**. Further, the wiring **183***g* is bent in the direction D**2** and extends in the direction D**2** after straddling the seal ring **160**. Further, the wiring **183***g* extends in the direction D**2**, and then bends in the direction D**3** and extends in the direction D**3**. Further, the wiring **183***g* extends in the direction D**3** and is configured to overlap the seal ring **160** and straddle the seal ring **160**. Further, the wiring **183***g* is bent in the direction D**2** and extends in the direction D**2** after straddling the seal ring **160**. The other end of the wiring **183***g* extending in the direction D**2** is electrically connected to the terminal E.

[0332] As shown in FIG. **37**, the wiring **183***j* and the wiring **183***o* overlapping the seal ring **160** are formed on the same plane and are parallel-formed along the direction D**3**.

[0333] As shown in FIG. **39**, a substantially coiled portion of a two-turn inductor 196 nm is formed by the wiring **183***j*, the via **184***j*, the wiring **156***j*, the via **155***j*, the wiring **154***j*, the via **153***j*, the wiring **152***j*, the via **151***j*, the through-hole electrode **131***j*, the wiring **196***j*, the wiring **183***nj*, the via **184***nj*, the wiring **156***nj*, the via **155***nj*, the wiring **154***nj*, the via **153***nj*, the wiring **152***nj*, the via **153***nk*, the wiring **154***nk*, the via **155***nk*, the wiring **156***nk*, the via **184***nk*, the wiring **183***nk*, the wiring **196***nk*, the through-hole electrode **131***k*, the via **151***k*, the wiring **152***k*, the via **153***k*, the wiring **154***k*, the via **155***k*, the wiring **156***k*, the via **184***k*, the wiring **183***k*, the via **184***m*, the wiring **156***m*, the via **155***m*, the wiring **154***m*, the via **153***m*, the wiring **152***m*, the via **151***m*, the through-hole electrode **131***m*, the wiring 196 nm, the wiring 183 nm, the via 184 nm, the wiring 156 nm, the via 155 nm, the wiring 154 nm, the via **155***no*, the wiring **156***no*, the via **184***no*, the wiring **183***no*, the wiring **196***no*, the through-hole electrode **131***o*, the via **151***o*, the wiring **152***o*, the via **153***o*, the wiring **154***o*, the via **155***o*, and the wiring **156***o*.

[0334] That is, similar to the inductor **372***f*, the inductor **372***g* is composed of the wiring **183***j* and the wiring **183***o* overlapping the seal rings **160***n* and **160***n*+1 and alternately provided in the same layer, the wiring **183***k* that is arranged in the same layer as the wiring **183***j* and the wiring **183***o* and does not overlap the seal rings **160***n* and **160***n*+1, and a plurality of wirings that are arranged in the same layer as the wirings which form the seal rings **160***n* and **160***n*+1 and are arranged across two memory chips (**110***n* and **110***n*+1).

[0335] Further, as shown in FIG. **37** and FIG. **39**, a substantially coiled part of the two-turn inductor **372***g* is not exposed to the first side surface **146**, similar to the end part **166***c* of the signal transmission wiring **166***b*.

[0336] By forming the substantially coil part of the inductor **372***g* arranged over the two memory chips on the first side surface **146** side by the wiring forming the seal ring **160**, the wiring **183***j* and the wiring **183***o* connected to the substantially coil part of the inductor **372***g* can overlap the seal rings **160***n* and **160***n*+1, and can straddle the seal rings **160***n* and **160***n*+1.

[0337] Consequently, the two-turn inductor **372***g* configured as described above is insulated from the seal rings **160***n* and **160***n*+1 by the insulating layers **151***a* to **182** provided in the two memory chips **110***n* and **110***n*+1, respectively, and is not shorted with the seal rings **160***n* and **160***n*+1.

[0338] In addition, although one inductor **372***g* is shown in FIG. **37** to FIG. **40**, the memory chips **110***n* and **110***n*+1 include a plurality of inductors **372***g*. For example, the plurality of inductors **372***g* is arranged side by side along the direction D**2**.

[0339] The semiconductor module **10**J includes the inductor **372***g* as described above, and has the same advantages as the semiconductor module **10**J including the inductor **372**.

11. Eleventh Embodiment

[0340] Referring to FIG. **41** to FIG. **46**, inductors **372***h* to **372***j* included in a semiconductor module **10**K according to an eleventh embodiment will be described. In a method for forming the inductor **372** included in the semiconductor module **10**K, two sides of the inductors **372***h* to **372***j* are formed in the memory cube **100**, and straight sides of the inductors **372***h* to **372***j* are formed in the rewiring layers **300** of the memory cube **100**. Straight sides of the inductors **372***h* to **372***j* formed on the rewiring layer **300** are formed on the first side surface **146**. For example, since other configurations and functions are the same as the configurations and functions described in the first embodiment, the second embodiment, or the tenth embodiment, a detailed description thereof will be omitted. The method for forming the inductor **372** according to the eleventh embodiment can be appropriately combined with or replaced with the configurations described in the first to tenth embodiments as long as they do not contradict each other.

[11-1. Manufacturing Method of One-Turn Inductor **372***h*]

[0341] Referring to FIG. **41** and FIG. **42**, an example of a manufacturing method of the one-turn inductor **372***h* included in the semiconductor module **10**K will be described. FIG. **41** is a plan view showing an example of the manufacturing method of the inductor **372***h* included in the semiconductor module **10**K, and a side view of the semiconductor module **10**K along a line F**1**-F**2**. FIG. **42** is a plan view showing the example of the manufacturing method of the inductor **372***h* shown in FIG. **41** and a cross-sectional view showing a cross section of the semiconductor module **10**K along a line F**3**-F**4**. The same or similar configurations as those in FIG. **1** to FIG. **40** will not be described here.

[0342] The manufacturing method of the semiconductor module **10**K shown in FIG. **41** and FIG. **42** includes that two sides of the one-turn inductor **372***h* are formed using a wiring **383** of the memory chip **110***n*, and that after the two sides of the inductor **372***h* are formed, one straight side of the inductor **372***h* is formed using a side wiring **361** of the rewiring layer **300**.

[0343] In addition, in the exemplary embodiments shown in FIG. **41** and FIG. **42**, it is assumed that inductors **372***h* included in the memory chip **110***n*+1 and the rewiring layer **300** and inductors **372***h* included in the memory chip **110***n* and the rewiring layer **300** are alternately arranged along the direction D**2**. Therefore, in the case where a certain region of the semiconductor module **10**K is enlarged, as shown in FIG. **41** and FIG. **42**, there are regions in which the memory chip **110***n* and the rewiring layer **300** include the inductor **372***h*, and there are regions in which the memory chip **110***n*+1 and the rewiring layer **300** do not include the inductor **372***h*. Further, although not shown, there are regions in which the memory chip **110***n*+1 and the rewiring layer **300** include the inductor **372***h*, and there are regions in which the memory chip **110***n* and the rewiring layer **300** do not include the inductor **372***h*. The memory chips **110** and the rewiring layer **300** other than the memory chips **110***n* and **110***n*+1 and the rewiring layer **300** included in the semiconductor module **10**K also have the same configuration as the memory chips **110***n* and **110***n*+1 and the rewiring layer **300**.

[0344] As shown in FIG. **41** and FIG. **42**, the memory cube **100** included in the semiconductor module **10**K includes memory chips **110***n* and **110***n*+1 and rewiring layers **300**. The memory chip **110***n* and the rewiring layer **300** include inductors **372**. The inductor **372***h* shown in FIG. **41** and FIG. **42** is, for instance, the one-turn inductor. In the method for forming the memory cube **100** including the one-turn inductor **372***h*, the memory cube **100** has two sides of the inductor **372***h* of the memory chip **110***n* formed by using the wiring **383**, and the wiring **383** forming the two sides of the inductor **372***h* is exposed to the first side surface **146**.

[0345] After the two sides of the one-turn inductor **372** are formed using the wiring **383** of the memory chip **110***n*, as shown in FIG. **42**, the side wiring **361** of the rewiring layer **300** is formed on the first side surface **146** of the memory chip **110***n*. That is, the side wiring **361** of the rewiring layer **300** is formed on the first side surface **146** so as to overlap the wiring **383** forming the two sides exposed to the first side surface **146**. As a result, the side wiring **361** is electrically connected to the wiring **383** forming the two sides. As shown in FIG. **42**, a wiring width of the side wiring **361** around the wiring **383** is increased so as to surround the cross section of the wiring **383**. As a result, the side wiring **361** is securely connected to the wiring **383**. In addition, the side wiring **361** around the wiring **383** may be referred to as an electrode pad, and may be individually formed as an electrode pad.

[0346] The inductor **372***h* according to the eleventh embodiment is formed using the wiring **383** and the side wiring **361** that differs from the wiring **383**. The side wiring **361** is included in the rewiring layer **300** and is formed on the first side surface **146** of the memory cube **100**. That is, a portion of the inductor **372***h* is included in the rewiring layer **300** by using the method for forming the inductor **372***h*. Consequently, a distance along the direction D**3** between the inductor **372***h* and the inductor **272** in the corresponding logic chip **200** on a one-to-one basis can be shorter than a distance along the direction D**3** between the inductor **372***h* included in the memory cube **100** and the inductor **272** in the logic chip **200**. Consequently, the inductor communication between the inductor **372***h* and the inductor **272** can be improved.

[11-2. Manufacturing Method of Three-Turn Inductor **372***i*]

[0347] Referring to FIG. **43** and FIG. **44**, an example of the manufacturing method of the three-turn inductor **372***i* included in the semiconductor module **10**K will be described. FIG. **43** is a plan view showing an example of a manufacturing method of the inductor **372***i* included in the semiconductor module **10**K, and a side view of the semiconductor module **10**K along a line F**5**-F**6**. FIG. **44** is a plan view showing an exemplary manufacturing method of the inductor **372***i* shown in FIG. **43** and a cross-sectional view showing a cross section of the semiconductor module **10**K along a line F**7**-F**8**. The same or similar configurations as those in FIG. **1** to FIG. **42** will be omitted here.

[0348] As shown in FIG. **43** and FIG. **44**, the semiconductor module **10**K may include the three-turn inductor **372***i*. The manufacturing method of the semiconductor module **10**K shown in FIG. **43** and FIG. **44** includes that two sides of each of the first turn to the third turn of the inductor **372***i* are formed using the wiring **383** of the memory chip **110***n*, that after the two sides of each of the first turn to the third turn of the inductor **372***i* are formed, a side wiring **361***a* corresponding to one side of the straight line of the first turn, a side wiring **361***b* corresponding to one side of the straight line of the second turn, and a side wiring **361***c* of the rewiring layers **300** corresponding to one side of the straight line of the third turn are formed.

[0349] In addition, in the examples shown in FIG. **43** and FIG. **44**, as in the examples shown in FIG. **42** and FIG. **43**, it is assumed that the inductors **372***i* included in the memory chip **110***n*+1 and the rewiring layer **300** and the inductors **372***i* included in the memory chip **110***n* and the rewiring layer **300** are alternately arranged along the direction D**2**. Therefore, in the examples shown in FIG. **43** and FIG. **44**, description of the same configuration and function as those of the examples shown in FIG. **42** and FIG. **43** will be omitted.

[0350] As shown in FIG. **43**, in the inductor **372***i*, if the innermost side is the first turn, two sides constituting the first-turn inductor, two sides constituting the second-turn inductor, and two sides constituting the third-turn inductor are formed by the wiring **383**. Therefore, the cross sections of the six wirings **383** forming the two sides constituting the first-turn inductor, the two sides constituting the second-turn inductor, and the two sides constituting the third-turn inductor are exposed to the first side surface **146**.

[0351] Similar to forming the one-turn inductor **372***h*, after the two sides of each of the first to third turns of the three-turn inductor **372***i* are formed using the wiring **383** of the memory chip **110***n*, as shown in FIG. **44**, the side wiring **361***a* corresponding to one side of the straight line of the first turn of the inductor **372***i*, the side wiring **361***b* corresponding to one side of the straight line of the second turn of the inductor **372***i*, and the side wiring **361***c* of the rewiring layer **300** corresponding to one side of the straight line of the third turn of the inductor **372***i* are formed on the first side surface **146** of the memory cube **100**.

[0352] For example, the side wiring **361***a* is formed on the first side surface **146** of the memory chip **110***n* so as to overlap the wiring **383** forming the second side of the first turn exposed on the first side surface **146**, and the side wiring **361***a* is electrically connected to the wiring **383** forming the second side of the first turn. For example, the side wiring **361***b* is formed on the first side surface **146** of the memory chip **110***n* and the first side surface **146** of the memory chip **110***n*+1 so as to overlap the wiring **383** forming the second side of the second turn exposed on the first side surface **146**, and the side wiring **361***b* is electrically connected to the wiring **383** forming the second side of the second turn. For example, the side wiring **361***c* is formed on the first side surface **146** of the memory chip **110***n* in the same manner as the side wiring **361***a*, and the side wiring **361***c* is electrically connected to the wiring **383** forming the second side of the third turn.

[0353] Similar to the one-turn inductor **372***h*, wiring widths of the side wirings **361***a* to **361***c* around the cross section of the wiring **383** of the three-turn inductor **372***i* are increased so as to surround the cross section of the wiring **383**.

[0354] The inductor **372***i* according to the eleventh embodiment is formed using the wiring **383** and the side wirings **361***a* to **361***c* that differ from the wiring **383**. The side wirings **361***a* to **361***c* are included in the rewiring layer **300** and are formed on the first side surface **146** of the memory cube **100**. As a consequence, similar to the inductor **372***h*, the inductor communication between the inductor **372***i* and the inductor **272** can be improved.

[11-3. Manufacturing Method of Three-turn Inductor **372***j*]

[0355] FIG. **45** and FIG. **46** show an exemplary method for manufacturing the three-turn inductor **372** included in the semiconductor module **10**K. FIG. **45** is a plan view showing an exemplary manufacturing method of the inductor **372** included in the semiconductor module **10**K, and a side view of the semiconductor module **10**K along a line F**9**-F**10**. FIG. **46** is a plan view showing an exemplary manufacturing method of the inductor **372** shown in FIG. **45**, and a cross-sectional view showing a cross section of the semiconductor module **10**K along a line F**11**-F**12**. Descriptions of the same or similar configurations as those in FIG. **1** to FIG. **44** will be omitted.

[0356] As shown in FIG. **45** and FIG. **46**, the semiconductor module **10**K may include an inductor **372***j* having a plurality of turns formed over a plurality of memory chips **110** (**110***n*-**2**, **110***n*-**1**, **110***n*, and **110***n*+1). Specifically, the semiconductor module **10**K includes the three-turn inductor **372***j* including three-turn coils formed in the rewiring layers **300** parallel to the first side surface **146** and a surface formed in the direction D**1** and the direction D**2**.

[0357] In addition, although not shown in the drawings, a plurality of inductors **372***j* is arranged along the direction D**2** in the same manner as the respective embodiments. Further, in the example shown in FIG. **45** and FIG. **46**, although two wirings **383** corresponding to two terminals of the inductor **372***j* are formed in the memory chip **110***n*, the two wirings **383** corresponding to the two terminals of the inductor **372***j* may be formed in any one of the memory chip **110***n*+1, the memory chip **110***n*-**1**, and the memory chip **110***n*-**2**. In addition, among the two wirings **383** corresponding to the two terminals of the inductor **372***j*, the wiring **383** corresponding to one terminal may be formed in any one of the memory chip **110***n*+1, the memory chip **110***n*-**1**, and the memory chip **110***n*-**2**, and the wiring **383** corresponding to the remaining terminals may be formed in a memory chip that differs from the memory chip in which the wiring **383** corresponding to the one terminal is formed.

[0358] The manufacturing method of the semiconductor module **10**K shown in FIG. **45** and FIG. **46** includes that a wiring connected to two terminals of a three-turn inductor **372***j* is formed using two wirings **383** of the memory chip **110***n*, and that after the two wirings **383** corresponding to the two terminals of the inductor **372***j* are formed, a side wiring **361***d* of the rewiring layer **300** corresponding to a three-turn coil is formed.

[0359] As shown in FIG. **45**, a wiring connected to the two terminals of a three-turn inductor **372***j* is formed by two wirings **383** of the memory chip **110***n*. Thus, a cross section of each of the two wirings **383** is exposed to the first side surface **146**.

[0360] After the two wirings **383** of the inductor **372***j* are formed, as shown in FIG. **46**, the side wiring **361***d* of the rewiring layer **300** corresponding to the coils of the three-turn inductor **372***j* are formed on the first side surface **146** of the memory cube **100**. That is, the side wiring **361***d* of the rewiring layer **300** is formed on the first side surface **146** of the memory cube **100** over the plurality of memory chips **110***n*-**2**, **110***n*-**1**, **110***n*, and **110***n*+1.

[0361] For example, the side wiring **361***d* is formed on the first side surface **146** of the memory cube **100** so as to overlap each of the two wirings **383** exposed on the first side surface **146** corresponding to the side surface of the memory chip **110***n*, and the side wiring **361***d* is electrically connected to each of the cross sections of the two wirings **383**.

[0362] Further, for example, the side wiring **361***d* of the three-turn inductor **372***j* shown in FIG. **46** includes wirings of first to third turns from the inside toward the outside. The first-turn wiring is formed over the memory chip **110***n* and the memory chip **110***n*-**1**, the second-turn wiring is formed over the memory chip **110***n*+1, the memory chip **110***n*, the memory chip **110***n*-**1**, and the memory chip **110***n*-**2**, and the third-turn wiring is formed over the memory chip **110***n*+1, the memory chip **110***n*, the memory chip **110***n*-**1**, and the memory chip **110***n*-**2**.

[0363] Similar to the inductors **372***h* and **372***i*, the side wiring **361***d* around the cross section of the wiring **383** of the three-turn inductor **372***j* is wider so as to surround the cross section of the wiring **383**.

[0364] The inductor **372***j* according to the eleventh embodiment is formed using the wiring **383** and the side wiring **361***d* that differs from the wiring **383**. The side wiring **361***d* is included in the rewiring layer **300** and is formed on the first side surface **146** of the memory cube **100**. Consequently, as with the inductors **372***h* and **372***i*, the inductor communication between the inductor **372***j* and the inductor **272** can be improved.

12. Twelfth Embodiment

[0365] A method for manufacturing the memory cube **100** according to a twelfth embodiment will be described with reference to FIG. **47** to FIG. **53**. FIG. **47** is a flowchart showing the method for manufacturing the memory cube **100**. FIG. **48** is a plan view showing the method for manufacturing the memory cube **100**. FIG. **49** is a plan view showing step S**110** (STEP **110**) of the manufacturing method of the memory cube **100**, and a cross-sectional view showing a cross section of a memory substrate **800***a* along a line G**1**-G**2**. FIG. **49** is an enlarged view of a memory chip region **804**. FIG. **50** is a plan view showing step S**120** (STEP **120**) of the manufacturing method of the memory cube **100**, and a cross-sectional view showing a cross section of the memory substrate **800***a* along a line G**1**-G**2**. FIG. **51** is a cross-sectional view showing step S**130** (STEP **130**) of the manufacturing method of the memory cube **100** and showing a cross section of the memory substrate **800***a* along a line G**1**-G**2**. FIG. **51** is a cross-sectional view showing step S**140** (STEP **140**) in the manufacturing method of the memory cube **100**. FIG. **52** is a cross-sectional view showing step S**170** (STEP **170**) and step S**180** (STEP **180**) of the manufacturing method of the memory cube **100**. FIG. **53** is a cross-sectional view showing step S**190** (STEP **190**) of the manufacturing method of the memory cube **100**. Descriptions of the same or similar configurations as those in FIG. **1** to FIG. **46** will be omitted.

[0366] As shown in FIG. **47**, the method for manufacturing the memory cube **100** includes step S**110** (STEP **110**) to step S**190** (STEP **190**).

[0367] When the manufacturing method of the memory cube **100** is started, the memory substrate **800***a* is prepared.

For example, as shown in FIG. **48**, a second surface **101**b of the memory substrate **800**a overlaps a peripheral region **802** and the memory chip region **804**. The memory chip region **804** overlaps the plurality of memory chips **110** and a groove **806**, and includes the plurality of memory chips **110** and the groove **806** (see FIG. **49**). The peripheral region **802** is a region around the memory chip region **804**. For example, the plurality of memory chips **110** includes the same configuration as that of the first embodiment. Therefore, a configuration similar to that of the first embodiment in the memory chip **110** will be described as necessary. STEP **110** is the step of using laser grooving to form the groove **806** that defines each of the plurality of memory chips **110**. The groove **806** of the memory substrate **800**a shown in FIG. **49** is formed by a laser with reference to alignment markers (not shown). The groove **806** is formed in the insulating layer **185** included in the wiring layer **150** and the insulating layer **184** included in the transistor layer **130**. In addition, for example, the wiring layer **150** includes a wiring **183**p, the transistor layer **130** includes a wiring **178**a, and the end part (side surface) of the wiring **183**p and the end part (side surface) of the wiring **178**a are exposed in the groove **806**. In addition, a portion of the groove **806** may be formed in the substrate **173** in accordance with the output of the laser or the like.

[0368] Step S**120** (STEP **120**) is a step of forming an insulating layer **807** and an insulating layer **808** on a substrate (the insulating layer **185**). The insulating layer **807** and the insulating layer **808** are formed in the groove **806** by STEP **120**, and the groove **806** is filled with the insulating layer **807** and the insulating layer **808**. That is, as shown in FIG. **50**, the insulating layer **807** is formed on upper and side surfaces of the insulating layer **185**, a side surface of the wiring **183**p, a side surface of the insulating layer **184**, a side surface of the wiring **178**a, and a portion of the surface of the substrate **173** exposed by the groove **806**, and the insulating layer **808** is formed on the insulating layer **807**. For example, a CVD device can be used to form the insulating layer **807** and the insulating layer **808**. For example, the insulating layer **807** may include SIN, SiON or the like as an insulating film, and the insulating layer **808** may include SiO_2 or the like as an insulating film. In addition, the insulating layer **185** and the insulating layer **184** include a low dielectric constant film having a dielectric constant lower than that of the insulating film **190** (see FIG. **5**A).

[0369] Step S**130** (STEP **130**) is a step of thinning the surface (insulating layer **808**) of the memory substrate **800**a. STEP **130** planarizes the surface of the memory substrate **800**a. For example, as shown in FIG. **51**, the surface of the memory substrate **800**a (the memory substrate **800**a) is polished. Consequently, as shown in FIG. **51**, a thickness THI**5** of the memory substrate **800**a is thinned to a thickness THI**6**. For example, a surface (insulating layer **808**) of a memory substrate **800**b is polished in the same manner as the memory substrate **800**a, and the thickness THI**5** of the memory substrate **800**b is thinned to a thickness THI**7**. For example, the thickness THI**6** is smaller than the thickness THI**5**, and the thickness THI**7** is smaller than the thickness THI**6**. For example, a CMP device can be used for polishing. In the embodiment shown in FIG. **51**, although the surface of the memory substrate **800**a and the surface of the memory substrate **800**b after thinning include the insulating layer **808**, the surface of the memory substrate **800**a and the surface of the memory substrate **800**b after thinning may not include the insulating layer **808** and the insulating layer **807** may be exposed as needed.

[0370] Step S**140** (STEP **140**) is a step of stacking the memory substrate. For example, the surface of the memory substrate **800**a where the insulating layer **808** is exposed and the surface of the memory substrate **800**b where the insulating layer **808** is exposed are F2F bonded by STEP **140** shown in FIG. **51**. STEP **140** may be replaced with STEP **130**.

[0371] Step S**150** (STEP **150**) is a step of determining whether or not a predetermined number (required number) of memory substrates have been stacked. If the required number of substrates are not stacked (NO in STEP **150**), the process returns to STEP **130**, and STEP **130** and STEP **140** are executed again. If the required number of substrates are stacked (YES in STEP **150**), step S**160** (STEP **160**) is executed.

[0372] STEP **160** is a step in which a plurality of substrates stacked in STEP **150** are attached to a DAF **810**. By performing STEP **160**, the required number of stacked substrates can be fixed to the DAF **810**. That is, the required number of stacked substrates are supported by the DAF **810**. In addition, an example of a manufacturing method of the memory cube **100** according to the twelfth embodiment is an example in which two memory substrates **800**a and **800**b are stacked. The DAF **810** may be an adhesive film, an adhesive layer, or the like as long as it can bond the substrate and fix the substrate to the film.

[0373] Step S**170** (STEP **170**) is a step of singulating the required number of stacked substrates to which the DAF **810** is attached in STEP **160**. A stacked chip **811**a is formed by performing STEP **170** shown in FIG. **52**. In particular, the required number of stacked substrates to which the DAF **810** is attached is singulated into individual stacked chips **811**a using plasma dicing (Plasma Dicing) with reference to alignment markers (not shown).

[0374] Step S**180** (STEP **180**) is a step of measuring a thickness of the singulated stacked chip **811**a in STEP **170**. For example, as shown in FIG. **52**, a thickness STH**1** of the stacked chip **811**a is measured.

[0375] Step S**190** (STEP **190**) is a step of stacking a plurality of stacked chips whose thickness is measured in STEP **180** to form the memory cube **100**. For example, the stacked chip **811**a is stacked with a stacked chip **811**b by performing STEP **190** shown in FIG. **53**, and the memory cube **100** is formed. For example, the thickness of the stacked chip **811**a is the thickness STH**1**, and a thickness of the stacked chip **811**b is a thickness STH**2**. In addition, the stacked chip **811**a includes the memory chips **110**n and **110**n+1, and the stacked chip **811**b includes the memory chips **110**n+2 and 110n+3. For example, the thickness of the memory chip **110**n is a thickness THI**6**, the thickness of the memory chip **110**n+1 is a thickness THI**7**, the thickness of the memory chip **110**n+2 is a thickness THI**8**, and the thickness of the memory chip **110**n+3 is a thickness THI**9**. The example of a manufacturing method of the memory cube **100** according to the twelfth embodiment is an example in which two stacked chips **811**a and **811**b are stacked.

[0376] As described above, the memory cube **100** is formed.

[0377] The manufacturing method of the memory cube **100** according to the twelfth embodiment includes forming the groove **806** defining each of the plurality of memory chips **110** on the memory substrate **800**a, forming insulating layers **807** and **808** in the groove **806**, and plasma dicing each of the plurality of memory chips **110** along the groove **806**, prior to stacking the plurality of memory chips **110**. Since the groove **806** is filled by the insulating layers **807** and **808**, the first side surface **106**, the second side surface **105**, the third side surface **108**, and the fourth side surface **107** of each memory chip **110** are covered by the insulating layers **807** and **808**. Therefore, a low dielectric constant film having high hygroscopicity is not exposed from the first side surface **106**, the second side surface **105**, the third side surface **108**, and the fourth side surface **107** of each memory chip **110**. As a result, the memory cube **100** can suppress corrosion, deterioration, and the like of each device due to moisture absorption and intrusion of impurities and the like. Therefore, the reliability of the semiconductor module can be maintained without impairing the long-term reliability of the semiconductor module.

[0378] Further, the manufacturing method of the memory cube **100** according to the twelfth embodiment includes forming the groove **806** defining each of the plurality of memory chips **110** on the memory substrate **800**a prior to stacking the plurality of memory chips **110**. Therefore, in the method for manufacturing the memory cube **100** according to the twelfth embodiment, it is possible to delete (remove) the wiring layers (metal layers) between the plurality of memory chips **110** before the singulation of the memory chip **110** by the plasma dicing. As a result, in the plasma dicing of the method for manufacturing the memory cube **100** according to the twelfth embodiment, since it is not necessary to dice the metal layer, the memory chip **110** can be easily singulated.

[0379] In addition, the memory cube **100** may not include the seal ring **160** because corrosion, deterioration, and the like of each device due to moisture absorption and intrusion of impurities and the like can be suppressed. As a result, the method for manufacturing the memory cube **100** according to the twelfth embodiment can simplify the design and manufacturing of the seal ring **160**.

[0380] In addition, the method for manufacturing the memory cube **100** according to the twelfth embodiment includes stacking a plurality of memory chips having different thicknesses. For example, as shown in FIG. **53**, as the mean thickness of the memory chip is a thickness THI_{ave}, the manufacturing method of the memory cube **100** according to the twelfth embodiment includes stacking the memory chip **110** thicker than the thickness THI_{ave} and the memory chip **110** thinner than the thickness THI_{ave}. Consequently, in the manufacturing method of the memory cube **100** according to the twelfth embodiment, the mean thickness of the memory chip can be set to a thickness $\mathrm{THI}_{ave}\pm5$ μm. Specifically, the mean thickness THI_{ave} of the memory chip included in the memory cube **100** shown in FIG. **53** can be expressed by the following Formula (1) using a thickness of the DAF **810** as a thickness DT. That is, in the method for manufacturing the memory cube **100** according to the twelfth embodiment, it is possible to reduce variations in the thicknesses of the plurality of memory cubes **100**.

$$\frac{(TH16+TH17+TH18+TH19-2\times DT}{4}\leq THI_{ave}\pm5\ \mu\mathrm{m} \tag{1}$$

13. Thirteenth Embodiment

[0381] A method of manufacturing the memory cube **100** according to a thirteenth embodiment will be described with reference to FIG. **54** to FIG. **56**. FIG. **54** is a flowchart showing a method of manufacturing the memory cube **100**. FIG. **55** is a cross-sectional view showing step S**210** (STEP **210**) of the manufacturing method of the memory cube **100**. FIG. **56** is a cross-sectional view showing step S**220** (STEP **220**) in the method of manufacturing the memory cube **100**, and a cross-sectional view showing step S**230** (STEP **230**) in the method of manufacturing the memory cube **100**. The same or similar configurations as those in FIG. **1** to FIG. **53** will not be described here.

[0382] As shown in FIG. **54**, the method for manufacturing the memory cube **100** includes steps S**210** (STEP **210**) to S**230** (STEP **230**). For example, the plurality of memory chips **110** includes the same configuration as that of the first embodiment. Therefore, a configuration similar to that of the first embodiment in the memory chip **110** will be described as necessary.

[0383] For example, when the manufacturing method of the memory cube **100** is started, the memory chips **110**n and **110**n+1 are prepared. For example, as shown in FIG. **55**, the memory chip **110**n includes the concave first surface **102** and the convex second surface **104** with the chip warped. For example, a thickness of the memory chip **110**n is a thickness THI10+a, and a distance (thickness) between the concave first surface **102** and the convex second surface **104** is a thickness THI**10**. Similar to the memory chip **110**n, the memory chip **110**n+1 includes the concave first surface **102** and the convex second surface **104**, and the chip is warped. Further, for example, a thickness of the memory chip **110**n+1 is a thickness THI10+b, and a distance (thickness) between the concave first surface **102** and the convex second surface **104** is a thickness THI**10**.

[0384] STEP **210** is a step of stacking and temporarily fixing a plurality of memory chips. As shown in FIG. **55**, STEP **210** includes that the convex second surface **104** of the memory chip **110**n is temporarily bonded (F2F bonded) with the convex second surface **104** of the memory chip **110**n+1, that the convex second surface **104** of the memory chip **110**n+2 is temporarily bonded (F2F bonded) to the convex second surface **104** of the memory chip **110**n+3, and that the temporarily bonded concave first surfaces **102** of the memory chips **110**n and **110**n+1 are temporarily bonded to the concave first surfaces **102** of the memory chips **110**n+2 and 110n+3. In addition, in STEP **210** surfaces may be temporarily bonded by mechanically adjusting the positions of the memory chips using alignment markers (not shown) provided in the memory chips **110**n to **110**n+3. Further, the position of each memory chip may be adjusted with reference to a chip outer shape by using a surface tension (not shown) of a liquid such as water applied to the chip surface in advance, and then temporarily bonded. In addition, an example of how to manufacture the memory cube **100** according to the thirteenth embodiment is an example in which four memory chips **110**n to **110**n+3 are stacked and temporarily fixed.

[0385] As shown in FIG. **56**, STEP **220** is a step of sealing the temporarily bonded memory chips **110**n to **110**n+3 with resin. For example, the resin is a molding material **812**. For example, STEP **220** is a step of the present bonding. In addition, a plurality of memory chips that are temporarily bonded can be collectively bonded (collectively molded) by

STEP **220**. Since each of the memory chips **110***n* to **110***n*+3 has a warp, a space is included between adjacent memory chips in the temporarily bonded memory chips **110***n* to **110***n*+3. Therefore, the resin can be injected into the respective spaces by STEP **220**, and the temporarily bonded memory chips **110***n* to **110***n*+3 can be sealed with the resin.

[0386] STEP **230** is a step of manufacturing the memory cube **100** by polishing and shaping the memory chips **110***n* to **110***n*+3 sealed by STEP **220**. A method for manufacturing the memory cube **100** includes polishing the memory chips **110***n* to **110***n*+3, planarizing the first side **142** and the second side **144** corresponding to an outermost surface along the direction D**1** of the memory cube **100**, planarizing the second side **145** and the fourth side **147** corresponding to an outermost surface along the direction D**2** of the memory cube **100**, and planarizing the first side **146** and the third side **148** corresponding to an outermost surface along the direction D**3** of the memory cube **100**, thereby forming the memory cube **100**. In the manufacturing method of the memory cube **100** according to the thirteenth embodiment, surplus resin of the memory chips **110***n* to **110***n*+3 sealed by STEP **220** can be removed.

[0387] In addition, although not shown, the method includes forming a flat second side surface **145** (see FIG. **1**) and a flat fourth side surface **147** (see FIG. **1**). For example, the CMP device can be used for polishing.

[0388] As described above, the memory cube **100** is formed.

[0389] Since the memory chip has inherent warpage, in the method for manufacturing the memory cube to which the method for manufacturing the memory cube **100** according to the thirteenth embodiment is not applied, when the number of memory chips to be stacked increases, the stress of each memory chip is integrated. As a result, in the method for manufacturing the memory cube to which the method for manufacturing the memory cube **100** according to the thirteenth embodiment is not applied, there is a possibility that the memory cube may be damaged. Also, if each memory chip is bonded using an adhesive, the long-term reliability of the memory cube may be affected depending on the thickness of the adhesive.

[0390] Meanwhile, the memory cube **100** according to the thirteenth embodiment includes STEP **210** to STEP **230**. In the method for manufacturing the memory cube **100** according to the thirteenth embodiment, the temporary bonding and the main bonding can be performed while maintaining the inherent warpage of the memory chip. As a result, in the method for manufacturing the memory cube **100** according to the thirteenth embodiment, even if the number of memory chips to be stacked is increased, it is possible to suppress integration of the stress of each memory chip and suppress the defect caused by the integration of the stress of each memory chip. Further, for example, the manufacturing method of the memory cube **100** according to the thirteenth embodiment includes stacking a plurality of memory chips **110** by F2F bonding, and includes resin sealing between the stacked memory chips **110**. Therefore, the method for manufacturing the memory cube **100** according to the thirteenth embodiment includes sealing each memory chip by filling a gap caused by warpage with resin, so that the thickness of the resin can be minimized.

14. Fourteenth Embodiment

[0391] A method for manufacturing the memory cube **100** according to a fourteenth embodiment will be described with reference to FIG. **57** to FIG. **60**. FIG. **57** is a flowchart showing a method for manufacturing the memory cube **100**. FIG. **58** is a perspective view showing step S**310** (STEP **310**) of the manufacturing method of the memory cube **100**. FIG. **59** is a perspective view showing step S**320** (STEP **320**) of a manufacturing method of the memory cube **100** and a cross-sectional view showing a cross section taken along a line H**1**-H**2**. FIG. **60** is a cross-sectional view showing step S**330** (STEP **330**) to step S**350** (STEP **350**) in the manufacturing method of the memory cube **100**. Descriptions of the same or similar configurations as those in FIG. **1** to FIG. **56** will be omitted.

[0392] As shown in FIG. **57**, the method for manufacturing the memory cube **100** include step S**310** (STEP **310**) to S**350** (STEP **350**). For example, the plurality of memory chips **110** includes the same configuration as that of the first embodiment. Therefore, a configuration similar to that of the first embodiment in the memory chip **110** will be described as necessary.

[0393] For example, when the manufacturing method of the memory cube **100** is started, the memory chips **110***n* to **110***n*+3 are prepared. An example of the manufacturing method of the memory cube **100** according to the fourteenth embodiment is an example in which four memory chips **110***n* to **110***n*+3 are stacked. In order to explain the manufacturing method of the memory cube **100** according to the fourteenth embodiment in an easy-to-understand manner, the first surface **102** and the second surface **104** of each of the memory chips **110***n* to **110***n*+3 shown in FIG. **58** and FIG. **59** are flat surfaces. In practice, each memory chip is warped, as shown in a cross-section along a line H**1**-H**2** shown in FIG. **59** and in each cross-section in FIG. **60**.

[0394] STEP **310** is a step of applying an adhesive **814** to the memory chip **110***n*. For example, as shown in FIG. **58**, the adhesive **814** is applied over the first side **102** of the memory chip **110***n*. In addition, as shown in FIG. **59**, the adhesive **814** is also applied on the second surface **104** of the memory chip **110***n*+1 and on the first surface **102** of the memory chip **110***n*+2. An exemplary method for manufacturing the memory cube **100** according to the fourteenth embodiment includes applying the adhesive **814** to five locations on respective corresponding surfaces of the memory chips **110***n* to **110***n*+2. In addition, positions of the adhesive **814** on the corresponding surfaces of the memory chips **110***n* to **110***n*+2 are not limited to those shown in FIG. **58**.

[0395] STEP **320** is a step of stacking a plurality of memory chips. As shown in FIG. **59**, the memory chip **110***n*+1 is bonded to the memory chip **110***n* such that the first surface **102** of the memory chip **110***n*+1 is bonded to the adhesive **814** on the first surface **102** of the memory chip **110***n*. That is, the memory chip **110***n*+1 is stacked on the memory chip **110***n* such that the first surface **102** of the memory chip **110***n*+1 faces the first surface **102** of the memory chip **110***n*. In addition, the memory chip **110***n*+2 is bonded to the memory chip **110***n*+1 so that the second surface **104** of the memory chip **110***n*+2 is bonded to the adhesive **814** on the second surface **104** of the memory chip **110***n*+1. That is, the memory chip **110***n*+2 is stacked on the memory chip **110***n*+1 such that the second surface **104** of the memory chip **110***n*+2 faces the second surface **104** of the memory chip **110**_n_+1. Similarly, the memory chip **110**_n_+3 is bonded to the memory chip **110**_n_+2 such that the first surface **102** of the memory chip **110**_n_+3 is bonded to the adhesive **814** on the first surface **102** of the memory chip **110**_n_+2. That is, the memory chip **110**_n_+3 is stacked on the memory chip **110**_n_+2 such that the first surface **102** of the memory chip **110**_n_+3 faces the first surface **102** of the memory chip **110**_n_+2.

[0396] STEP **330** is a step of temporarily bonding the plurality of stacked memory chips **110**_n_ to **110**_n_+3. Specifically, as shown in STEP **330** of FIG. **60**, STEP **330** includes sealing an end part **148**_a_ that ultimately corresponds to the third side surface **148** of the memory cube **100** with a UF agent **816**.

[0397] An exemplary manufacturing method of the memory cube **100** according to the fourteenth embodiment can be temporarily bonded by STEP **330** while maintaining the inherent warpage of the memory chip. STEP **330** may be performed in a vacuum environment.

[0398] As shown in FIG. **59**, STEP **340** is a step of sealing the memory chips **110**_n_ to **110**_n_+3 which are temporarily bonded with resin. For example, an example of a method for manufacturing the memory cube **100** according to the fourteenth embodiment includes injecting resin into a gap between the memory chips from the first side surface **146**, the second side surface **145** (see FIG. **1**), and the fourth side surface **147** (see FIG. **1**). For example, the resin is a UF agent **817**.

[0399] STEP **350** is a step of manufacturing the memory cube **100** by polishing and shaping the memory chips **110**_n_ to **110**_n_+3 sealed by STEP **340**. The manufacturing method of the memory cube **100** includes, as in the thirteenth embodiment, polishing the memory chips **110**_n_ to **110**_n_+3, planarizing the first surface **142** and the second surface **144** corresponding to an outermost surface along the direction D**1** of the memory cube **100**, planarizing the second side surface **145** and the fourth side surface **147** corresponding to an outermost surface along the direction D**2** of the memory cube **100**, and planarizing the first side surface **146** and the third side surface **148** corresponding to an outermost surface along the direction D**3** of the memory cube **100**, thereby forming the memory cube **100**. In the manufacturing method of the memory cube **100** according to the fourteenth embodiment, the memory cube **100** can be formed by removing surplus resin of the sealed memory chips **110**_n_ to **110**_n_+3 by STEP **350**.

[0400] In addition, although not shown, the method includes forming a flat second side surface **145** (see FIG. **1**) and a flat fourth side surface **147** (see FIG. **1**). For example, the CMP device can be used for polishing.

[0401] As described above, the memory cube **100** is formed.

[0402] Since the memory chip has inherent warpage, in the method for manufacturing the memory cube to which the method for manufacturing the memory cube **100** according to the fourteenth embodiment is not applied, when the number of memory chips to be stacked increases, the stress of each memory chip is integrated. As a result, in the method for manufacturing a memory cube to which the method for manufacturing the memory cube **100** according to the fourteenth embodiment is not applied, there is a possibility that the memory cube may be damaged. Also, if each memory chip is bonded using an adhesive, the long-term reliability of the memory cube may be affected depending on a thickness of the adhesive.

[0403] Meanwhile, the manufacturing method of the memory cube **100** according to the fourteenth embodiment includes STEP **310** to STEP **350**. Manufacturing methods of the memory cube **100** according to the fourteenth embodiment include applying the adhesive **814** to a plurality of locations on a surface such as the memory chip **110**_n_, bonding and laminating a plurality of memory chips such as the memory chip **110**_n_ using the adhesive **814**, applying and temporarily bonding the UF agent **816** to the end part **148**_a_ of a plurality of memory chips such as the memory chip **110**_n_, and injecting and sealing the UF agent **817** from each side surface of the memory cube **100** after the temporary bonding. As a result, in the method for manufacturing the memory cube **100** according to the fourteenth embodiment, it is possible to temporarily bond and permanently bond the memory chips while maintaining the inherent warpage of the memory chips. As a result, in the method for manufacturing the memory cube **100** according to the fourteenth embodiment, even if the number of stacked memory chips is increased, it is possible to suppress the integration of the stress of each memory chip and suppress the defect caused by the integration of the stress of each memory chip. In addition, since the manufacturing method of the memory cube **100** according to the fourteenth embodiment includes sealing the memory chips by filling the gaps caused by the warpage with the UF agents **816** and **817**, a thickness of the resin can be minimized.

15. Fifteenth Embodiment

[0404] A method for manufacturing the memory cube **100** according to the fifteenth embodiment will be described with reference to FIG. **61** to FIG. **64**. FIG. **61** is a flowchart showing the method for manufacturing the memory cube **100**. FIG. **62** is a cross-sectional view showing step S**410** (STEP **410**) of the manufacturing method of the memory cube **100** and a plan view of the memory chip **110**. FIG. **63** is a plan view showing a plurality of markers used in the method of manufacturing the memory cube **100**. FIG. **64** is a cross-sectional view showing an example of step S**420** (STEP **420**) and an example of step S**430** (STEP **430**) in the manufacturing method of the memory cube **100**. The same or similar configurations as those in FIG. **1** to FIG. **60** will not be described here.

[0405] As shown in FIG. **61**, the method for manufacturing the memory cube **100** include step S**410** (STEP **410**) to step S**430** (STEP **430**). For example, each of the plurality of memory chips **110**_n_ to **110**_n_+7 (see FIG. **62**) includes the same configuration as the memory chip **110** described in the first embodiment. Further, stacking of the plurality of memory chips **110**_n_ to **110**_n_+7 includes the same configuration as the configuration of the plurality of memory chips **110**_n_ to **110**_n_+5 stacked by F2F bonding described in the first embodiment. Therefore, a configuration similar to that of the first embodiment in the memory chip **110** and the like will be described as necessary.

[0406] Since the memory chips **110**_n_ to **110**_n_+7 include the same configuration, the configuration of the memory chip **110**_n_ will be described here, and the configurations of the memory chips **110**_n_+1 to 110n+7 will be described as needed. As shown in FIG. **62**, the memory chip **110**_n_ includes a marker **818**, a plurality of markers **820**, and a plurality of marker groups **844** and **846**. Although not shown, for example, the marker **818**, the plurality of markers **820**, and the plurality of marker groups **844** and **846** are formed in the same layers as the power wiring **164**, the ground wiring **165**, the signal transmission wiring **166** (see FIG. **3**, FIG. **7**A, FIG. **7**B, and the like), and the wiring **183** (see FIG. **11**B and the like).

[0407] First, with reference to FIG. **62** and FIG. **63**, the marker **818**, the plurality of markers **820**, and the plurality of marker groups **844** and **846** will be described in detail.

[0408] For example, the marker **818** and the marker **820** are markers for adjusting a position of a laser irradiation device **822** and a stacked chip **849**. For example, the marker groups **844** and **846** are markers for adjusting positions between the memory chips and polishing amounts of the memory chips and detecting the positions between the memory chips when the stacked chip **849** is planarized using the laser irradiation device **822**. Further, for example, the marker groups **844** and **846** are markers for adjusting positions between the memory chips and polishing amounts of the memory chips and detecting the positions between the memory chips when the stacked chip **849** is planarized using the CMP device.

[0409] As shown in FIG. **62**, for example, the marker **818** is arranged at a corner portion where the second side surface **105** and the third side surface **108** intersect, and is formed to be parallel to the direction D**2** and the direction D**3**. The marker **820** is arranged at a corner portion at which the third side surface **108** and the fourth side surface **107** intersect, a corner portion at which the fourth side surface **107** and the first side surface **106** intersect, and a corner portion at which the first side surface **106** and the second side surface **105** intersect, and is formed parallel to the direction D**2** and the direction D**3**. Three marker groups **844** are arranged in parallel along the third side surface **108**. Three marker groups **846** are arranged in parallel along each of the first side surface **106**, the second side surface **105**, and the fourth side surface **107**.

[0410] The marker **818**, the plurality of markers **820**, and the plurality of marker groups **844** and **846** are configured as shown in FIG. **63**. The markers **818** and **820** may be referred to as alignment markers, and the marker groups **844** and **846** may be referred to as detection markers.

[0411] For example, the marker **818** has a bordered cross-shaped component shape, and the marker **820** has a cross shape. In addition, the shapes of the markers **818** and **820** shown in FIG. **63** are merely examples, and the shapes of the markers **818** and **820** may be any shapes as long as the positions of the laser irradiation device **822** and the stacked chip **849** can be adjusted. For example, the marker group **844** includes a marker **848**, a plurality of markers **850**, and a plurality of markers **852**. For example, the marker group **844** includes seven markers **850** and two markers **852**. The marker **848** is a rectangular marker having a height H**1** and a width W**1**, marker **850** is a rectangular marker having a height H**2** and a width W**2**, and marker **852** is a rectangular marker having a height H**3** and a width W**1**. A distance between the marker **848** and the marker **850**, a distance between the markers **850**, a distance between the marker **850** and the marker **852**, and a distance between the markers **852** are distances S. A distance between a bottom of the marker **848** and a bottom of the marker **850** is a distance L**1**, and a distance between a bottom of the marker **850** and a bottom of the adjacent marker **850** is a distance L**2**. In addition, for example, a width of the marker group **844** is a width WG**1**, and a height of the marker group **844** is a height HG**1**. In addition, a configuration of the marker group **844** shown in FIG. **63** is an example, and is not limited to the example shown here. For example, the number and shape of each marker can be changed according to the configuration, specification, application, and the like of the memory cube **100**.

[0412] For example, marker group **846** has fewer markers **852** than marker group **844**. Further, a width of the marker group **846** is a width WG**2**, and a height of the marker group **846** is the height HG**1**. Other configurations of the marker group **846** are the same as those of the marker group **844**, and description thereof will be omitted.

[0413] In the marker groups **844** and **846** according to the fifteenth embodiment, for example, the height H**1** is 10 μm, the width W**1** and the width W**2** are 5 μm, the height H**2** is 2 μm, the height H**3** is 11 μm, the width WG**1** is 120 μm, the height HG**1** is 30 μm, and the width WG**2** is 110 μm.

[0414] Next, a method for manufacturing the memory cube **100** will be described in detail with reference to FIG. **61** to FIG. **64**. In addition, although an example of the method for manufacturing the memory cube **100** with reference to FIG. **61** to FIG. **64** includes polishing the stacked chip **849** and manufacturing the memory cube **100** using the laser irradiation device **822**, the method for manufacturing the memory cube **100** is not limited to the example with reference to FIG. **61** to FIG. **64**. For example, the method for manufacturing the memory cube **100** may include polishing the stacked chip **849** using the CMP device (not shown) to manufacture the memory cube **100**, and may include polishing the stacked chip **849** using the laser irradiator device **822** and the CMP device to manufacture the memory cube **100**.

[0415] For example, when the manufacturing method of the memory cube **100** is started, step S**410** (STEP**410**) is executed. STEP **410** is a step of stacking a plurality of memory chips. As shown in FIG. **62**, an example of the manufacturing method of the memory cube **100** according to the fifteenth embodiment is an example including F2F bonding and stacking of eight memory chips **110***n* to **110***n*+7 and forming the stacked chip **849**. In order to explain an example of the method for manufacturing the memory cube **100** according to the fifteenth embodiment in an easy-to-understand manner, reference signs of the first surface and the second surface of each memory chip are omitted.

[0416] STEP **420** is, for example, a step of adjusting positions of the laser irradiator device **822** and the stacked chip **849** with reference to the marker **818** and the plurality of markers **820** included in the stacked chip **849**. Adjusting the positions of the laser irradiation device **822** and the stacked chip **849** may be referred to as, for example, alignment between the laser irradiation device **822** and the stacked chip **849**.

[0417] STEP **430** is a step of irradiating the stacked chip **849** with a laser and polishing the stacked chip **849** according to the positions of the laser irradiator device **822** and the stacked chip **849** adjusted by STEP **420**. For example, as shown in FIG. **64**, STEP **430** includes, based on a reference surface **824**, irradiating the stacked chip **849** with a laser from the memory chip **110***n* by the laser irradiation device **822**, irradiating the stacked chip **849** with the laser from the laser irradiation device **822** while moving the laser irradiation device **822** along the direction D**2**, and irradiating the stacked chip **849** with the laser from the laser irradiation device **822** while moving the laser irradiation device **822** along the direction D**3** from above the stacked chip **849** toward the reference surface **824**.

[0418] In this case, STEP **430** includes irradiating the stacked chip **849** with a laser from the laser irradiation device **822** while detecting the markers **848**, **850**, and **852** included in the plurality of marker groups **844**, respectively. Consequently, for example, in the case where a marker 850 g of the plurality of marker groups **844** is exposed from the third side surface **148**, it can be seen that the laser irradiation device **822** polishes the stacked chip **849** by 11 μm or more, and in the case where a marker **850*f*** is exposed from the third side surface **148**, it can be seen that the laser irradiation device **822** polishes the stacked chip **849** by 13 μm or more. That is, in the manufacturing method of the memory cube **100** according to the fifteenth embodiment, it is possible to detect mutual positional deviations along the direction D**1**, the direction D**2**, and the direction D**3** of the memory chips **110*n*** to **110*n***+7 by detecting the markers of each of the memory chips exposed from the third side surface **148**.

[0419] Further, for example, STEP **430** includes, based on a reference surface **826**, irradiating the stacked chip **849** with the laser from the memory chip **110*n*** by the laser irradiation device **822**, irradiating the stacked chip **849** with the laser from the laser irradiation device **822** while moving the laser irradiation device **822** along the direction D**2**, and irradiating the stacked chip **849** with the laser from the laser irradiation device **822** while moving the laser irradiation device **822** along the direction D**3** from below the stacked chip **849** toward the reference surface **826**. As a consequence, in the manufacturing method of the memory cube **100** according to the fifteenth embodiment, similar to the third side surface **148**, by detecting the markers of the memory chips exposed from the first side surface **146**, it is possible to detect mutual positional deviations along the direction D**1**, the direction D**2**, and the direction D**3** of the memory chips **110*n*** to **110*n***+7.

[0420] Further, for example, STEP **430** includes, based on the reference surfaces **824** and **826**, irradiating the stacked chip **849** with the laser from the memory chip **110*n*** by the laser irradiation device **822**, and irradiating the stacked chip **849** with the laser from the laser irradiation device **822** while moving the laser irradiation device **822** along the direction D**3** and the direction D**2**. As a consequence, in the manufacturing method of the memory cube **100** according to the fifteenth embodiment, similar to the third side surface **148**, by detecting the markers of the memory chips exposed from the second side surface **145**, it is possible to detect mutual positional deviation along the direction D**1**, the direction D**2**, and the direction D**3** of the memory chips **110*n*** to **110*n***+7. In the manufacturing method of the memory cube **100** according to the fifteenth embodiment, similar to the second side surface **145**, by detecting the markers of the respective memory chips exposed from the fourth side surface **147**, it is possible to detect mutual positional deviation along the direction D**1**, the direction D**2**, and the direction D**3** of the memory chips **110*n*** to **110*n***+7.

[0421] In addition, in the case where the CMP device is used, STEP **420** is a step to install the stacked chip **849** in the CMP device. STEP **430** of using the CMP device is to polish the stacked chip **849** while adjusting a polishing amount of the memory chip with reference to the marker groups **844** and **846**. For example, STEP **430** includes polishing the third side surface **148** with the CMP device while detecting the markers **848**, **850**, and **852** included in each of the plurality of marker groups **844** so as to have a polishing amount corresponding to the reference surface **824**. Further, for example, STEP **430** includes polishing the first side surface **146** by the CMP device while detecting the markers **848**, **850**, and **852** included in each of the plurality of marker groups **844** so as to have a polishing amount corresponding to the reference surface **826**. STEP **430** includes polishing the second side **145** and the fourth side **147** with the CMP device, as well as the third side **148** and the first side **146**. As a consequence, the manufacturing method of the memory cube **100** according to the fifteenth embodiment includes the use of the CMP device, and the same advantageous effects as those obtained by using the laser irradiation device **822** can be obtained.

[0422] Further, in the case where the stacked chip **849** is polished and the memory cube **100** is manufactured using the laser irradiation device **822** and the CMP device, first, STEP **420** and STEP **430** are executed using the CMP device to roughly polish the stacked chip **849**, then STEP **420** and STEP **430** are executed using the laser irradiation device **822**, and the stacked chip **849** is polished to finely adjust the polishing amount. First, STEP **420** and STEP **430** may be performed using the laser irradiator **822** to polish the stacked chip **849**, and then STEP **420** and STEP **430** may be performed using the CMP device to polish the stacked chip **849**. As a consequence, the manufacturing method of the memory cube **100** according to the fifteenth embodiment includes the use of the laser irradiation device **822** and the CMP device, and can achieve the same advantageous effects as those obtained in the case where the laser irradiation device **822** is used or in the case where the CMP device is used.

[0423] As described above, the memory cube **100** is formed.

[0424] When a plurality of memory chips are stacked, positional deviation occurs between the plurality of memory chips along the direction D**1**, the direction D**2**, and the direction D**3**. For example, in the method for manufacturing a memory cube to which the method for manufacturing the memory cube **100** according to the fifteenth embodiment is not applied, if the memory cube is polished using the CMP device, there is a possibility that a side surface of the memory cube is flattened so as to have a slope, and therefore, there is a possibility that a predetermined flatness of the memory cube cannot be obtained. Further, for example, in the method for manufacturing a memory cube to which the method for manufacturing the memory cube **100** according to the fifteenth embodiment is not applied, it is difficult to detect positions of the memory chips included in the memory cube with respect to each other, and if the memory cube is polished using the CMP device, the memory cube may be polished exceeding a predetermined polishing amount.

[0425] Meanwhile, the memory cube **100** according to the fifteenth embodiment includes the markers **818** and **820** for adjusting the positions of the device for polishing and the memory chip (for example, the memory chip **110*n***), and the marker groups **844** and **846** for adjusting the positions between the memory chips and the polishing amounts of the memory chips and detecting the positions between the memory chips. The memory cube **100** according to the fifteenth embodiment includes STEP **410** to STEP **430**. The manufacturing method of the memory cube **100** according to the fifteenth embodiment includes stacking the plurality of memory chips **110**_n_ to **110**_n_+7, adjusting a position of a device for polishing and the memory chip (for example, the memory chip **110**_n_), polishing each side surface of the stacked chip **849** based on preset reference surfaces **824** and **826**, exposing each side surface of the plurality of memory chips **110**_n_ to **110**_n_+7, and detecting a position of the plurality of memory chips **110**_n_ to **110**_n_+7 with respect to each other using the plurality of marker groups **844** and **846** exposed from each side surface of the plurality of memory chips **110**_n_ to **110**_n_+7. As a result, in the manufacturing method of the memory cube **100** according to the fifteenth embodiment, since the positions of the plurality of memory chips **110**_n_ to **110**_n_+7 included in the memory cube **100** can be adjusted based on the plurality of markers, the memory cube **100** having a flat predetermined side surface can be manufactured. Further, in the manufacturing method of the memory cube **100** according to the fifteenth embodiment, the positions of the plurality of memory chips **110**_n_ to **110**_n_+7 can be adjusted and the positions of the plurality of memory chips **110**_n_ to **110**_n_+7 can be detected as three-dimensional data based on the direction D**1** to the direction D**3**.

[0426] Therefore, polishing of the side surfaces of the memory cube **100** (the plurality of memory chips **110**_n_ to **110**_n_+7) can be stopped at a predetermined polishing amount. That is, the method for manufacturing the memory cube **100** according to the fifteenth embodiment can be configured such that each side surface of the memory cube **100** according to the fifteenth embodiment has a predetermined flatness. Further, since the memory cube **100** according to the fifteenth embodiment has high flatness on each side surface, positional deviation at the time of connection between the memory cube **100** according to the fifteenth embodiment and the logic chip **200** (see FIG. **1**) is suppressed.

16. Sixteenth Embodiment

[0427] A semiconductor module **10**L and a memory cube **100**A included in the semiconductor module **10**L according to the sixteenth embodiment will be described referring to FIG. **65** to FIG. **68**. The semiconductor module **10**L and the memory cube **100**A included in the semiconductor module **10**L includes a configuration in which the plurality of memory chips **110** are sandwiched by dummy chips **828**_a_ and **828**_b_. For example, since other configurations and functions are the same as the configurations and functions described in the first embodiment and the like, detailed description thereof will be omitted here. In addition, the semiconductor module **10**L can be appropriately combined with or replaced with the configurations described in the first to fifteenth embodiments as long as they do not contradict each other.

[16-1. Configuration of Memory Cube **100**A]

[0428] An exemplary configuration of the memory cube **100**A will be described referring to FIG. **65** and FIG. **66**. FIG. **65** is a side view (plan view) showing a side view of the memory cube **100**A viewed from a second side **145**_a_ of the memory cube **100**A. FIG. **66** is a side view (plan view) showing a side view of the memory cube **100**A viewed from a third side **148**_b_ of the memory cube **100**A. Descriptions of the same or similar configurations as those in FIG. **1** to FIG. **64** will be omitted.

[0429] The memory cube **100**A includes a configuration in which a plurality of memory chips **110** are stacked, and includes a configuration in which a plurality of memory chips **110** are sandwiched by the dummy chips **828**_a_ and **828**_b_. The memory cube **100**A also includes the first surface **142** parallel to the direction D**2** and the direction D**3** and the second surface **144** opposite the first surface **142** and parallel to the first surface **142** with respect to the direction D**1**. In addition, the memory cube **100**A includes a first side surface **146**_a_ perpendicular to the first surface **142** and the second surface **144**, a second side surface **145**_a_ adjacent to the first side surface **146**_a_, a third side surface **148**_b_ adjacent to the second side surface **145**_a_, and a fourth side surface (not shown) adjacent to the third side surface **148**_b_ and the first side surface **146**_a_. The first side surface **146**_a_ of the memory cube **100**A contacts a second surface **304** (see FIG. **68**) of the rewiring layers **300** (see FIG. **68**). The first side **146**_a_ is in contact with the rewiring layer **300**, the memory cube **100**A is electrically connected to the rewiring layer **300**, and the memory cube **100**A is arranged on the second surface **204** of the logic chip **200** (see FIG. **68**). In addition, as described in the first embodiment, the logic chip **200** includes the plurality of TCI-IO **212** and the plurality of logic modules **211**, but for example, the plurality of TCI-IO **212** and the plurality of logic modules **211** may be formed on different substrates, and the plurality of TCI-IO **212** formed on different substrates and the plurality of logic modules **211** may be connected to each other using wirings, vias, and the like, and may be stacked.

[0430] In the memory cube **100**A, the first side surface **146**_a_ is an outermost surface on the side where the logic chip **200** is arranged in the direction D**3**, and the third side surface **148**_b_ is an outermost surface on the side opposite the first side surface **146**_a_ in the direction D**3**. The second side surface **145**_a_ and the fourth side surface opposite the second side surface **145**_a_ are the two outermost surfaces of the memory cube **100**A in the direction D**2**.

[0431] The dummy chip **828**_a_ contacts the first surface **142** and is arranged on the outermost side of the memory cube **100**A in the direction D**1**, and includes an outermost surface. The dummy chip **828**_b_ contacts the second surface **144** and is arranged on the outermost side of the memory cube **100**A in the direction D**1**, and includes an outermost surface. Also, the corner portions of the outermost surfaces of the dummy chips **828**_a_ and **828**_b_ include an R surface **830**. For example, the dummy chips **828**_a_ and **828**_b_ do not include the wiring layers **150**, transistors, and the like. For example, the dummy chips **828**_a_ and **828**_b_ are a Si substrate and a Si-wafer.

[0432] The memory cube **100**A also includes a configuration similar to the memory cube **100** described with respect to FIG. **8**A and FIG. **8**B. Specifically, the memory cube **100**A includes a configuration in which the side power wirings **162** are formed on the plurality of power wirings **164** exposed on the third side surface **148**_b_, and the side ground wirings **163** are formed on the plurality of ground wirings **165**. For example, the side power wiring **162** is arranged so as to contact a portion of the side surface and the upper surface of the insulating film **168** (see FIG. **8**B), the four power supply wirings **164** exposed on the first side surfaces **106**_n_+2 to 106n+5 of the memory chips **110**_n_+2 to

110n+5, and the first side surfaces **106**_n_+2 to 106n+5 of the memory chips **110**_n_+2 to 110n+5, and overlaps the dummy chips **828**_a_ and **828**_b_. Further, similar to the side power wiring **162**, the other side power wiring **162** is arranged so as to be in contact with a portion of the side surface and the upper surface of the insulating film **168**, the two power wiring wirings **164** exposed on the first side surfaces **106**_n_ to **106**_n_+1 of the memory chips **110**_n_ to **110**_n_+1, and the first side surfaces **106**_n_ to **106**_n_+1 of the memory chips **110**_n_ to **110**_n_+1, and overlaps the dummy chips **828**_a_ and **828**_b_.

[0433] In addition, for example, the side ground wiring **163** has the same configuration and function as the side power wiring **162**.

[0434] In addition, the side power wirings **162** and the side ground wirings **163** extend in the direction D**1** and are alternately arranged along the direction D**2** in the direction D**1**, for example, and overlap the dummy chips **828**_a_ and **828**_b_.

[0435] The memory cube **100**A includes the configuration described above. For example, if the corner portions of the outermost surface of the memory cube do not include an R surface, the corner portions of the memory cube are easily chipped. On the other hand, the outermost part of the memory cube **100**A includes the dummy chips **828**_a_ and **828**_b_ including the R surface **830**. Therefore, the memory cube **100**A can protect the circuit inside the memory cube **100** and the devices such as transistors from an impact from the outside of the memory cube **100**A. Consequently, long-term reliability of the memory cube **100**A is higher. Although an example is shown in which an outermost surface of a memory cube according to an embodiment of the present invention is an R surface, the outermost surface of the memory cube according to the embodiment of the present invention is not limited to the R surface. A corner portion of a memory cube according to an embodiment of the present invention may be configured so as not to be easily chipped. For example, an outermost part of a memory cube according to an embodiment of the present invention may be a C surface, a thread surface, or a corner cut surface, and the R surface may be called a C surface, a thread surface, a corner cut surface, or the like.

[0436] Further, since thermal conduction of the dummy chips **828**_a_ and **828**_b_ is better than that of the memory chips **110**_n_ to **110**_n_+7, the memory cube **100**A can suppress thermal expansion toward the memory chips **110**_n_ to **110**_n_+7.

[16-2. Configuration of Memory Cube **100**B]

[0437] An exemplary configuration of a memory cube **100**B will be described referring to FIG. **67**. FIG. **67** is a side view (plan view) showing a side view of the memory cube **100**B viewed from a third side surface **148**_c_ of the memory cube **100**B. Descriptions of the same or similar configurations as those in FIG. **1** to FIG. **66** will be omitted.

[0438] The memory cube **100**B does not include the dummy chips **828**_a_ and **828**_b_ as compared to the memory cube **100**A, and includes the dummy chips **828**_c_ to **828**_f_. The memory cube **100**B is configured to surround the first surface **142**, the second surface **144**, the fourth side surface **147**, and the second side surface **145** with the dummy chips **828**_c_ to **828**_f_. In the memory cube **100**B, the third side surface **148**_c_ facing the first side surface **146**_a_ and the first side surface **146**_a_ of the plurality of stacked memory chips **110** are not surrounded by the dummy chips. The remaining configuration of the memory cube **100**B is similar to the configuration of the memory cube **100**A, and the configuration of the memory cube **100**A will be described as needed.

[0439] In the memory cube **100**B, the first side surface **146**_a_ is an outermost surface on the side where the logic chip **200** is arranged in the direction D**3**, and the third side surface **148**_c_ is an outermost surface on the side opposite the first side surface **146**_a_ in the direction D**3**. The first side surface **146**_a_ is in contact with and electrically connected to the rewiring layer **300**, and the memory cube **100**B is arranged on the second side **204** of the logic chip **200** (see FIG. **68**).

[0440] The dummy chip **828**_c_ contacts the first surface **142** and is arranged on the outermost side of the memory cube **100**B in the direction D**1**, and includes an outermost surface. The dummy chip **828**_d_ is arranged on the outermost side of the memory cube **100**B in the direction D**1** while contacting the second surface **144**, and includes an outermost surface.

[0441] The dummy chip **828**_e_ contacts the fourth side surface **147**, an end part of the dummy chip **828**_c_ on a side of the fourth side surface **147**, and an end part of the dummy chip **828**_d_ on a side of the fourth side surface **147**, and is arranged on the outermost side of the memory cube **100**B in the direction D**2**, and includes an outermost surface. The dummy chip **828**_f_ includes the second side surface **145**, an end part of the dummy chip **828**_c_ on a side of the second side surface **145**, and an end part of the dummy chip **828**_d_ on a side of the second side surface **145**, and an outermost surface arranged on the outermost side of the memory cube **100**B in the direction D**2**. Also, the corner portions of the outermost surfaces of the dummy chips **828**_e_ and **828**_f_ include R surfaces **830**.

[0442] In addition, the dummy chips **828**_c_ to **828**_f_ do not include the wiring layers **150**, transistors, and the like, similar to the dummy chips **828**_a_ and **828**_b_. For example, the dummy chips **828**_c_ to **828**_f_ are a Si substrate and a Si-wafer similar to the dummy chips **828**_a_ and **828**_b_.

[0443] In addition, the memory cube **100**B may include a configuration in which, similar to the memory cube **100**A, the side power wiring **162** is formed on the plurality of power wirings **164** exposed on the third side surface **148**_c_, and the side ground wirings **163** are formed on the plurality of ground wiring wirings **165**.

[0444] Further, when the memory cube **100**B includes the side power wiring **162** and the side ground wiring **163** similar to those of the memory cube **100**A, the side power wirings **162** and the side ground wirings **163** extend in the direction D**1**, are alternately arranged along the direction D**2** in parallel to the direction D**1**, overlap the dummy chips **828**_c_ and **828**_d_, and do not overlap the dummy chips **828**_e_ and **828**_f_.

[0445] Further, the memory cube **100**B includes the configuration described above. The outermost part of the memory cube **100**B includes the dummy chips **828**_c_ and **828**_d_ including the R surface **830**, and the dummy chips **828**_e_ and **828**_f_. That is, except for the first side surfaces **146**_a_ and the third side surfaces **148**_c_ of the plurality of memory chips **110** in the memory cube **100**B, the first surface **142**, the second surface **144**, the second side surface **145**, and the fourth side surface **147** are surrounded by the dummy chips **828**_c_ to **828**_f_.

[0446] Therefore, the memory cube **100**B can protect the circuit inside the memory cube **100**B and devices such as transistors from an external impact of the memory cube **100**B. Consequently, the long-term reliability of the memory cube **100**B is higher.

[0447] In addition, since thermal conductivity of the dummy chips **828***c* to **828***d* is better than that of the memory chips **110***n* to **110***n*+7, the memory cube **100**B can suppress thermal expansion toward the memory chips **110***n* to **110***n*+7 except for the first side surface **146***a* and the third side surface **148***c*.

[16-3. Configuration of Semiconductor Module **10**L]

[0448] A mounting example of the semiconductor module **10**L will be described referring to FIG. **68**. FIG. **68** is a cross-sectional view showing the mounting example of the semiconductor module **10**L. The same or similar configurations as those in FIG. **1** to FIG. **67** will be omitted here.

[0449] The semiconductor module **10**L replaces the memory cube **100** of the semiconductor module **10**D with the memory cube **100**A as compared to the semiconductor module **10**D shown in FIG. **17**, and includes a conductive layer **832** and a heat spreader **834**. The remaining configuration of the semiconductor module **10**L is the same as that of the semiconductor module **10**D, and the same configuration as that of the semiconductor module **10**D will be described as needed. In addition, the semiconductor module **10**L may include a heat sink (not shown), and may include the heat spreader **834** on which the heat sink is arranged. The memory cube **100**A may be the memory cube **100**B. The heat spreader **834** may be referred to as a heat conductive member.

[0450] The side power wiring **162** is electrically connected to the conductive layer **832**. For example, the conductive layer **832** is solder. The heat spreader **834** is electrically connected to the conductive layer **832**, and is electrically connected to the side power wiring **162** via the conductive layer **832**.

[0451] The semiconductor module **10**L includes the configuration described above.

[0452] For example, an upper limit T_{jmaxm} of an operating temperature of the plurality of memory chips **110** is lower than an upper limit T_{imaxr} of an operating temperature of the logic chip **200**. For example, the upper limit T_{jmaxm} is approximately 85° C. and the upper limit T_{jmaxr} is approximately 125° C. That is, if heat is transferred to the memory cube **100**A due to the heat generated by the logic chip **200**, there is a risk that the memory cube **100**A may not operate normally.

[0453] The memory cube **100**A and the logic chip **200** of the semiconductor module **10**L can transmit signals to each other by inductor communication. That is, the memory cube **100**A and the logic chip **200** the semiconductor module **10**L do not transmit signals using physical wirings. Therefore, the memory cube **100**A and the logic chip **200** can suppress generation of cracks due to warpage of the chip. Consequently, the semiconductor module **10**L is excellent in long-term reliability.

[0454] In addition, since the memory cube **100**A is sandwiched between the dummy chips **828***a* and **828***b*, the heat generated in the memory cube **100**A can be discharged to the outside of the semiconductor module **10**L via the dummy chips **828***a* and **828***b*. Further, since the logic chip **200** is electrically connected to the wiring substrate **400**, heat generated in the logic chip **200** can be discharged to the outside of the semiconductor module **10**L via the wiring substrate **400**. Further, the semiconductor module **10**L includes a configuration in which the side power wiring **162** and the heat spreader **834** are electrically connected, and power for driving the semiconductor module **10**L is supplied to the side power wiring **162** through the heat spreader **834**. The heat spreader **834** supplies power to the semiconductor module **10**L and can diffuse heat generated in the semiconductor module **10**L.

[0455] Therefore, the memory cube **100**A and the logic chip **200** of the semiconductor module **10**L can suppress the electric interference and thermal interference from each other. As a consequence, the memory cube **100**A and the logic chip **200** of the semiconductor module **10**L can be stably operated by suppressing thermal malfunctions. Further, since the semiconductor module **10**L suppresses malfunctions due to heat, thermal designs of the respective members constituting the respective circuits and the respective devices in the semiconductor module **10**L can be simplified.

17. Seventeenth Embodiment

[0456] A semiconductor module **10**M and a memory cube **100**C included in the semiconductor module **10**M according to the seventeenth embodiment will be described referring to FIG. **69** and FIG. **70**. The semiconductor module **10**M and the memory cube **100**C included in the semiconductor module **10**M include dummy chips **838** including a configuration in which the plurality of memory chips **110** is sandwiched by the dummy chips **828***a* and **828***b*, and an opening **840** and a bottom opening **841** between any chips of the plurality of memory chips **110**. For example, since other configurations and functions are the same as those described in the sixteenth embodiment and the like, a detailed description thereof will be omitted. In addition, the semiconductor module **10**M according to the seventeenth embodiment can be appropriately combined with or replaced with the configurations described in the first to sixteenth embodiments as long as they do not contradict each other.

[17-1. Configuration of Memory Cube **100**C]

[0457] An exemplary configuration of the memory cube **100**C will be described referring to FIG. **69**. FIG. **69** is a side view (plan view) showing a side surface of the memory cube **100**C from the second side **145***a* of the memory cube **100**C, and a plan view of the dummy chip **838**. The same or similar configurations as those in FIG. **1** to FIG. **68** will not be described here.

[0458] The memory cube **100**C includes the plurality of dummy chips **838** shown in FIG. **69** as compared to the memory cube **100**A. The remaining configuration of the memory cube **100**C is similar to the configuration of the memory cube **100**A, and the configuration of the memory cube **100**A will be described as needed.

[0459] For example, the dummy chip **838** is arranged between two memory chips **110**. The plurality of memory chips **110** are arranged between two dummy chips **838**. For example, the plurality of memory chips **110** may be two, three, four, or more, and may include both a configuration in which two memory chips **110** are stacked and a configuration in which three memory chips **110** are stacked.

[0460] In addition, the dummy chip **838** includes a substrate **842**, and the opening **840** and the bottom opening **841** formed in the substrate **842**. The substrate **842** is a Si substrate and a Si-wafer similar to the dummy chips **828***a* to **828***f*. The opening **840** is arranged adjacent to the bottom opening **841**.

[17-2. Configuration of Semiconductor Module **10**M]

[0461] A mounting example of the semiconductor module **10**M will be described referring to FIG. **70**. FIG. **70** is a cross-sectional view showing the mounting example of the semiconductor module **10**M. The same or similar configurations as those in FIG. **1** to FIG. **69** will not be described here.

[0462] The semiconductor module **10**M replaces the memory cube **100**A with the memory cube **100**C as compared to the semiconductor module **10**L shown in FIG. **68**. The remaining configuration of the semiconductor module **10**M is the same as that of the semiconductor module **10**L, and the same configuration as that of the semiconductor module **10**L will be described as needed.

[0463] The semiconductor module **10**M includes the plurality of dummy chips **838**. For example, since the plurality of dummy chips **838** includes the openings **840** and the bottom openings **841**, the memory cube **100**C includes a plurality of cavities **843** along the direction D**2**.

[0464] The semiconductor module **10**M includes the configuration described above.

[0465] For example, the upper limit T_{jmaxm} of the operating temperature of the plurality of memory chips **110** is lower than the upper limit T_{imaxr} of the operating temperature of the logic chip **200**. For example, the upper limit T_{jmaxm} is approximately 85° C. and the upper limit T_{imaxr} is approximately 125° C. That is, if heat is transferred to the memory cube **100**C due to the heat generated by the logic chip **200**, there is a risk that the memory cube **100**C may not operate normally.

[0466] The semiconductor module **10**M and the memory cube **100**C include configurations similar to those of the semiconductor module **10**L and the memory cube **100**B. Therefore, the semiconductor module **10**M and the memory cube **100**C have the same advantageous effects as those of the semiconductor module **10**L and the memory cube **100**B. That is, the semiconductor module **10**M and the memory cube **100**C can suppress generation of cracks due to warpage of the chip, and are excellent in long-term reliability. In addition, the semiconductor module **10**M and the memory cube **100**C can diffuse the heat generated in the semiconductor module **10**M through the dummy chips **828***a* and **828***b* and the heat spreader **834** and discharge the heat to the outside of the semiconductor module **10**M.

[0467] In addition, for example, the semiconductor module **10**M may be used in a coolant. For example, the coolant may comprise fluorine and may comprise isobutane. Since the semiconductor module **10**M includes the cavity **843**, the cavity **843** functions as a flow path of the coolant. Consequently, the semiconductor module **10**M can dissipate heat through the coolant passing through the cavity **843**.

[0468] Therefore, the semiconductor module **10**M can be cooled by liquid immersion in the case where the semiconductor module **10**M is immersed in the coolant. In addition, the cavity **843** increases a surface area of the semiconductor module **10**M, so that the semiconductor module **10**M is a highly cooling-efficient module.

18. Eighteenth Embodiment

[0469] A method for manufacturing the memory cube **100** according to an eighteenth embodiment will be described with reference to FIG. **48**, FIG. **71** to FIG. **76**. FIG. **71** is a flowchart showing the method for manufacturing the memory cube **100**. FIG. **48** is a plan view showing the method for manufacturing the memory cube **100**. FIG. **72** is a plan view showing a manufacturing method of the memory cube **100** and a cross-sectional view showing a cross section of the memory chip region **804** along a line J**1**-J**2**. FIG. **72** is an enlarged view of the memory chip region **804**. FIG. **73** is a cross-sectional view of the memory substrates **800***a* and **800***b* taken along a line J**1**-J**2**, showing step S**510** (STEP **510**) in the manufacturing method of the memory cube **100**. FIG. **74** is a cross-sectional view showing step S**520** (STEP **520**) and step S**530** (STEP **530**) of the manufacturing method of the memory cube **100**. FIG. **75** is a cross-sectional view showing step S**540** (STEP **540**) and step S**550** (STEP **550**) in the manufacturing method of the memory cube **100**. FIG. **76** is a cross-sectional view showing step **560** (STEP **560**) of the manufacturing method of the memory cube **100**. The same or similar configurations as those in FIG. **1** to FIG. **70** will not be described here.

[0470] As shown in FIG. **71**, the method for fabricating the memory cube **100** includes STEP **510** to STEP **560**.

[0471] When the manufacturing method of the memory cube **100** is started, the memory substrate **800***a* is prepared as in the twelfth embodiment. For example, the memory substrate **800***a* includes the configuration described with respect to FIG. **48**. Here, the configuration of the memory substrate **800***a* shown in FIG. **48** will be described as needed. Further, for example, although the plurality of memory chips **110** includes the same configuration as that of the first embodiment, the same configuration as that of the first embodiment in the memory chip **110** will be described as necessary.

[0472] As shown in FIG. **72**, the memory substrate **800***a* (FIG. **48**) includes a memory chip region **804**, and the memory chip region **804** includes a dicing region **854**. The dicing region **854** is a region for stacking a plurality of substrates and dicing the stacked substrates. For example, as shown in the cross-sectional view along a line J**1**-J**2** in FIG. **72**, the dicing region **854** for separating memory chips **110***a* and **110***b* does not include a wiring **183***q* and a wiring **178***b*. That is, the dicing region **854** includes the insulating layers **184** and **185** and the like, and does not include a layer including a metal such as a wiring.

[0473] The STEP **510** is a step of laminating a memory substrate. For example, by performing STEP **510** shown in FIG. **73**, the first surface **102** of the memory substrate **800***a* and the first surface **102** of the memory substrate **800***b* are F2F bonded to each other to form a stacked substrate **856**. In this case, as shown in FIG. **73**, ideally, the memory substrates **800***a* and **800***b* overlap each other, and the dicing regions **854** are grouped together. In practice, however, it goes without saying that there are variations in the bonding.

[0474] STEP **520** is a step of thinning the stacked substrate **856**. By performing STEP **520** shown in FIG. **74**, the substrate **173** of the memory substrate **800***b* is thinned, and the stacked substrate **856** is planarized. For example, as shown in FIG. **74**, the surface of the substrate **173** of the memory substrate **800***b* is polished, and a thickness STH**3** of the stacked substrate **856** is thinned to a thickness STH**4**.

[0475] STEP **530** is a step of forming a groove **858** that defines each of a plurality of memory cubes including memory cubes **100**D and **100**E. For example, by performing STEP **530** shown in FIG. **74**, the grooves **858** are formed in the dicing region **854** using plasma dicing. The groove **858** ultimately includes a sidewall **860** that is part of the side surface of the memory cubes **100**D and **100**E, as well as a bottom part **862**. The sidewall **860** includes portions of the side surfaces of the transistor layers **130** and the wiring layers **150** of the memory substrates **800***a* and **800***b*, respectively. The bottom part **862** is formed on the substrate **173** of the memory substrate **800***a*. In addition, the plasma dicing in STEP **530** is sometimes referred to as half-cutting.

[0476] STEP **540** is a step of forming insulating layer **864** in the groove **858**. By performing STEP **540** shown in FIG. **75**, the insulating layers **864** are formed on the grooves **858** and the thinned substrates **173***a* and **173***b*. For example, materials and devices for forming the insulating layer **864** may be materials and devices for forming the insulating layer **807**.

[0477] In STEP **550**, the stacked substrate **856** on which the insulating layer **864** is formed in STEP **540** is attached to an adhesive layer **866**. For example, the insulating layer **864** is attached to the adhesive layer **866** by performing STEP **550** shown in FIG. **75**. The stacked substrate **856** is fixed to the adhesive layer **866** by bonding the stacked substrate **856** to the adhesive layer **866**. That is, the adhesive layer **866** has a function of fixing and supporting the stacked substrate **856**. Consequently, stable singulation can be achieved in the step of singulation (S**560**) after STEP **550**. For example, the adhesive layers **866** may be adhesive films and may be back grinding tapes (Back Grinding Tape, BGT). For example, the BGT is used in a back grinding process of a semiconductor substrate, and is an adhesive tape for protecting the semiconductor substrate.

[0478] Step **560** (STEP **560**) is a step of forming a single layer of the stacked substrate **856** to which the adhesive layers **866** are attached in STEP **550**. For example, by performing STEP **560** shown in FIG. **76**, the stacked substrate **856** is polished using the CMP device, and part of the substrate **173** and the bottom part **862** of the memory substrate **800***b* are removed. By performing STEP **560**, the substrate **173** of the memory substrate **800***b* is separated into substrates **173***c* and **173***d* corresponding to the individual memory cubes, and the thickness STH**4** of the stacked substrate **856** is thinned to a thickness STH**5**. As a result, as shown in FIG. **76**, the stacked substrate **856** can be singulated into individual memory cubes. For example, the individual memory cubes (memory cube **100**D and **100**E) singulated from the adhesive layers **866** can be removed.

[0479] As described above, the memory cube **100** is formed.

[0480] A method of manufacturing the memory cube **100** according to the eighteenth embodiment includes laminating a plurality of memory substrates to form a stacked substrate, thinning the stacked substrate, forming a groove including a bottom part along a dicing region of the stacked substrate, forming an insulating layer on the formed groove and a surface of the stacked substrate, attaching a surface including the insulating layer to the adhesive layer, and singulating the stacked substrate by polishing a surface opposite to the adhesive layer of the stacked substrate.

[0481] The dicing region **854** of the stacked substrate **856** according to the eighteenth embodiment does not include a metal such as a wiring. As a result, in the plasma dicing of the method for manufacturing the memory cube **100** according to the eighteenth embodiment, since it is not necessary to dice the metal layer, the plurality of memory cubes **100** can be easily singulated.

[0482] Further, since the groove **858** is filled with the insulating layer **864**, each side surface of each memory cube **100** is covered with the insulating layer **864**. Therefore, a low dielectric constant film having high hygroscopicity is not exposed from each side surface of each memory cube **100**. As a result, the memory cube **100** according to the eighteenth embodiment can suppress corrosion, deterioration, and the like of each device due to moisture absorption and intrusion of impurities and the like. Therefore, by using the method for manufacturing the memory cube **100** according to the eighteenth embodiment, it is possible to maintain the reliability of the semiconductor module without impairing the long-term reliability of the semiconductor module.

[0483] Further, by using the method for manufacturing the memory cube **100** according to the eighteenth embodiment, since each side surface of the memory cube **100** is protected by the insulating layer, moisture absorption and impurities do not enter from each side surface during manufacturing. Therefore, by using the method for manufacturing the memory cube **100** according to the eighteenth embodiment, it is possible to suppress corrosion, deterioration, and the like of each device due to moisture absorption, impurities, and the like at the time of manufacturing, so that the memory cube **100** does not need to include the seal ring **160**.

[0484] Various configurations of the semiconductor module and the semiconductor module manufacturing method shown as an embodiment of the present invention can be appropriately combined or replaced as long as they do not conflict with each other, and technical matters common to the embodiments are included in the embodiments without explicit description. Further, based on the semiconductor module and the manufacturing method of the semiconductor module disclosed in the present specification and the drawings, any addition, deletion, or design change of components by a person skilled in the art, or addition, omission, or change of conditions of a process, is included in the scope of the present invention as long as it comprises the gist of the present invention.

[0485] It is to be understood that the present invention provides other effects that are different from the effects provided by the aspects of the embodiments disclosed herein, and effects that are obvious from the description herein or that can be easily predicted by a person skilled in the art.

(Additional Notes)

[0486] In addition, the present invention is not limited to the embodiments described above, and can be appropriately modified without departing from the scope of the present invention. For example, one embodiment according to the present invention may have the following configuration.

(Additional Note **1**)

[0487] A semiconductor module in which a plurality of memory chips includes memory cubes stacked along a first direction in parallel to a plane formed by a second direction orthogonal to the first direction and a third direction orthogonal to the second direction, wherein

[0488] the memory cube includes a bottom surface which is an outermost surface on a side where a semiconductor chip is arranged in the third direction,

[0489] each of the plurality of memory chips includes a substrate and a wiring layer stacked on the substrate,

[0490] the wiring layer includes a first internal wiring, a second internal wiring and a first inductor,

[0491] a first insulating film is included between the first internal wiring and the first inductor and the bottom surface,

[0492] the second internal wiring is exposed to the bottom surface,

[0493] the first internal wiring, the second internal wiring, and the first inductor are spaced apart from each other,

[0494] a length of the first internal wiring extending along the second direction is longer than a length of the first inductor extending along the second direction,

[0495] a length of the first internal wiring extending along the third direction is shorter than lengths of the first inductor and the second internal wiring extending along the third direction, and

[0496] part of the first inductor overlaps the first internal wiring in a cross-sectional view.

(Additional Note **2**)

[0497] The semiconductor module according to additional note **1**, wherein

[0498] the first inductor has a plurality of turns in a plan view,

[0499] the first inductor includes a third internal wiring and a fourth internal wiring,

[0500] the third internal wiring is arranged in the same layer as the first internal wiring and the fourth internal wiring, and

[0501] the first internal wiring, the third internal wiring, and the fourth internal wiring are spaced apart from each other, and the third internal wiring, the first internal wiring, and the fourth internal wiring are arranged in order from the bottom surface.

(Additional Note **3**)

[0502] The semiconductor module according to additional note **1**, wherein

[0503] the first inductor has a plurality of turns and is arranged over at least two memory chips of the plurality of memory chips in a cross-sectional view,

[0504] the first inductor includes a fifth internal wiring, and

[0505] the fifth internal wiring is arranged in the same layer as the first internal wiring, is spaced apart from the first internal wiring, and is arranged closer to the bottom surface than the first internal wiring.

(Additional Note **4**)

[0506] The semiconductor module according to additional note **1**, wherein

[0507] each of the plurality of memory chips includes a seal ring arranged on an outer peripheral portion of each of the plurality of memory chips, and

[0508] the seal ring includes the first internal wiring.

(Additional Note **5**)

[0509] The semiconductor module according to additional note **1**, wherein

[0510] the semiconductor module further includes a rewiring layer arranged on the bottom surface,

[0511] the rewiring layer includes a wiring, and

[0512] the wiring is electrically connected to the second internal wiring.

(Additional Note **6**)

[0513] The semiconductor module according to additional note **5**, wherein

[0514] the internal wiring includes at least one power wiring exposed to the bottom surface, and

[0515] the at least one power wiring includes the second internal wiring.

(Additional Note **7**)

[0516] The semiconductor module according to additional note **5**, wherein

[0517] the semiconductor module further includes a semiconductor chip for driving the memory cube, and includes a first surface and a second surface parallel to the first surface and a second inductor,

[0518] the rewiring layer is arranged between the second surface and the bottom surface, and

[0519] the second inductor communicates with the first inductor in a non-contact manner.

(Additional Note **8**)

[0520] A semiconductor module in which a plurality of memory chips includes memory cubes stacked along a first direction in parallel to a plane formed by a second direction orthogonal to the first direction and a third direction orthogonal to the second direction, wherein

[0521] the memory cube includes a bottom surface which is an outermost surface on a side where the semiconductor chip is arranged in the third direction,

[0522] each of the plurality of memory chips includes a substrate and a wiring layer stacked on the substrate,

[0523] the wiring layer includes a first internal wiring, a second internal wiring and part of a first inductor,

[0524] a first insulating film is included between the first internal wiring and the bottom surface,

[0525] the second internal wiring is exposed to the bottom surface,

[0526] the first internal wiring, the second internal wiring, and the part of the first inductor are spaced apart from each other,

[0527] a length of the first internal wiring extending along the second direction is longer than a length of the part of the first inductor extending along the second direction,

[0528] a length of the first internal wiring extending along the third direction is shorter than lengths of the first inductor and the second internal wiring extending along the third direction, and

[0529] the part of the first inductor overlaps the first internal wiring in a cross-sectional view.

(Additional Note 9)

[0530] The semiconductor module according to additional note 8, wherein

[0531] each of the plurality of memory chips includes a seal ring arranged on an outer peripheral portion of each of the plurality of memory chips, and

[0532] the seal ring includes the first internal wiring.

(Additional Note 10)

[0533] The semiconductor module according to additional note 8, wherein

[0534] the semiconductor module further includes a rewiring layer arranged on the bottom surface,

[0535] the rewiring layer includes a first side surface wiring and a second side surface wiring,

[0536] the first side surface wiring is electrically connected to the second internal wiring, and

[0537] the second side surface wiring is electrically connected to the part of the first inductor, and constitutes the first inductor.

(Additional Note 11)

[0538] The semiconductor module according to additional note 10, wherein

[0539] the first inductor has a plurality of turns in a plan view,

[0540] the part of the first inductor includes a third internal wiring and a fourth internal wiring,

[0541] the rewiring layer further includes a third side surface wiring spaced apart from the first side surface wiring and the second side surface wiring,

[0542] the third internal wiring is arranged in a layer different from the first internal wiring, is arranged in the same layer as the second internal wiring and the fourth internal wiring, and is spaced apart from the second internal wiring and the fourth internal wiring,

[0543] the third internal wiring is electrically connected to the second side surface wiring, and

[0544] the fourth internal wiring is electrically connected to the third side surface wiring.

(Additional Note 12)

[0545] The semiconductor module according to additional note 10, wherein, the first inductor has a plurality of turns and is arranged over at least two memory chips of the plurality of memory chips in a cross-sectional view.

(Additional Note 13)

[0546] The semiconductor module according to additional note 8, wherein

[0547] each of the plurality of memory chips includes a seal ring arranged on an outer peripheral portion of each of the plurality of memory chips, and

[0548] the seal ring includes the first internal wiring.

(Additional Note 14)

[0549] The semiconductor module according to additional note 10, wherein

[0550] the internal wiring includes at least one power wiring exposed to the bottom surface, and

[0551] the at least one power wiring includes the second internal wiring.

(Additional Note 15)

[0552] The semiconductor module according to additional note 10, wherein

[0553] the semiconductor module further includes a semiconductor chip for driving the memory cube, and having a first surface, a second surface parallel to the first surface, and a second inductor,

[0554] the rewiring layer is arranged between the second surface and the bottom surface, and

[0555] the second inductor communicates with the first inductor in a non-contact manner.

(Additional Note 16)

[0556] A method for manufacturing a semiconductor module including a memory cube in which a plurality of memory chips are stacked along a first direction in parallel with a plane formed by a second direction orthogonal to the first direction and a third direction orthogonal to the second direction, comprising:

[0557] in each of a first memory substrate including a plurality of memory chips and a second memory substrate including a plurality of memory chips, using laser grooving to form a groove partitioning the plurality of memory chips into respective memory chips, forming a first insulating layer in the groove, forming a second insulating layer on the first insulating layer, and polishing the second insulating layer;

[0558] laminating the first memory substrate having the polished second insulating layer and the second memory substrate having the polished second insulating layer, and forming a stacked substrate;

[0559] dicing the stacked substrate along the groove of the stacked substrate, and singulating the stacked substrate into a plurality of stacked chips;

[0560] measuring a thickness of each of the plurality of stacked chips; and

[0561] stacking at least two stacked chips of the plurality of stacked chips to form the memory cube.

(Additional Note 17)

[0562] The method for manufacturing the semiconductor module according to additional note 16, further comprising, after forming the stacked substrate, attaching an adhesive layer to the stacked substrate.

(Additional Note 18)

[0563] The method for manufacturing the semiconductor module according to additional note 16, wherein

[0564] forming the stacked substrate includes that the polished second insulating layer of the first memory substrate overlaps and is stacked on the polished second insulating layer of the second memory substrate, and

[0565] a thickness of the first memory substrate on which the second insulating layer is polished is different from a thickness of the second memory substrate on which the second insulating layer is polished.

(Additional Note 19)

[0566] The method of manufacturing the semiconductor module according to additional note 16, wherein

[0567] each of the first memory substrate and the second memory substrate includes a transistor layer including a transistor for driving the first wiring and the memory chip, and a wiring layer including a second wiring and contacting the transistor layer along the first direction, and

[0568] the first insulating layer is in contact with a side surface of each of the first wiring and the second wiring exposed in the groove.

(Additional Note **20**)

[0569] The method for manufacturing the semiconductor module according to additional note **16**, wherein a thickness of each of the plurality of memory chips included in the memory cube is an average thickness of the memory chip −5 μm or more and an average thickness of the memory chip +5 μm or less.

(Additional Note **21**)

[0570] A method for manufacturing a semiconductor module in which a plurality of memory chips having warps includes memory cubes stacked along a first direction in parallel to a plane formed by a second direction orthogonal to the first direction and a third direction orthogonal to the second direction, comprising:

[0571] bonding a plurality of memory chips having the warps; and

[0572] injecting a resin into at least a portion of a space between each of a plurality of memory chips having the bonded warps and sealing the plurality of memory chips having the bonded warps with a resin.

(Additional Note **22**)

[0573] The method for manufacturing the semiconductor module according to additional note **21**, further comprising polishing the plurality of memory chips having the sealed warp to form the memory cube.

(Additional Note **23**)

[0574] The method of manufacturing a semiconductor module according to additional note **22**, wherein

[0575] the polishing comprises:

[0576] planarizing first surfaces of outermost surfaces along the first direction and second surfaces opposite the first surfaces in the plurality of memory chips having the bonded warpage;

[0577] planarizing second side surfaces of outermost surfaces along the second direction and fourth side surfaces opposite the second side surfaces in the plurality of memory chips having the bonded warp; and

[0578] planarizing first side surfaces of outermost surfaces along the third direction and third side surfaces opposite the first side surfaces in the plurality of memory chips having the bonded warp.

(Additional Note **24**)

[0579] The method for manufacturing the semiconductor module in which a plurality of memory chips having warps includes memory cubes stacked along a first direction in parallel to a plane formed by a second direction orthogonal to the first direction and a third direction orthogonal to the second direction, wherein

[0580] the plurality of warped memory chips includes at least a first memory chip and a second memory chip,

[0581] the method for manufacturing the semiconductor module includes:

[0582] applying adhesive layers to a plurality of locations on the surface of the first memory chip;

[0583] attaching the second memory chip on the adhesive layers,

[0584] applying a resin to first end parts of the first memory chip and the second memory chip along a third direction; and

[0585] injecting a resin into at least a portion of a space between each of the plurality of memory chips from side surfaces of the first memory chip and the second memory chip, and joining the resin to form the memory cubes.

(Additional Note **25**)

[0586] The method for manufacturing the semiconductor module according to additional note **24**, further comprising polishing the first memory chip and the second memory chip bonded to each other to form the memory cube.

(Additional Note **26**)

[0587] The method of manufacturing a semiconductor module according to additional note **24**, wherein the polishing comprises:

[0588] planarizing a first surface of an outermost surface along the first direction and a second surface opposite the first surface in the first memory chip and the second memory chip;

[0589] planarizing a second side surface of an outermost surface along the second direction and a fourth side surface opposite the second side surface in the plurality of memory chips having the bonded warp; and

[0590] planarizing a first side surface of an outermost surface along the third direction and a third side surface opposite the first side surface in the plurality of memory chips having the bonded warp.

(Additional Note **27**)

[0591] A method for manufacturing a semiconductor module including a memory cube in which a plurality of memory chips are stacked along a first direction in parallel with a plane formed by a second direction orthogonal to the first direction and a third direction orthogonal to the second direction, wherein

[0592] each of the plurality of memory chips includes an alignment marker for adjusting positions of a device for polishing and a memory chip, and a detection marker group for detecting positions between the plurality of memory chips and polishing amounts of the plurality of memory chips, and

[0593] the method for manufacturing the semiconductor module includes:

[0594] stacking the plurality of memory chips to form a stacked chip;

[0595] adjusting the positions of the plurality of memory chips and the device for polishing;

[0596] polishing a plurality of side surfaces of the stacked chip based on a predetermined reference surface of the stacked chip; and

[0597] detecting positions of the plurality of memory chips with respect to each other using a plurality of marker groups exposed from the plurality of side surfaces of the stacked chip by the polishing.

(Additional Note **28**)

[0598] The method for manufacturing the semiconductor module according to additional note **27**, wherein the device for polishing is a laser irradiation device.

(Additional Note **29**)

[0599] The method for manufacturing the semiconductor module according to additional note **28**, wherein

[0600] the polishing includes:

[0601] irradiating a laser from at least one memory chip side by the laser irradiation device;

[0602] irradiating the laser to the stacked chip while moving the laser irradiation device along the second direction; and

[0603] irradiating the laser to the stacked chip while moving the laser irradiation device along the third direction from above the stacked chip toward the reference surface.

(Additional Note **30**)

[0604] The method for manufacturing the semiconductor module according to additional note **27**, wherein

[0605] the detection marker group includes a plurality of markers having different sizes, and

[0606] the polishing includes irradiating the laser to the stacked chip while detecting at least one of the plurality of markers.

(Additional Note **31**)

[0607] A method for manufacturing a semiconductor module including a memory cube in which a plurality of memory chips are stacked along a first direction in parallel with a plane formed by a second direction orthogonal to the first direction and a third direction orthogonal to the second direction,

[0608] each of the plurality of memory chips includes a detection marker group for detecting positions between the plurality of memory chips and polishing amounts of the plurality of memory chips, and

[0609] the method for manufacturing the semiconductor module includes:

[0610] stacking the plurality of memory chips to form a stacked chip;

[0611] polishing a plurality of side surfaces of the stacked chip based on a predetermined reference surface of the stacked chip; and

[0612] detecting positions of the plurality of memory chips with respect to each other using a plurality of marker groups exposed from the plurality of side surfaces of the stacked chip by the polishing.

(Additional Note **32**)

[0613] The method for manufacturing the semiconductor module according to additional note **31**, wherein the device for polishing is a CMP device.

(Additional Note **33**)

[0614] A semiconductor module including a memory cube in which a plurality of memory chips are stacked along a first direction in parallel with a plane formed by a second direction orthogonal to the first direction and a third direction orthogonal to the second direction, wherein

[0615] the memory cube sandwiches the plurality of memory chips in the first direction, and includes a first substrate and a second substrate including outermost surfaces of the memory cube in the first direction, and a bottom surface and an upper surface opposite the bottom surface on a side where a semiconductor chip is arranged in the third direction,

[0616] each of the first substrate and the second substrate does not include a wiring layer,

[0617] each of the plurality of memory chips includes a substrate and a wiring layer stacked on the substrate,

[0618] the wiring layer includes a first internal wiring, a second internal wiring, and a first inductor,

[0619] the first internal wiring is exposed to the upper surface,

[0620] the second internal wiring is exposed to the bottom surface,

[0621] the first internal wiring, the second internal wiring, and the first inductor are spaced apart from each other, and

[0622] the semiconductor module includes a first conductive layer in contact with the first internal wiring, the upper surface, and an upper surface along a third direction of the first substrate, a second conductive layer electrically connected to the first conductive layer, and a heat conductive member electrically connected to the second conductive layer.

(Additional Note **34**)

[0623] The semiconductor module of additional note **33**, wherein each of the outermost surface of the first substrate and the outermost surface of the second substrate includes an R surface.

(Additional Note **35**)

[0624] The semiconductor module according to additional note **33**, wherein

[0625] the memory cube includes a third substrate including a plurality of openings and a plurality of bottom openings,

[0626] the third substrate is arranged so as to be in contact with any one of the plurality of memory chips and a memory chip different from any one of the memory chips, and

[0627] the third substrate does not include a wiring layer.

(Additional Note **36**)

[0628] The semiconductor module according to additional note **33**, wherein

[0629] the memory cube further includes a third substrate and a fourth substrate sandwiching the plurality of memory chips in the second direction and including an outermost surface of the memory cube in the second direction, and

[0630] each of the outermost surface of the third substrate and the outermost surface of the fourth substrate includes an R surface.

(Additional Note **37**)

[0631] The semiconductor module according to additional note **33**, wherein

[0632] the internal wiring includes a first power wiring exposed on the upper surface and a second power wiring exposed on the bottom surface,

[0633] the first power wiring includes the first internal wiring, and

[0634] the second power wiring includes the second internal wiring.

(Additional Note **38**)

[0635] The semiconductor module according to additional note **33**, wherein

[0636] the semiconductor module further includes a rewiring layer arranged on the bottom surface,

[0637] the rewiring layer includes a wiring, and

[0638] the wiring is electrically connected to the second internal wiring.

(Additional Note **39**)

[0639] The semiconductor module of additional note **38**, wherein

[0640] the semiconductor module has a first surface, a second surface parallel to the first surface, and a second inductor, and further includes a semiconductor chip for driving the memory cube,

[0641] the rewiring layer is arranged between the second surface and the bottom surface, and

[0642] the second inductor communicates with the first inductor in a non-contact manner.

(Additional Note **40**)

[0643] The semiconductor module according to additional note **39** further comprising a plurality of metal pillars arranged on the first surface, wherein

[0644] the semiconductor chip is directly electrically connected to an external substrate using the plurality of metal pillars.

(Additional Note **41**)

[0645] A method for manufacturing a semiconductor module including a memory cube in which a plurality of memory chips are stacked along a first direction in parallel with a plane formed by a second direction orthogonal to the first direction and a third direction orthogonal to the second direction,

[0646] bonding each first surface of a first memory substrate and a second memory substrate including the first surface, second surface, and a plurality of memory chips to form a stacked substrate by stacking the first memory substrate and the second memory substrate,

[0647] polishing the second surface of the second memory substrate of the stacked substrate,

[0648] forming grooves partitioned into respective memory cubes on the stacked substrate so as to reach a portion of the substrate included in the first memory substrate,

[0649] forming an insulating layer on the groove and the polished second surface,

[0650] attaching an adhesive layer for fixing the stacked substrate to the polished second surface, and

[0651] removing the portion of the substrate and polishing the substrate to expose the groove, thereby forming the memory cube.

(Additional Note **42**)

[0652] The method for manufacturing the semiconductor module according to additional note **41**, wherein the groove is formed between wiring layers included in the respective memory cubes.

(Additional Note **43**)

[0653] The method for manufacturing the semiconductor module according to additional note **41**, wherein the adhesive layer is a background tape.

(Additional Note **44**)

[0654] The method for manufacturing the semiconductor module according to additional note **41**, wherein the first surface of a third memory substrate including a first surface, a second surface, and a plurality of memory chips is bonded to the second surface of the second memory substrate to form a stacked substrate made of the first to third memory substrates, polishing the second surface of the third memory substrate of the stacked substrate, forming grooves partitioned into respective memory cubes in the stacked substrate so as to reach a portion of a substrate included in the first memory substrate, forming an insulating layer on the grooves and the polished second **10** surface, attaching an adhesive layer for fixing the stacked substrate to the polished second surface, and removing the portion of the substrate and polishing so as to expose the grooves to form the memory cubes.

What is claimed is:

1. A semiconductor module comprising a memory cube in which a plurality of memory chips is stacked along a first direction parallel to a plane formed by a second direction perpendicular to the first direction and a third direction perpendicular to the second direction, wherein

the memory cube includes a first outermost surface, a third outermost surface opposite the first outermost surface arranged on a side on which a semiconductor chip is arranged in the third direction, and a second outermost surface and a fourth outermost surface opposite the second outermost surface, the second outermost surface and the fourth outermost surface being the two outermost surfaces in the second direction,

each of the plurality of memory chips includes a substrate and a wiring layer stacked on the substrate,

the substrate includes a first side surface along the first outermost surface, a second side surface along the second outermost surface, a third side surface along the third outermost surface and a fourth side surface along the fourth outermost surface, a first insulating film is included between the first outermost surface and the first side surface, between the second outermost surface and the second side surface, between the third outermost surface and the third side surface, and between the fourth outermost surface and the fourth side surface,

the wiring layer includes an internal wiring,

the internal wiring is connected to an electrode provided on at least one of the second outermost surface, the third outermost surface, and the fourth outermost surface, and

the memory cube is configured to transmit signals while being spaced apart from the semiconductor chip.

2. The semiconductor module according to claim **1**, wherein

the plurality of memory chips includes at least one memory controller chip.

3. The semiconductor module according to claim **1**, wherein

each of the memory chips includes a seal ring arranged on an outer periphery of the memory chip,

the seal ring is arranged in the wiring layer,

the seal ring extends parallel to the substrate, extends in the third direction, and is arranged on the substrate, in a cross-sectional view, and

the internal wiring extends in the third direction, overlaps the seal ring, and is arranged outside the seal ring along the third direction, in the cross-sectional view.

4. The semiconductor module according to claim **1**, wherein

the internal wiring includes a plurality of power supply wirings and a plurality of ground wirings,

the plurality of power supply wirings and the plurality of ground wirings are exposed to either the second outermost surface or the fourth outermost surface, or, both the second outermost surface and the fourth outermost surface.

5. The semiconductor module according to claim **4**, wherein

parts of the second outermost surface and the fourth outermost surface are covered with a conductive film.

6. The semiconductor module according to claim **1**, further comprising a second insulating film,

wherein

the second outermost surface, the third outermost surface, and the fourth outermost surface are sealed with the first insulating film or the second insulating film.

7. The semiconductor module of claim **6**, further comprising a semiconductor chip having a first surface and a second surface parallel to the first surface, for driving the memory cube.

8. The semiconductor module of claim **7**, further comprising a plurality of metal pillars arranged on the first surface,

wherein

the semiconductor chip is directly electrically connected to an external substrate using the plurality of metal pillars.

9. The semiconductor module according to claim **1**,

wherein

the plurality of memory chips includes a plurality of memory chips of different thicknesses, and

the memory cube has a total thickness determined by stacking the plurality of memory chips of different thicknesses.

10. The semiconductor module according to claim **1**,

wherein

the wiring layer includes a first internal wiring, a second internal wiring, and a first inductor, and

the first internal wiring, the second internal wiring, and the first inductor are arranged apart from each other.

11. The semiconductor module according to claim **10**,

wherein

the first inductor includes a plurality of turns and is provided across at least two or more memory chips of the plurality of memory chips, in a cross-sectional view.

12. The semiconductor module according to claim **4**, further comprising:

a conductive film provided on either or both of the second outermost surface and the fourth outermost surface; and

a pressure bonding mechanism including a power supply pin;

wherein

the pressure bonding mechanism is connected to the conductive film via the power supply pin and configured to supply power from outside to the memory cube via the power supply pin.

13. The semiconductor module according to claim **12**, wherein

the pressure bonding mechanism includes a power supply pin support part and a plurality of power supply pins,

the power supply pin support part includes the plurality of power supply pins and includes a function of supporting the plurality of power supply pins, and

each of the plurality of power supply pins is a connector composed of at least a terminal part, a cylindrical member connected to the terminal part, and an elastic member inserted into the cylindrical member.

14. The semiconductor module according to claim **12**, wherein

the pressure bonding mechanism includes a plurality of power supply pins, each of the plurality of power supply pins is pressure-bonded to the conductive film and connected to the conductive film, and is configured to supply a voltage from the outside to the memory cube via the plurality of power supply pins.

15. The semiconductor module according to claim **12**, wherein

the conductive film includes a conductor made of a metal, and

the conductor is composed of one of copper, nickel, and gold, or a combination of copper, nickel, and gold.

16. A semiconductor module comprising a memory cube in which a plurality of memory chips is stacked along a first direction parallel to a plane formed by a second direction perpendicular to the first direction and a third direction perpendicular to the second direction,

wherein

the memory cube includes a first substrate, a second substrate, a bottom, and an upper surface opposite to the bottom surface,

the first substrate and the second substrate sandwich the plurality of memory chips in the first direction, the bottom surface is an outermost surface on a side where the semiconductor chips are arranged in the third direction,

each of the first substrate and the second substrate does not include a transistor layer including a transistor,

each of the memory chips includes a substrate, and the transistor layer and a wiring layer stacked on the substrate,

the wiring layer includes a first internal wiring, a second internal wiring, and a first inductor,

the first internal wiring, the second internal wiring, and the first inductor are arranged apart from each other, and

the semiconductor module includes the first internal wiring, a first conductive layer in contact with the upper surface and an upper surface of the first substrate along the third direction, a second conductive layer electrically connected to the first conductive layer, and a heat conductive member electrically connected to the second conductive layer.

17. The semiconductor module according to claim **16**, wherein

each of the outermost surface of the first substrate and the outermost surface of the second substrate includes an R surface.

18. The semiconductor module according to claim **16**, wherein

the memory cube includes a third substrate and a fourth substrate,

the third substrate and the fourth substrate sandwich the plurality of memory chips in the second direction, and

each of the outermost surface of the third substrate and the outermost surface of the fourth substrate includes an R surface.

19. The semiconductor module according to claim **16**, wherein

the first internal wiring includes a first power supply wiring exposed on the upper surface,

the first power supply wiring includes the first internal wiring, and

the second power supply wiring includes the second internal wiring.

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